19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

CASE 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

CASE 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service

EVIDENTIARY HEARING

DATE: February 24, 2020

LOCATION: 19th Floor Hearing Room
Agency Building Three,
Empire State Plaza
Albany, New York

ALJ JAMES A. COSTELLO, DPS

ALJ MAUREEN F. LEARY, DPS

APPEARANCES:

NATIONAL GRID

By: Patric O'Brien, Assistant General Counsel/Director
Jeremy Euto
Carlos Gavilondo
Phillip Decicco
Kenneth Maloney
James Molloy
40 Sylvan Road
Waltham, MA 02451

PUBLIC UTILITY LAW PROJECT
BY: Richard Berkley
Laurie Wheelock
90 South Swan Street, Suite 305
Albany, NY 12210

ENVIRONMENTAL DEFENSE FUND
BY: Martin C. Rothfelder, Esq.
Erin Murphey
407 Greenwood Avenue, #301
Trenton, NJ 08609

CITY OF NEW YORK
BY: Robert Loughney, Esq.
Justin Fung, Esq.
Couch and White LLP
540 Broadway,
Albany, NY 12201

NEW YORK STATE DEPARTMENT OF STATE, UTILITY INTERVENTION UNIT
BY: Jillian D. Kasow, Esq.
Kathleen O'Hare
99 Washington Avenue
Albany, NY 12231

NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE
Brandon Goodrich, Esq.
Raquel Parks, Esq.
3 Empire State Plaza
Albany, NY 12224

Nicholas Forst, Esq.
125 E. Bathpage Road
Plainview, NY 11803

SANE ENERGY PROJECT:
BY: Lee Ziesche
Kim Fraczek
232 E. 11th Street
New York, NY 10003

AARP NEW YORK:
BY: Saul Rigberg, Esq.
93 Fernbank Avenue
Delmar, NY 12054

BOB WYMAN
203 W 85th Street, Apt PH2
New York, NY 10024

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

INDEX OF PROCEEDINGS

WITNESSES;

TIM GRAHAM, ROSS TURRINI, ANNETTE SAXMAN:

Direct Examination by Mr. Decicco 3669

Cross Examination by Ms. Murphy 3843

ELIZABETH ARRANGIO:

Direct Examination by Mr. Goodrich 3860

Cross Examination by Ms. Murphy 3947

DAVE MAIORIELLO, JENNIFER PARK:

Direct Examination by Mr. Goodrich 4006

RANNY CHOW, JONATHAN MERCURIO, MICHAEL PASNOW, SURESH THOMAS:

Direct Examination by Mr. Goodrich 4025

Cross Examination by Mr. Decicco 4125

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

EXHIBIT INDEX

Exhibit 671 3848
Response
Exhibit 672 3947
Confidential version of Exhibit 5SS
Exhibit 673 4164
Fitch Affirms Brooklyn Union and KeySpan Gas East IDRs at Triple B Plus, Outlook Stable," dated 2-20-2020
Exhibit 674 4192
Panels response NG 47, 11-2-2019
Exhibit 675 4195
NG 49 11-22-2019

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

(On the record at 9:03 a.m.)

A.L.J. COSTELLO: We're here on the eighth day of the evidentiary hearings on cases 19-G-0309 and 19-G-0310. Going to start with appearances and we'll start with the companies.

MR. DECICCO: Good morning, Your Honors. For the companies Phillip Decicco, Patric O'Brien, and Kenneth Maloney from the law firm of Cullen and Dykman.

A.L.J. COSTELLO: Department of Public Service staff.

MR. FORST: From the Department of Public Service staff you have Brandon Goodrich, Raquel Parks, and Nicholas Forst.

A.L.J. COSTELLO: Utility Intervention unit.

MS. KASOW: Department of State Utility Intervention Unit we have Jillian Kasow and Katie O'Hare.

A.L.J. COSTELLO: The Public Utility Law Project.

MR. BERKLEY: For the Public Utility Law Project, Richard Berkley, and Laurie Wheelock will be joining us shortly.

A.L.J. COSTELLO: City of New York.

MR. CONWAY: For the City of New York Adam Conway and Justin Fung from the law firm of Couch White.

A.L.J. COSTELLO: Okay. Do we have someone here from the Environmental Defense Fund?

MS. MURPHY: Yes. For Environmental Defense Fund you have Erin Murphy and Marty Rothfelder with Rothfelder Stern on behalf of EDF.

A.L.J. COSTELLO: Okay. Good morning. Is there anyone else making an appearance here this morning who I have not called? Not hearing any. Okay. We'll get started. So we're starting today with the company's gas safety panel.

And before we get started, I just want to ask the witnesses just to remember to state your name before you answer questions, respond to questions. You don’t have to do it every time if -- but, you know, when you start just state your name until the court reporter gets -- gets used to who it is. I'm going to ask the gentleman from the furthest -- seated from -- seated the furthest from me to state your name and business address for the record.

MR. GRAHAM: My name is Tim Graham. My

business address -- I'm sorry. My name is Tim Graham. My business address is 300 Erie Boulevard, Syracuse, New York.

A.L.J. LEARY: Your microphones are not on.

MR. TURRINI: Okay.

MS. SAXMAN: Oh, they're not?

MR. TURRINI: So we'll just talk loud.

A.L.J. LEARY: No. You've just got to go like this and make sure your green light is on.

MR. TURRINI: Oh, we just need to turn them on.

MR. GRAHAM: Oh, we've got to push them on. Okay sorry.

A.L.J. COSTELLO: There we go.

MR. TURRINI: Ross Turrini and my business address is 25 Hub Drive, Melville, New York.

MS. SAXMAN: Annette Saxman. My business address is 300 Erie Boulevard, Syracuse, New York.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

PANEL: I do.

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

TIM GRAHAM; SWORN

ROSS TURRINI; SWORN

ANNETTE SAXMAN; SWORN

A.L.J. COSTELLO: Thank you. You may be seated.

DIRECT EXAMINATION

BY MR. DECICCO:

Q. Good morning, Panel. Phil Decicco for the company. Panel, you have before you a document entitled the direct testimony of the Gas Safety Panel on behalf of the Brooklyn Union Gas Company, or KEDNY, and KeySpan Gas East Corporation, or KEDLI, dated April 2019, consisting of 64 pages, a cover sheet and a table of contents. Is that correct?

A. (Panel) Yes, correct.

Q. You also have before you a document entitled the corrections and updates testimony of the Gas Safety Panel on behalf of KEDNY and KEDLI dated July 3, 2019, consisting of five pages, a cover sheet and a table of contents. Is that correct?

A. (Panel) Yes, that's correct.

Q. Finally, you have before you a document entitled the rebuttal testimony of the Gas

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas Safety Panel on behalf again of both KEDNY and KEDLI dated September 18, 2019, consisting of 64 pages, a cover sheet and a table of contents. Is that correct?

A. (Panel) Yes, that's correct.

Q. Panel, was this testimony prepared by you or under your direction and supervision?

A. (Panel) Yes, it was. That's correct. Yes.

Q. Is the information contained in these testimonies true and correct to the best of your knowledge?

A. (Panel) Yes, it is. Yes.

Q. And if the questions contained in these documents were asked today, would your answers be the same?

A. (Panel) Yes, they would. Yes.

Q. And finally, do you adopt these documents as your sworn testimony in these proceedings?

A. (Panel) Yes, I do. Yes.

MR. DECICCO: Your Honors, I ask that the Panel's prefiled direct Corrections and Updates and Rebuttal Testimony be admitted into the record in

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

these proceedings.

A.L.J. COSTELLO: Okay. The request is granted. And at this point in the record, the court reporter should insert the following files: KEDNY-KEDLI Gas Safety Panel Direct Testimony, KEDNY-KEDLI Gas Safety Panel Corrections and Updates Testimony and KEDNY-KEDLI Gas Safety Panel Rebuttal Testimony.

#

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Gas Safety Panel

Ross W. Turrini
Timothy S. Graham
Annette M. Saxman

Dated: April 2019

Table of Contents

I. Introduction 1
II. Gas Safety and Compliance Programs 11
A. Plastic Fusion Programs 14
B. Inside Service Line Inspections Program 17
C. Enhanced Contractor Safety Inspections Program 20
D. Enhanced Inactive Accounts Program 22
E. I&R Programs 24
F. Enhanced Pipeline Safety Management and Damage Prevention Programs 30
G. New Materials Test Lab 34
H. Buried Vent Lines Program 35
I. Low Pressure Main Valve Installation Program 36
J. Proactive Pre-1974 High Definition Polyethylene ("HDPE") Services Replacement Program 37
K. Single Meter Regulator Inspections Program 38
L. Gas Control SOP Training Program 39
M. KEDLI Meter Pits Program 40
N. Enhanced First Responder and Field Training 41
O. Expanded Residential Methane Detection Program 43
P. Enhanced High Emitter Methane Detection Program 45
Q. Restrictions for Elevated Gas Infrastructure Program 46
III. Gas Safety and Compliance Measures 47
A. LPP Replacement 49
B. Leak Backlog 53
C. Emergency Response 58
D. Damage Prevention 59
E. Gas Regulations Performance Measure 61

I. Introduction

Q. Please introduce the members of the Gas Safety Panel.

A. The Panel members are Ross W. Turrini, Timothy S. Graham, and Annette M. Saxman.

Q. Mr. Turrini, please state your name and business address.

A. My name is Ross W. Turrini. My business address is 25 Hub Drive, Melville, New York 11747.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company"), a subsidiary of National Grid USA ("National Grid"), as the Group and US Chief Gas Engineer. I oversee approximately 800 employees and more than $9.5 billion of gas infrastructure assets serving over 3.6 million customers in New York, Massachusetts, and Rhode Island.

National Grid owns and operates three gas distribution companies in New York that provide retail gas service to more than 2.4 million customers in three service territories: Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk") serves areas of eastern and central New York; The Brooklyn Union Gas

Company d/b/a National Grid NY ("KEDNY") serves Brooklyn, Staten Island, and parts of Queens in New York City; and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (KEDNY and KEDLI are collectively referred to as the "Companies") serves customers on Long Island and the Rockaway Peninsula in Queens. I am responsible for all aspects of the performance of National Grid's New York gas networks, including emergency response, gas engineering, construction activities, operation and maintenance of gas transmission and distribution facilities, gas network strategy issues, including the gas capital investment strategy, and gas pipeline safety management.

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Science in Civil Engineering from the United States Military Academy at West Point in 1985. I have worked for National Grid and its predecessor companies (the Long Island Lighting Company and KeySpan Corporation) for 25 years in various roles in engineering, operations, and procurement. Prior to joining National Grid, I spent five years as an Officer in the United States Army Corps of Engineers and three years in engineering and construction roles at Brown & Root Services Corporation, an international engineering, procurement, and construction company.

Q. Have you previously testified before the New York State Public Service Commission ("Commission")?

A. Yes. I testified before the Commission on behalf of the Companies in Cases 16-G-0058 and 16-G-0059 (the "2016 KEDNY and KEDLI Rate Cases"). I also testified most recently on behalf of Niagara Mohawk in Case 17-G-0239 (the "2017 NMPC Rate Case").

Q. Mr. Graham, please state your full name and business address.

A. My name is Timothy S. Graham. My business address is 300 Erie Boulevard West, Syracuse, New York 13212.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Vice President for Gas Field Operations. I oversee approximately 2,300 employees and am responsible for the maintenance of KEDNY, KEDLI, and Niagara Mohawk's gas distribution systems. This includes gas metering, construction, and field activities to carry out gas mandated programs, leak response, leak repair, instrumentation and regulation, damage prevention, and inspection programs.

Q. Please describe your educational background and experience.

A. I received a Bachelor of Engineering in Electrical Engineering from the State University of New York at Buffalo. I also have a Master's in Business Administration from Lemoyne College in Syracuse, New York. I have worked for National Grid and its predecessor company (Niagara Mohawk) for 32 years in various roles in engineering and operations.

Q. Have you previously testified before the Commission?

A. No.

Q. Ms. Saxman, please state your full name and business address.

A. My name is Annette M. Saxman. My business address is 300 Erie Boulevard West, Syracuse, New York 13212.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as the Director of Gas Pipeline Safety and Compliance. I am responsible for issues related to the safe operation of National Grid's New York gas systems, including compliance with federal and state pipeline safety regulations. My responsibilities include working as the regulatory liaison on gas safety and compliance issues.

Q. Please describe your educational background and experience.

A. I received a Bachelor of Arts in Physics from the State University of New York at Potsdam in 1988 and a Bachelor of Science in Civil and Mechanical Engineering from Clarkson University in 1988. I have worked for National Grid and its predecessor company (Niagara Mohawk) for 27 years in various technical positions, including gas engineering, gas field operation, customer meter services, resource planning, and process and performance excellence.

Q. Have you previously testified before the Commission?

A. Yes. I submitted pre-filed testimony in the 2016 KEDNY and KEDLI Rate Cases.

Q. What is the purpose of the Panel's testimony?

A. The Panel's testimony addresses programs and performance metrics related to the safe operation of the Companies' gas systems during the 12-month period ending March 31, 2021 ("Rate Year"), and the three subsequent 12-month periods ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (Data Year 1, Data Year 2, and Data Year 3 are collectively referred to as the "Data Years"). The Panel also presents a forecast for the 15-month period ("Bridge Period") between December 31, 2018, which is the end of the Companies' Historic Test Year, and the beginning of the Rate Year.

Specifically, the Panel's testimony addresses the Companies' proposals for (i) enhanced and incremental gas safety and compliance programs, including programs providing additional training, safety inspections, and contractor oversight; (ii) gas safety performance metrics; and (iii) an accelerated procedure for addressing inactive accounts.

Q. Before discussing the specific programs presented in this rate filing, please discuss the Companies' commitment to safety and compliance.

A. The Companies strive for continuous improvement in safety performance by identifying, assessing, and managing risks on the gas system – and making prudent and necessary investments in gas infrastructure to address these risks. As discussed by KEDNY and KEDLI's respective Gas Infrastructure and Operations Panels ("GIOP"), KEDNY will invest more than $3.8 billion in gas infrastructure in the Rate Year and Data Years to enhance gas pipeline safety, meet increasing regulatory requirements, and maintain and improve the reliability of its gas distribution system, including the retirement of more than 300 miles of leak prone pipe ("LPP") during that period. Similarly, KEDLI is investing more than $2.3 billion in its gas network to support safe operations and enhance system performance, including the retirement of more than 600 miles of LPP during the Rate Year and Data Years. The Companies are also investing in pipeline and transmission station integrity management programs, automatic shut off valves, and

other programs to enhance public safety. To ensure sufficient oversight and quality control over this expanding portfolio of capital investments, the Companies are proposing to implement an enhanced inspection program that will ultimately deploy dedicated field inspectors to oversee and inspect the work of every construction crew.

Importantly, investments in the gas system must be supported by technology and processes that enable the Companies to manage their assets and workforce and meet heightened safety and regulatory expectations. Accordingly, as discussed in the testimony of Companies' Witness Christopher J. Connolly, National Grid has undertaken a comprehensive effort to evaluate the technology and business processes that support its gas businesses (referred to as the Gas Business Enablement ("GBE") Program). Among the core investments of the GBE Program are new asset and work management, scheduling, geographic information system, and field mobility solutions designed to improve the Companies' gas safety and operational performance.

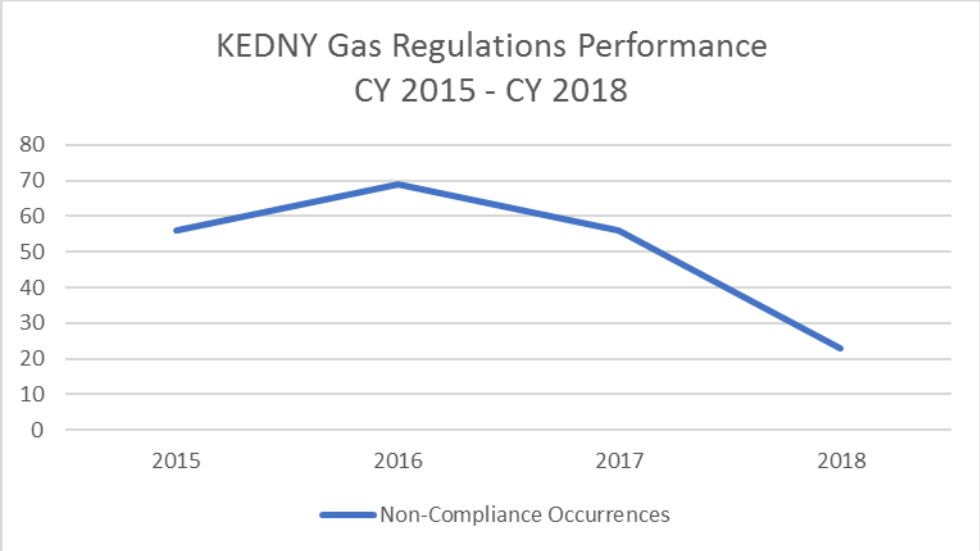
Q. Please describe the Companies' safety and compliance performance.

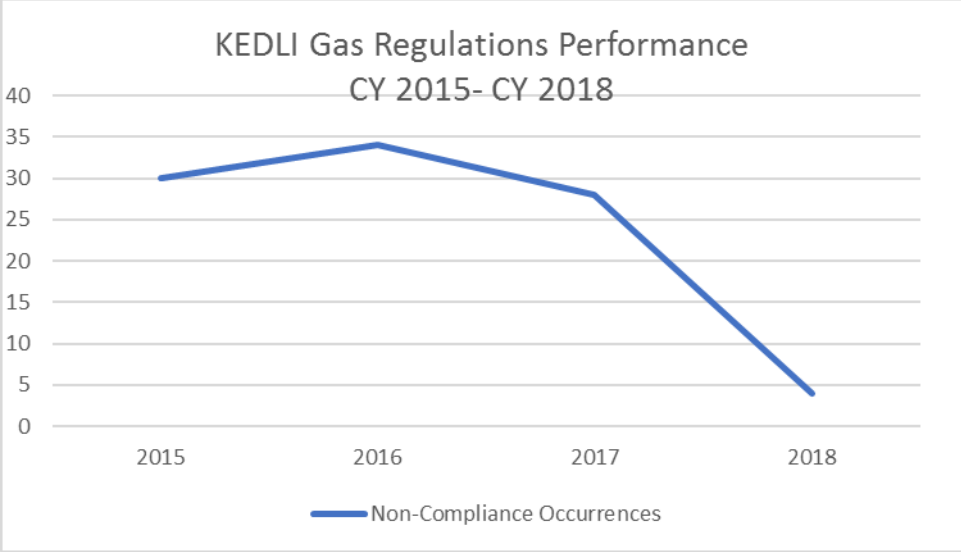
Despite the need for technology investments and increasing work plan requirements, KEDNY and KEDLI have historically performed well and met all safety performance targets. Each year, the Department of Public Service Staff

("Staff") issues an annual report on local distribution companies' ("LDCs") gas reliability performance in relation to established metrics. Highlights of KEDNY and KEDLI's performance, as reported by Staff's reports through 2017 include: (i) an overall reduction in the damages due to excavator error since 2015; (ii) a reduction in emergency response times to gas leak and odor calls within 30 minutes every year since 2015; and (iii) a reduction in the backlog of total leaks every year since 2013. The Staff report for 2018 will not be released until later in 2019; however, the Companies recently reported their performance against safety metrics to the Commission for calendar year ("CY") 2018, as part of the annual gas safety performance metric filing, and have again outperformed every safety metric target (*e.g.*, LPP Removal, Leak Management, Damage Prevention, Emergency Response).

Each year, Staff audits the Companies' compliance with gas safety regulations in the field, as well as the Companies' records and documentation. Since 2016, KEDNY and KEDLI have shown year-over-year improvements with respect to performance against gas safety compliance requirements, as shown in Table 1.

Table 1: KEDNY & KEDLI Compliance Trends (Field and Records Audits)





While KEDNY and KEDLI have demonstrated significant progress in this area, the Companies acknowledge that further improvement is needed and, therefore, are advancing various compliance programs in these rate filings, including a

comprehensive pipeline safety management system, enhanced training and inspection programs, and enhanced compliance monitoring and documentation capabilities through the GBE program.

Q. What safety and compliance improvements have the Companies made over the last several years?

Following an independent assessment ("IA") of National Grid's safety and compliance processes by an external pipeline safety consultant several years ago, the Companies have worked toward systematically improving their safety and compliance performance. Over the past several years, the Companies have made significant progress in implementing the recommendations of the IA, including additional field inspections, damage prevention enhancements, investments in the GBE Program, and efforts to enhance training and simplify procedures and practices in the field. As set forth in more detail below, the Companies are now in the process of carrying these efforts forward through the implementation of the American Petroleum Institute's ("API") recommended pipeline safety management standards (Recommended Practice 1173 ("RP 1173")), which supports a comprehensive framework for identifying and managing risks on the Companies' gas systems, and is aligned with the recommendations of the IA. Additionally, National Grid's recent reorganization of its Gas Business Unit provides organizational support for further improvements in the areas of gas safety and

compliance by providing clear, end-to-end accountability for gas safety performance, as well as enabling the implementation of best practices across the organization. The Companies are confident that implementation of API RP 1173, along with the organizational enhancements and new programs and projects set forth below, will advance KEDNY and KEDLI's efforts to enhance compliance performance going forward.

II. Gas Safety and Compliance Programs

Q. Please describe the Companies' proposed gas safety and compliance programs.

A. As the gas industry has intensified the focus on safe operations over the last decade, KEDNY and KEDLI have increased their efforts to assess system risks and deliver programs designed to enhance safety and increase the public's awareness of gas safety issues. For example, following the 2016 KEDNY and KEDLI Rate Cases, the Companies implemented enhanced quality assurance and quality control ("QA/QC"), damage prevention, and first responder training programs. The Companies intend to continue to advance such efforts during the Rate Year and Data Years. In addition, the Companies remain committed to enhancing and expanding upon its public awareness and public engagement efforts with respect to natural gas safety, such as through its residential methane detector program that will

distribute roughly 10,000 residential methane detectors ("RMDs") and gas safety brochures to customers between CY 2017 and the end of 2019.

Moreover, in response to gas safety incidents in the industry, including the recent incident in Merrimack Valley, Massachusetts, the Companies recognize that a number of their pipeline safety programs must evolve to address the lessons learned from these events. Indeed, additional gas safety investments are necessary to both comply with emerging safety requirements and implement best practices resulting from these incidents. Specifically, the Companies propose the following gas safety programs and enhancements:

- Plastic Fusion Programs, including a QA/QC Re-Digs Program (mandated) and a supplemental In-Process Inspections Program
- Inside Service Line Inspections Program (mandated)
- Enhanced Contractor Safety Inspection Program
- Enhanced Inactive Accounts Program
- Instrumentation and Regulation ("I&R") programs that address over-pressurization and a new I&R testing and training lab
- Enhanced Pipeline Safety Management and Damage Prevention Programs
- New Materials Test Lab
- Buried Vent Lines Program
- Low Pressure Main Valve Installation Program

- Proactive Replacement of Pre-1974 High Definition Polyethylene ("HDPE") Services Program
- Single Meter Regulator Inspections Program
- KEDLI Meter Pits Program
- Gas Control Standard Operator Procedure ("SOP") Training Program
- Enhanced First Responder and Field Training
- Expanded Residential Methane Detection Program
- Enhanced High Emitter Methane Detection Program
- Restrictions for Elevated Gas Infrastructure Program

Each of these programs is described in detail below. The cost information for these programs is set forth in Exhibit __ (GIOP-1) and Exhibit __ (GIOP-5) for capital investments and KEDLI's Exhibit __ (GIOP-7) and KEDNY's Exhibit __ (GIOP-8) for incremental operating and maintenance ("O&M") expenditures. KEDLI's Exhibit __ (GIOP-8) and KEDNY's Exhibit __ (GIOP-9) sets forth the incremental full-time equivalents ("FTEs") required to support these programs, and the costs for the incremental FTEs are provided in Exhibit___(RRP-3), Schedule 27 and supported by the Revenue Requirements Panel ("RRP").

Q. Are the Companies proposing to fund any of its incremental gas safety programs using prior period negative revenue adjustments ("NRAs")?

A. Yes. The Joint Proposal in the 2016 KEDNY and KEDLI Rate Cases provides that NRAs incurred for gas safety compliance violation will be deferred and may be used to fund future gas safety programs. KEDNY and KEDLI currently carry regulatory liabilities of approximately $21.2 million and $13.9 million, respectively, for NRAs relating to prior performance periods.

The Companies support using NRAs to fund the following safety programs: (i) the Residential Methane Detector Program and (ii) the incremental I&R Control Line Survey and GPS Mapping Program. These programs are described in more detail below. While the costs for these two programs are currently included in the revenue requirements, the Companies are agreeable to funding them through NRAs to help mitigate the rate request in these proceedings. Additionally, the Companies would consider funding additional safety programs through NRAs, such as a program to provide handheld methane detection equipment to first responders (that is not currently included in the Companies' proposals below).

A. <u>Plastic Fusion Programs</u>

Q. Please describe the Plastic Fusion Programs.

A. There are two plastic fusion safety programs in the Companies' capital investment plans, the Plastic Fusions QA/QC Program and a new Plastic Fusion In-Process Inspections Program. Both programs are in response to increased focus on the risk of plastic fusion failures following the 2014 East Harlem, New York incident and the Commission's new, expansive requirements regarding plastic fusion compliance.

The QA/QC Program is mandated by the Commission's "Order Adopting Further Improvements in Plastic Fusion Practices on Natural Gas Systems" issued on May 18, 2018 in Case 14-G-0212 ("Plastic Fusion Order"). Under this program, the Companies conduct re-dig inspections of a statistically significant number of plastic fusions on completed construction jobs within 30 days of installation. The Plastic Fusion QA/QC Program requires capital investments of approximately $3.3 million in the Rate Year, $3.4 million in Data Year 1, $3.5 million Data Year 2, and $3.5 million in Data Year 3 for KEDNY. For KEDLI, the program requires capital investments of $1 million in the Rate Year and each of the Data Years. The costs for the Plastic Fusion Program are shown separately in Exhibit __ (GIOP-1) for transparency, but will ultimately be accounted for in the total unit costs of the various main and service replacement programs.

The Plastic Fusion In-Process Inspection Program is a supplemental program to enhance the safety and reliability of plastic fusion work. Under this program, construction crews performing plastic fusions will send photographs of the fusion to a contractor, ControlPoint, to perform a remote, real-time inspection. The inspection includes evaluation of stab depth, surface clearing, scraping, alignment, fusion time, and temperature compensation. The findings from the inspection are communicated to the construction crews prior to completion of the project, allowing for any corrections to be made in real-time and before a project is backfilled. A National Grid affiliate company in the United Kingdom recently adopted an in-process inspection program using the same contractor, and the program has proved effective in detecting potential issues and enabling corrective actions prior to work being completed. In-process inspections are not required by the Plastic Fusion Order but have been shown to be a cost-effective means to reduce risk related to plastic fusions. Because of the significant benefits of this program, the Companies are beginning to conduct in-process inspections during fiscal year ("FY") 2020. The capital investment plan includes approximately $0.3 million for KEDNY in each of the Rate Year and Data Years and $0.6 million for KEDLI in each of the Rate Year and Data Years (adjusted for inflation). In addition, there are incremental non-labor O&M costs of $0.102 million for KEDNY and $0.203 million for KEDLI in the Rate Year, inflated through Data Year 3. The costs for the In-process Inspection Program are shown separately in Exhibit __ (GIOP-1) for

transparency, but will ultimately be accounted for in the total unit costs of the various main and service replacement programs.

Q. Do any of the Companies' other gas safety proposals help to lower risk related to plastic fusions?

A. Yes. The Companies' proposals to enhance their general inspection, QA/QC, and Damage Prevention programs will enable additional real-time inspections of plastic fusions and other construction work. Additionally, the Companies' proposal for a new Materials Test Lab supports enhanced incident analysis of failed fusions and improvements to plastic fusion processes. These programs are described in detail below.

B. <u>Inside Service Line Inspections Program</u>

Q. Please describe the Inside Service Line Inspection Program and what is required to implement the new mandated inside service line inspection intervals.

A. In 2015, the Commission issued an Order that adopted a new definition of "service line" in New York (Case 14-G-0357). Under the former definition in New York's pipeline safety regulations (16 NYCRR section 255.3(29)), the service line extended "to the first accessible fitting" inside the customer's building where a meter is located. Under the new definition, the service line now extends "to the

outlet of the customer meter or at the connection to a customer's piping, whichever is further downstream where a meter is located within the building." While the Commission adopted the new service line definition in 2015, it stayed implementation pending development of a plan to address the new leak survey and corrosion inspection requirements for inside facilities. In April 2017, the Commission issued its "Order Establishing Statewide Inspection Schedules and Procedural Requirements" in Case 15-G-0244 ("Inside Service Inspection Order") that established mandated leak survey and corrosion inspection intervals for inside services. Citing the 2015 explosion on 2nd Avenue in Manhattan, which involved unauthorized tampering with utility gas facilities on the inside of a building, the Inside Service Line Inspection Order set an aggressive schedule for utilities to complete baseline leak surveys and inspections, and shorter continuous inspection intervals thereafter. For example, for inside services located in business districts, leak surveys were to be completed by the end of CY 2018 and re-inspections are required to be completed once every year thereafter. For non-business districts, baseline leak surveys are to be completed by April of 2020, and re-inspections must occur every three years thereafter. Baseline corrosion inspections, regardless of location, also must be completed by April of 2020, and must be repeated on three-year intervals. The Companies began to ramp-up implementation of the Inside Service Line Inspection Program during CY 2017 and CY 2018. Additional work is required to timely complete baseline and ongoing inspections.

Q. What are the projected incremental costs to implement the Inside Service Line Inspection Program for KEDNY and KEDLI in the Rate Year and Data Years?

A. More frequent leak surveys and corrosion inspections are required in the Rate Year and Data Years compared to the Historic Test Year to meet the compliance requirements in the Inside Service Line Inspection Order. This requires incremental O&M expense for additional inspections and documentation requirements, incremental FTEs (field inspectors) to manage inspection and quality control of contractor work, and a modest reactive capital program to address service replacements required when the surveys and inspections reveal substandard conditions. The impact of these changes is greater in KEDNY's service territory because it is more urban with more apartment buildings and other multi-story buildings with meters located inside. Based on the schedule of work to be completed each year, KEDNY requires non-labor O&M expense of approximately $5.900 million in the Rate Year, inflated through Data Year 3. In addition, 3.5 incremental field inspector FTEs are needed in each of the Rate Year and Data Years. KEDLI requires incremental non-labor O&M of approximately $0.135 million in the Rate Year, inflated through Data Year 3. In addition, 1.5 incremental FTE field inspectors are needed in each of the Rate Year and Data Years.

The capital forecast is based on the level of repairs required to date to address issues identified during inside service inspections, extrapolated to the level of inspections that will occur during the Rate Year and Data Year. The Companies' capital forecasts are shown on KEDNY and KEDLI's respective Exhibit __ (GIOP-1) as "Atmospheric Corrosion Inside Inspections." KEDNY and KEDLI's respective capital forecasts are approximately $0.7 million in the Rate Year, and $0.1 million in each of the Data Years, adjusted for inflation.

C. <u>Enhanced Contractor Safety Inspections Program</u>

Q. Please describe the Companies' Enhanced Contractor Safety Inspections Program.

A. The Companies' existing Contractor Safety Inspector Program deploys safety and compliance inspectors to perform on-site inspections and oversight of contractor crews, including field verification of operator qualifications, jobsite and safe work practices, work methods and construction standards/procedure compliance, and work scope documentation. Currently, the program's scope does not cover every contractor crew, and inspectors are assigned to cover multiple contractor crews by rotating through work schedules. The Companies propose to enhance the Contractor Safety Inspections Program by assigning one inspector to every contractor crew. To achieve this goal, KEDNY requires a total of 125 incremental FTEs (110 field inspectors and 15 supervisors), and KEDLI requires a total of 74

FTEs (65 inspectors and 9 supervisors). To facilitate the hiring of such a significant number of FTEs, the Companies propose a phased approach to ramp-up the program over three years.

Q. What are the forecast costs for the Enhanced Contractor Safety Inspection Program?

A. The Companies' hiring plans for this program are shown in the data sheet provided in Exhibit __(GIOP-5). KEDNY's program requires capital investments of approximately $5.4 million in the Rate Year, $16.4 million in Data Year 1, $27.8 million in Data Year 2, and $28.2 million in Data Year 3. KEDLI's program requires capital investments of approximately $3.6 million in the Rate Year, $11.1 million in Data Year 1, $18.5 in Data Year 2, and $18.8 million in Data Year 3. Both programs also require incremental O&M expense to cover vehicle costs related to the time the FTEs will charge to expense (15 percent KEDNY, 10 percent KEDLI), small tools and supplies, and facilities costs related to the program. KEDNY's O&M costs are approximately $0.228 million in the Rate Year, $0.660 million in Data Year 1, $1.075 million in Data Year 2, and $1.025 million in Data Year 3. KEDLI's O&M costs are approximately $0.123 million in the Rate Year, $0.362 million in Data Year 1, $0.589 million in Data Year 2, and $0.575 million in Data Year 3. The costs for the Enhanced Contractor Safety Inspection Program are shown separately in Exhibit __ (GIOP-1) for transparency, but will ultimately be

accounted for in the total unit costs of the various main and service replacement programs.

D. <u>Enhanced Inactive Accounts Program</u>

Q. Please describe the Companies' improvements to the inactive accounts process over the past several years.

A. The Companies have implemented inactive account process improvements that considerably mitigate the risks of inactive accounts (*i.e.*, accounts without an active customer of record). Over the last several years, the Companies have implemented various process improvements for addressing inactive accounts, including a revised end-to-end process that provides a more structured approach to accessing unlocked meters. The Companies have also enhanced their call center procedures, scripts, and training; dedicated additional resources to support the inactive account process; made investments in the Companies' web site and customer and work management systems to support the inactive account process; established a legal replevin process to secure access to gas facilities through court orders; and improved customer communications regarding inactive accounts (*e.g.*, occupant/owner letters, post cards, door stickers). Significantly, the Companies perform a site analysis on every visit to determine whether conditions, such as trespassing at a vacant premises or other observable unsafe condition, warrant the Companies immediately securing the service line. In such instances, the service is cut within 24 hours.

As a result of these process improvements, over the past three years, KEDNY has resolved approximately 47 percent of inactive accounts within the first 15 days of inactivity, and KEDLI has resolved approximately 89 percent of accounts within the first 15 days. Excluding controlled accounts that are referred to the legal replevin process to gain access, between day 45 and day 60, approximately five percent of KEDNY's accounts and 1.6 percent of KEDLI's accounts remain inactive.

Q. Are the Companies proposing to modify their inactive accounts processes?

A. Yes. The Companies' current process provides that all inactive accounts will be addressed (*e.g.*, locked or cut in the street) no later than 60 days after the account becomes inactive. In response to Staff's recommendations for further process improvements, the Companies propose to implement a 30-day inactive account process. To mitigate the significant costs of this process change, the Companies intend to take a phased approach to shorten the end-to-end inactive account process to 45 days by the end of the Rate Year, and to 30 days by the end of Data Year 2.

Implementation of this process change requires additional resources in field operations, CMS, Call Center, Legal, and back office, including additional FTEs. In KEDNY's service territory, significantly more resources are required due to

population density and a higher percentage of transitory customers in rental properties. For KEDNY, the program requires incremental O&M expenses of approximately $17.083 million in each of Data Years 1 through 3, plus a total of 133 FTEs that will be hired between FY 2020 and Data Year 1. For KEDLI, the program requires incremental O&M expenses of approximately $0.412 million in each of Data Years 1 through 3, plus a total of three FTEs that will be hired in Data Year 1.

E. I&R Programs

Q. Please describe I&R and the need for the Companies' I&R safety proposals.

A. I&R involves the safe and reliable operation of critical gas infrastructure used to maintain appropriate gas system pressures, including pressure regulating stations, heaters, gas quality and odorization equipment, pressure control and pressure monitoring, and system automation equipment. The recent incident in Merrimack Valley, Massachusetts involved an over pressure event on the utility's gas distribution system. Following that incident, at the direction of Staff, New York gas utilities immediately reviewed their pressure regulation and overpressure protection assets and procedures to identify system risks, identify mitigation actions, and consider best practices, including those recommended by Staff. The Companies' review identified areas where best practices were already in place, and recommended improvements and enhancements to improve the safety and

reliability of I&R assets. As a result, the Companies are proposing a suite of I&R safety programs designed to improve procedures, information availability and standardization, knowledge transfer, employee competencies, public safety, and over pressure risk reduction. Specifically, the Companies propose two capital investment programs – the Distribution Station Over Pressure Protection Program and the I&R Training and Test Lab – and three O&M programs – the I&R Control Line Survey and GPS Mapping Program, the I&R Site Specific Procedures Program, and the I&R Field Trainers Initiative.

Q. Please describe the Distribution Station Over Pressure Protection Program.

A. This program will directly address over pressurization risk by installing override pilots on all low-pressure stations. If there is an over pressure condition, the override pilot operates automatically to limit the amount of outlet pressure from the station and prevent the pipeline from over pressurizing until the Companies are able to respond to the situation. The Companies are beginning to install override pilots in FY 2020, and the program will continue through the Rate Year and Data Years to address all stations. The capital investment required for this program is approximately $0.9 million in the Rate Year and $0.3 million in each of the Data Years for KEDNY, and approximately $0.5 million in the Rate Year and $0.3 million in each of Data Years 1, 2, and 3 for KEDLI.

Q. Please describe the proposed capital investment in an I&R Training and Test Lab.

A. I&R employees and supervisors are responsible for operating and maintaining complex gas pressure reducing facilities. In the absence of a dedicated training and test lab, I&R employees learn the finer details of equipment maintenance and operation on fully functioning in-service stations during on-the-job training under the supervision of senior experienced employees.

The I&R Training and Test Labs proposed by the Companies are functioning regulator stations that are not connected to the Companies' live gas distribution systems and operate on compressed air instead of gas. The training and test labs can be set up to simulate any type of operating condition, including failure modes, and can provide an opportunity for repeat training on simulated seasonal and weather conditions. In this way, employees can learn to recognize and react to abnormal operating conditions in a sandbox environment without risking damage to the Companies' facilities. The proposed training and test labs also will allow employees to learn in a controlled environment while crews working on live facilities can fully focus on their work rather than on supervising trainees. Moreover, these stations facilitate employee competency assessments in simulated operating conditions and provide for consistent training.

The Companies' assessment of downstate training needs determined that a facility in each territory is required due to the number of downstate employees and variation of station conditions and training needs between the two Companies. There is currently no I&R training and test lab in KEDNY's service territory. KEDLI currently has an I&R training and test lab, but the lab is not sized appropriately for its training needs, and the components that are there need to be upgraded to ensure the facilities continue to be fit for purpose. Therefore, the Companies propose capital investments to (i) construct a new I&R training and test lab in KEDNY's service territory and (ii) expand KEDLI's I&R training and test lab and refurbish existing equipment. KEDNY's program requires a capital investment of $0.8 million in the Rate Year and $1 million in Data Year 1. KEDLI's program requires a capital investment of $0.8 million in the Rate Year and $0.4 million in Data Year 1. The I&R Training and Test Lab will be managed by the Companies' I&R department as a direct capital investment program.

The Companies each require a small yearly O&M expense program, approximately $0.075 million in the Rate Year and Data Year 1, increasing to $0.085 million in Data Year 2 and Data Year 3. The program cost is to address wear and tear of parts and maintenance of the associated air compressor equipment.

Q. Please describe the I&R Control Line Survey and GPS Mapping Program.

A. Control lines are specific piping systems at pressure regulation systems that allow the pressure regulating equipment to sense the amount of gas pressure in the downstream distribution system. Damage to control lines can cause connected pressure regulators to malfunction that could, under some circumstances, result in an over pressure condition on the downstream distribution system. The Companies' risk assessment determined that legacy documentation of control line facilities could be enhanced using updated technologies to mitigate the risk of third-party damages. Indeed, a similar survey and mapping program already underway in Niagara Mohawk's service territory revealed control line facilities that have shifted over time due to ground conditions, such that documents no longer reflect correct locations, confirming the need for this program. Thus, the Companies propose a program to survey and map the I&R control lines, using GPS technology and licensed land surveyors, to obtain exact locations and GPS coordinates. The costs of surveying, excavating (where necessary to locate facilities), and mapping regulator station schematics are based on Niagara Mohawk's surveying project. The program requires incremental non-labor O&M in the amount of approximately $1.496 million in the Rate Year for KEDNY and $0.500 million in the Rate Year for KEDLI. The program will be completed by the end of the Rate Year; thus, there is no spending required in the Data Years. As discussed above, while the costs for this program are included in the revenue requirements, the Companies would be

willing to use the NRAs incurred for gas compliance violations to fund these programs.

Q. Please describe the I&R Site Specific Procedures Program.

A. I&R operates about 550 pressure reducing stations requiring periodic inspection, maintenance, and repair in KEDNY and KEDLI's service territories. The procedures currently utilized by I&R technicians require updating. Specifically, many of the procedures lack step-by-step instructions and station-specific detail about individual pieces of equipment and individual station configurations. The Site Specific Procedure Program will develop new and update existing procedures that are specific to tasks, equipment, and managing emergencies at each individual gas regulator station. Updated site specific procedures will enhance operations safety, support mutual aid, create troubleshooting guides, improve training, and provide for the retention of institutional technical knowledge when employee turn-over occurs.

This initiative will require two incremental I&R engineers (one FTE for each company) and one analyst (one-half of an FTE for each company) in each of the Rate Year and Data Years to build, document, maintain, and update the procedure library.

Q. Please describe the I&R Field Trainer Initiative.

A. The I&R Field Trainer initiative requires two FTEs (one each for KEDNY and KEDLI) that will work in the field with I&R Mechanics and Technicians providing technical assistance and training while performing day to day activities. The Field Trainers will possess significant I&R station operations experience and knowledge and will provide on the spot training and guidance. Field Trainers will also be involved in using the I&R training and test labs and will also serve as subject matter experts during classroom training sessions with National Grid Academy instructors.

F. <u>Enhanced Pipeline Safety Management and Damage Prevention Programs</u>

Q. How do the Companies plan to improve overall gas pipeline safety management and damage prevention?

A. A cornerstone of the Companies' long-term plan to effectuate gas pipeline safety improvements is the Companies' commitment to implement the API RP 1173, which is a holistic approach to pipeline safety that applies key process safety principles to existing risk control programs. Currently, the Companies' QA/QC Programs conduct field inspections and assessments of traditional mains and service work performed by in-house and contractor crews. Inspections can occur on-site during construction activities or occur through a "re-dig" program whereby newly constructed underground facilities are exposed and inspected to evaluate workmanship and compliance. This program is distinct from the Contractor Safety

Inspections because it addresses in-house crews as well as contractors. QA/QC is an integral component in assessing compliance with safety related procedures and work methods, pipeline safety, and regulatory requirements. Adoption of RP 1173 involves conducting more QA/QC inspections, such as more on-site inspections of plastic fusions, and expanding the scope of the Companies' existing QA/QC programs for gas construction activities across additional programs executed by field operations and customer meter services, including leak management, corrosion, and damage prevention.

The Companies propose to add 15.5 Pipeline Safety Management FTEs (10.4 dedicated to KEDNY and 5.1 dedicated to KEDLI) to enable the Companies to execute this important expansion of QA/QC.

Q. Are the Companies making additional improvements to its Damage Prevention Program?

A. Yes. To complement RP 1173 implementation, the Companies are also expanding their existing Damage Prevention Program. Damage to gas facilities from third parties digging near pipelines is a major cause of gas accidents and a contributing factor to some of the worst incidents in the history of the natural gas industry. In addition to the detrimental impacts on safety and reliability, utilities spend millions of dollars every year repairing damage caused by these incidents. Over the last

several years, the Companies have implemented improvements to their Damage Prevention programs, including:

- Implementing a ticket risk assessment ("TRA") system that utilizes an algorithm to evaluate and prioritize one call tickets based on the probability of damage to the Companies' underground facilities and enables the Companies to respond to the tickets with the highest risk of a damage occurring;
- creating a Damage Prevention Advisor Program whereby the Companies' Damage Prevention Advisors meet with and educate municipal personnel, excavators, and contractors on damage prevention awareness and other pipeline safety issues; and
- expanded use of Damage Prevention Advisors to perform regular damage prevention patrols of the Companies' entire transmission system pursuant to code requirements and proactively implementing weekly patrols of transmission lines in proximity to populated areas and daily weekday patrols of the most critical transmission lines.

In line with the Companies' wholistic approach to gas pipeline safety improvement, the Companies intend to continue to focus on damage prevention. The volume of one-call tickets has increased annually and is expected to continue increasing in the Rate Year and Data Year due to the success of the NY 811 program outreach and

education and increasing construction and development. As one-call tickets increase, the Companies need to increase the number of Damage Prevention Advisors to enable the Companies to visit at least 15 percent of one call tickets daily. At current staffing levels, KEDNY's Damage Prevention Advisors can visit only the top five percent of riskiest tickets consistently, and KEDLI's Damage Prevention Advisors can visit only the top ten percent of the riskiest tickets consistently. The Companies propose the addition of two Damage Prevention supervisors (1 FTE dedicated to KEDNY and 1 FTE dedicated to KEDLI) in each of the Rate Year and Data Years. Additional contractor resources for expanding the scope of the Damage Prevention Advisor Program are needed and require incremental non-labor O&M expense in the amount of $1.404 million in the Rate Year, $1.542 million in Data Year 1, $1.798 million in Data Year 2, and $1.948 million in Data Year 3 for KEDNY; and $0.779 million in the Rate Year, $0.894 million in Data Year 1, $1.115 million in Data Year 2, and $1.241 million in Data Year 3 for KEDLI.

Additionally, the expected increase in one-call tickets requires additional contractor resources to perform mark outs and operate the TRA system. To manage the expected increased volume of work, KEDNY requires incremental non-labor O&M expense in the amount of approximately $0.980 million in the Rate Year, $1.233 million in Data Year 1, $1.561 million in Data Year 2, and $1.986 million in Data

Year 3; and KEDLI requires approximately $0.225 million in the Rate Year, $0.325 million in Data Year 1, $0.443 million in Data Year 2, and $0.584 million in Data Year 3.

Q. Does the expected increase in damage prevention mark out tickets have any impact on field operations?

A. Yes. Mark outs that contractor crews are unable to complete due to field conditions, such as lack of a tracer wire or discrepancies between field conditions and the Companies' records, are referred to field operations for completion. These mark outs are known as "turn backs." As requests for mark outs increase, so do the number of mark out turn backs referred to field operations. Because mark outs are anticipated to increase in KEDLI's service territory during the Rate Year and Data Years, KEDLI forecasts incremental non-labor O&M expense in the amount of approximately $0.442 million in the Rate Year, $0.734 million in Data Year 1, $1.083 million in Data Year 2, and $1.501 million in Data Year 3 for field operations to address additional turn backs.

G. <u>New Materials Test Lab</u>

Q. Please describe the need for and safety benefits of the proposed new Materials Test Lab.

A. As stated in KEDLI's GIOP testimony, KEDLI's facility plans include a new Materials and Testing Lab to be located in KEDLI's service territory for the benefit of both KEDNY and KEDLI. The current materials test lab located in Hicksville, New York, is undersized, and under-equipped for the Companies' current and future needs. Generally, the Materials Test Lab provides support for gas pipeline safety through testing of new and existing equipment, and to support incident analysis of failed equipment. An increase in capacity and capability is required to: (i) address increasing safety inspections and requirements, including the testing of plastic fusion joints; and (ii) enable testing of larger diameter plastic pipe and support for the Companies' cured in place lining and CISBOT large diameter programs. The existing lab has no capability to accommodate large diameter pipe and is not sized sufficiently to add this capability. To support this program, the Companies propose adding a new position, as shown on Exhibit___(GIOP-9), with 50 percent of time charged to O&M. The costs are shown in Exhibit___(RRP-3), Schedule 27 and supported by the RRP.

H. <u>Buried Vent Lines Program</u>

Q. Please explain the Buried Vent Line Program.

A. The Companies each estimate that approximately five percent of their respective high-pressure services with inside meter sets have regulator vent lines that are buried underground. Buried vent lines are more susceptible to corrosion and

damage because the vent pipes are not coated and do not have cathodic protection. Corroded and/or damaged vent lines pose an increased risk for gas migration into a building or regulator over-pressurization caused by water intrusion and/or freeze-ups caused by extreme weather. In February 2016, KEDNY experienced an incident on its system that involved a buried vent line that was found to be corroded and could have contributed to an over pressure condition due to freezing water in the regulator. To mitigate the risk of buried vent lines, the Companies propose a capital program to systematically address buried vent lines through either re-routing of the vent line to above-grade, relocation of the meter set, and in locations where neither of those solutions is possible, relocation of the meter set. Buried vent lines will be identified through the inside meter inspection program. The capital forecast for this program is approximately $0.1 million for KEDNY and $0.3 million for KEDLI in each of the Rate Years and Data Years, adjusted for inflation.

I. Low Pressure Main Valve Installation Program

Q. Please describe the Low-Pressure Main Valve Installation Program.

A. The Companies are proposing a safety enhancement program to install shut-off valves approximately every 1,000 feet on low-pressure main in conjunction with the Companies' retirement of LPP. The valves can be operated by first responders to enable faster, and more effective reaction to an incident and are an important component toward improving the overall safety of low-pressure systems. During

2018, KEDNY conducted a pilot to test installation design for shut off valves through installation of six shut-off valves on the low-pressure system between 64th Street and 66th Street in Brooklyn. Using lessons learned from the pilot program, the Companies modified the installation design to reduce costs. Based on the Companies' annual target mileage for LPP retirement, the program includes installation of between 290 and 330 valves on KEDNY's system each year between CY 2020 and CY 2023, and approximately eleven valves per year on KEDLI's system during the same period. The capital investment plan includes a forecast of approximately $2.5 million for KEDNY and $0.05 million for KEDLI for each of the Rate Year and Data Years, adjusted for inflation, respectively, and adjusted for a slight increase in valve installations in the Data Years for KEDNY. The costs for the Low-Pressure Main Valve Installation Program are shown separately in Exhibit __ (GIOP-1) for transparency, but will ultimately be accounted for in the total unit costs of the various main and service replacement programs.

J. <u>Proactive Pre-1974 High Definition Polyethylene ("HDPE") Services Replacement Program</u>

Q. Why are the Companies proposing to proactively replace pre-1974 HDPE services?

A. Since 2012, the Companies have noted an increase in identified leaks occurring on pre-1974 HDPE services. Accordingly, these services should be removed in

conjunction with the retirement of associated LPP. There are estimated to be less than 800 pre-1974 services in KEDLI's service territory; thus, retirement of the services in conjunction with proactive LPP retirement does not require additional funding. In KEDNY's service territory, there are approximately 9,000 pre-1974 HDPE services. KEDNY anticipates replacing approximately 400 of these services per year in conjunction with LPP retirements over the 20 years remaining in the Proactive Main Replacement Program, requiring a capital investment of approximately $2.5 million in each of the Rate Year and Data Years, adjusted for inflation. The costs for Proactive HDPE Replacement are shown separately in Exhibit __ (GIOP- 1) for transparency, but will ultimately be accounted for in the total unit costs of the various main and service replacement programs.

K. <u>Single Meter Regulator Inspections Program</u>

Q. Please describe the Single Meter Regulator Inspection Program.

A. The Companies currently follow an inspection program for regulators supplying multi-meter headers ("multi-meter regulators"), but regulators supplying single meters ("single meter regulators") are inspected at the time of periodic meter change for active services under the Companies' meter test program, which is based on a sample population of meters each year. The Companies propose the Single Meter Inspection Program to ensure that the entire single meter regulator population is inspected at least once every 20 years, like multi-meter regulator inspections, to

ensure continued safe and reliable operation. The program requires incremental non-labor O&M to conduct additional inspections in the amount of approximately $0.780 million for KEDNY and $4.000 million in KEDLI in each of the Rate Year and Data Years, respectively. Incremental customer meter services FTEs are also required to perform this work. KEDNY requires three additional technicians in each of the Rate Year and Data Years, and KEDLI requires 24 additional technicians, and one management supervisor in each of the Rate Year and Data Years, due to the higher number of single meters and broader geographic area covered by KEDLI's service territory.

L. Gas Control SOP Training Program

Q. Please explain the Gas Control SOP Training Program.

A. Similar to the I&R safety programs described above, this program directly responds to recommendations coming out of the Merrimack Valley incident. Following that incident, Staff recommended that New York gas utilities develop job-specific procedures for certain activities, including high/low pressure tie-ins, abandonment, and regulator station operation verification. Staff further recommends that employees and contractors should be trained on the new procedures, and that training should include job-specific abnormal operating conditions and responsive actions to be taken. The Companies already had in place job-specific procedures, known internally as System Operator Process ("SOP") procedures, that covered

most areas of concern; however, the Companies made some changes and improvements to its SOPs in light of the Companies' engineering review and Staff's recommendations. The result of the Companies' SOP improvements is that more field work is covered by SOPs, requiring increased training and field oversight.

The Gas Control SOP Training Program is a further safety improvement to implement recommended best practices, including supplemental field-based training on SOPs and enhanced field QA/QC oversight of SOP execution and inspection of work. The proposal is for 2.3 Gas Control Center FTEs for KEDNY and 2.2 Gas Control Center FTEs in KEDLI in each of the Rate Year and Data Years to conduct field training and oversight.

M. <u>KEDLI Meter Pits Program</u>

Q. Please explain the KEDLI Meter Pits Program.

A. In KEDLI's service territory, some customer meters are located in below-grade meter pits that KEDLI has found can experience water intrusion and related corrosion. This program funds remediation of the entire population of meter pits. KEDLI intends to remove meter pits and relocate meters to above-grade where possible, and where meter relocation is not possible, KEDLI will address corrosion issues and will make improvements to the meter pits to decrease the possibility of water intrusion. The Companies' capital investment plan includes approximately

$1.1 million per year in each of the Rate Year and Data Years, adjusted for inflation, for this program.

N. <u>Enhanced First Responder and Field Training</u>

Q. Please describe the Companies' proposals to enhance first responder and field training.

A. The Companies propose to implement enhancements to first responder training and field training, similar to successful programs that were recently adopted in Niagara Mohawk's service territory. Niagara Mohawk collaborated with first responders to: (i) create standard operating procedures so firefighters are better informed prior to an incident regarding what actions to take or avoid and how to operate certain field devices; (ii) expand the content of its Fire Fighter 1st Class training program, as well as other training curriculum, to include additional information and protocols regarding natural gas; and (iii) implement a "train the trainer" program to teach designated first responders gas safety protocols so that they could then train others in their departments.

Similarly, Niagara Mohawk revised its field training curriculum, expanded the scope of field training, and incorporated new technologies. Recent issues relating to operator qualifications of contractors highlight the importance of robust training

programs, particularly field training. Field training involves assessing contractor training and performance, including:

- Developing standardized assessment criteria;
- Identifying training program strengths and weaknesses;
- Evaluating each contractor training program to ensure that it is consistent with the Companies' guidelines and that personnel have construction skills meeting National Grid's standards;
- Continuously monitoring and evaluating each contractor's training program to ensure corrective actions are effectively implemented;
- Working as a partner with contractor trainers to share best practices and communicate policies and procedure changes that will impact training modules; and
- Ensuring that contractor personnel operator qualifications are current.

The Companies propose to incorporate these improvements and add four First Responder Trainers (two FTES dedicated to KEDNY and two FTEs dedicated to KEDLI) and four additional Field Trainers (two FTEs dedicated to KEDNY and two FTEs dedicated to KEDLI) in the Rate Year and Data Years. In addition, incremental non-labor O&M expense in the amount of approximately $0.050 million in each of the Rate Year and Data Years for each company is needed to support first responder training, and $0.050 million in each of the Rate Year and

Data Years for each company is needed to support field training. Non-labor costs are based on the costs to implement the Niagara Mohawk program improvements.

O. Expanded Residential Methane Detection Program

Q. Please explain the Residential Methane Detection Program.

A. In the 2016 KEDNY and KEDLI Rate Cases, KEDNY implemented a Residential Methane Detection Pilot Program to purchase and distribute commercially available RMDs and gas safety brochures to customers living in apartments with inside meters (room sets) when crews are on-site to perform mandated inside service line inspections. In CY 2018, Niagara Mohawk implemented a similar Residential Methane Detection Program to distribute over 50,000 RMDs during service line inspections and other customer interfacing opportunities (such as the New York State Fair). Niagara Mohawk's program also included an enhanced customer education component to develop a new and improved gas safety brochure for distribution to customers along with the RMDs that includes information related to use of RMDs and gas safety in general. KEDNY is incorporating the new brochure into its pilot program for CY 2019 and is continuing to make improvements to the brochure to address additional topics including moving inside meters to outside.

Both KEDNY's and Niagara Mohawks' Residential Methane Detection Programs have been very well-received by customers. RMDs are low cost, and easy-to-use

plug in devices, similar to smoke and carbon monoxide detectors, and are an effective means to add a layer of protection against the risk of leaking gas. The program also has shown an ancillary benefit of encouraging customers to allow access to inside facilities for required inspections. For these reasons, the Companies propose to expand the scope of KEDNY's Residential Methane Detection Program and to incorporate this program in KEDLI's service territory. In particular, the Companies propose to distribute approximately 54,000 RMDs in KEDNY's service territory and approximately 36,000 RMDs in KEDLI's service territory, including enhancements to customer education and outreach, beginning in the Rate Year and continuing through the Data Years. The Companies intend to continue to leverage the inside service inspections and will seek other customer facing opportunities for distribution across all customer classes, and specifically targeting room set meter customers, low income customers, multi-use residences, and customers served by smaller diameter (two and four inch) cast iron piping. This program requires incremental non-labor O&M of approximately $1.896 million for KEDNY and $1.264 million for KEDLI in each of the Rate Year and Data Years, respectively. As discussed above, while the costs for this program are currently included in the revenue requirements, the Companies would be willing to fund this program through the NRA deferrals to help mitigate the rate request in these proceedings.

P. Enhanced High Emitter Methane Detection Program

Q. Please explain the Enhanced High Emitter Methane Detection Program.

A. In the 2016 KEDNY and KEDLI Rate Cases, the Companies committed to collaborating with the Environmental Defense Fund ("EDF") to implement a methane detection pilot program to gather and analyze leak flow data to prioritize system investments and leak repairs. During CYs 2017 and 2018, EDF conducted a cavity ring-down spectroscopy ("CRDS") pilot program using a Google Car outfitted with CRDS technology to drive on various streets in KEDLI's service territory. The results were analyzed by Colorado State University to attempt to quantify detected leaks for methane emissions, and leaks characterized through this methodology as high emitters were communicated to the Companies.

During CY 2019, the Companies are continuing to assist Colorado State with its next phase of analysis of the 2018 CRDS testing results. The Companies are providing access to and guidance on their gas system GIS mapping to enable the University to conduct geospatial analysis. The Companies will continue to support these efforts and will consider the results and possible expanded application of the CRDS technology.

While those efforts are ongoing, the Companies propose an incremental O&M program to continue to explore limited use of CRDS to enhance the leak repair

programs. Specifically, the Companies are proposing to hire a contractor to operate a Google car with CRDS technology for use in areas previously identified as possible high leak concentration areas to aid in prioritizing for repair possible large emitters within those areas. This program requires incremental non-labor O&M in the amount of approximately $0.250 million for KEDNY and $0.350 million for KEDLI in each of the Rate Year and Data Years, respectively. The program also requires an incremental gas distribution engineer FTE, as shown on Exhibit___(GIOP-9), to work with procurement and contracts strategy to develop a contract for leasing the CRDS technology, manage the contractor, and monitor progress, and analyze and report results.

Q. <u>Restrictions for Elevated Gas Infrastructure Program</u>

Q. Please describe the Restrictions for Elevated Gas Infrastructure Program.

A. This program would install fencing or other physical deterrents to restrict and deter public access to elevated gas facilities (four feet or higher above the ground or across a body of water). The purpose of this program is to reduce the risk to the public of climb and fall injuries or fatalities. The program is in part in reaction to an accident that occurred in the service territory of the Companies' U.K. affiliate involving a teenager who climbed an elevated gas facility and fell. Niagara Mohawk is currently undertaking a similar program that was approved in Niagara Mohawk's recent rate case. The program requires capital investments of

approximately $0.4 million for KEDNY and $0.5 million for KEDLI in each of the Rate Year and Data Years.

III. Gas Safety and Compliance Measures

Q. Are the Companies proposing revisions to their current gas safety and compliance measures?

A. Yes. The Companies' gas safety and compliance measures were set in the 2016 KEDNY and KEDLI Rate Cases with performance targets established for CYs 2017, 2018, and 2019, continuing until modified by the Commission. As discussed below, the Companies propose revisions to: (i) establish more stretching performance targets for certain metrics, such as KEDNY's LPP retirement mileage targets, that are challenging, but achievable, and will ensure the Companies are improving their performance; (ii) simplify and streamline the mechanics of certain metrics by collapsing categories and layers of measurement; (iii) reflect and reward performance against metrics by redistributing potential NRAs among the various performance measures while maintaining the same overall exposure; and (iv) align metrics with those of other utilities, including Orange & Rockland ("O&R") and Niagara Mohawk. The Companies also propose limited incentives to encourage outperformance and productivity improvements.

It should be noted that the gas safety and compliance measures are measured on a CY basis. As such, references to adjustments in performance targets or implementation of new metrics and incentives are proposed to begin January 1, 2020.

Q. Please summarize the Companies' proposed gas safety and compliance measure proposal.

A. To continue to encourage performance against targets, the Companies propose to maintain the existing cap of 150 pre-tax basis points at risk per year for failure to meet targets that was established in the 2016 KEDNY and KEDLI Rate Cases. To more fairly allocate the 150 at risk basis points across the categories of performance targets, the Companies propose reducing the total at risk for gas regulations performance measure from 100 basis points per year to 75 basis points per year, and distributing the 25 basis points among the other performance measure categories. This approach is aligned with the approach recently adopted by the Commission for O&R's safety and compliance performance measures. The Companies' proposal is illustrated in Table 2. Specific targets and measures for each category are set forth in more detail below.

Table 2: Summary of Safety and Compliance Targets and NRA Exposure for KEDNY and KEDLI

Performance Measure	Basis Points at Risk CY 2017 - 2019	Basis Points at Risk CY 2020 – CY 2023
LPP Retirement	8	15
Leak Backlog	12	15
Damage Prevention	18	20
Emergency Response Time	12	25
Gas Regulations Performance Measure	100	75
Total at Risk	**150**	**150**

A. LPP Replacement

Q. What are KEDNY and KEDLI's proposal regarding LPP retirement mileage targets and incentives?

A. As discussed in KEDNY and KEDLI's respective GIOP testimony, KEDNY proposes to continue its five-mile per year acceleration of LPP retirement during the Rate Year and Data Years because leak rates in KEDNY's service territory have continued to increase. KEDLI proposes to maintain its CY 2019 LPP retirement level of 155 miles per year during the Rate Year and Data Year because leak rates in KEDLI's service territory have stabilized in a relatively low range.

The Companies propose to continue the existing LPP retirement metric structure with a minimum annual threshold and a cumulative rate plan target. Specifically, as

shown in Table 3, the Companies propose an NRA of 15 basis points should they fail to remove from service a minimum target of miles of LPP in CY 2020, CY 2021, CY 2022, and CY 2023 or a cumulative four-year total of LPP by the end of CY 2023.

Table 3: LPP Replacement Proposal (15 Basis Points at Risk)

KEDNY	Current Target 2019	Calendar Year 2020	Calendar Year 2021	Calendar Year 2022	Calendar Year 2023
Minimum Removal Targets	60	65	70	75	310 (cumulative CY20-CY23)

KEDLI	Current Target 2019	Calendar Year 2020	Calendar Year 2021	Calendar Year 2022	Calendar Year 2023
Minimum Removal Targets	150	150	150	150	620 (cumulative CY20 – CY23)

Q. Are the Companies proposing to continue a LPP mileage replacement incentive?

A. Yes. The Companies propose to continue the existing positive revenue adjustment of two basis points for each full mile of LPP removed above a set incentive threshold, capped at ten basis points per year, with one adjustment to the existing

structure. The existing incentive threshold is set at one full mile above the rate allowance mileage target. To align with the positive incentive mechanism recently adopted for Niagara Mohawk, the incentive would be earned for every full mile of retired LPP above the rate allowance threshold, thereby simplifying the mechanism and aligning National Grid's New York companies. The positive revenue adjustment would continue to be recovered through the Gas Safety and Reliability Surcharge ("GSRS"). The GSRS and the LPP incentive mechanism is illustrated in Exhibit ___ (GIOP-3).

Table 4: LPP Mileage Replacement Incentive

KEDNY	Calendar Year 2020	Calendar Year 2021	Calendar Year 2022	Calendar Year 2023
Incentive Threshold – Incremental Mileage	70	75	80	310 (cumulative CY20-CY23)
Incentive (capped at 10 basis points annually)	2 basis points per mile above threshold	2 basis points per mile above threshold	2 basis points per mile above threshold	2 basis points per mile above threshold

KEDLI	Calendar Year 2020	Calendar Year 2021	Calendar Year 2022	Calendar Year 2023
Incentive Threshold – Incremental Mileage	155	155	155	620 (cumulative CY21 – CY23)
Incentive (capped at 10 basis points annually)	2 basis points per mile above threshold	2 basis points per mile above threshold	2 basis points per mile above threshold	2 basis points per mile above threshold

Any mile collected through the GSRS (*i.e.*, above the rate allowance) will not count against the target in subsequent years.

Q. How do the Companies propose to recover the costs to retire additional LPP?

The Companies propose to continue to recover costs of retired LPP incremental to the level funded in base rates through the GSRS mechanism that was approved in the 2016 KEDNY and KEDLI Rate Cases. The GSRS allows the Companies to recover a return on investment and depreciation expense associated with prudent investment in LPP replacement incremental to the level funded in base rates and is illustrated in the Companies' respective Exhibit __ (GIOP-3).

Q. Are the Companies proposing a LPP replacement incentive for improved productivity?

A. Yes. The Companies propose to continue the incentive mechanism associated with improved productivity that was adopted in the 2016 KEDNY and KEDLI Rate

Cases. Specifically, the productivity incentive allows the Companies to earn a positive revenue adjustment of up to ten basis points if the Companies can meet specific unit cost targets. The incentive mechanism and proposed targets are set forth in the Companies' respective GIOP testimonies.

Q. Please summarize the Companies' LPP proposal.

A. Overall, the Companies' LPP proposal ensures they collectively will replace at least 215 miles of LPP each year and a cumulative 930 miles over four years (with an associated negative revenue adjustment for failing to achieve the targets), while providing incentives to accelerate LPP replacement each year. The proposal provides flexibility to target additional replacements when resources are available and other opportunities present to complete the work more cost effectively. The surcharge mechanism ensures that KEDNY and KEDLI will recover LPP replacement costs only to the extent it is successful in delivering its program.

B. <u>Leak Backlog</u>

Q. Please describe the Companies' current leak backlog performance metric, incentives, and performance against targets.

A. The current leak backlog performance metrics and incentives were established in the 2016 KEDNY and KEDLI Rate Cases. The performance metrics set specific annual leak backlog targets for workable leaks (Types 1, 2, and 2A) and total leaks

and imposed NRAs for failure to meet targets, as shown on Table 5. KEDNY's total leaks target is reduced by 150 leaks each year, and KEDLI's total leaks target is reduced by 750 leaks each year.

Table 5: CY 2017 – CY 2019 Leak Backlog Metric

KEDNY Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	8 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 17	3,550	4 basis points
	CY 18	3,400	
	CY19	3,250	

KEDLI Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	8 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 17	10,000	4 basis points
	CY 18	9,250	
	CY19	8,500	

Additionally, the Companies are incentivized to repair incremental Type 3 leaks that have been ranked as high emitters. The Companies are permitted to earn a positive revenue adjustment of one basis point for each 50 such leaks eliminated, up to a maximum of five basis points per year. The Companies must achieve both annual backlog reduction targets to qualify for this incentive.

Q. Please explain the Companies' proposal for the leak backlog metrics and incentives.

A. The Companies propose to retain the annual reduction of the leak target (KEDNY reduction of 150 leaks and KEDLI reduction of 750 leaks), and further reduce the annual target based on the prior year's performance in eliminating incremental high emitter Type 3 leaks. The Companies similarly propose to retain the existing incentive mechanism to repair incremental Type 3 high emitter leaks and the existing GSRS cost recovery mechanism for incremental leak repairs that is demonstrated in Exhibit __ (GIOP-3). The proposed metrics and incentives for each company are presented in Table 6.

Table 6: Proposed Leak Backlog and Incentive Structure (15 basis points at risk)

KEDNY Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	3,100*	5 basis points
	CY 21	2,950 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	2,800 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	2,650 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 150 reduction from CY19 target

KEDLI Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	7,750*	5 basis points
	CY 21	7,000 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	6,250 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	5,500 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 750 reduction from CY19 target

KEDNY & KEDLI # High Emitting Type 3 Leaks Repaired	Positive Revenue Adjustment	Total Leak Target Reduction Subsequent Year
0- 49	None	0
50-99	1 basis point	50
100-149	2 basis points	100
150-199	3 basis points	150
200-249	4 basis points	200
>=250	5 basis points	250

To avoid the situation where leaks identified in the final days of the year frustrate the Companies' ability to meet their year-end targets, the Companies propose that they will be deemed as having met the leak targets if achieved at any time between December 21st and December 31st of each CY.

Q. Would the Companies be eligible for incentives for repairing high emitter leaks in years that the leak backlog target is not met?

A. No. If the Companies fail to meet the leak backlog target in any given year, the Companies would not be eligible to earn any incentive on repairs of high emitter leaks. Also, there would be no additional reduction in the subsequent year leak target based on the high emitter total leaks repaired. The subsequent year leak target would still be reduced by the fixed annual reduction (*i.e.*, 150 leaks at

KEDNY). An illustration of the incentive mechanism is provided in Exhibit __ (GIOP-3).

Q. Have the Companies made any adjustments to account for the incremental leaks repaired during the Historic Test Year?

A. Yes. The Companies adjusted the Historic Test Year to remove incremental leaks repaired for which the Companies received cost recovery through the GSRS. This adjustment is reflected in the RRPs' testimonies, Exhibit __ (RRP-3), Schedule 2.

C. Emergency Response

Q. What are the Companies' proposals for the emergency response metrics?

A. The Companies propose to maintain the existing metric that is based on statewide performance targets established for this performance measure and the existing mechanism for excluding instances of 20 or more odor calls in a two-hour period from a mass odor issue not caused by the Companies, contingent on Staff's consent. Specifically, the Companies propose the following metrics:

Table 7: Emergency Response Proposal (25 basis points at risk)

Required Response Time ("RRT")	Calls that must be responded to within RRT	NRA for failure to meet RRT
30 minutes	75%	12 basis points
45 minutes	90%	8 basis points
60 minutes	95%	5 basis points

Q. Are the Companies proposing an emergency response incentive?

A. Yes. The Companies propose to add an incentive that is the same as Niagara Mohawk's and O&R's emergency response incentive. The Companies would earn a positive revenue adjustment of up to six basis points for meeting leak and odor call response targets as set forth in Table 8.

Table 8: Emergency Response Incentive

Emergency Response Incentive	Response within 30 minutes		
Response Rate	86% - 88%	> 88% - 90%	> 90%
Positive Revenue Adjustment	2 basis points	4 basis points	6 basis points

D. Damage Prevention

Q. What are the Companies' proposals for the damage prevention metrics and incentives?

A. The Companies' existing metrics for damage prevention include targets for three categories of damages: total damages, mis-marks, and company plus company/contractor. The Companies propose to set the metric based on a single category, "overall damages," that would include mismarks, company plus company/contractor, third party damages, and no calls. The proposed metric maintains challenging performance levels, but simplifies its administration and

removes uncertainty with regard to what types of damages should be included in each of the former three categories. The Companies also propose adding a damage prevention incentive for a positive revenue adjustment of up to six basis points if certain targets are met. The proposed metrics and incentive targets are presented in Table 9.

Table 9: Proposed Damage Prevention Metric and Incentives

(20 basis points at risk)

KEDNY Overall Damages Per 1,000 One-Call Tickets					
NRA/PRA	**Basis Points**	**CY20**	**CY21**	**CY22**	**CY23**
NRA	20	>1.75	<1.25	>1.70	>1.6
NRA	9	1.5-1.75	<1.25	1.5-1.7	1.4-1.6
	0	<1.6	<1.25	<1.5	<1.4
PRA	3	1.25-1.5	<1.25	1.25-1.5	1.25-1.5
PRA	6	<1.25	<1.25	<1.25	<1.25

KEDLI Overall Damages Per 1,000 One-Call Tickets					
NRA/PRA	**Basis Points**	**CY20**	**CY21**	**CY22**	**CY23**
NRA	20	>2.25	>2.25	>2.0	>1.9
NRA	9	1.85-2.25	1.77-2.25	1.71-2.0	1.65-1.9
	0	<1.85	<1.77	<1.71	<1.65
PRA	3	1.25-1.5	1.25-1.5	1.25-1.5	1.25-1.5
PRA	6	<1.25	<1.25	<1.25	<1.25

E. Gas Regulations Performance Measure

Q. Please describe how the Companies currently measure performance against safety regulations.

A. KEDNY and KEDLI are currently subject to a Gas Safety Regulations Performance Metric that exposes the Companies to significant NRAs (up to 100 pre-tax basis points) based on violations of certain Commission's rules and regulations contained in 16 NYCRR Parts 255 (Transmission and Distribution of Gas) and 261 (Piping Beyond the Meter) that are set forth in Appendix 9 of the 2016 KEDNY and KEDLI Rate Case Joint Proposal, as determined by Staff's annual field and records

audits. Specifically, the metric imposes a NRA for each occurrence of a violation identified during these audits in accordance with specific targets. The number of occurrences of non-compliance with each "high risk" and "other risk" regulation is capped at ten per year.

Q. What are the Companies' proposals regarding the Gas Regulations Performance Measure?

A. KEDNY and KEDLI take compliance with the Commission's regulations, especially regulations affecting the safe operation of the Companies' gas system, very seriously. The Companies also acknowledge that the cap of ten occurrences per year of non-compliance with each regulation that was put in place in the 2016 KEDNY and KEDLI Rate Cases helped to alleviate the overly punitive nature of the prior metric that allowed for unlimited occurrences. However, based on the Companies record of improvement in compliance since 2015, the Companies believe that a further adjustment to the targets is appropriate. In particular, the Companies propose that the NRAs be spread between larger tiers, in line with the structure that was approved for Niagara Mohawk. The Companies also propose that the total NRA per year be capped at 75 basis points. The Companies' proposal is reflected in Table 10.

Table 10: Proposed Compliance Regulations Performance Metric

CY 2021 – CY 2024 (75 basis points at risk)

Audit	Type	Occurrence	Basis Points
Records	High Risk	1-10	0
		11-40	0.5
		41+	1
	Other Risk	1-30	0
		31+	0.3
Field	High Risk	1-40	0.5
		41+	1
	Other Risk	All	0.3

The Companies propose to maintain the other features of the metric that are currently in place, including the requirement for a compliance improvement plan submission to Staff and development of a mitigation plan in the event of more than ten occurrences with a particular high risk or other regulation and procedures following audit results, and deferral of NRAs for future use to fund gas safety and compliance improvement programs. The Companies also propose that if an alleged occurrence of non-compliance with a high risk or other risk regulation is the subject of a separate penalty proceeding by the Commission, the occurrence will not count toward the performance measure, and that that any non-compliance event that encompasses more than one code section (*e.g.*, a violation of a gas safety regulation

and a company operating procedure under 16 NYCRR 255.603(d)) will count as one occurrence for purposes of calculating a negative revenue adjustment.

Q. Do the Companies intend to continue to report on performance against each of the above categories of gas safety and compliance measures?

A. Yes. The Companies will report annual performance in each of the areas set forth above to the Commission's Secretary by April 1 following the end of each CY.

Q. Does this conclude your testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of the

Gas Safety Panel

Ross W. Turrini
Timothy S. Graham
Annette M. Saxman

Date: July 3, 2019

TABLE OF CONTENTS

I. Introduction....1
II. Corrections and Updates....2
A. Inside Service Line Inspections Program....2
B. Enhanced Inactive Accounts Program....2
C. I&R Distribution Station Overpressure Protection and Control Line Survey and GPS Mapping Programs....3
D. KEDNY Damage Prevention Metric....4

I. Introduction

Q. Please introduce the members of the Gas Safety Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, and Annette Saxman.

Q. Is this the same Panel that previously submitted direct testimony in these proceedings?

A. Yes. The terms defined in the Panel's direct testimony have the same meanings here.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The purpose of the Panel's corrections and updates testimony is to identify and explain the updates and corrections to the capital investment and/or O&M forecasts for the Inside Service Line Inspections Program, the Enhanced Inactive Accounts Program, and the I&R Distribution Station Overpressure Protection and Control Line Mapping Programs. Updates to the forecasts for these programs are described below and are included in KEDNY and KEDLI's respective Exhibits __ (GIOP-1CU) (updated capital investment plan), KEDNY Exhibit __ (GIOP-8CU) and KEDLI Exhibit __ (GIOP-7CU) (updated incremental O&M expenditures), and KEDNY Exhibit __ (GIOP-9CU) and KEDLI Exhibit __ (GIOP-8CU) (updated incremental FTE positions). These exhibits were prepared by or under the supervision and direction of one or more members of the Panel.

The Panel also corrects a typographical error in KEDNY's Damage Prevention Metric table.

II. Corrections and Updates

A. Inside Service Line Inspections Program

Q. Please describe the update to the forecast of costs for the Inside Service Line Inspections Program.

A. The forecast of costs for the Inside Service Inspections Program has been updated to include non-labor O&M costs to seal points of entry (inside wall penetrations) in cases where no inside seal is observed during an inside service line inspection. This is a program enhancement developed in collaboration with Staff. The annual costs for point-of-entry sealing are approximately $0.73 million and $0.65 million for KEDNY and KEDLI, respectively, during the Rate Year and Data Years.

B. Enhanced Inactive Accounts Program

Q. Please describe the update to the Enhanced Inactive Accounts Program forecast.

A. As described in the Panel's direct testimony, the Companies proposed a phased approach to shorten the end-to-end process for addressing inactive accounts to a 45-day process by the end of the Rate Year and to 30 days by the end of Data Year 2. The Companies re-evaluated current resources and the enhanced program requirements and determined that KEDNY's initial forecast of incremental FTEs was overstated. Specifically, KEDNY's incremental FTE forecast of 95 CMS representatives is being reduced by 25 to a total of 70 CMS representatives. The revised KEDNY FTE hiring plan is provided in KEDNY Exhibit __ (GIOP-9CU) and Exhibit__(RRP-11) Workpapers to Exhibit (RRP-3), Schedule 27, Workpaper 2.

Q. Are there any changes to the non-labor forecast for the Enhanced Inactive Accounts Program?

A. Yes. The Companies' non-labor forecasts failed to include costs of uniforms and fleet expenses for incremental FTEs. The non-labor forecasts for the Enhanced Inactive Accounts Program have been corrected to include approximately $0.3 million and $0.04 million in non-labor expense in the Rate Year for KEDNY and KEDLI, respectively. There are no non-labor adjustments for this program in the Data Years.

C. I&R Distribution Station Overpressure Protection and Control Line Survey and GPS Mapping Programs

Q. Please describe the corrections to the forecast for the I&R Distribution Station Overpressure Protection and Control Line Survey and GPS Mapping Programs.

A. The Companies inadvertently excluded costs for these two programs related to enhancing overpressure protection and safety at distribution stations. First, KEDLI's capital forecast for the Distribution Over Pressure Protection Program did not include the costs of adding relief valves at distribution stations of approximately $1.2 million in each of the Rate Year, Data Year 1, and Data Year 2. There are no costs added for Data Year 3 because relief valve installation will be complete by the end of Data Year 2. Installation of relief valves will serve as an additional layer of overpressure protection for distribution stations.

Second, KEDNY and KEDLI's non-labor O&M forecasts for the Control Line Survey and GPS Mapping programs are being adjusted to include the costs of reviewing as-built transmission station drawings against the physical facilities to ensure the accuracy of the

station documentation in coordination with survey and mapping of the control lines. The costs to review transmission station as-built drawings are approximately $1.1 million and $0.7 for KEDNY and KEDLI, respectively, in each of the Rate Year and Data Years 1 and 2. There are no costs in Data Year 3 because the document review will be complete by the end of Data Year 2.

Both programs are important components of the overall I&R gas safety program enhancements that will improve the Companies' ability to respond to abnormal operating conditions, such as an over pressurization. As discussed in the Panel's direct testimony, while these costs are included in the revised revenue requirement, the Companies are willing to use the NRAs incurred for gas compliance violations to fund these programs.

D. <u>KEDNY Damage Prevention Metric</u>

Q. Is there a correction to KEDNY's Damage Prevention Metric?

A. Yes. KEDNY's Damage Prevention Metric that was presented in the Panel's direct testimony on page 60 as Table 9 included a typographical error where the same value was repeated in each row under the "CY21" column. The correct version of Table 9, the proposed KEDNY Damage Prevention metric targets and incentives, is as follows:

Table 9CU: Proposed Damage Prevention Metric and Incentives

(20 basis points at risk)

KEDNY Overall Damages Per 1,000 One-Call Tickets					
NRA/PRA	**Basis Points**	**CY20**	**CY21**	**CY22**	**CY23**
NRA	20	>1.75	<1.75	>1.70	>1.6
NRA	9	1.5-1.75	1.55-1.75	1.5-1.7	1.4-1.6
	0	<1.6	<1.55	<1.5	<1.4
PRA	3	1.25-1.5	1.25-1.5	1.25-1.5	1.25-1.5
PRA	6	<1.25	<1.25	<1.25	<1.25

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Gas Safety Panel

Ross W. Turrini
Timothy S. Graham
Annette M. Saxman

Case 19-G-0309
Case 19-G-0310

September 18, 2019

TABLE OF CONTENTS

I. Introduction 1
II. Companies' Record of Continuous Improvement in Gas Safety 3
A. Commitment to Overall Improvements in Gas Safety 3
B. Improved Performance Against Gas Safety Regulation Measures 6
C. Achievement of 2018 Damage Prevention Metric 7
III. Gas Safety Measures Performance Targets and Incentives 13
A. Goals of Gas Safety Performance Targets and Incentives 13
B. Distribution of NRA Exposure Among Various Metrics 15
C. Use of NRA Deferral Balances to Fund Safety Programs 16
IV. Safety Metric Recommendations 19
A. Leak Prone Pipe Retirement 19
i. LPP Targets and Prioritization 19
ii. LPP Incentives 23
B. Leak Backlog Metric 26
i. Leak Backlog Metric Targets and Incentives 26
ii. Leak Survey Techniques 33
C. Damage Prevention Metric 35
D. Emergency Response Metric 39
E. Gas Regulations Performance Metric 42
V. Proposed Adjustments to the Companies' Incremental Safety Programs 44
A. Plastic Fusions QA/QC Re-Digs Program 44
B. Inside Service Line Inspection Program 46
C. Enhanced Contractor Inspection Program 47
D. Enhanced Inactive Accounts Program 48
E. Buried Vent Lines Program 50
F. Low Pressure Main Valve Installation Program 51
G. Expanded Residential Methane Detector ("RMD") Program 52
H. Enhanced High Emitter Methane Detection Program 54
VI. Other Recommendations 56
A. Gas Safety Reporting 56
B. Local Law 152 56
C. Inside Meter Relocations 59
D. SEP's Concerns Regarding Radon 63

I. Introduction

Q. Please identify the members of the Gas Safety Panel.

A. The Gas Safety Panel consists of Ross W. Turrini, Timothy S. Graham, and Annette M. Saxman.

Q. Is this the same Panel that testified previously in this proceeding?

A. Yes. Capitalized terms defined in the Panel's direct and corrections and updates testimony have the same meanings here.

Q. What is the purpose of the Panel's rebuttal testimony?

A. The purpose of the Panel's rebuttal testimony is to respond to certain gas safety performance and program recommendations set forth in the prepared testimony of the Department of Public Service Staff ("Staff") Gas Infrastructure and Operations Panel ("SGIOP"), the Staff Pipeline Safety Panel ("SPSP"), the Staff Policy Panel ("SPP"), the Staff Efficiency and Sustainability Panel ("SESP") (collectively, "Staff"), as well as the New York City Gas Infrastructure and Safety Panel ("NYCGISP") and New York City Policy Panel ("NYCPP") (collectively, "NYC"), the Environmental Defense Fund's witness Joseph Von Fischer ("EDF"), and the Sane Energy Project's witnesses Kim Fraczek and Lee Ziesche ("SEP").

Specifically, the Panel's rebuttal testimony will address:

- The Companies' record of continuous improvement in gas safety performance since 2015;
- The benefits of performance incentives in promoting further improvements to gas safety;
- Staff's and other parties' recommendations regarding the Companies' gas safety metrics and incentives;
- Staff's and other parties' recommendations regarding the Companies' incremental gas safety programs and enhancements; and
- Staff's and other parties' other recommendations regarding gas safety, including: gas safety reporting, NYC's Local Law 152, inside meter relocations, and SEP's radon concerns.

Q. Does the Panel sponsor any exhibits as part of its rebuttal testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared under its direction and supervision:

- Exhibit __ (GSP-1R): Relevant Information Request ("IR") responses in this proceeding;
- Exhibit __ (GSP-2R): 2018 Damage Prevention Metrics Annual Report and Most Recent Damage Prevention Analysis Quarterly Report Transmittal Letter;

- Exhibit __ (GSP-3R): Companies' Damage Prevention IR Responses in Case 13-G-0314; and
- Exhibit __ (GSP-4R): Relevant portions of the Companies' Implementation Plan filed on May 16, 2016, and the Companies' Updated Plan filed on March 8, 2018, in Case 13-G-0314.

II. Companies' Record of Continuous Improvement in Gas Safety

A. Commitment to Overall Improvements in Gas Safety

Q. Before discussing the SGSP's and other parties' specific proposals on safety performance measures and the Companies' incremental safety programs, would the Panel like to comment generally on the Companies' commitment to safety and compliance?

A. Yes. The Companies have worked diligently to develop a culture where safety and compliance are paramount. The Companies have focused on people, processes, systems, and technology to reinforce the National Grid commitment to safety and compliance. As is stated in the Companies' direct testimony, these commitments are reflected in improvements the Companies have made, including: additional field inspections; damage prevention enhancements; investments in technology solutions through the Gas Business Enablement Program; efforts to enhance training and simplify procedures and practices in the field; and efforts to implement the American Petroleum Institute's

recommended comprehensive risk identification and management framework (Recommended Practice 1173 ("RP 1173")).

The Companies' efforts have resulted in improvements in safety performance since 2015, and the Companies are committed to further progress. The Companies' first priority is the safety of our customers, communities, and employees, and we are continuously seeking opportunities to enhance our pipeline safety management processes. To that end, the Companies' proposals set forth in the Panel's direct testimony, including incremental safety programs, performance metrics that are challenging, and incentives that encourage the Companies to exceed regulatory requirements, will help the Companies to drive continuous performance improvements over the next several years.

Q. Please respond to NYC's comments questioning the Companies' ability to safely and prudently manage its system and capital budget.

A. NYC questions the Companies' "ability to safely and prudently manage" their gas systems and capital budgets based on a recent Order to Show Cause issued by the Commission in Case 17-G-0317 ("Order to Show Cause"). (NYCGISP at 75). NYC uses the mere issuance of the Order to Show Cause to support its argument that the Companies should not be allowed to adjust their Gas Safety Performance Measures and posits that increased penalty exposure "could be

justified in light of the allegations in the Order to Show Cause." (NYCGISP at 74, 76).

Q. Do the Companies believe that the Order to Show Cause should be considered when reviewing the capital budgets and gas safety metrics subject to these rate proceedings?

A. No. It would be premature and inappropriate to consider the Order to Show Cause in these proceedings. The Order to Show Cause was issued on July 12, 2019, a little over two months ago. The proceeding has not been developed further than the Commission's Order to Show Cause and Staff's list of alleged violations. The Companies intend to respond to the Order to Show Cause within the next month to rebut certain allegations and demonstrate why administrative penalty and prudency proceedings are not warranted. Therefore, it would be inappropriate to draw inference from the alleged violations in the present rate cases as the Companies have not yet responded to the allegations and the allegations have not been fully vetted. Rather, the Companies' actual record of safety performance and compliance improvements that is presented in the Panel's direct testimony and discussed further below should be the basis of recommendations in this proceeding.

B. Improved Performance Against Gas Safety Regulation Measures

Q. Please discuss the Companies' recent compliance performance efforts.

A. The Companies have demonstrated continuous improvement through a steady decline of audit violations over the past several years, as is shown in the Panel's direct testimony at pages 7-8. Indeed, pursuant to Staff's recently-completed review of 2017 field audit and 2017 records audit results shown in Exhibit __(SGIOP-7) at 2 and 11, KEDLI incurred only one High Risk violation, and KEDNY incurred 15. Moreover, based on the level of improved performance during 2016, KEDNY earned a 100 percent reduction to the remaining negative revenue adjustments ("NRAs") that resulted from the 2013 and 2014 audits under a mechanism agreed in the 2016 KEDNY and KEDLI Rate Cases. The Companies are committed to continuing these trends.

Q. Please explain the reduction to KEDNY's NRA balance that has been earned.

A. Section IV.8.6 of the Joint Proposal adopted in the 2016 KEDNY and KEDLI Rate Cases included a mechanism "to promote compliance improvements" to allow a downward adjustment of $1.2 million in NRAs incurred from the 2013 and 2014 audits if KEDNY met certain reductions in high-risk violations in the 2016 field audits and 2017 audits of the Companies' 2016 records. As is indicated by Staff's August 21, 2019 letter (Exhibit __ (SGIOP-6) pages 1-2),

the final determination of the 2016 audits results was 16 high risk violations. Pursuant to the mechanism, KEDNY earned a reduction of the entire $1.2 million NRA.

C. Achievement of 2018 Damage Prevention Metric

Q. Is the SPSP correct that the Companies did not achieve their Damage Prevention Metric targets in 2018 and earned NRAs as a result?

A. No. The SPSP is incorrect. KEDNY and KEDLI met their 2018 Damage Prevention Metric targets that were established in the Joint Proposal in the 2016 KEDNY and KEDLI Rate Cases.

The Companies' results against the targets appear in Table 1, below, and were provided in the response to IR DPS-668 that is included in Exhibit __ (GSP-1R).

Table 1: 2018 Damage Prevention Metric Performance Against Targets

KEDNY 2018	PSC Performance Targets	National Grid Final CY Performance
Mismarks	0.42	0.34
Co. & Co. Contractors	0.09	0.01
Total Damages	1.87	1.17

KEDLI 2018	PSC Performance Targets	National Grid Final CY Performance
Mismarks	0.60	0.49
Co. & Co. Contractors	0.13	0.09
Total Damages	2.48	1.70

Q. Is NYC correct that the Companies misrepresented or artificially deflated their 2018 Damage Prevention Metric Results?

A. No. KEDNY and KEDLI reported their Damage Prevention Metrics results properly to Staff as required by the 2016 KEDNY and KEDLI Rate Cases.

Q. Why do the SPSP and NYC believe that the Companies did not achieve their Damage Prevention Metric targets?

A. Both the SPSP and NYC are reviewing the wrong damage prevention data and are applying the wrong data to the Companies' Damage Prevention Metric targets.

Q. Please explain.

A. All three of National Grid's New York gas local distribution companies ("LDCs") – KEDNY, KEDLI, and Niagara Mohawk – report *two sets* of

damage prevention data to the Commission for two different purposes, each of which is calculated differently, as explained below:

(i) Damage Prevention Performance Metric: This metric is set forth in the Joint Proposal adopted by the Commission in the 2016 KEDNY and KEDLI Rate Cases and is calculated pursuant to the historic methodology used to set the targets in those proceedings. As explained in the direct testimony of the Companies' Gas Safety Panel in the 2016 KEDNY and KEDLI Rate Cases (at 49-50), the metric excludes "retransmit" tickets and includes coating damage, both of which resulted in an adjustment to the damage prevention targets. The metric does not, however, reflect any changes to the Companies' historic practice of *including* "relocate" tickets in the ticket count used to set the targets for the metric.

(ii) Quarterly Damage Prevention Analysis: This metric is calculated pursuant to the recommendations from the Case 13-M-0314 operational audit ("Data Audit") that are set forth in Staff's December 11, 2015 letter (Exhibit __ (SPSP-8)). This data is submitted quarterly to Staff for analysis purposes only and is the data Staff included in its 2019 Gas Safety Performance Measures Report in Case 19-G-0298. In contrast to the Damage Prevention Performance Metric, this metric excludes "relocate" tickets. Additionally, it also includes forms of damages that have historically not been included in the Damage

Prevention Performance Metric (damages with no ticket by homeowners and non-company contractors digging with hand tools).

Exhibit __ (GSP-2R) includes: (i) the Companies' 2018 Annual Gas Safety Performance Metrics Report that includes the Companies' performance against the Damage Prevention Metric, and (ii) the most recent quarterly Damage Prevention Analysis submission transmittal letter. The Analysis data is provided to Staff in zip files along with the transmittal letter, and the transmittal letters clearly state that the information provided is different from the data included in the Companies' Damage Prevention Performance Metric. (Exhibit __ (GPS-2) at page 8). The Damage Prevention Analysis data that was provided for 2018 is set forth in the Companies' response to IR DPS-134 that is included in Exhibit __ (GSP-1R).

Q. Are Staff and the Commission aware of the differences in how these two metrics are calculated?

A. Yes. The Companies explained the differences in calculating these two metrics in its Implementation Plan filed in Case 13-M-0314. Specifically, the Implementation Plan, relevant portions of which are included in Exhibit ___ (GSP-4R), states: "KEDNY and KEDLI's Quarterly Performance Filing excludes retransmits from the ticket count, and the proposed damage prevention

targets in the 2016 Rate Cases reflect this exclusion as well." However, "[r]elocate tickets are included in the one-call ticket count for purposes of measuring KEDNY and KEDLI's damage prevention performance." The Commission approved the Implementation Plan in its *Order Approving Implementation Plans* in Case 13-M-0314, directing the Companies to implement the plan as approved.

At the request of Staff, the Companies agreed to exclude relocates for purposes of the Quarterly Damage Prevention Analysis. Specifically, in the response to IR DPS-025 in the Data Audit and the Companies' Updated Implementation Plan in the Data Audit, the Companies indicated their agreement to exclude "relocates" from the one-call ticket count for purposes of the Quarterly Damage Prevention Analysis, beginning with the first quarter report in calendar year 2018, but made clear that, for purposes of the Damage Prevention Performance Metric, the Companies would continue to include "relocates" in the one-call ticket count, consistent with the methodology used to establish the targets in the Joint Proposal. Copies of the response to IR DPS-025 and the Companies' Updated Implementation Plan are included in Exhibit ___ (GSP-3R) and Exhibit ___ (GSP-4R), respectively.

For these reasons, Staff's assertion is simply wrong, and the Companies' approved Implementation Plans belie Staff's claims. The data in the Quarterly Damage Prevention Analysis is reported by the Companies to the Commission for analysis purposes only and has never been adopted for purposes of the Damage Prevention Performance Metrics for any of National Grid's three New York gas LDCs. As is stated in the Companies' responses to IRs DPS-157 and DPS-668, copies of which are included as Exhibit __(GSP-1R), both KEDNY and KEDLI achieved their Damage Prevention Performance Metric targets for the past five years, including 2018.

Q. If the Commission were to require the Companies to exclude "relocates" in the one-call ticket count for purposes of calculating the Damage Prevention Performance Metric, would the targets proposed by the Companies in these rate filings require modifications?

A. Yes. The baseline used to calculate the targets included "relocates" in the ticket count. Adjusting the baseline to exclude them would necessitate changes in the targets and would require additional resources. As discussed in section IV(C), below, the Companies do not believe that "relocates" should be excluded from the metric.

III. Gas Safety Measures Performance Targets and Incentives

A. Goals of Gas Safety Performance Targets and Incentives

Q. What is the purpose of gas safety metrics and performance targets?

A. The Companies' current rate plans establish certain metrics and performance targets to ensure that the Companies meet their regulatory requirements to provide safe and reliable service to customers. The metrics measure the Companies' performance across various activities, ranging from LPP removal to emergency response, and are designed to ensure accountability to customers. Metrics may include positive revenue adjustments ("PRAs") or NRAs if the Companies' performance varies from established levels. NRAs serve to penalize the Companies for underperformance, while PRAs incent the Companies to overachieve base expectations and deliver incremental improvements. Inclusion of corresponding adjustments is important to drive performance and support public policy objectives.

Q. Please explain why inclusion of PRAs and incentives are appropriate.

A. PRAs do not simply reward the Companies for basic operations or business as usual activities. PRAs incentivize the Companies to perform additional work and provide enhanced services, whether it be incremental leak repair or LPP removal, that ultimately benefit customers and the community and further align the Companies' performance with public policy objectives. Appropriately

challenging PRAs support continuous improvement and superior performance with the goal of exceeding customer expectations and industry standards.

Q. What is the Companies' position on the SPP's proposal for an overall safety and reliability capital investment productivity metric in lieu of the LPP unit cost productivity incentive?

A. The Companies believe the LPP unit cost productivity incentive is an appropriate mechanism to encourage better cost management for the proactive LPP retirement program, assuming the targets are set appropriately based on current unit costs. The Companies are interested in the SPP's proposal for a broader capital program metric that could complement the existing net plant tracker. The Companies do not agree that the LPP productivity incentive should be removed; an overall capital program productivity metric could be designed to exclude the LPP retirement program productivity to avoid any overlap.

Q. Are metrics appropriate for a one-year rate plan?

A. No. The gas safety performance metrics are designed to drive performance improvements over time. To that end, the Companies' proposals for performance metrics contemplate a multi-year rate plan. The Companies do not agree with the inclusion of performance metrics in a one-year litigated case. Indeed, the Companies submit that their consistently high level of performance

should provide the Commission with sufficient comfort that service will not be diminished in a one-year rate plan.

B. Distribution of NRA Exposure Among Various Metrics

Q. Does the SPSP support the Companies' proposal to better allocate the NRA basis points across the categories of performance targets?

A. Yes. Staff supports the Companies' proposal to re-allocate 25 of the 100 basis points previously assigned to the Gas Regulations Performance metric across the other metrics (LPP replacement, Leak Management, Damage Prevention, and Emergency Response). This leaves a total of 75 basis points still assigned to the Gas Regulations Performance Metric. (SPSP at 62-63).

Q. Please respond to NYC's opposition to redistributing the basis points.

A. NYC's general concerns seem to stem from the Order to Show Cause that, as noted above, is procedurally in its infancy and is not reflective of the Companies' actual performance and recent improvements in the areas of gas safety compliance. As SPSP notes, redistributing the basis points at risk appropriately balances the Companies' exposure among the safety performance measures and is consistent with what has been approved for other gas utilities. (SPSP at 62-63). The Gas Regulations Performance Metric would still carry the greatest exposure for NRAs; the highest exposure from any other metric would be 25

basis points. Thus, the Companies' proposal carries substantial weight to continue to drive improvements in this area, and, based on the Companies' recent performance record against this metric, there is no basis to increase penalty exposure.

C. Use of NRA Deferral Balances to Fund Safety Programs

Q. Please explain the Companies' NRA deferral balances.

A. The Companies' current rate plans contain several gas safety metrics to ensure the provision of safe and reliable service to customers. The Companies' performance relative to those metrics is tracked annually, and failure to meet the metrics can result in NRAs, which are deferred for future customer benefit.

Q. How have the Companies historically used NRA deferral balances?

A. The Companies have historically used NRA balances to fund designated safety programs for the benefit of customers. Such treatment is consistent with how other utilities across New York State, including Consolidated Edison and Niagara Mohawk, have used such deferral balances. In the 2016 KEDNY and KEDLI Rate Cases, the Joint Proposal approved by the Commission specifically provides that "[a]ny Gas Safety Performance Metrics negative revenue adjustments incurred by KEDNY [and KEDLI] will be deferred for future use in incremental safety programs." For example, Niagara Mohawk is

currently using NRA deferral balances to fund its Residential Methane Detection program and for GPS mapping of gas facilities.

Q. What are the SPSP's recommendations for the use of KEDNY and KEDLI's NRA deferral balances?

A. The SPSP recommends that the Companies use the NRA deferral balances to fund the Companies' proposed Expanded Residential Methane Detector, I&R Control Line Survey and GPS Mapping, and Low Pressure Valve Installation Programs and, in consultation with Staff, that the Companies submit a plan for using NRA balances to fund additional incremental safety programs. (SPSP at 60-61).

Q. Do the Companies agree with the SPSP's recommendations?

A. As is stated in the Panels' direct testimony at page 14, the Companies support using portions of the deferral balances to fund the Residential Methane Detector Program and the Incremental I&R Control Line Survey and GPS Program. The Companies maintain that, for a one-year rate plan, these programs should remain in rates. The Companies address the Low Pressure Valve Installation Program separately below.

Q. Do any other parties have specific proposals regarding the Companies' current NRA balances?

A. Yes. NYC recommends that the Companies' immediately return their full historic NRA balances to customers with interest, arguing that the balances should be returned to customers because of their age and that NRAs should not be conditioned on specific capital programs. (NYCGIOP at 77).

Q. Do the Companies agree with NYC's position?

A. No. The Companies believe the practice of deferring gas safety related NRAs for future investment in gas safety programs should continue. Use of NRA deferrals for gas safety programs allows the Companies to fund important safety programs for the benefit of the customers and communities served, while also mitigating increased rate requests that would normally accompany such incremental programs. This allows the Companies to enhance the benefits provided to customers without increasing customer bill impacts. Immediate return of the full balance of the NRA deferrals, as NYCGISP suggests, would merely provide a one-time credit on customer bills without providing the funding required for, or benefits of, the incremental gas safety programs proposed.

The NYCGISP testimony also appears to draw a negative inference from the fact that the Companies are not proposing to utilize the full NRA balances at this time. Immediate use of the full balances, however, would be short-sighted as the gas industry, and the safety regulations that govern it, are constantly evolving. The Companies need the ability to react quickly and implement programs in response to these evolving regulatory and customer needs. By leaving a portion of NRA balances in deferrals, the Companies are positioning themselves to respond timely to future regulatory mandates and implement corresponding programming that will directly benefit customers. Therefore, the Companies propose to continue the accounting treatment and use of NRA funds in the manner established in prior rate cases.

IV. Safety Metric Recommendations

A. Leak Prone Pipe Retirement

i. LPP Targets and Prioritization

Q. Does the SPSP agree with the Companies' proposed annual LPP retirement targets and prioritization methodology for retirement of LPP?

A. Yes, the SPSP supports KEDNY's proposal to continue its five-mile per year acceleration of LPP retirement and KEDLI's proposal to maintain its CY 2019 LPP retirement level of 155 miles per year, though the SPSP does not present an opinion on cumulative targets for a multi-year rate plan. (SPSP at 20). The

Companies reiterate that a cumulative target for a multi-year rate plan is appropriate to allow for construction flexibility and efficient use of resources and is aligned with the LPP retirement target metric in place for the Companies' upstate affiliate, Niagara Mohawk.

Moreover, the SPSP supports KEDNY's risk-based method for prioritizing the highest-risk LPP for retirement based on multiple risk factors. (SPSP at 15-16).

Q. Please discuss NYC's recommendation that the Companies accelerate retirement of low pressure LPP that is located in FEMA 500 and 100-year floodplains.

A. NYC recommends that the Companies be required to replace all LPP located in FEMA 500 and 100-year floodplains within 10 years. While NYC expresses opposition to incentives for all other gas safety metrics, including incentives for the general retirement of additional LPP, NYC proposes an incentive to encourage accelerated LPP retirement in floodplains. (NYCGISP at 25-26).

Q. Do the Companies currently replace LPP in floodplains?

A. Yes, as NYC notes, the Companies have made a modification to their main prioritization algorithm, procedure ENG04030 "Identification, Evaluation and Prioritization of Distribution Main Segments for Replacement" that assigns a

higher weight to LPP mains in floodplains. This will help replace the LPP mains in floodplains earlier but will not guarantee completion of these replacements in ten years. In addition, the Companies have identified several pockets of low-pressure LPP in floodplains that the Companies intend to replace to uprate the network to high pressure over the next five years. The Companies also note that all of the LPP and low-pressure main has been replaced west of Bay 32nd Street in the Rockaways, which was an area hit particularly hard during Superstorm Sandy. Therefore, the Companies maintain that increased acceleration to replace LPP in floodplains is unnecessary.

Q. How would further acceleration of LPP retirement in the floodplains impact the Companies' proactive LPP retirement program?

A. To specifically target and accelerate more retirements in the floodplains, the Companies may need to defer or delay the replacement of higher-risk pipe in other regions. This is not recommended. The Companies' acknowledge NYC's concern that water intrusion caused by flooding can result in shutoffs and relights that can take time (NYCGISP at 23); however, the risks of possible failures on riskier segments of pipe experiencing condition issues, or of older vintages than those in floodplains, outweigh re-light concerns. Moreover, the Companies' program to install automatic flood-sensing shutoff valves in the floodplains over the next several years will provide instant information of the

number of customers impacted by flooding that can help inform and make more efficient the Companies' service restoration and re-light processes.

In short, the Companies believe that their existing risk-based prioritization model is the best approach for meeting the goal of retiring all LPP within floodplains.

Q. Is NYC correct in its belief that low pressure LPP retirement in the floodplains could be accelerated and replaced within 10 years at no additional cost?

A. No. NYC reasons incorrectly that the pace of replacements in the floodplains over the past several years would be representative and that the Companies would therefore easily meet NYC's targets within the proposed program budgets. First, some of the Companies' low-pressure main that is located in floodplains are part of extensive low-pressure networks that have no high-pressure mains nearby to facilitate easy conversion and, therefore, would be challenging to address over a shorter period of time from a resource perspective. Second, because the Companies' prioritization model is risk-based and the work plan is established each year for the following year, there is no guarantee that the historic pattern of replacements in floodplains will repeat in future periods. Riskier segments may be more concentrated outside the floodplain in

any given year depending on the totality of circumstances. The Companies would lose efficiencies of coordinating LPP replacements with municipal projects and other company work across the Companies' territory if they were required to target floodplain work over other work, which could increase costs. Third, because of the volume of work that would need to be done in these more concentrated areas, municipal permitting stipulations could slow the pace of the work and require additional resources to enable the Companies to meet targets. For these reasons, the Companies believe that NYC's proposal would require incremental resources.

ii. LPP Incentives

Q. Please explain Staff's position regarding the Companies' proposals to retain the current PRA for LPP removal adopted in the 2016 KEDNY and KEDLI Rate Cases.

A. The SPSP recommends enhancing the NRAs and removing the PRAs for LPP removal. (SPSP at 20-21). The SPP cites Governor Cuomo's newly enacted Climate Leadership and Community Protection Act to support its recommendation, noting that the incentives provided to natural gas utilities in the areas of gas safety and reliability should be "appropriate and produce desired outcomes." (SPP at 35-36). The SPP also cites the increased costs in

the Companies' capital programs associated with LPP removal in support of its recommendation.

Q. Do any other parties make recommendations regarding the Companies' LPP incentives proposals?

A. Yes. NYC opposes the LPP removal PRA, claiming the Companies have not provided justification for rewarding additional main replacement or established "a compelling need" to increase the costs to customers to fund the PRA in this area. (NYCGISP at 53).

Q. Do the Companies agree with the recommendations of the SPSP and NYC?

A. No. The recommendations to remove the LPP removal PRA is entirely counterintuitive to the policy and safety goals discussed throughout the Staff and NYC testimonies.

Q. Please explain why the Companies believe retention of the PRA for LPP removal is appropriate.

A. PRAs are intended to incentivize companies to engage proactively in work that will ultimately benefit customers. It is undisputed that prioritizing LPP removal is Commission policy and required across the Companies' service territories to not only provide a safer, more resilient system for customers, but also advance

clean energy objectives, including methane reduction. Indeed, the SPSP acknowledges that removal of LPP is important to reduce the environmental risks and potential safety hazards created by leaks on the gas system. (SPSP at 13). The SPSP outlines additional benefits of LPP removal that result in cost savings to customers, including reduction in O&M and emergency response costs required to respond to leaks. (SPSP at 15). Despite extensively detailing the array of benefits associated with LPP removal, however, Staff and NYC are unwilling to fund incentives supporting incremental, proactive targets to further advance LPP removal.

This inherent contradiction in Staff and NYC's positions is further highlighted in Staff's discussion regarding complete LPP removal. Current statewide goals target complete elimination of LPP from each natural gas utilities' system. The SPSP notes that, at current removal rates, the Companies' LPP removal programs would be completed by approximately 2038. (SPSP at 19). The SPSP also notes that the "costs to customers for removal in later years will certainly be more than the removal of that same pipe now." (SPSP at 20). If complete LPP removal is truly a focus of all parties, it would seem more prudent and cost effective to proactively address more LPP now, when the costs of LPP removal are less than the inflated rates projected for future years, through PRAs and incentives rather than wait to address LPP removal in subsequent years. The

Companies are already proposing to increase their LPP removal targets by five miles per year and additional incentives to encourage the Companies to stretch beyond those targets will help to further advance LPP removal across the system in furtherance of safety and policy goals.

B. Leak Backlog Metric

i. Leak Backlog Metric Targets and Incentives

Q. What are the SPSP's proposals for modification of the Leak Backlog Metric?

A. The SPSP recommends the following changes to the Companies' proposed Leak Backlog Metric:

- Baseline target: Reduce the baseline leak backlog targets for CY 2020 based on averages of leak repairs for the past five years;
- Annual reduction: Retain flat target reductions each year of 150 leaks for KEDNY and 750 leaks for KEDLI, and remove additional target reductions based on repairs of high-emission Type-3 leaks;
- Workable leaks: Reduce annual target from 25 to 20 each year;
- Year-end target range: Set the year-end target range to December 26th – December 31st (in lieu of Companies' proposal of December 21st – December 31st); and

- Incentives: Ability to earn PRAs is based on total leak repairs, removing existing incentive for repairing high-emitter Type-3 leaks.

(SPSP at 28-30).

Q. Do the Companies agree with the SPSP's proposals?

A. No. The Companies believe the Leak Backlog Metrics they have proposed are challenging and strike the correct balance to drive reductions of the overall leak backlog while strategically targeting the highest emission Type-3 leaks to maximize methane emissions reductions from this program. The Companies respond specifically as follows:

- Baseline targets: The current Leak Backlog metric reduces the targets by 150 and 750 leaks each year for KEDNY and KEDLI, respectively. It is reasonable, therefore, to set CY 2020 targets based on a 150 and 750 reduction from the 2019 targets as the Companies have proposed. Leak repair is reactive and unpredictable. Even as LPP is replaced with new main, the LPP that remains on the system is aging, and leaks are continuing to rise in KEDNY's service territory. The Companies' proposal is appropriately challenging, especially given the additional downward adjustment annually based on high-emitter Type-3 leak repairs.

- Annual reduction: The Companies support a fixed annual reduction. The proposal to further reduce the targets each year based on high-emitter Type-3 leak repairs was intended to work in conjunction with the Companies' proposed incentive structure, so that as the Companies earned incentives for repairing high-emitter Type-3 leaks, the metric would become more challenging the following year.
- Workable leaks: Past year-end backlogs are not necessarily representative of workable leaks that might arise. Workable leaks typically require more time and resources to repair, and, if too many arose near the end of the year, the Companies could struggle to meet a lower metric. The existing target of 25 leaks per year is challenging and reasonable.
- Year-end target range: The SPSP gives no justification for reducing the Companies' proposed 11-day target window to a 6-day target window. Again, new leaks can arise at any time and increase as the weather becomes colder. The Companies' leak repair performance could be excellent for nearly a year, yet the Companies would be at risk of missing their targets if too many leaks arise in the last month of the year without sufficient time to make repairs. Moreover, the Commission accepted a December 21st – December 31st target period recently for

Orange and Rockland Utilities, Inc. ("Orange and Rockland") in Case 18-G-0068.

- Incentives: The Companies strongly believe that the Leak Backlog Metric should continue to incentivize and provide cost recovery for repair of high-emitter Type-3 leaks. High-emission Type-3 leaks are typically harder to access and more expensive to repair but can have a more significant impact toward reducing overall methane emissions than repair of lower emitting non-hazardous leaks. The Companies' proposal works in conjunction with its proposal for an Enhanced High Emitter Methane Detection Program that the SPSP supports and that will deploy advanced leak detection technology to help the Companies identify the highest-emitting leaks. The Companies also believe that incentives for high-emitter Type-3 leaks as part of the Leak Backlog Metric help to address EDF's concerns about the structure of a backlog reduction metric.

Q. Should re-checks of Type-3 leaks be required to count Type-3 leak repairs in the metric?

A. No. As the SPSP notes in its testimony, although 16 NYCRR §255.819(a) requires a re-check inspection for Types 1, 2, or 2A leaks, there is no follow-up inspection requirement for Type-3 leaks. The SPSP has provided no

justification for requiring a follow-up inspection for Type-3 leak repairs, and this requirement would require the Companies to add considerable O&M resources that are not currently reflected in the proposed revenue requirements. Requiring deployment of crews back to every Type-3 leak repair adds costs without commensurate benefits.

Q. Do the Companies agree with NYC's proposal to eliminate the entire leak backlog in five years?

A. No. The Commission's safety regulations include deadlines for repairing Type 1, 2, and 2A leaks. The Companies re-iterate that Type-3 leaks are defined by regulation as non-hazardous and are reasonably expected to remain so. There are not sufficient benefits for removing all non-hazardous NYC leaks to justify the significant additional resources that would be required. NYC raises concerns regarding the costs of the Companies' proposed LPP retirement program; however, retirement of LPP is a far more effective strategy for reducing leaks, increasing public safety, and reducing methane emissions than repair of non-hazardous leaks. Yet NYC proposes acceleration of leak repair that would raise costs for customers without providing the benefits of the LPP retirement program.

Q. Please summarize EDF's proposal to change the leak backlog metric to an emissions reduction metric.

A. EDF's proposal is to change fundamentally the structure of the leak management from a leak backlog measure to an overall methane reduction measure. EDF recommends that the Companies use advanced leak detection technology (essentially the cavity ring-down spectroscopy ("CRDS") technology mounted on a Google car) to conduct a baseline survey of the Companies' entire service territory to establish a baseline methane emissions level and then to establish a target, such as a 50 percent reduction of emissions, to be achieved over a fixed period, such as five years. EDF acknowledges that a baseline survey could take time and recommends that the existing metric structure stay in place until such time as the new metric could be developed. (EDF at 30-35).

Q. Do the Companies agree with EDF's proposal?

A. No, not at this time. The Companies are prioritizing overall methane reductions, as is demonstrated by the Future of Heat Panel's testimony, their commitments to retire their remaining LPP inventory, and enhanced identification and targeting of high emission Type-3 leak repairs.

The Companies maintain, however, that the advanced leak detection technology is not yet sufficiently reliable to form the basis for a new metric for reasons that are discussed in further detail in section B(2) below. The Companies' agree that they should *begin* to integrate CRDS technology into their arsenal of tools to combat methane emission, via their proposed Enhanced High Emitter Methane Detection Program. This program provides for deployment of Google car-mounted CRDS technology operated by contractors in each of KEDNY's and KEDLI's service territories to prioritize high emission leaks for repair in limited areas already know to have a high leak concentration. Even if the Companies believed that the technology could produce a reliable baseline measure, to achieve a full baseline survey of the Companies entire service areas that EDF proposes would require significantly more contractor resources to cover each territory and to conduct multiple passes required to obtain a sufficient data set, plus additional incremental Company full-time equivalents ("FTEs") to process and analyze data, and manage the program. Given all factors including the limitations of the technology and costs, the Companies believe a more measured approach during the Rate Year and Data Years is more reasonable than rushing toward a new metric. The Companies' proposed Enhanced High Emitter Methane Detection Program supports use of the Google Car/CRDS technology in a more reasonable way and can provide an opportunity to further evaluate this technology.

ii. Leak Survey Techniques

Q. Can advanced leak detection technology such as CRDS replace the Companies' existing leak survey methods?

A. Not at this time. Importantly, the Commission has not approved CRDS or any other advanced leak detection technology for use to conduct required leak surveys. At best, from a regulatory standpoint, the Companies can begin to deploy these technologies to supplement and enhance their traditional programs, as the Companies have proposed via their Enhanced High Emitter Methane Detection Program.

Additionally, although the Companies acknowledge that advancements have been made and utilities are beginning to integrate these technologies, the Companies' findings during recent pilots of the technology did not result in a high level of confidence in their accuracy. Indeed, EDF's testimony notes one of the flaws that the Companies identified – that the number of leaks detected by the technology can vary greatly in the same location depending on the weather. (EDF at 15). The Companies' findings are set forth in detail in the Niagara Mohawk's Methane Reduction Efforts Report for 2018 that was submitted on March 15, 2019, in Case 17-G-0239. As is stated above in response to EDF's metric proposal, and in the Companies' response to IR UIU-51, part (e), a copy of which is included in Exhibit __(GSP-1R), the Companies

have proposed to expand use of these technologies, but they are not a substitute for traditional leak survey.

Q. Do the Companies agree with NYC's recommendation that the Companies should adopt the same leak survey techniques and frequencies in their New York City service areas as Consolidated Edison Company of New York, Inc. ("Con Edison") deploys?

A. No. Con Edison has chosen to deploy more frequent walking surveys and monthly low-speed surveys that exceed regulatory leak survey requirements. KEDNY and KEDLI have also deployed strategic leak survey techniques that exceed regulatory requirements to enhance customer safety, but in a different manner than Con Edison. For example, the Companies have an enhanced leak survey procedure for the winter months when leaks are more likely to occur. The procedure includes:

- more frequent patrols of all cast iron distribution mains (an average of 14 days and not to exceed 20 days) beginning in either late December or early January regionally depending on weather patterns and cast-iron break history; and
- additional surveillance when the seven-day frost degree days (daily average temp over seven consecutive days) is below 27 degrees Fahrenheit.

The fact is, Con Edison, KEDNY, KEDLI, and other utilities across the state deploy various and differing safety enhancement programs. Although the utilities in the state often collaborate and share best practices through organizations like the Northeast Gas Association ("NGA"), to *require* uniformity across neighboring service territories for programs that exceed regulatory requirements would be burdensome and costly and it would disincentivize innovation and chill the pursuit of discretionary safety enhancements.

C. <u>Damage Prevention Metric</u>

Q. Setting aside the SPSP's misunderstanding of how the Companies' current Damage Prevention Performance Metric is calculated, what are the SPSP's recommendations for adjusting the metric targets?

A. The SPSP supports the Companies' proposal to revise the metrics to measure overall damages rather than splitting the metric into three categories. (SPSP at 41). The SPSP recommends adjusting the targets to levels that are higher than the targets the Companies proposed for both KEDNY and KEDLI. The SPSP's targets are also uniform for KEDNY and KEDLI, whereas the Companies had proposed generally lower, more challenging targets for KEDNY. (SPSP at 42-43).

Q. What is the Companies' recommendation for the Damage Prevention Performance Metric?

A. The Companies recommend retaining the current metric structure and calculation methodology, which includes relocates and excludes damages that have always been excluded from the Companies' Damage Prevention Metric – damages by homeowners with no ticket and damages by non-company contractors digging with hand tools with no ticket – with the more challenging targets proposed by the Companies. This is the most balanced measure of damage prevention results that exposes the Companies to NRAs.

Although the Companies acknowledge that they are able to reduce instances of third party (non-company or company contractor excavators digging with hand tools) damages, there will always be some third parties who will excavate without calling 811 to generate a ticket. It is useful to review data regarding the number of damages occurring without a ticket to inform improvements to the State one-call systems and programs and drive improvements to external outreach, which is why the Companies have agreed in the past to provide this data for analysis purposes. It is unfair, however, to penalize the Companies financially for the actions of third parties that are outside the direct control of the Companies.

Additionally, the Companies should receive credit for relocates because relocates are treated like a new ticket for purposes of Companies' responses (as if the first ticket was never issued). Indeed, relocates are generated by the same requesting party at the same location, but could be generated months after the first ticket was issued. For the downstate New York One-Call Notification System, a "relocate" occurs when the One-Call Center issues a ticket and KEDNY and KEDLI respond and mark-out their facilities. Thereafter, the excavator contacts the One-Call Center to issue a second ticket, or a "relocate," for KEDNY and KEDLI to re-mark their facilities (*e.g.*, where the excavation is delayed to the point that the original mark-out is no longer reliable or valid (*i.e.*, beyond the ten-day limit) or conditions such as weather or construction cause the original markout to fade or disappear). Although the ticket has the same requesting party and location, this is not a "retransmit" because KEDNY and KEDLI must respond again and relocate and re-mark their facilities as if they had not been located and marked in the first instance. Indeed, from a safety standpoint, treating relocates as new tickets is a best practice, and exclusion of these from the metrics could create a perverse incentive for the Companies to focus less on relocate responses. Because the Companies' response to relocates is the same as for all other tickets, and, to encourage further improvements in damage prevention, they should continue to be counted in the Companies' metric.

Q. Would excluding relocates from the KEDNY's and KEDLI's Damage Prevention Performance Metric help to drive uniformity in how the metric is measured in New York?

A. No. The two One-Call Centers in New York, Dig Safely New York (covering upstate New York), and New York 811 (covering downstate New York) do not have consistent rules and classify tickets differently. A "relocate" ticket in upstate New York is not the same as in downstate. Analysis of statewide data shows that "relocates" make up more than 50 percent of the downstate ticket volume, and yet, none of the upstate utilities come anywhere close to that same percentage, which demonstrates that this is not an apples-to-apples comparison. Therefore, the SPSP's recommendations to attempt to treat relocates the same statewide for purposes of performance metrics is flawed and unfair as there is no standardized way the two One-Call Centers receive, classify, and then send the 811 tickets to their members.

KEDNY and KEDLI should continue to be judged by how they improve year-over-year from their own previous performance, not by how the Companies perform compared to other gas operators in New York. The only fair way to continue to do that and apply smart, measurable, and attainable performance measure goals for the Rate Year and Data Years is to continue to allow the

Companies to follow the same metric calculation they have for over a decade, but make the targets for that metric more challenging. This is precisely what the Companies have done in their metric proposals.

Q. If the Companies adopted Staff's proposed changes to the damage prevention metric, would the Companies require additional resources to meet the targets?

A. Yes. The amounts included in the Companies' rate plan, including the Companies' forecasts for the Enhanced Damage Prevention Program that is described on pages 31-34 of the Panel's direct testimony, assume the current metric structure. If the structure were to change, the Companies would need to devote additional incremental resources to these programs to meet Staff's proposed targets.

D. <u>Emergency Response Metric</u>

Q. Does the SPSP support the Companies' proposed Emergency Response Metric and the addition of incentives to the Metric?

A. Yes. The SPSP supports the Companies' proposal to maintain the existing Emergency Response targets, including the exclusion of mass odor calls from the metric, and to add incentives to the metric. The SPSP notes that incentives for emergency response have been approved for other utilities including Niagara

Mohawk, Central Hudson Gas and Electric Corporation, and Orange and Rockland. (SPSP at 50-51).

Q. Does NYC present alternative recommendation for the Emergency Response Metric?

A. Yes. NYC recommends an alternate target of 80 percent for the 30-minute response (a five percent increase in the target). NYC's basis for this adjustment is that Con Edison's emergency response 30-minute response performance has exceeded KEDNY's and KEDLI's in the past several years, and the Companies should be held to the standard of actual performance that Con Edison has set. (NYCGISP at 56-57).

Q. Do the Companies agree with NYC's alternate metric for the 30-minute response?

A. No. The Companies' Emergency Response Metric targets are aligned with Niagara Mohawk's and other New York gas utilities' targets – the targets are uniform statewide. It is inappropriate to continuously ratchet up gas safety metric targets based on performance that exceeds targets. This would create a perverse incentive for utilities to aim only to meet targets to avoid greater exposure to NRAs. The Companies' Emergency Response Metric Target is reasonable and appropriately challenging.

Q. Is NYC correct in its belief that the Companies could achieve a higher target for the 30-minute response without the need for additional funding?

A. No. Emergency response is reactive and is driven by leaks that are unpredictable from year to year. To ensure that the Companies were able to meet stricter targets, the Companies would need to add resources.

Q. Does NYC provide any other recommendations regarding the Emergency Response Metric?

A. Yes. NYC recommends that the Commission examine whether the metric should be measured based on when company personnel who are qualified to actually make a repair without calling for additional assistance arrives on the scene, rather than from when a company First Responder arrives on the scene. (NYCGISP at 58-59).

Q. Please comment on this recommendation.

A. The Commission has already determined that, for the purpose of reporting on the Emergency Response Metric, the response times are measured from the time the notification is sent from the company dispatch to the time of arrival of "qualified company personnel" at the location, and that "qualified company personnel" are defined as "company representatives who are properly trained and equipped to investigate leak, odor, and emergency reports in accordance

with accepted company procedures and 16 N.Y.C.R.R. §255.604; operator qualification requirements." *In the Matter of Staff's Analysis of Local Distribution Company Performance Related to the Pipeline Safety Measures*, Case 19-G-0298, 2018 Pipeline Safety Performance Measures Report at 24 n10 (June 13, 2019) (underlined emphasis added). The Companies' First Responders are Customer Meter Services ("CMS") operator qualified technicians who meet these requirements for emergency response.

If the metric was changed to measure from the time company personnel who are trained and equipped <u>to repair</u> leaks arrives on the scene, as NYC suggests, the Companies would be required to dispatch a Gas Field Operations crew to respond to every reported leak. This would require significant incremental resources and is inefficient because most reported leaks do not actually require an immediate action. For example, in CY 2018 and CY 2019 thus far, only approximately 15 percent of total leaks reported were reported as an odor outside. Of the total leaks called in, only approximately 0.41 percent resulted in a classification of a Type-1 leak requiring an immediate action.

E. <u>Gas Regulations Performance Metric</u>

Q. Does the SPSP support the Companies' proposal that the NRAs for this metric be spread among larger tiers?

A. The SPSP does not support this proposal and instead recommends that the metric's tiers remain as they are currently established. (SPSP at 63-64).

Q. Do the Companies agree with the SPSP's recommendation for the metric tiers?

A. No. Based on the Companies' record of improved performance, and because the broader tiers the Companies are proposing were recently adopted for Niagara Mohawk's Gas Regulations Performance Metric, the Companies' proposal should be accepted.

Q. Do the Companies agree with the SPSP's recommendation to remove the existing 10-violation cap for violations of a single code section identified during field audits?

A. No. Currently, the Companies' Gas Safety Regulations Performance Metric includes a cap of 10 violations of a single code section for both records audit violations and field audit violations. Removal of the cap for field audits is overly punitive, not justified based on the Companies' record of performance against this metric, and unnecessary in light of the implementation plan requirements for violations that exceed the cap. The SPSP states its basis for its recommendation is that it expects that the Companies' performance should be at its best while being observed; therefore, any violation committed in front of

Pipeline Safety Staff could indicate a lack of competency in performing the action or task. (SPSP at 67). The SPSP's statement amounts to a pre-judgment on the root cause of potential violations before they even occur. The 10-violation cap was agreed-to in the Companies' Joint Proposal in the 2016 KEDNY and KEDLI Rate Cases to ensure that the metric did not create a scenario where violations with a high number of occurrences could result in an overly punitive penalty without focusing attention properly on the underlying compliance issue. To re-balance the metric, a provision was included that requires the Companies to conduct a root cause analysis for violations that exceed the cap and file an implementation plan and to develop a mitigation plan that is acceptable to Staff. The Companies' recommendation is to continue this mechanism in the metric for both field and records audits, rather than removing the 10-violation cap for field audits. This mechanism allows for the fair evaluation of the root cause of field violations and a mechanism to mitigate root causes yet still provides a strong incentive to achieve full compliance with gas safety regulations.

V. Proposed Adjustments to the Companies' Incremental Safety Programs

A. Plastic Fusions QA/QC Re-Digs Program

Q. Please describe the SGIOP's recommended adjustment to the Plastic Fusions QA/QC Re-Digs Program.

A. The SGIOP recommends a downward adjustment to the capital forecast for this program in the amount of approximately $0.25 million for KEDNY and $0.31 million for KEDLI. Staff notes that the Companies' response to IR DPS-711 indicates that the Companies' forecast is based on the estimated number of incremental inspections times the unit price per inspection. The SGIOP does not question the Companies' estimated number of inspections or unit prices, but states that the total forecast includes additional costs that were not justified, which Staff proposes to remove.

Q. Do the Companies agree with the adjustment?

A. No. The Companies inadvertently neglected to include the basis for the entire forecast in their response to IR DPS-711. The forecast is based on the number of inspections times the unit price, plus the capital labor required to conduct the inspections that is not included in the unit price. The capital labor portion of the forecast is what would be removed by the SGIOP's adjustment. The labor portion of the forecast is required to enable the Companies to meet their regulatory obligations to conduct additional inspections of plastic fusions; therefore, the capital forecast for this program should remain as proposed by the Companies.

B. Inside Service Line Inspection Program

Q. Does the SPSP make recommendations for the Companies' Inside Service Line Inspection Program?

A. Yes, to help manage costs, the SPSP recommends that (i) the Companies coordinate inspection work with other work when possible, (ii) automatically impose non-access penalties upon one refusal or two missed appointments, and (iii) use revenues collected through non-access penalties to offset program costs.

Q. Do the Companies agree with the SPSP's recommendations?

A. Yes, in part. The Companies continuously seek opportunities to coordinate work and to perform work more efficiently across programs. The Companies' also have incorporated the non-access penalty for one refusal or two missed appointments into its tariff and into its O&M procedures for inside service line inspections. It is not clear what the SPSP means by "automatic" assessment of a penalty; however, the Companies intend to utilize the penalty effectively as a tool to gain access for inspections. The process for assessing the penalty should ensure that the Companies have adequate records of one refusal or two missed appointments and that customers have been provided adequate notice of the penalty before it is assessed. The Companies will consider a mechanism for using penalty revenues to offset the costs of the Inside Service Line Inspection Program.

C. Enhanced Contractor Inspection Program

Q. Please address the SPSP's recommendation regarding the Companies' proposed Enhanced Contractor Inspection Program.

A. The Enhanced Contractor Inspection Program adds inspectors and a small amount of associated capital labor so that the ratio of inspectors to contractor crews will be one-to-one. The SPSP (at 91) supports the program but recommends a reduction to the Companies' proposed number of FTEs to reflect the estimated workload. The Companies do not agree with the SPSP's five percent reduction in the FTEs associated with the program and believe the number of FTEs included in its rebuttal filing are necessary to maintain the one-to-one ratio of inspectors to contractor crews in the Rate Year and Data Years. Notwithstanding the foregoing, the Companies' rebuttal filing has been adjusted to reflect the assumption that the NESE project will not be completed within the timeframe of the rate plan in this case. As a result, the Companies have reduced the number of FTEs required for the Enhanced Contractor Inspection Program by 0.4 FTEs for KEDNY and 3.0 FTEs for KEDLI. Modification to the number of FTEs required for this program has a flow-through impact on both the operating and capital expense portions of the program.

Additionally, the Companies note that the adjustments to the Companies' FTE proposal for the program in Staff's exhibits do not accurately reflect a five

percent reduction. For example, Staff's Exhibit___(SRRP-2) reflects a reduction of two FTEs for KEDNY when a five percent reduction would be equal to 1.25 FTEs based on KEDNY's Correction and Updates filing. Should any FTE adjustments for the program be recommended in this proceeding, the corresponding impact on operating expense and capital expense should be recalculated to reflect the adjustments appropriately.

D. Enhanced Inactive Accounts Program

Q. Please address the SPSP's comments regarding response to inactive accounts.

A. The SPSP notes that inactive accounts should be addressed immediately and that the Companies should also address meters found not to be readily accessible. As was stated in the Panel's direct testimony, under the Companies' current inactive accounts process, the Companies act immediately to perform a site analysis on every site to determine whether conditions, such as a vacant premises or other observable unsafe condition, warrant immediately securing the line. In such instances, the service is cut within 24 hours. Assuming there is not an urgent circumstance, the process is designed to resolve inactive accounts within 60 days. In practice, however, 47 percent of KEDNY's inactive accounts and 89 percent of KEDLI's inactive accounts are resolved within the first 15 days, and only approximately 5 percent of KEDNY's and 1.6 percent of KEDLI's

accounts remain inactive between days 45 and 60 of the current process. The Companies' proposal to achieve a 30-day process by the end of Data Year 2 will expedite the entire process.

Regarding meter locations, at the time of original installation, the Companies install meters in readily accessible locations (*i.e.*, locations not obstructed and able to be accessed to perform inspections and maintenance), regardless of whether the meter is physically located on the outside or inside of a customer's building. Over the past several years, the Companies have discovered meters that are obstructed or inaccessible because of subsequent customer alterations or obstructions installed around the meter. The Companies' current practice is to advise customers of obstructions and access issues upon discovery. If required to resolve an imminent safety concern, the Companies would terminate service. Outside of imminent safety conditions, however, the Companies have encountered several impediments that prevent or delay immediate removal of the obstructions, including customer refusals, local building codes and regulations, exterior or interior obstacles, and other various safety reasons. These challenges are further compounded in the City of New York, where KEDNY has been unable to relocate any meters outside of the building because of code interpretation issues with the NYC Department of Buildings ("DOB"). Accordingly, although the Companies will continue their efforts to remove

obstructions and overcome the impediments noted above as quickly and efficiently as possible, Staff's recommendation that the Companies take "immediate and continuous" action to remedy these conditions is simply unachievable for reasons beyond the Companies' control. The Companies acknowledge that improvements in this area will require the involvement of third parties, including municipalities, and invite affected stakeholders to provide comments and work with the Companies towards effective resolutions.

E. <u>Buried Vent Lines Program</u>

Q. What is the SGIOP's recommendation regarding the Companies' proposed program to remove or remediate buried vent lines?

A. The SGIOP recommends that the Buried Vent Lines Program should be removed from the Companies' capital plans and should instead be funded by the Companies' shareholders because the SGIOP believes the Companies should already have addressed buried vent lines. (SGIOP at 95).

Q. Do the Companies agree with the SGIOP's proposed adjustment?

A. No. The Companies have identified and addressed buried vent lines for years during the regular course of operations and maintenance. In the past, this work was done in connection with various programs, such as regulator inspections and meter changes, and procedures for ensuring the safe operation of buried vent

lines and performing corrective actions when necessary are included in the Companies' CMS procedures. Since the Companies began conducting inside service line inspections in 2017, the pace of identification of buried vent lines has increased such that the Companies have proposed a distinct capital program with centralized funding to enable the Companies to timely address these issues. This program is properly funded through rates.

F. <u>Low Pressure Main Valve Installation Program</u>

Q. Do the Companies agree with the SGIOP's proposal for the Low-Pressure Main Valves Installation Program to be funded through NRAs rather than through rates?

A. No. This program is an incremental safety program that will be ongoing because the valves will be installed on new services as well as on existing main. This is an ongoing program with long-term benefits to customers that justify inclusion in rates. There will also be future investment required to ensure the valves remain operational over time. The costs of this program and the nature of this program render it inappropriate to fund through the limited pool of NRA dollars that are available. This program should be funded through rates, and NRAs should be reserved for shorter-term programs that do not require ongoing spending.

G. Expanded Residential Methane Detector ("RMD") Program

Q. Please respond to the SPSP's and NYC's recommendations that the Companies consider development of advanced RMDs.

A. The Companies support the development of utility detection devices with advanced technologies for residential methane detection, similar to the Con Edison device, that can automatically communicate information, including methane emissions, to the Companies' systems. The Companies note that Con Edison's device is not compatible with the Companies' existing network and that there is no device currently on the market that would be compatible. Notwithstanding, the Companies have ongoing research and development ("R&D") efforts underway and actively communicate with all major manufacturers regarding potential development of a product that meets the Companies' technical specifications.

The Companies also support the advancement and improvement of RMDs for customers' personal use, similar to smoke detectors, of the type that the Companies plan to distribute via the proposed Expanded Residential Methane Detection Program. The Companies' efforts are described in their responses to IR DPS-671, a copy of which is included in Exhibit __(GSP-1R), and in KEDLI's Settlement Agreement Implementation Update report that was filed on August 22, 2019, in Cases 14-G-0058 and 14-G-0060.

The Companies intend to continue these efforts during the Rate Year and Data Years. The Companies note that Staff's proposed reduction of the Companies' incremental FTE to support R&D efforts that is noted in the GIOP's rebuttal testimony is incompatible with the SPSP's recommendation to dedicate additional R&D resources toward the advancement of RMDs.

Q. Do the Companies agree that NYC's recommendations that the Companies should install RMDs and provide enhanced tracking and support to customers as part of its Expanded Residential Methane Detection Program?

A. No. The Companies' Expanded Residential Methane Detection Program is a program to put commercially-available consumer RMD devices in the hands of customers for free for their own personal use. The devices are simple for customers to install and to operate, similar to common smoke detectors and carbon monoxide detectors that most customers already have in their homes. As is shown response to IR CNY-75, a copy of which is included in Exhibit __(GSP-1R), Niagara Mohawk's similar program tracks which customers have been provided with RMDs, and there have thus far been no odor calls that have originated from a Niagara Mohawk customer who was provided with a device. The Companies intend to similarly track which customers receive RMDs and can use this data in conjunction with leak calls to track whether leak calls are

reported from customers who have an RMD device. As noted in the response, Niagara Mohawk is tracking this type of information and, to date, has received no odor or leak calls resulting from an RMD alarm. Installation and support for using these simple devices would be an inefficient use of Companies' resources that should instead be dedicated toward the advancement of advanced technologies for advanced utility methane detection devices and advanced consumer RMDs.

H. <u>Enhanced High Emitter Methane Detection Program</u>

Q. What are the SPSP's recommendations regarding the Companies' proposed Enhanced High Emitter Methane Detection Program?

A. The SPSP supports the Companies' program, which will deploy a Google car mounted with CRDS technology to prioritize repair of high emission leaks in high leak concentration areas. The SPSP describes the program as a "pilot program" and recommends that the program be funded using the Companies' balance of NRAs. The SPSP does not support the Companies' proposed addition of one FTE to support this program. (SPSP at 87-88).

Q. Please comment on the SPSP's recommendations.

A. The Companies had not intended this program as a pilot program. The Companies' proposal was for an ongoing program that should be funded in rates.

The program is not a leak detection program but is a supplement to the Companies' leak management program that will help the Companies prioritize repair of high emission leaks in areas known to have a high leak concentration. The Companies, however, agree that part of the purpose of the program is to continue to test and evaluate the technology to explore possible future uses and expanded integration into the Companies' leak management programs. As such, the Companies could consider moving forward with this program as a pilot program in a multi-year rate plan. As is stated above, the Companies believe that its incremental safety programs for a one-year case should be funded in rates.

Q. Does EDF make recommendations regarding the Enhanced High Emitter Methane Detection Program?

A. Yes. EDF recommends that this program be expanded to incorporate a complete emissions survey of both Companies' entire service territory. For the same reasons stated above regarding EDF's recommendations for revision of the Leak Backlog metric, the Companies do not believe that expanded deployment of the CRDS technology is appropriate at this time but will continue to work with EDF on the advancement and evaluation of this technology.

VI. Other Recommendations

A. Gas Safety Reporting

Q. Do the Companies support the SPSP's recommendations for capital reporting?

A. For the purposes of a one-year rate plan, the Companies support capital reporting that is aligned with the reporting structure adopted in the 2016 KEDNY and KEDLI Rate Cases. The Companies are concerned that the enhanced reporting recommended for LPP retirements may be burdensome to prepare without commensurate benefits for enabling evaluation of the program; however, the Companies are open to discussing modifications to existing reporting requirements for a multi-year rate plan.

B. Local Law 152

Q. Does the SPSP discuss Local Law 152 ("LL152")?

A. Yes. As stated by the SPSP, LL152 is a New York City Council law that requires building owners in New York City to conduct inspections including atmospheric corrosion and leak surveys of the natural gas piping inside buildings, including the same segments that the Companies are required to inspect pursuant the inside service line definition change. (SPSP at 76).

The SPSP is correct that New York gas LDCs had hoped to coordinate LL152 inspections with utility inside service line inspections to eliminate overlapping inspections of the same facilities and reduce the burden and costs for both building owners and utility customers. The Companies, Con Edison, and the NGA worked with the DOB for nearly a year prior to issuance of the proposed rules implementing the law to provide guidance and information on how coordination could be achieved. When the DOB issued its proposed rules on November 30, 2018, the Companies were disappointed that the rules were not written in a manner that would allow for coordination of inspections to meet both the utilities and LL152 requirements, and the Companies filed joint comments with Con Edison in the rulemaking proceeding.

Q. What are the SPSP's recommendations regarding LL152?

A. The SPSP has three recommendations: (i) that the Companies continue to coordinate with DOB to encourage modifications to the rule that would enable coordinated inspections; (ii) if the rules are issued without modification, that the Companies file a plan within 60 days of the issuance of DOB's final rule or within 60 days of the Commission's Order in these proceedings detailing how the Companies will coordinate with building owners to encourage them to use plumbers qualified to perform the utility's inspections, including technology sharing and an educational campaign; and (iii) that the amounts the Companies

can collect in rates should be a "placeholder" with a reconciliation to a new estimate once the final rules are issued.

Q. Please comment on the SPSP's recommendations.

A. The Companies are willing to continue to work with DOB to attempt to align LL152 with the utilities' service line inspection requirements; however, it should not be assumed that these efforts will be successful. The Companies do not agree with the remaining recommendations. The SPSP's recommendations hinge on the concept that building owners, if properly educated and provided with data recording tools, could voluntarily hire plumbers qualified to conduct the inspections on behalf of the utilities so that the utilities could count the inspections as completed for their regulatory obligations. The misalignment of the inspection cycle times (LL152 rules are every five years but the utilities are one- and three-year cycles), however, and the borough-by-borough approach in the LL152 rules as proposed, would render even voluntary coordination impossible. There are also other impediments caused by the rules as proposed including the lack of a requirement to inspect room set meter locations. Even if the cycle times aligned, it would be difficult for the Companies to manage their compliance program, resource needs, etc. around an unknown portion of inspections occurring voluntarily in any given year. The Companies remain hopeful that their comments in the rulemaking proceeding will be given due

consideration and that the final rules adopted will allow for coordination; however, in the event they are passed as proposed, the Companies strongly believe that the most cost-efficient way to proceed for the Companies' customers is for the Companies to manage their required inside service line inspections independently.

Q. Do the Companies agree that the amounts collected in rates for inside service line inspections should be temporary pending issuance of DOB's final rules?

A. No. The Companies' Inside Service Line Inspection Program for the Rate Year will be conducted independently by the Companies. In the event that the LL152 rules are passed with sufficient modifications to allow for coordination of inspections, it will take the Companies time to develop coordination plans that ensure the Companies can meet their regulatory requirements. The Companies would not anticipate realization of coordination benefits in the Rate Year, but this could be covered by an exogenous cost provision.

C. <u>Inside Meter Relocations</u>

Q. Please explain the inside meter relocation practices that were established in the 2016 KEDNY and KEDLI Rate Cases and the SPSP's recommendations regarding relocation of inside meters going forward.

A. Pursuant to the Joint Proposal adopted in the 2016 KEDNY and KEDLI Rate Case, beginning in CY 2018, the Companies were to begin to relocate gas meters located inside premises to outside the premises when replacing or relocating gas service lines as part of the LPP retirement program, except where (i) the customer refuses permission to relocate the meter outside; (ii) local building codes or similar regulations prohibit outside meters; (iii) exterior or interior obstacles prevent meter relocation; or (iv) the gas meters should not be moved outside for safety reasons. The SPSP recommends that the Companies continue to relocate inside meters to outside and expand their efforts to perform this work not only in coordination with the LPP replacement programs, but also whenever the Companies perform service transfers or move an existing service line to a newly installed pipeline. (SPSP at 82).

Q. Can the Companies support continued and/or expanded inside meter relocations?

A. The Companies agree with the safety benefits of relocating meters from inside to outside. Meter relocations would also eventually reduce costs of performing required inspections of inside service lines. At this time, however, KEDNY is not able to make specific commitments regarding relocating meters from inside to outside for reasons that are explained in more detail below. KEDLI supports continued inside meter relocations in coordination with the LPP replacement

program, assuming the existing exceptions would still apply. In the event of a multi-year rate plan, KEDLI is open to considering expansion of these efforts to other work where KEDLI is replacing the service from the building to the curb. KEDLI notes that it would not be appropriate to require meter relocations to be coordinated with service replacement work from the main to the curb.

Q. Please explain why KEDNY is unable to commit to meter relocations at this time.

A. KEDNY's meter relocation strategy assumed that it could perform this work using its crews that were on-site performing the main and service replacements using its standard processes and procedures for replacing meters. This is the process followed by KEDLI in its service territory. Early in CY 2018, KEDNY became aware that the DOB considered part of the work that is required to relocate the meters–specifically, the replacement of a small section of pipe inside the premises where the meter was located to the new connection with the meter outside the building–to be interior house pipe work falling under the requirements of the New York City Building Code and Fuel Gas Code ("Codes"). These requirements are varied and complex; some examples include:

- three-pound pressure testing beyond the installation to the entire premises that could compromise internal piping that otherwise

operates safely at lower usage pressures, particularly on older buildings common in NYC ;

- multiple inspections by DOB including a close out inspection such that the work could not be completed in one day and would require customers to be out of service multiple days or longer; and
- work could not be performed by KEDNY's crews, requiring additional resources to contract for master plumber services and would require coordination of KEDNY's service line replacement crews with additional plumber crews.

Taken together, the Code requirements prevent KEDNY from performing the meter relocation work.

During 2018 and to date, KEDNY has suspended its meter relocation efforts to pursue ongoing discussions with the NYC Mayor's Office staff and the DOB regarding whether there is a path forward that would enable KEDNY to perform this work. KEDNY has also updated Staff on this matter throughout the past year.

Q. What is KEDNY's recommendation regarding meter relocations?

A. KEDNY will continue ongoing discussions with DOB; however, at this time there is no clear path forward such that KEDNY can commit to meter

relocations. At best, KEDNY can look for opportunities to relocate meters in coordination with customer-planned work on inside house piping where the customer would be responsible for the replacement of the house-piping on the inside of the building where the meter is removed.

The Companies also note that KEDLI's ability to continue relocations outside of New York City may be similarly impacted by municipal building codes as the program moves forward.

D. SEP's Concerns Regarding Radon

Q. Please address SEP's recommendation that the Companies should provide radon testing for its customers.

A. The Companies agree that customer safety is paramount; however, SEP has presented no credible evidence to suggest that the gas transported by the Companies causes harmful radon emissions when burned inside homes for heating, cooking, and appliances. Respectfully, SEP's witnesses are not scientists or chemical engineers, and SEP has not presented testimony by any witnesses who are experts in the fields of chemical engineering, radon, or the composition of natural gas. SEP cites only to a publicly-available article that discusses increasing radon levels in Pennsylvania, where fracking occurs, to support its belief that there could be unacceptable levels of radon emitting from

natural gas far from fracking sites. Absent compelling evidence that radon emissions should be addressed by New York utilities, the Companies cannot support spending customer dollars to address an unproven risk.

Moreover, the Companies adhere to all currently applicable regulatory requirements and standards for testing of the quality of natural gas entering their systems. To the extent that a scientific consensus is ultimately formed that raises a legitimate concern over radon in the northeastern natural gas supply, such an issue should be addressed statewide and for all local gas utilities. SEP notes that that legislation was proposed in 2013 in the New York State Legislature related to radon concerns, but that legislation appears to have died in committee and never passed. The Companies' rate proceedings are not an appropriate forum to address this issue.

Q. Does that conclude your testimony?

A. Yes.

MR. DECICCO: Thank you, Your Honor.

BY MR. DECICCO: (Cont.):

Q. Panel, with regard to the exhibits, for your rebuttal testimony did you sponsor four exhibits that were identified as GPSR1 through GPSR4, which have been premarked for identification as Exhibits 336 through 339.

A. (Panel Cont.) Yes, we do. Yes.

Q. Were these exhibits prepared by you or under your supervision and direction?

A. (Panel) Yes, they were. Yes.

Q. Are these exhibits complete and accurate to the best of your knowledge?

A. (Panel) Yes, they are. Yes.

MR. DECICCO: Thank you. Your Honors, the panel is available for cross examination.

A.L.J. COSTELLO: Thank you. Department of Public Service Staff.

MR. GOODRICH: I believe we had said that we had no cross, but I can't recall if I had actually sent that around or not at this point this morning. So we don’t have any cross, though.

A.L.J. COSTELLO: I may not have seen it but that's --

A.L.J. LEARY: Did you send us an email?

MR. GOODRICH: At this point this morning, I honestly cannot recall if I did or not.

A.L.J. LEARY: I have you with one hour on damage prevent -- damage prevention issues.

MR. GOODRICH: I'm sorry. We don't -- we don't have cross.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Okay fine.

A.L.J. COSTELLO: Environmental Defense Fund.

MS. MURPHY: Yes. We do have cross. Is this microphone working?

A.L.J. LEARY: Is the green light on?

MS. MURPHY: It is.

A.L.J. LEARY: You really have to speak very closely.

MS. MURPHY: I am. I don't think it's on.

A.L.J. LEARY: I think that's one of the broken ones, so if you can grab the other one and see if that goes.

MS. MURPHY: How's this one?

A.L.J. LEARY: That sounds good.

CROSS EXAMINATION

BY MS. MURPHY:

Q. So good morning to the panel. I'd like to start by focusing on your direct testimony starting at page 54, which discusses the companies leak backlog performance metric. So at page 56 -- oh, I'm sorry. I'd actually like to start at page 56. At page 56 of your direct testimony, table six displays the proposed leak backlog and incentive structure and there are two tables there, one for KEDNY and one for KEDLI. Is that correct? Can I get a verbal response please?

A. (Turrini) Yes. That's correct.

Q. So let's talk about this incentive structure. Is it accurate to say that the leak metric is structured to incentivize the companies to decrease their leak backlog each year such that if they fail to decrease their leak backlog they would suffer a negative revenue adjustment?

A. Yes. That is correct.

A.L.J. LEARY: Speak into the microphone.

BY MS. MURPHY: (Cont.)

A. I'm sorry. Ross Turrini. Yes. That is correct.

Q. And so we can see from the tables that KEDLI has a larger leak backlog and a larger annual leak reduction target than KEDNY. Is that correct?

A. Yes. That is correct.

Q. So other than that numerical difference, the performance metric for the two companies is structured the same way. Is that correct?

A. Yes. That is correct.

Q. So is it accurate to say that the leak backlog performance metric is structured to incentivize each of the companies to maintain a low leak backlog of known leaks?

A. Yes. That is correct.

Q. And is it accurate to say that the leak backlog is a backlog of known or identified leaks?

A. Yes. That is correct.

Q. And so I'm referring now to page 55 line nine of your direct testimony, which states the companies propose to retain the annual reduction

of the leak target KEDNY reduction of 150 leaks and KEDLI reduction of 750 leaks and further reduce the annual target based on the prior year's performance in eliminating incremental high emitter type three leaks. Is that correct?

A. Yes. That is correct.

Q. And so I understand that under the company's proposal that means that KEDNY'S backlog of identified leaks is targeted to decrease by 150 leaks per year meaning that the company would repair or replace -- I guess repair, which could include through pipe replacement at least 150 leaks more than it found and added to the backlog during that year. Is that correct?

A. Yes. That is correct.

Q. And can you explain what you mean by the statement that the companies will further reduce the annual target based on the prior year's performance in eliminating high emitter type three leaks?

A. So could you just repeat the question please?

Q. Sure. I asked if you could explain what you mean by the statement that the

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

companies will further reduce the annual target based on the prior year's performance for eliminating the high emitter type three leaks?

A. Yeah. So what that means is the targets that are laid out on the chart on slide -- on page 56 I'll use KEDNY as an example. So the target would be 2950 for the total leak backlog and it would be reduced in addition by the number of high emitter leaks.

Q. And so since we're referring to that table, just for clarity, where in the table for both KEDNY and KEDLI for calendar years 2021 through 2023 you state the target and then it says the -- less the total leak reduction adjustment factor from the previous calendar year. Is that the same reduction that you're referring to?

A. Yes. That is.

Q. Thank you for clarifying. So let's talk about how leak repair works on the ground. My understanding is that the companies are continuously identifying and repairing leaks throughout the year. Is that correct?

A. Yes. That's correct.

Q. And could you provide an estimate of how many leaks on average the company might identify in a given year?

A. Yes. It's based -- we build a work plan every year, which includes an estimate on the number of leaks that we're going to receive and the number of leaks that we currently have and then the number of leaks that we're targeting to repair that year.

Q. And I'm just looking for a rough number here. I mean do you recall what that estimate was for last year for the number of leaks to be identified during the course of the year?

A. So could you just repeat the question please?

Q. Sure. I'm asking if you could provide an estimated number on the number of leaks that the company might identify in a given year?

A. I don’t have that information with me right now. We -- we could provide that though.

Q. Sure. No problem. We're going to pass out an exhibit. This is not a previously-marked exhibit. And this document is the company's response

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas to information request number DPS-116. The response was issued on April 30, 2019.

A.L.J. COSTELLO: That will be marked for identification as Exhibit 671.

BY MS. MURPHY: (Cont.)

Q. Has the panel taken a look at the exhibit?

A. Yes, we have.

Q. So we have two charts here each demonstrating the total number of type 1, 2A, 2 and 3 leaks discovered on the system during each of the previous five calendar years for 2014 through 2018. And if you look at the first chart for KEDNY, which I'm mostly going to talk about KEDNY just for purposes of an example throughout this cross, you can see in the totals there that the company has identified roughly somewhere between 4000 at the lowest and as high as a little over 5800 leaks per year at the highest over the last couple of years. Is that correct?

A.L.J. LEARY: For KEDNY. Is that correct?

BY MS. MURPHY: (Cont.)

A. Yes. That's correct.

A.L.J. LEARY: Just make sure that you identify which company rather than saying the company.

MS. MURPHY: Sure. I will do that.

BY MS. MURPHY: (Cont.)

Q. And so I'm just going to refer to 4000 as I'm going to do an example in a few minutes and I'm going to refer to the idea that the company might identify 4000 leaks per year based off of this chart as an example. Is that a fair example to provide based on this chart?

A. So, when you take a look at leak receipts, they vary year over year based on a number of factors. So one factor they're going to vary on is the areas that we conduct leak survey in. So we split the system basically into a third, a third, a third and some areas that we leak survey are higher leak rate areas. So, in those years, the leaks will be up. An additional factor in leak receipts is temperature. The colder the temperature typically the higher amount of leaks that we see. So you will see variability over the years on the number of leak receipts that we have, but if -- we can work with the example of 4000.

Q. Thank you. And so, you know, we can see here that the company is of course identifying thousands of leaks every year and therefore the company is also continuously repairing thousands of leaks every year. Is that right?

A. Yes. That's correct.

A.L.J. LEARY: Again.

MS. MURPHY: For KEDNY. I'm sorry.

A.L.J. LEARY: That's okay.

BY MS. MURPHY: (Cont.)

Q. So now that we've talked broadly about the leak backlog, I'd like to take for example calendar year 2021 for KEDNY. And I am referring again to page 56 of your direct testimony, table six. So according to that table, if the company was on target for its goal in 2020, then KEDNY would start out 2021 with a leak backlog of 3100 known leaks. Is that correct?

A. Yes. That's correct.

Q. And KEDNY'S target is to reduce its leak backlog to 2950 by the end of calendar year 2021. Is that correct?

A. Yes. That's correct.

Q. So working off of the example that I referenced earlier, if KEDNY were to discover 4000 new leaks during 2021, then the company would have to repair 4150 leaks during the year to achieve its targeted leak backlog. Is that correct?

A. Yes. That's correct.

Q. And then if KEDNY were to repair additional type three leaks, then you would subtract those repaired leaks as well so that the leak backlog number at the end of 2021 could be lower than 2950, correct?

A. Yes. If you're referring back to our earlier testimony about the high emitter leaks.

Q. Yes. Thank you. So I'm going to walk through a hypothetical now continuing to discuss this same leak backlog metric. I will say this might sound silly perhaps to the panel whose very familiar with this. I found it helpful to visualize the leak backlog as a jar of marbles and marbles are being added to the jar throughout the year as leaks are discovered and removed from the jar as leaks are repaired. And then, at the end of the year, the number that you're reporting is the number of marbles in the jar or the current leak backlog. I'm going to

dive into this hypothetical. So let's say that in addition to the current leak detection methods that the company is employing the company started using a different technology that resulted in more identified natural gas leaks. And let's say that KEDNY deploys this technology in 2021, so that instead of identifying 4000 new leaks throughout the year as we discussed in the previous example, the company identified 7000 new leaks during the course of 2021. Are you following the hypo?

A. Yes.

Q. Thank you. So as we discussed earlier, if everything is on target, KEDNY would start 2021 with a backlog of 3100 known leaks as reflected in table six on page 56. And if KEDNY were to discover an additional 7000 leaks that would be a total universe of 10,100 leaks during 2021 for the company to consider. Does that make sense -- for KEDNY to consider?

A. Yes.

Q. And so let's say that KEDNY repaired the same number of leaks that they did in the previous example, which was 4150 leaks during the course of 2021. So at the end of year under this

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

hypothetical KEDNY'S leak backlog would be 5950 leaks. Is that correct?

A. I'll trust your math.

Q. Thank you. So under this hypothetical and following the same metric in table six that we've been discussing KEDNY would be subject to a negative revenue adjustment of five basis points for calendar year 2021. Is that correct?

A. Yes. That's correct.

Q. And I'm going to refer now to page 55 of your direct testimony starting at line 12. So your testimony states that the company similarly proposed to retain the existing incentive mechanism to repair incremental type three high emitter leaks and the existing GSRS cost recovery mechanism for incremental leak repair that is demonstrated in an attached exhibit. Is that correct?

A. Yes. That's correct.

Q. And page 55 line one describes that existing incentive mechanism and states additionally the companies are incentivized to repair incremental type three leaks that have been ranked as high emitters. The companies are permitted to earn a positive revenue adjustment of one basis point for

each 50 such leaks eliminated up to a maximum of five basis points per year. Is that correct?

A. Yes. That's correct.

Q. And one more quote, sorry. Just continuing from where we just stopped at line four on page 55 of your direct your testimony states that the companies must achieve both annual backlog reduction targets to qualify for this incentive. Is that correct?

A. Yes. That's correct.

Q. So turning now to page 57 of your direct testimony at the top of the page. This chart identifies how the companies could achieve positive revenue adjustments as we've discussed and under this framework I'm going to focus on KEDNY. So if the company repaired 150 leaks as a baseline to reduce its backlog and then it repaired another 250 type three leaks, which would be a total of 400 repaired leaks in one year on top of any repairs of -- of newly-identified leaks then KEDNY would receive a positive revenue adjustment of five basis points. Is that correct?

A. Yes. That's correct.

Q. So I'm going to return to our new technology hypothetical, which I can restate for clarity. So we're discussing if in addition to the current leak detection methods the company started using a different technology that resulted in more identified natural gas leaks and KEDNY deployed that technology in 2021, so that instead of identifying 4000 new leaks throughout the year the company identified 7000 new leaks throughout the year. Let's now say that the company repaired 4400 leaks throughout the year. Under that framework, the company's leak backlog would be 5700 known leaks at the end of 2021. Do you -- is that correct, my math?

A. Again, I'll trust your math.

Q. Thank you. So, under that hypothetical, KEDNY would not be able to earn the positive revenue adjustment for repairing the high-emitting leaks. Is that correct?

A. Yes. That's correct.

Q. And so if the commission were to adopt the company's proposed leak backlog performance metric, the companies could be subject to negative revenue adjustments for identifying significantly

more leaks on their system and adding -- that would be added to the leak backlog. Is that correct?

MR. DECICCO: Is that -- I'm sorry. Is that under your hypothetical situation or is that -- you're asking that in normal course?

MS. MURPHY: That is not limited to the hypo.

MR. DECICCO: Okay.

MS. MURPHY: That is a broader question. I can restate.

BY MS. MURPHY: (Cont.)

A. So could you restate the question, please?

Q. So, if the commission were to adopt the company's proposed leak backlog performance metric, the companies could be subject to negative revenue adjustments for identifying additional leaks on their system that would be added to the leak backlog. Is that correct?

A. So that would be correct, but currently the company uses the approved technology in New York State for leak detection, right? Any change in that technology or approval of a new technology in the adopt -- and the adoption of a new technology

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas would require a change if we're going to see a significant change. And I'm not convinced one way or the other whether we would see a change, right? But if -- if we felt that technology was going to see a change in the way that leaks were received -- we received leaks then we would have to revisit the structure of the chart to be fair to the company. The other complicating factor would be if there's going to be a significant lamp -- ramp up in leak repair we would have to train more people to do that leak repair, right? So, you know, all in all right now this chart is based on the current leak detection technology that's approved in New York State. The company this is a really -- it's a combination program for us. It's leak reduction from a risk management standpoint and gas safety standpoint, but we've also expanded its use over the last few years because we do believe in methane reduction as -- as part of climate change. So we have leveraged it from both perspectives, but any -- any change in technology or change would require the way this metric is structured. And just one last piece to add, you know it's our job to find as many leaks as possible because that's how we reduce the -- the risk

on the system. So -- so the way we've structured our -- our leak survey programs is to use the existing approved technology to best find the amount of leaks that we have on a system.

Q. Thank you for that explanation. And understanding the -- the approval process for technology that you use I would just finally ask and I -- well before I refer to your testimony, I'll ask broadly are you aware that the company began I believe in 2016 using what ADF has referred to as Advanced Leak Detection? I believe the company refers to it as cavity-down spectrometry.

A. So yes, I am and I understand that's a mouthful to say. So I don’t try to say it either, but we actually do plan to use that technology in -- in this case. So where we think it's appropriate is trying to find high-emitter leaks on our most leak-prone pipe segments so that we can add another factor into our main replacement program, which is primarily based on the risk. But, over the last few years, we've actually sort of advanced our main replacement program to start to make some inroads on methane reduction from the high emitters. So the first thing we did was a few years ago we

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

added the class three leaks in as a factor to determining which main segments that we should use. And, in this case, we're proposing to use that technology on our highest leak rate areas to identify the leak segments with the highest leak rates from a high emitter standpoint and then in the following year we would work those into our main replacement program to take another tool in order to reduce methane emissions.

Q. Thank you. That concludes my cross.

A.L.J. COSTELLO: Do you want to take a moment?

MR. DECICCO: No Your Honor. I think we are good. No redirect.

A.L.J. COSTELLO: Okay. Panel, thank you very much for your testimony and you're excused. We'll go off the record.

Could you please state your name and business address for the record?

MS. ARRANGIO: Yes. It's Elizabeth Arrangio and my business address is 40 Sylvan Road, Waltham, Massachusetts.

A.L.J. COSTELLO: Please stand and

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MS. ARRANGIO: Yes, I do.

ELIZABETH ARRANGIO; SWORN

A.L.J. COSTELLO: Thank you. You may be seated. Mr. Maloney.

MR. MALONEY: Thank you, Your Honor. For the record, Kent Maloney of behalf of the companies.

DIRECT EXAMINATION

BY MR. MALONEY:

Q. Good morning, Ms. Arrangio. Ms. Arrangio, do you have before you a document consisting of 48 pages including 45 pages of questions and answers under a cover entitled The Brooklyn Union Gas Company doing business as National Grid New York and KeySpan Gas East Corporation doing business as National Grid Direct Testimony of Elizabeth D. Arrangio dated April 2019?

A. (Arrangio) Yes, I do.

Q. Do you also have before you a document consisting of 11 pages including eight pages of questions and answers also labeled The Brooklyn Union Gas Company doing business as National Grid New

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas York and KeySpan Gas East Corporation doing business as National Grid Supplemental Direct Testimony of Elizabeth D. Arrangio dated June 11, 2019?

A. Yes, I do.

Q. Do you also have before you a document consisting of six pages including four pages of questions and answers again labeled -- under a coversheet labeled The Brooklyn Union Gas Company doing business as National Grid New York and KeySpan Gas East Corporation doing business as National Grid Corrections and Updates Testimony of Elizabeth D. Arrangio dated July 3, 2019?

A. Yes, I do.

Q. Do you also have before you a document consisting of 13 pages including 11 pages of questions and answers under a coversheet labeled The Brooklyn Union Gas Company doing business as National Grid New York and KeySpan Gas East Corporation doing business as National Grid Rebuttal Testimony of Elizabeth D. Arrangio dated September 18, 2019?

A. Yes, I do.

Q. And finally, do you also have before you a document consisting of six pages including four pages of questions and answers under a

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas coversheet labeled The Brooklyn Union Gas Company doing business as National Grid New York and KeySpan Gas East Corporation doing business as National Grid Second Supplemental Testimony of Elizabeth D. Arrangio dated December 13, 2019?

A. Yes, I do.

Q. And was this testimony, these various pieces of testimony, prepared by you or under your supervision and direction in connection with the company's filings and these proceedings -- the companies' filings and these proceedings?

A. Yes, they were.

Q. And to your knowledge is the information contained there true and correct to the best of your knowledge?

A. Yes, it is.

Q. And if I were to ask you the questions appearing in your various pieces of testimony today would your answers be the same?

A. Yes, they would.

Q. And do you adopt your prepared testimony as your sworn testimony in these proceedings?

A. Yes, I do.

MR. MALONEY: Your Honor, at this time I'd ask that the direct supplemental corrections and updates rebuttal and second supplemental testimony of Ms. Arrangio be copied into the record as if orally given.

A.L.J. COSTELLO: Okay. That request is granted. And at this point, the court reporter should insert the following files KEDNY-KEDLI Elizabeth D. Arrangio Direct Testimony, KEDNY-KEDLI Elizabeth D. Arrangio Supplemental Direct Testimony, KEDNY-KEDLI Elizabeth D. Arrangio Corrections and Updates Testimony, KEDNY-KEDLI Elizabeth D. Arrangio rebuttal testimony, KEDNY-KEDLI Elizabeth D. Arrangio Redacted Second Supplemental Testimony. And in a separate confidential file transcript, the file should go in KEDNY-KEDLI Elizabeth D. Arrangio Confidential Second Supplemental Testimony. #

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY

And

KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID

Direct Testimony

of

Elizabeth D. Arangio

Dated: April 2019

Table of Contents

I. Introduction and Qualifications 1

II. Purpose of Testimony 2

III. Gas Supply Portfolio 4

A. Indigenous and Renewable Natural Gas 11

B. Northeast Supply Enhancement Project 14

IV. Portfolio Management 17

V. Marginal Cost Studies 30

VI. Customer Choice Program 31

VII. Gas Cost Volatility Management 36

VIII. Conclusion 45

Testimony of Elizabeth D. Arangio

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Please describe your business position and responsibilities.

A. I am the Director of Gas Supply Planning with responsibility for the gas supply resource portfolios of National Grid USA's ("National Grid") local gas distribution companies in New York, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively the "Companies"). In addition to the New York portfolios, I am responsible for planning the gas resource portfolios of National Grid's New England subsidiaries. I also manage National Grid's gas Customer Choice programs.

Q. Please summarize your educational background and your professional experience.

A. I graduated from the University of Massachusetts in 1991 with a Bachelor of Business Administration. In 1995, I graduated from Bentley College with a Master of Business Administration. From 1991 to 1994, I worked as a Gas Accounting Analyst in the Marketing Operations Department at Algonquin Gas Transmission Company. In 1994, I joined Boston Gas Company as a Gas Supply Analyst. In 1997, I was promoted to Group Leader Transportation Services, with

responsibility for managing all activities associated with the customer choice program. In 1998, I was promoted to Director of Gas Acquisition and Transportation Services and assumed responsibility for the administration of Boston Gas's gas resource portfolio and customer choice program in Massachusetts and, in 2000, the resource portfolio of EnergyNorth Natural Gas, Inc. in New Hampshire. In February 2004, I assumed the additional responsibility for gas supply planning for the Companies' resource portfolios. Following the acquisition of KeySpan Corporation by National Grid, I was named to my current position and assumed added responsibility for National Grid's gas resource portfolios in Upstate New York and Rhode Island. In August 2018, I assumed the added responsibility of managing National Grid's gas Customer Choice programs.

II. Purpose of Testimony

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to:

(i) describe the Companies' efforts to purchase gas supply, pipeline transportation, and storage services on a reliable, least-cost basis in the twelve months ended December 31, 2018 ("Historic Test Year"), the twelve months ending March 31, 2021 ("Rate Year"), the twelve months ending March 31, 2022 ("Data Year 1"), the twelve months ending March 31, 2023 ("Data Year 2") and the twelve months ending March 31, 2024 ("Data Year 3," and together with Data Years 1 and 2, "Data Years");

(ii) discuss the Companies' efforts to supplant pipeline capacity with additional indigenous and renewable gas supplies;

(iii) discuss the existing gas supply constraints in the downstate New York area and the potential impact on the Companies' ability to meet forecast demand, the Companies' efforts to alleviate the supply shortage by supporting additional pipeline capacity to the region, and the potential need to impose a moratorium on new or additional gas service if constraints are not resolved;

(iv) present the Companies' forecast of gas costs for the Rate Year;

(v) describe the results of the marginal cost gas supply studies for the Rate Year;

(vi) discuss the Companies' Gas Cost Volatility Program; and

(vii) discuss the Companies' Customer Choice Program.

Pursuant to the New York State Public Service Commission's ("Commission") "Order Authorizing Combined Gas Portfolios" issued October 28, 2005 in Case 05-G-0903, as of November 2005, the Companies have combined the planning and dispatching of their gas supply portfolios to provide the Companies' customers enhanced reliability of supply and lower costs. Therefore, my testimony addresses the combined portfolios of the Companies and the material I present is applicable to both KEDNY and KEDLI.

Q. Does your testimony include any exhibits?

A. Yes. My testimony includes the following exhibits that were prepared under my supervision and direction:

Exhibit __ (EDA-1) KEDNY & KEDLI Portfolio Schematics;

Exhibit __ (EDA-2) KEDNY & KEDLI Pipeline Transportation Contracts;

Exhibit __ (EDA-3) KEDNY & KEDLI Storage Contracts;

Exhibit __ (EDA-4) KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage;

Exhibit __ (EDA-5) KEDNY/KEDLI Purchased Gas Expense for the Twelve Months Ending ("TME") December 31, 2018;

Exhibit __ (EDA-6) KEDNY/KEDLI Forecast of Variable Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;

Exhibit __ (EDA-7) KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;

Exhibit __ (EDA-8) KEDNY & KEDLI Estimated Marginal Commodity Cost of Gas;

Exhibit __ (EDA-9) KEDNY & KEDLI Estimated Annualized Marginal Capacity Cost of Gas;

Exhibit __ (EDA-10) KEDNY & KEDLI Non-Migration Capacity Release Revenues; and

Exhibit __ (EDA-11) KEDNY & KEDLI OSS Transaction Revenues.

III. <u>Gas Supply Portfolio</u>

Q. Please describe the Companies' gas distribution systems.

A. KEDNY's gas distribution system serves Brooklyn, Staten Island, and portions of Queens, all located within New York City. KEDLI's gas distribution system serves a portion of Queens not served by KEDNY, as well as Nassau and Suffolk

counties on Long Island. KEDNY and KEDLI are parties to an agreement with Consolidated Edison Company of New York, Inc. ("Con Edison") concerning the ownership and operation of the New York Facilities System ("NYFS"), the high-pressure gas transmission system serving the three downstate New York distribution companies. This agreement permits the parties to contract for the transportation and receipt of gas from various interstate pipelines that interconnect with the NYFS, including Transcontinental Gas Pipeline Company LLC ("Transco"), Texas Eastern Transmission LP ("Texas Eastern"), Iroquois Gas Transmission System LP ("Iroquois"), and Tennessee Gas Pipeline Company, LLC ("Tennessee"). KEDNY and KEDLI contract for service from each of these pipelines as well as various other upstream pipelines and storage service providers.

Q. Please describe the Companies' gas supply planning process.

A. Typically, in the spring of each year, the Gas Supply Department develops plans to meet the Companies' gas supply obligation for the annual period from November 1 of that year through October 31 of the following year. This planning process begins with an updated ten-year demand forecast that provides the foundation for customer requirements that ultimately determine incremental pipeline, storage, or peaking needs.

Q. What are the Companies' combined forecast design day requirements for the Rate Year and Data Years?

A. The Companies' design day requirements are as follows:

- Rate Year = 2,894 MDth (thousand dekatherms)/day
- Data Year 1 = 2,912 MDth/day
- Data Year 2 = 2,954 MDth/day
- Data Year 3 = 2,987 MDth/day

Q. What is the basis for the Companies' city gate requirements?

A. The primary firm demand (*i.e.*, core customer load forecast) forms the basis for the Companies' gas supply portfolio. The primary firm demand is the demand imposed on the Companies by their firm customers, regardless of whether they purchase gas commodity from the Companies or energy service companies ("ESCOs"). Pipeline and storage capacity, along with peaking assets, are used to satisfy the primary firm demand. An annual load duration curve or similar approach is utilized to structure capacity contracts to best meet the shape and frequency of the anticipated loads and to assure the Companies' ability to meet those loads. The Companies do not incorporate any reserve margin assumptions when developing their design weather forecasts and capacity requirement determinations.

Q. What contracts or assets are included in the Companies' existing portfolio?

A. Exhibit __ (EDA-1) sets forth schematics of the Companies' gas portfolios and provides their maximum delivery entitlements from various sources of supply, including underground storage contracts.

Q. Please describe Exhibit __ (EDA-2) – Pipeline Transportation Contracts.

A. Exhibit __ (EDA-2) summarizes the firm pipeline transportation capacity and bundled peaking assets in the Companies' gas supply portfolio for the 2018-2019 winter season (November 1, 2018 to March 31, 2019). Listed for each contract is information concerning the service provider (pipeline or supplier), tariff rate schedule, contract volume, and contract expiration date.

Q. Please describe Exhibit __ (EDA-3) – Storage Contracts.

A. Exhibit __ (EDA-3) summarizes the Companies' firm storage contracts and the transportation contracts used to deliver storage withdrawal volumes to the city gate for the 2018-2019 winter season. Listed for each contract is information concerning the storage service provider, tariff rate schedule, contract volume, and contract expiration date.

The Companies source gas supply at the following liquid points:

- Dawn, Ontario
- Transco, zone 4
- Dominion, South Point
- Tx. Eastern, M-3
- Transco, zone 6 N.Y.

- Transco, zone 6 non-N.Y.
- Transco, zone 6 non-N.Y. North
- Transco, zone 3
- Iroquois, receipts (Waddington)
- Texas Eastern, M-2 receipts
- Tennessee, zone 4-300 Leg (Marcellus)
- Iroquois, zone 2
- Transco, zone 1
- Millennium, East receipts
- Dominion, North Point
- Leidy Hub
- Transco, zone 2
- Transco, Leidy Line receipts
- Tennessee, zone 5 White Plains (no index)

Q. What is the role of underground storage in satisfying customer requirements?

A. Approximately 31 percent of the Companies' normal winter supply obligation and 28 percent of their design day demand requirement are met by deliveries of gas withdrawn from storage. Under the Companies' storage contracts, storage deliverability typically declines as inventory decreases (known as "withdrawal ratchets"). Once reached, these ratchets cannot be reversed until the following year. Therefore, the Companies establish a storage withdrawal plan prior to the winter season to maintain inventories at levels that allow sufficient storage deliverability to meet forecast winter peak conditions (storage rule curve).

Market area storage provides the Companies with services that cannot be easily duplicated with other assets. The most important attribute of storage assets is flexibility, which is vital in serving changing customer requirements. The Companies' Transco storage service contracts provide end-of-day balancing that

minimize the risk of over/under run penalties. Whereas typical supply purchases must be dispatched at the same volume for each day of a weekend or holiday, storage allows enhanced flexibility by allowing intraday adjustments to meet changing conditions. In addition, storage improves the load factor of flowing pipeline assets and is critical in meeting design weather conditions. This value cannot be duplicated by replacing storage with Marcellus production. The Companies utilize a least-cost dispatch to fill storage. Where Marcellus shale supplies present an opportunity to displace long haul supplies for storage refill, the Companies take advantage of the economically priced supply.

The Companies also have approximately 20 billion cubic feet ("Bcf") of Gulf Coast storage capacity. These storage fields are made available to ESCOs as part of the optional Gulf Coast storage release that occurs every April. Assuming no ESCOs opt-in for the Gulf Coast storage release, approximately 4.5 Bcf of the total capacity is reserved for *force majeure* purposes. The remaining 15 Bcf of capacity, usually the entire KEDNY Washington Storage Service ("WSS") field, is utilized for off-system sales ("OSS"). If all ESCOs opt-in, the capacity utilized for OSS would be reduced accordingly.

Q. What is the role of liquefied natural gas ("LNG") in the Companies' portfolio?

A. The Companies maintain two on-system LNG facilities in Greenpoint, Brooklyn and Holtsville, Long Island. The Greenpoint LNG facility allows KEDNY to

store approximately 1.6 Bcf of gas and has peak day vaporization capability of approximately 291,000 dekatherms (Dth) per day. The Holtsville LNG facility allows KEDLI to store approximately 0.6 Bcf of gas and has vaporization capability of 103,000 Dth/day. Collectively, the Greenpoint and Holtsville LNG facilities provide the Companies with approximately 10 percent of their peak day gas supplies.

LNG provides the Companies with on-system services that cannot be easily duplicated with other assets. Because these resources can be brought on line quickly, the LNG plants can be used to meet hourly fluctuations in demand, maintain deliveries to customers, and balance pressures across portions of the distribution system during periods of high demand. Most importantly, these resources are vital in preserving delivery pressures in the event that an off-system resource becomes unavailable.

Q. What changes are planned related to swing services, peak shaving or winter peaking assets, facilities, or operations for the Rate Year through a five-year planning horizon?

A. As further discussed in the direct testimony of the Gas Infrastructure and Operations Panel ("GIOP"), the Companies plan to take the Holtsville LNG facility out of service for critical maintenance repairs in April 2022. This planned capital work, however, is contingent on the availability of additional supply expected to be provided by the Northeast Supply Enhancement ("NESE") project

by December 2020. There are no other planned changes related to swing services, peak shaving or winter peaking assets, facilities or operations through a five-year planning horizon.

Q. What are the projected monthly beginning and ending volumes and dollar balances for gas stored through the end of the Rate Year and Data Years?

A. Exhibit __ (EDA-4) provides the projected monthly volume and cost of injections and withdrawals for the Companies' underground and LNG storage facilities for the Rate Year and Data Years, summarized by market area, Gulf Coast, and LNG storage.

A. Indigenous and Renewable Natural Gas

Q. Do the Companies purchase supply from any local natural gas production?

A. Currently, the Fresh Kills Landfill ("Landfill"), located in Staten Island, is the Companies' only source of local production and accounts for less than one percent of the total system throughput. The Landfill is owned by the City of New York. KEDNY has a supply agreement with the Landfill for up to 6,500 Dth/day, but there is no contractual minimum daily quantity. Landfill supply is directly fed into the distribution system.

The Companies' use of local production has not changed over the past several years; the Companies continue to purchase any supply the Landfill can provide. There is no forecast of Landfill production, but the Gas Control department

communicates with the Landfill regularly and is notified of maintenance work and outages. When the Landfill is unable to sell supply, the Companies replace planned volumes with other resources in the portfolio.

The table below shows the Landfill's production for the last three years:

Year	Total (Dth)
2016	510,838
2017	406,245
2018	448,990

Additionally, as discussed in the direct testimony of the Future of Heat Panel, the Companies expect a new bio-gas facility at Newtown Creek to be placed into service by November 2019. The new facility is expected to produce approximately 275,000 Dth/year (750 Dth/day) of renewable natural gas ("RNG").

Q. What steps are the Companies taking to capture the benefits of connecting additional indigenous and RNG gas supplies directly to their distribution systems?

A. KEDNY and KEDLI recently updated their Gas Transportation Operating Procedures ("GTOP") manual to include a "Renewable Natural Gas Engineering Services Agreement" template and "Gas Sales (Interconnect) Agreement" template. These documents define the respective responsibilities of the developer and the Companies in connecting new RNG projects. The agreements also list the

billing and payment terms and conditions, gas quality specifications, and other relevant provisions for interconnecting to the Companies' distribution systems. The "Gas Sales (Interconnect) Agreement" template is modeled after the existing agreement with the Landfill.

Further, the Companies recently collaborated in the development of the "NYS Interconnect Guide for RNG" document along with the Gas Technology Institute ("GTI"), Northeast Gas Association ("NGA"), and other gas distribution companies in New York State. The document establishes a clear process for both project developers and utilities to successfully connect RNG projects. Please refer to the testimonies of GIOP and Future of Heat Panel for further discussion.

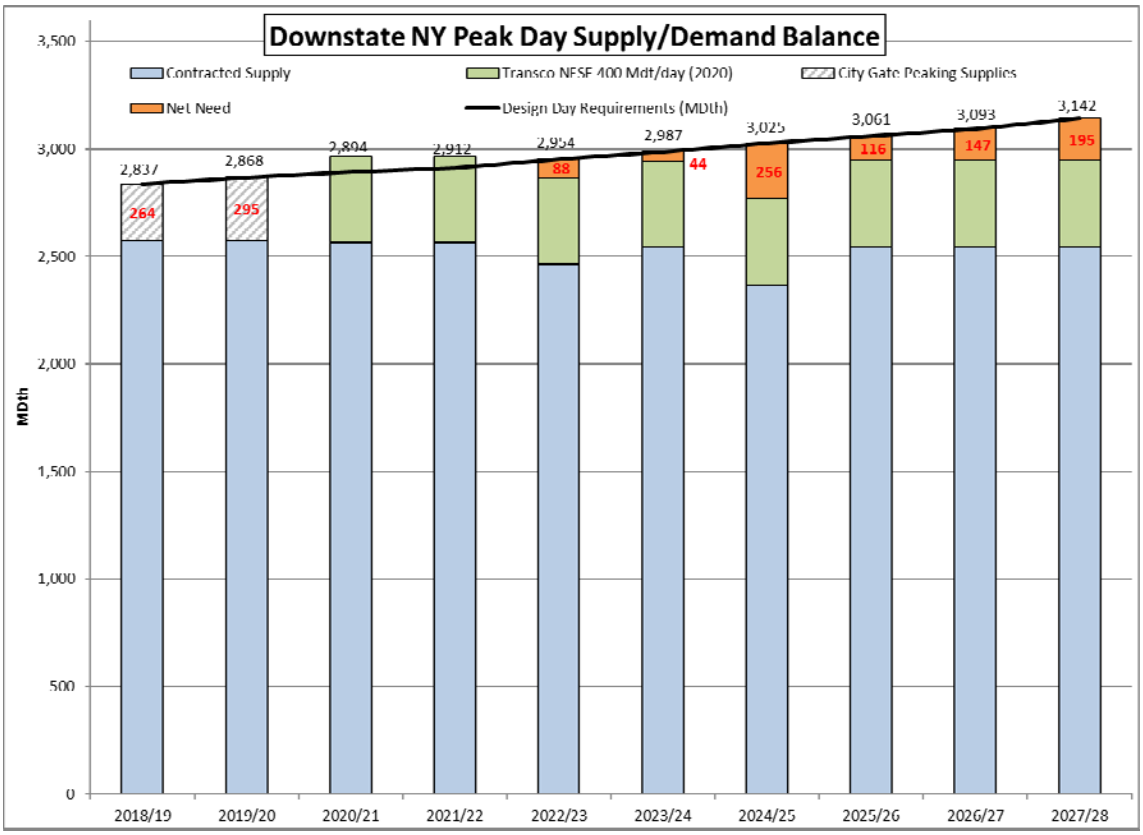
Q. What efforts are the Companies making to reduce their pipeline capacity?

A. Pipeline capacity is an integral component of the Companies' gas supply portfolio to deliver sufficient natural gas to serve our customers. While pipeline capacity cannot be entirely displaced, the Companies continue to explore ways to reduce reliance on pipeline capacity. As discussed in the Future of Heat Panel's testimony, the Companies are pursuing various decarbonization initiatives including renewable gas projects to reduce their carbon footprint while meeting customer needs.

B. Northeast Supply Enhancement Project

Q. Do the Companies plan to add any incremental pipeline capacity in the next five years to meet forecast design day load requirements?

A. Yes. Due to continued customer interest in conversions from oil to natural gas for space and water heating, as well as increased demand from existing customers and new construction in their service territories, the Companies have continued to experience growth in the demand for natural gas. The Companies' ten-year load forecasts show that demand for gas will continue to grow. KEDNY and KEDLI expect the demand for gas to grow at an annual rate of more than 1.3 percent and 1.0 percent for the next ten years, respectively. The graph below shows the balance between the forecast demand and the Companies' supply portfolio for the ten-year period ending 2027/28. The graph shows that without significant additional pipeline capacity, as further described below, supply falls short of the demand.



In response to the forecast growth, KEDNY and KEDLI signed precedent agreements with Transco to deliver an additional 400,000 Dth of natural gas per day to the downstate New York area. To provide the additional supply, Transco will expand its existing pipeline system along Pennsylvania, New Jersey, and New York to connect to the Companies' system in the Rockaway Peninsula. The NESE project will provide access to abundant and economical gas supplies, thereby relieving supply constraints in the Companies' service territories. Furthermore, NESE will include a fully looped pipeline segment to interconnect with the existing delivery lateral in New York Bay, further enhancing reliability of the system. The Companies have contracted for 100 percent of the incremental pipeline capacity that will become available via NESE.

Q. What will be the impact to the Companies' ability to meet forecast demand if the NESE project is not completed in time?

A. In the absence of additional pipeline capacity, the Companies cannot continue to add new gas load without creating an unacceptable risk of significant supply shortfalls and corresponding drops in system pressure to below minimum thresholds. Such conditions will jeopardize the reliability of service and public safety for the Companies' existing firm customers.

As can be seen in the graph above, the Companies' current supply planning assumes that the NESE project will be completed in time to meet the forecast growth in demand for natural gas. Currently, NESE is awaiting permit approval from New York Department of Environmental Conservation. If the project does not become available by the 2020/21 winter season, the Companies will not be able to prudently satisfy new or additional service requests without jeopardizing the Companies' ability to provide safe, reliable service to its existing firm customers. In that case, National Grid will have no choice but to impose a moratorium on new and additional gas service in affected areas to maintain system reliability.

The GIOP direct testimony addresses NESE's impact to the Companies' capital projects. Notably, as mentioned above, the Companies will not be able to take their LNG facilities out of service to complete necessary maintenance repairs if NESE is not completed in time.

IV. Portfolio Management

Q. Please describe the Companies' gas supply planning goals.

A. The Companies' primary gas supply planning goals are to:

(i) Dispatch the gas supply portfolio assets under a least-cost strategy to reliably meet projected primary firm demand;

(ii) Maintain a diverse portfolio of gas supply, storage, and transportation capacity contracts with varying terms and pricing provisions; and

(iii) Implement a formal hedging program to mitigate price volatility.

These goals are consistent with the Commission's "Statement of Policy Regarding Gas Purchasing Practices" issued in Case 97-G-0600 and updated by letter issued March 31, 2011. The Companies maintain a portfolio that meets requirements under design conditions while maintaining sufficient flexibility for mild winters.

The Companies monitor these goals with regular meetings (monthly supply plan, quarterly review, and annual RFP review). Pursuant to Recommendation IX-4 from the final audit report in the Commission's previous gas management audit (Case 13-G-0009), the Companies established a process for the quarterly review of gas supply procurement plans compared to actual purchases for a sample of days during the quarter. The review identifies variances in volumes and the use of storage and delivery pipelines caused by weather, market conditions, operational constraints, or other factors. Variances are reviewed for patterns and opportunities to improve the procurement process. The Companies' Energy

Procurement group has been conducting quarterly reviews since 2015. Reviews are attended by representatives from Gas Supply Planning, Gas Trading, and Origination and Price Volatility Management.

Q. Please describe the Companies' gas purchasing process.

A. The Companies contract for quantities of gas to ensure sufficient supply to reliably meet design conditions, as well as to account for daily and seasonal load variations. A combination of term and spot contracts provides necessary flexibility with respect to volume, which allows the Companies to respond to fluctuations in demand. Both term and spot contracts are firm to ensure reliability. The varying provisions in these contracts allow for pricing diversity as well.

Term Contracts

Term contracts have durations of longer than one month and less than one year. These contracts generally have a specified fixed daily base load quantity necessary to satisfy requirements under a "warm winter" scenario and may include some summer storage refill quantities. Other term contracts provide call options to buy bundled gas supplies delivered either to the Companies' city gates or other upstream supply points. KEDNY and KEDLI also utilize no-notice storage to manage variations in load that allow the Companies to automatically withdraw or inject gas at the end of the day to balance system load.

Term contracts are generally priced monthly. In addition to first-of-the-month (FOM) published indices, monthly prices for some term domestic supply contracts also use the New York Mercantile Exchange ("NYMEX") last day settle price plus a competitively bid location basis differential. Monthly pricing ensures the effectiveness of the NYMEX futures contracts entered as part of the Companies' hedging strategy.

Spot Contracts

Spot contracts are firm contracts with a term of one month or less. There are no specific contractual triggers that require the purchase of spot gas. These contracts are made throughout the year to supplement term contract supplies, manage demand variations due to weather, and maintain storage inventory targets. Spot contracts also provide pricing diversity (*i.e.*, daily index vs. monthly index).

The Companies purchase spot gas from a number of qualified and reliable suppliers who have North American Energy Standards Board ("NAESB") contracts to minimize risk and obtain competitive pricing. The amount of capacity available depends on the time of year and storage availability.

Daily spot purchases are priced either at reliable, daily published index prices or at a negotiated short-term (daily) fixed price.

Q. What is the current level of natural gas supply used by the Companies to meet normal customer requirements, and how will that amount change in the Rate Year?

A. Of the 230 Bcf of forecast normal customer requirements for November 2018 to October 2019, the Companies are required to purchase approximately 201 Bcf. The remainder will be procured by ESCOs. For the Rate Year, forecast normal customer requirements are approximately 221 Bcf, with 208 Bcf served by the Companies and the remainder by ESCOs.

Q. How do the Companies evaluate expiring contracts for gas supply?

A. As decision dates for contract extension/termination approach, the Companies determine the need to maintain and or modify (to the extent possible) each contract as part of the resource portfolio. The Companies use several criteria to assess the need for transportation and storage contracts including but not limited to: receipt point liquidity, reliability, complement to the existing portfolio, and economics.

Also, the Companies consider options to replace long-haul capacity with shorter-haul capacities. For example, as supplies from the Marcellus shale region became abundant and readily accessible, the Companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada.

While the current level of Transco long haul capacity is required to serve the Companies' peak day and peak season needs, it is not always the least cost option for supply. Recently, the Companies were able to reduce the path on Transco long-haul contracts to reduce fixed costs. Effective March 1, 2019, the Companies have agreed to turnback 40 percent (27,473 Dth/day) of their combined entitlements from Zone 1 (Sta 30). Transco will allow the remaining portion (41,210 Dth/day) to be turned back as early as October 2020. After the turnbacks, the Companies will still be able to reliably fill 100 percent of the original contract volumes at downstream points. The option to reduce capacity paths is not one typically offered by the pipelines, so, when the opportunities occur, the Companies will seek to take full advantage of such de-contracting providing such options do not have an adverse effect on the reliability and economics of the portfolio.

Q. Does the Companies' supply purchasing strategy enable them to benefit from the increased production from the Marcellus and Utica shale regions?

A. Yes. KEDNY and KEDLI purchased approximately 75.6 Bcf from the Northeast producing region during the 2017/18 winter. In addition, KEDNY and KEDLI purchased approximately 17.0 Bcf from the Gulf Coast and 3.7 Bcf from Canadian transportation paths.

The table below shows the Companies' actual purchases from November 2017 through March 2018. Over the last several years, the Companies have decreased

reliance on Gulf Coast and Canadian resources, while increasing reliance on points associated with Marcellus shale.

Firm Transportation	
Gulf Coast	11.0%
Canadian	2.0%
Marcellus Shale	50.0%
Total Firm Transportation	**64.0%**
Storage Withdrawals	34.0%
Peaking Supplies	1.6%
Local LNG Vapor	0.4%
Total	**100.0%**

For the Historic Test Year, the Companies met their city gate requirements as follows:

96.4% Domestic purchases and underground storage

3.4% Canadian purchases at Dawn and Waddington

<1% LNG and Landfill

Based on current and forecast prices, the Companies expect continued reliance on domestic supply purchases and underground storage going forward.

Q. Do the Companies engage in off-system sales, capacity release and other arrangements to reduce their total gas costs?

A. Yes. The Companies constantly monitor their gas resource portfolio to ensure that the appropriate mix of pipeline transportation capacity, market area storage, bundled city gate supply, and peaking resources are available to serve projected firm design requirements. To further minimize costs, the Companies seek to optimize portfolio assets when they are not being utilized for the benefit of firm customers. Except for Gulf Coast storage, for the Rate Year and beyond, the Companies have no plans to enter any pre-arranged off-system sales, capacity release, or streaming arrangements that would encumber upstream assets. As in years past, once the winter heating season begins, the Companies will actively pursue opportunities to sell available supply and/or capacity in a manner that does not diminish overall supply adequacy, reliability or operational flexibility to firm customers. For example, on warm days in shoulder months when pipeline capacity is not fully utilized, the Companies will seek to sell excess capacity to off-system customers. Under this type of transaction, the Companies will purchase supply and transport gas on available pipeline capacity and then assess the associated variable costs plus a negotiated margin to the off-system customers. The Companies will also look for opportunities to execute physical pipeline trades where available. Under such trades, the Companies would purchase firm supply delivered to the city gate on one interstate pipeline and sell a like amount of supply to another third party at a higher price on a different interstate pipeline.

The Companies hold approximately 20 Bcf of production area storage that is used to:

1. maintain supply reliability during force majeure supply outages in the Gulf coast;
2. balance on-system loads;
3. support the Companies' Price Volatility Management Program; and
4. make off-system production area sales.

KEDNY and KEDLI used approximately 15.4 Bcf of that capacity for production area off-system sales for the 2018/19 winter. The amount of Gulf Coast storage capacity released to marketers under the retail access program for 2018/19 was 12,920 Dth. At this time, the Companies do not propose to modify their OSS practices.

Q. Do the Companies enter into asset management agreements ("AMAs") to maximize the value of their supply portfolio and reduce their overall supply costs?

A. Yes. The Companies currently have eight AMAs in place for the 2018/19 period. These arrangements permit the Companies to benefit from the expertise of third-parties who are more familiar with market conditions and opportunities in particular regions or on particular pipeline systems while still maintaining access to essential firm supply sources. The terms of these arrangements do not exceed one year. The Companies will explore beneficial opportunities to enter asset optimization agreements in the future as current agreements expire.

Below are the list of current AMAs and a summary of the terms:

KEDNY:

- Counterparty: Nextera Energy Marketing, LLC
    - Key terms: Baseload or Daily Call (April 2019-October 2019); Transco Contract 1006500; Transco Z2/Z3 Gulf to Transco Z6 Leidy NNY; November 2018-October 2019; 10,688 Dth/day;
- Counterparty: J. Aron & Company LLC
    - Key terms: Daily Call (November 2018-March 2019); Transco Long-Haul Station 65 to Transco Z6 Narrows NY; November 2018-October 2019; 25,000 Dth/day;
- Counterparty: EQT Energy, LLC ("EQT")
    - Key terms: Daily Call (November 2018-March 2019); Transco Long-Haul Station 65 to Transco Z6 Narrows NY; November 2018-October 2019; 25,000 Dth/day; and
- Counterparty: Emera Energy Services, Inc. ("Emera")
    - Key terms: Daily Call (December 2018- April 2019); Union Dawn to Iroquois S. Commack; December 2018-October 2019; 57,498 Dth/day.

KEDLI:

- Counterparty: EQT
    - Key terms: Daily Call (November 2018-March 2019); Transco Long-Haul Station 65 to Transco Z6 Long Beach NY; November 2018-October 2019; 25,000 Dth/day;

- Counterparty: Emera
    - Key terms: Daily Call (December 2018- April 2019); Union Dawn to TransCanada Waddington; December 2017-October 2018; 25,357 Dth/day;
- Counterparty: Emera
    - Key terms: November Daily Call; Base-Load Winter Supplies (December 2018-March 2019); Off-Peak Season Daily Call on any 60 days during April 2019-October 2019; Additional Call on any Day during the Term; NE07 capacity; Millennium Corning to Iroquois S. Commack; November 2018-October 2019; 25,000 Dth/day; and
- Counterparty: Consolidated Edison Energy, Inc.
    - Key terms: November Daily Call; Base-Load Winter Supplies December 2018-March 2019; Off-Peak Season Daily Call on any 60 days during the period April 2019-October 2019; Additional Call on any Day during the Term; NE07 capacity; Millennium Corning to Iroquois S. Commack; November 2018-October 2019; 50,000 Dth/day.

The Companies regularly evaluate the contracts in their supply portfolio, taking into account market interest in the asset, whether the resource is utilized by the Companies to meet baseload or swing requirements on a seasonal or year-round basis, intraday flexibility of the asset, and whether the Companies believe the resource may be better managed by a third party. At this time, the Companies

determined that the existing AMAs are in the best interest of the firm gas customer but will consider additional opportunities in future years based on market conditions.

Q. Do the Companies have existing or previous gas supply arrangements with any affiliated marketing/trading organizations?

A. No.

Q. What are the revenues received over the last five fiscal years (April 2014 – January 2019) from releases to shippers other than on-system customers that have migrated from bundled sales to transportation service?

A. Exhibit __ (EDA-10) summarizes the revenues received from non-migration capacity releases. Over the five-year period (April 2014 – January 2019), the revenues received from such capacity releases totaled $37.5 million, of which $31.9 million (85 percent) was credited to customers and the remaining $5.6 (15 percent) was retained by the Companies.

Q. What are the revenues received over the last five fiscal years (April 2014 – January 2019) from OSS transactions, WSS transactions, and AMAs?

A. Exhibit __ (EDA-11) summarizes the revenues received from revenues received from off-system sales transactions. Over the five-year period (April 2014 – January 2019), the revenues received from off-system sales transactions, WSS transactions and AMAs totaled $284.2 million, of which $241.7 million (85

percent) was credited to customers and the remaining $42.5 million (15 percent) was retained by the Companies.

Q. Please describe Exhibit __ (EDA-5) – Purchased Gas Expense.

A. Exhibit __ (EDA-5) shows the Companies' purchased gas expense for the Historic Test Year. This expense includes the purchased cost of gas minus the cost of storage injections plus the cost of storage withdrawals, and all pipeline fixed and variable charges.

Q. Please describe Exhibit __ (EDA-6) – Forecast of Variable Gas Expense TME March 31, 2021, 2022, 2023 and 2024.

A. Exhibit __ (EDA-6) shows the projected commodity prices of the various natural gas supplies that are forecast to be purchased and delivered to the Companies for the Rate Year and the Data Years to serve the estimated requirements of the Companies' firm customers under the assumption of normal weather. This commodity price projection serves as the basis for the forecast of purchased gas expense developed for these periods. A least cost dispatch analysis was performed to determine the mix of flowing supplies and storage withdrawals that would be dispatched to the city gate each month to serve estimated normal firm customer demand.

Q. Please describe Exhibit __ (EDA-7) – Forecast of Purchased Gas Expense TME March 31, 2021, 2022, 2023 and 2024.

A. Exhibit __ (EDA-7) shows a forecast of purchased gas expense for the Rate Year and Data Years. This expense includes the purchased cost of gas minus the cost of storage injections plus the cost of storage withdrawals, and all pipeline fixed and variable charges. The forecast assumes the NESE project is available by the 2020/21 winter season. If it is not, the Companies will update the forecast in corrections and updates.

Q. How do the Companies review and monitor the costs associated with their pipeline capacity?

A. The Companies closely monitors and reviews all rate and tariff filings submitted to the Federal Energy Regulatory Commission ("FERC") by the interstate natural gas pipelines with whom the Companies contract for service. The Companies will intervene, and when necessary, actively participate in such proceedings to ensure the Companies continue to pay fair rates and receive just and reasonable service. In interstate pipeline base rate proceedings, the Companies often participate with other local distribution company customers in a shipper group, which will retain an expert witness to advance commonly-held positions and enhance the Companies' influence. The Companies also regularly engage in settlement discussions with gas pipelines and FERC Staff to settle tariff-related matters without the need for a formal and costly hearing process. More broadly, the Companies monitor all natural gas related activity at FERC to stay abreast of

and influence policy developments that could impact the Companies and their customers. The Companies track FERC orders, attend technical conferences convened by FERC to address major policy issues, and submit comments in response to notices of proposed rulemaking and notices of inquiry issued by FERC on policy matters impacting wholesale gas markets.

V. Marginal Cost Studies

Q. Please describe Exhibit __ (EDA-8) – Estimated Marginal Commodity Cost of Gas For Period: November 1, 2020 through March 31, 2021.

A. Exhibit __ (EDA-8) shows the projected marginal gas commodity costs for the period November 1, 2020 to March 31, 2021. By running two dispatch simulations, the marginal gas supply sources that would be dispatched to serve an incremental increase in customer demand were identified. First, a baseline dispatch on the simulation model was prepared to establish the least-cost mix of gas supplies that would be dispatched to serve firm sales customer demand under normal weather. Then, the simulation model was rerun with an increased customer demand of 1,000 Dth per day over the winter months (November through March) to identify those marginal supplies that would be dispatched to serve the increased demand. The exhibit reflects the average monthly commodity cost of the marginal supplies that were dispatched.

Q. Please describe Exhibit __ (EDA-9) – Estimated Annualized Marginal Capacity Cost of Gas for Period November 1, 2020 through March 31, 2021.

A. Exhibit __ (EDA-9) shows the projected annualized marginal gas capacity cost for the period November 1, 2020 to March 31, 2021. This study incorporates the fixed costs of new capacity that the Companies would acquire for the Rate Year to reliably meet projected design demand growth. Based on this calculation, the annualized marginal capacity cost was determined to be $1.12 per dekatherm, as set forth on Exhibit __ (EDA-9).

VI. Customer Choice Program

Q. Describe the Companies' Customer Choice Program.

A. The Companies' Customer Choice Program provides customers the option to purchase their supplies from Marketers. There are two service options, Daily and Monthly Balancing. The Companies currently assign to Marketers at maximum rates such interstate pipeline transportation and storage capacity as is necessary to meet migrating firm customers' load. This practice is consistent with the Commission's August 30, 2007 Order in Case 07-G-0299. City gate pipeline and storage capacity contracted for core customers is assigned to Retail Marketers for migrating customers in proportion to the anticipated design day load of the customers. All migrating firm customers are required to participate in the Companies' mandatory assignment program.

Q. What service classes are available to transportation customers?

A. For KEDNY, transportation customers can receive service under Service Classification ("SC") 17 Firm and SC-18 Non-firm. For KEDLI, transportation service is available under SC-5 Firm, SC-7 Non-Firm, and SC-13 Non-Firm.

Q. How are transportation customer imbalances tracked and reconciled?

A. When an ESCO customer's meter is read, the customer's account transaction is sent from the customer system to the Companies' Gas Transportation Information System ("GTIS"). GTIS then compares the ESCO's expected deliveries, adjusted for actual weather for the billing period, to the customer's actual usage for that same period. The usage is prorated into the applicable calendar month based on actual degree days. Thereafter, a cash-out index price is applied to the difference between the ESCO's deliveries and customer usage. Cancel and rebill activity will be captured in the month in which the cancel or the rebill occurred as far back as three years. There have been no imbalance penalties assessed to a customer in the last three years.

Q. Were there any problems or issues arising from implementation of the Commission's August 30, 2007 and March 28, 2008 Mandatory Capacity Release Orders (in Case 07-G-0299) and FERC Order 712?

A. The Companies have not experienced any issues from implementation of the Commission's Mandatory Release orders. The Companies no longer have any ESCO capacity that is grandfathered to the city gate.

Q. What are the pipelines and allocation percentages being utilized for the mandatory assignment of capacity?

A. The table below provides a list of the pipelines being utilized for mandatory assignment of capacity, and the allocated percentage of each, for Winter 2018-2019.

Transco Long Haul	29.7%
Transco Leidy	4.4%
Transco Leidy Seasonal	2.5%
Transco Rockaway	17%
Tetco Long Haul	7.1%
Tetco Leidy Short Haul	5.6%
Dominion Leidy	7.0%
Tennessee	2.6%
Northeast 07	15.1%
Dawn to Iroquois	9.0%

Q. What process, if any, is utilized to true-up any differences between the Companies' weighted average cost of capacity ("WACOC") and the charges paid by marketers and direct customers for released capacity?

A. For the 2017/2018 winter season (November 2017 through October 2018), the Companies' WACOC was between $0.8872 and $0.9006 per Dth. There is no true-up and/or reconciliation process in place to account for any differences in cost.

Q. Did the Companies implement any changes to their Customer Choice Programs as a result of the non-firm service collaborative in the 2016 KEDNY Rate Case?

A. Yes. In accordance with the "Order Approving Tariff Provisions and Directing Further Tariff Filings" issued February 7, 2019 following the non-firm service collaborative in the 2016 KEDNY Rate Case, the Companies modified their Customer Choice Programs to address concerns raised by ESCOs that participated in the collaborative.

First, the Companies modified their procedures for accepting month-ahead and day-ahead nominations to allow ESCOs more flexibility with the delivery points. Whereas ESCOs were previously required to deliver 100 percent of their supplies on Transco, they are now permitted to deliver gas supply on behalf of their temperature controlled ("TC") and interruptible ("IT") customers up to the following percentage of their TC and IT customers' total requirements:

- Transco – up to 100 percent
- Texas Eastern – up to 50 percent
- Iroquois – up to 50 percent
- Tennessee – up to 6 percent

These modifications do not apply to power generation customers who have negotiated points of receipt in their gas transportation agreements.

Second, the Companies modified their operating procedures to allow ESCOs serving non-firm loads and power generation customers to request access to any available city gate capacity on an intra-day basis. To request access to such capacity, ESCOs must notify the Companies by phone or email between 8:30 A.M. and 10:00 A.M. prior to the start of the effective Gas Day. Between 10:00 A.M. and 11:00 A.M., the Companies will allocate available capacity on a pro rata basis to each requesting ESCO and notify the ESCOs of the capacity they have received. If there is any remaining capacity after allocation, it will be made available on a first-come, first-served basis after 11:00 A.M.

Both changes to the Companies' Customer Choice Programs have been reflected in the latest version of the GTOPs.

Q. Did the Companies modify the TC/IT penalties as suggested in the non-firm service collaborative in Cases 16-G-0058 and 16-G-0059?

A. Yes. The Companies submitted revised tariff leaves for changing the current penalties from:

- the *higher* of (i) two times the sum of the market gas price plus the applicable IT or TC transportation rate; or (ii) nine times the applicable IT or TC sales rate, to
- the *lower* of (i) two times the sum of the market gas rate plus the applicable or IT or TC transportation rate; or (ii) nine times the applicable IT or TC sales rate

This modified TC/IT penalty language provides a strong incentive for customers to switch to their alternate fuel at the designated temperature thresholds.

Q. Are the Companies proposing to make any further changes to their Customer Choice Programs?

A. No.

VII. Gas Cost Volatility Management

Q. What steps do the Companies take to mitigate the impact of gas cost volatility on their customers?

A. The Companies mitigate volatility in the gas commodity markets in several ways. First, they maintain a balanced portfolio that includes contract storage. This allows the Companies to inject gas during the summer for withdrawal during the winter, which enables the Companies to mitigate price volatility during the winter when demand is greatest. Second, the Companies maintain a geographically diverse gas supply portfolio that helps to reduce exposure to volatility in any single supply region. Third, the portfolio incorporates pricing diversity that minimizes exposure to volatility at a single pricing point or market index. Finally, the Companies mitigate price volatility with a formal hedging program.

Q. Please describe the Companies' hedging program.

A. The Companies' overall pricing diversity and volatility mitigation plan is to protect prices for approximately 50 percent, but no more than 60 percent, of

forecast winter firm sales and to allow prices to float with the market for the remaining forecast winter sales. For this purpose, the "winter period" is defined as November through March. Prices are fixed through a combination of planned storage withdrawals, which provide a natural hedge at the average price of summer period injections, and financial hedging using NYMEX gas futures contracts or NYMEX Over the Counter ("OTC") financial settled swaps and options using bilateral master agreements. Beginning in the 2015-2016 winter, the Companies are also using basis hedges for forecast purchases in the Northeast producing region. Locational basis swaps in conjunction with NYMEX hedges are being used because of an observed disconnect between prices in the Marcellus and Utica supply basins and NYMEX prices. The use of these locational basis swaps ensures that the Companies are mitigating volatility in the markets where they purchase supplies.

To determine the quantity to be hedged, the Companies forecast firm sales for each month November through March, assuming normal winter weather conditions, and multiply the results by 50 percent (Step 1). Next, monthly storage withdrawals to meet system operational needs are forecast and subtracted from the result obtained in Step 1 (Step 2). The results from Step 2 equal the quantity of gas to be financially hedged to achieve the 50 percent target in each month.

Once the volume of gas to be financially hedged each month is known, a monthly hedging plan is created. The purchases are spread evenly over 16 months starting

18 months prior to the start of each November. Each financial transaction is settled against the expiring month's last day settlement price and the applicable Inside FERC "First of the Month" index price. The gross settlement payout or gain is included in the gas cost for that month.

The table below summarizes the percentage of forecasted gas supply hedged physically and financially for November 2017-March 2018:

Pricing Mechanism	Downstate NY
Physical (storage)	30%
Financial (swaps and options)	19%
Index	51%

The table below itemizes the number of financial swap (futures) and option contracts hedged for the November 2017 through March 2018 winter season. The units for the data below are in contracts; one contract equals 10,000 Dth.

Pricing Mechanism	Quantity
Swaps	1,815
Options	750

Q. Have the Companies' hedging practices changed in the past year?

A. Yes. The derivatives market at Dominion South Point has developed with enough liquidity so that the Companies can adequately acquire financially settled Dominion South Point options. The Companies have incorporated these options in the winter 2018-19 hedge portfolio. The Companies continue to monitor the

effectiveness of NYMEX and South Point for all receipt points in the forecast to determine the most appropriate hedging location.

Lower market volatility and market prices have translated to lower option premiums. Options provide superior downside protection versus swaps. As a result, "lessons learned" include favoring up to 100 percent of options for the financial hedge component of the portfolios for the next winter

Q. Please explain how the Companies calculate gas price volatility.

A. Gas price volatility is measured as the standard deviation of the lognormal of the ratio of the monthly hedged price change through the winter, November through March. This standard deviation is compared against a similar calculation for the NYMEX and South Point Natural Gas monthly settlement prices for the same November through March period.

Q. Discuss how the Companies determine the success or failure of their gas price mitigation program.

A. The success or failure of the gas price mitigation program is determined by comparing the physical and financial hedged prices with natural gas market price indices over the same period. The hedge program is expected to show lower month-to-month price volatility than that of the market index.

Q. Please discuss internal reporting, oversight, and the audit structure of the Companies' gas price mitigation program.

A. The Companies' hedging program reporting and oversight is documented in the US Energy Commodity Risk Management Policy. National Grid's US based energy procurement activities are managed under the direction of National Grid plc's Finance Committee. The Finance Committee authorized the Energy Procurement Risk Management Committee ("EPRMC") and delegated authority to focus on energy risk, metrics, energy strategies, financial impacts and other opportunities to the EPRMC. The EPRMC provides Vice President level review of strategies, with a focus on market risks, inclusive of price, credit, operational, liquidity and reputational risks and Policy compliance. The EPRMC serves to administer a consistent and comprehensive sanctioning process for such strategies and commitments across the organization. The EPRMC has delegated some of its authority to the Commodity Management Committee ("CMC"). The EPRMC appoints the CMC members and chairperson. The CMC provides detailed review of strategies, products and new opportunities with a focus on energy risk, metrics, financial impacts and opportunities. The CMC conducts research to understand market rules, regulatory requirements, customer needs, risks, barriers to entry, logistical requirements, competition, deal economics, risk measurement and valuation requirements, all related requirements and exposures (including credit, liquidity, tax and legal), and the impact of current and proposed strategies on existing controls and limits, while administering a consistent and comprehensive sanctioning process for such strategies and commitments across the organization.

The Companies review the hedging program and highlights any changes with the Commission on an annual basis. The Companies have a comprehensive internal approval process to authorize a hedge strategy and maintain oversight requirements. Prior to execution, a hedging strategy is documented with details that may include: risks to be hedged, volumetric targets, duration and cadence of hedge plan transactions, and types of instruments to be used. This documentation is then reviewed and approved by the CMC prior to going to the EPRMC for approval and authorization.

The Companies employs the industry standard best practice of an independent three-office model:

- The Front Office develops and executes the hedge strategies hedges in accordance with the prevailing policies and strategies by entering into transactions with counterparties to mitigate natural gas price volatility.
- Risk Management is part of the Middle Office and maintains the overall control environment and assesses compliance with the Companies' risk policy. The Middle Office reports through the Treasury organization and provides a significant level of control and independent policing of the Front Office's activities. The Middle Office confirms all trades with counterparties and monitors the risk exposures of the deals, as well as verifying approved strategies are executed in accordance with the plan.
- The Accounting Department is part of the Back Office whose functions include processes in support of the Front Office, such as accounting,

invoicing, check-outs, actualization, accounts receivable and payable, and financial reporting. The Back Office is also independent of both the Front and Middle Offices and reports up through the Finance organization.

Additionally, the Companies' internal auditing department has performed audits to ensure compliance, for example, with Sarbanes-Oxley.

Q. Please provide the actual price hedging performance versus planned price hedging performance for the last winter season.

A. A comparison of actual price hedging performance versus planned price hedging performance for winter 2017/2018, which includes separate quantities for each hedging instrument, is shown in the tables below:

Testimony of Elizabeth D. Arangio

Winter 2017-18 Purchasing Plan & Projected Prices

notes	Percent Hedged	Portfolio	Amount	Price
		Physical Hedges		
1	32%	Market Area Storage	38,619,000	$ 2.14
1	1%	Gulf Coast Storage	1,500,000	$ 2.14
		Fixed Price Contracts		
		Financial Hedges *		
	6%	NYMEX Correlated Futures or Swaps	7,860,000	$ 3.26
2	8%	Dom. SP Correlated Futures or Swaps	10,290,000	$ 2.80
2	0%	Collars		
2	6%	Calls	7,500,000	$ 2.73
		Puts		
		Flowing or Floating Price Gas		
	46%	NYMEX/Dom SP (18%/82%)	55,707,000	$ 2.52
		Spot/Daily Price		
	100%	**TOTAL**	121,476,000	

Winter 2017-18 Actual Purchases & Prices

notes	Percent Hedged	Portfolio	Amount	Price
		Physical Hedges		
	32%	Market Area Storage	41,587,000	$ 2.02
	1%	Gulf Coast Storage	1,500,000	$ 2.02
		Fixed Price Contracts		
		Financial Hedges *		
3	5%	NYMEX Futures or Swaps	6,700,000	$ 3.25
2,3	9%	Dom. SP Correlated Futures or Swaps	11,450,000	$ 2.81
2,3	0%	Collars	0	$ -
2,3	6%	Calls	7,500,000	$ 2.64
		Puts		
		Flowing or Floating Price Gas		
3	1%	Monthly Index	746,000	$ 2.46
3	**20%**	**Subtotal Monthly/Baseload**	**26,396,000**	**$ 2.86**
	44.0%	Spot (non-peaking)	57,855,000	**$ 4.10**
	2%	Peaking	2,500,000	**$ 22.40**
	46%	**Subtotal Spot/Daily Price**	60,355,000	$ 4.86
	100%	**TOTAL**	129,838,000	$ 3.51

note 1 In the ground volumes and WACOG

note 2 Financial hedges settled against Domionion SP index.

note 3 Represents total Baseload, financial and physical

Q. What percentage of the Companies' gas supply is physically hedged?

A. As a result of planned storage withdrawals, which are based on normal weather, approximately 30 percent of the forecasted November through March firm sales demand is physically hedged. The Companies do not hedge storage injections. The Companies do not have any physical supply contracts with fixed price terms.

Q. How do the Companies use swaps/futures?

A. The Companies use OTC swaps to execute fixed price hedged transactions. OTC swaps do not have any execution, transaction or commission fees. The Companies rely on credit thresholds in their bilateral master agreements to limit the amount and frequency of margin calls associated with the daily mark-to-market valuation of each hedge transaction. When the mark-to-market with each OTC counterparty exceeds the credit threshold, the Companies use their various credit facilities to meet the cash collateral margin calls.

Q. What types of options do the Companies use?

A. The Companies use calls, puts, and collars.

Q. Describe how the Companies decide which types of options to use.

A. When the underlying futures price is expected to fall, then call options are preferred over swaps. Collars may be purchased instead to reduce the premiums paid or when the underlying futures prices are expected to be stable.

Q. Do the Companies place a limit on what they spend on options in any year?

A. The Companies cap their option premiums at $13 million per year.

VIII. Conclusion

Q. Does this conclude your testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Supplemental Direct Testimony

of

Elizabeth D. Arangio

Date: June 11, 2019

Supplemental Direct Testimony of Elizabeth D. Arangio

Table of Contents

I. Introduction....1

II. Purpose of Testimony....1

III. Impact of the Unavailability of the NESE Project....4

IV. Conclusion....8

I. Introduction

Q. Please state your name and business address.

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Elizabeth D. Arangio who previously submitted direct testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively the "Companies") in these proceedings?

A. Yes, I am. The terms defined in my direct testimony have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of your supplemental direct testimony in these proceedings?

A. The purpose of my supplemental direct testimony is to discuss the impacts to the Companies' rate filings if KEDNY and KEDLI are unable to obtain incremental natural gas supplies from Transcontinental Gas Pipe Line LLC's ("Transco") Northeast Supply Enhancement ("NESE") project during the Rate Year and Data Years. Specifically, I will describe the recent

developments that have occurred with respect to the NESE project since the Companies' April 2019 rate filings in these proceedings and explain how the failure to place the project in service affects the Companies' proposals in these cases. While the Companies remain cautiously optimistic the NESE project will go forward, and therefore are not proposing any changes to their April 2019 filings at this time, I will address the impact that continued gas supply constraints would have on the Companies' forecast of gas supply costs, ability to add new load, and certain aspects of the Companies' forecast of capital additions. These and other impacts resulting from the unavailability of the NESE project are further discussed in the supplemental direct testimonies of the Revenue Requirements Panels, the Gas Infrastructure and Operations Panels ("GIOPs"), and Company Witness Theodore Poe, Jr.

Q. Does your supplemental direct testimony include any exhibits?

A. Yes. My supplemental direct testimony includes the following exhibits that were prepared or compiled under my direction and supervision:

Exhibit __ (EDA-1S) Forecast of Variable Gas Expense (without NESE) for the Twelve Months Ending ("TME") March 31, 2021, 2022, 2023, and 2024;

Exhibit __ (EDA-2S)	Forecast of Purchased Gas Expense (without NESE) for TME March 31, 2021, 2022, 2023, and 2024;
Exhibit __ (EDA-3S)	Estimated Marginal Commodity Cost of Gas (without NESE);
Exhibit __ (EDA-4S)	Estimated Annualized Marginal Capacity Cost of Gas (without NESE); and
Exhibit __ (EDA-5S)	The Companies' updated peak day supply/demand balance under the assumption that the NESE project will not be available.

Q. What recent developments have occurred that indicate that the NESE project may not be available during the Rate Years and Data Years?

A. On May 15, 2019 the New York State Department of Environmental Conservation ("DEC") denied Transco's application for a federal Clean Water Act Section 401 Water Quality Certificate ("WQC") for the NESE project. The denial was without prejudice to the resubmission of the application to address the deficiencies in Transco's application identified by DEC. Furthermore, on June 5, 2019, the New Jersey Department of Environmental Protection denied (without prejudice) Transco's application for necessary permits to construct the NESE project. While Transco has

already resubmitted its application for a WQC to the DEC and expects to resubmit the applications in New Jersey shortly, it is not clear when or whether the requests will be granted. Without the necessary approvals from both states, the NESE project cannot proceed.

Q. Did the Companies' April 2019 filings in these proceedings discuss the possibility that the NESE project might not be available and that this could affect the Companies' filings?

A. Yes. While the Companies' filings assumed the NESE project would be completed by the winter of 2020/2021, my direct testimony (at 14-16) identified the possibility that the NESE project would not be available as planned and explained that the Companies would not be able to prudently satisfy new or additional service requests if the NESE project were not placed in service.

III. <u>Impact of the Unavailability of the NESE Project</u>

Q. What actions have the Companies taken as a result of the DEC's decision to reject Transco's application for a WQC for the NESE project?

A. Effective May 15, 2019, consistent with the Commission's rules and regulations and the terms of the Companies' tariffs, and in consideration of

public safety, the Companies advised the Commission and customers that they will not fulfill applications for new firm service connections, or requests for additional firm load from existing customers, in the affected areas of their service territories until the current supply constraints are addressed. The Companies, however, will continue processing applications for non-firm service from qualifying customers to the extent the Companies are able to provide such service. The impact of these actions on the Companies' sales forecast is discussed by Company Witness Poe.

Q. Why was it necessary for the Companies to take such actions?

A. Yes. The Companies have an obligation to ensure they have access to supplies of gas that are sufficient to serve the peak day, winter season, and annual demands of their firm customers. While the Companies obtain a small portion of their total gas supply from local sources, the vast majority must be accessed in various supply basins and transported on a firm basis to the Companies' city gates to ensure reliability. Based on current forecasts, the Companies believe that without the NESE project they will not have access to quantities of reliable firm supplies sufficient to serve future increases in firm demand beyond those that the Companies have committed to as of May 15, 2019. Exhibit __ (EDA-5S), which updates a chart included on page 15 of my direct testimony, reflects the Companies'

peak day supply/demand balance based on the revised forecast by Company Witness Poe as well as the current supply portfolio.

Q. Have the Companies added incremental supplies to partially replace the supplies that will not be available if the NESE project is not placed in service?

A. Yes. The Companies have procured 261,000 dekatherms per day of incremental peaking supplies under contracts that provide for delivery at the Companies' city gates during the 2020/2021 winter period. These incremental peaking supplies will enable the Companies to provide service reliably to their existing firm customers, but are not sufficient to serve additional firm customer load in the affected areas of the Companies' service territories. These incremental peaking supplies are included in the revised peak day supply/demand balance set forth on Exhibit __ (EDA-5S).

Q. Will the unavailability of the NESE project have any impact on the Companies' projected capital spending for gas supply-related projects?

A. Yes. As discussed by the Companies' GIOPs, the Companies planned to undertake tank modernization projects at KEDNY's Greenpoint and KEDLI's Holtsville Liquefied Natural Gas ("LNG") facilities. These projects require the Companies to take each LNG facility out of service for

one winter period each, with Holtsville being taken out of service first, followed by Greenpoint. Because the supplies from the LNG facilities are essential components of the Companies' peak day supply portfolio, without access to incremental supplies from the NESE project, the Companies cannot take either tank out of service and cannot proceed with these projects. In addition, as further discussed by the GIOPs, the unavailability of the NESE project will affect the Companies' projected capital and related operating expenditures associated with new business during the Rate Year and Data Years.

Q. Will the unavailability of the NESE project affect the Companies' forecast of gas costs in the Rate Year and Data Years?

A. Yes. Illustrative forecasts of variable gas expense and purchased gas expense for the Rate Years and Data Years are set forth in Exhibits __ (EDA-1S) and __ (EDA-2S), respectively.

Q. Will the unavailability of the NESE project affect the Companies' estimated marginal commodity cost of gas and estimated annualized marginal capacity cost of gas?

A. Yes. Illustrative estimates of the estimated marginal commodity cost of gas and estimated marginal capacity cost of gas are set forth in Exhibits __ (EDA-3S) and __ (EDA-4S), respectively.

Q. Do the Companies plan to continue to provide updates in these proceedings concerning the status of the NESE project?

A. Yes. The Companies are cautiously optimistic that the NESE project will be approved. As the project's status is uncertain, the Companies will work with Staff and other parties throughout these proceedings and will supplement the record as further developments occur.

IV. Conclusion

Q. Does that conclude your supplemental direct testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Elizabeth D. Arangio

Dated: July 3, 2019

Corrections and Updates Testimony of Elizabeth D. Arangio

Q. Please state your name and business address.

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Elizabeth D. Arangio who previously submitted direct and supplemental testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes, I am. The terms defined in my direct testimony have the same meanings here.

Q. What is the purpose of your corrections and updates testimony?

A. The purpose of my testimony is to provide updates to certain exhibits based on an updated gas sales forecast provided by Company Witness Theodore Poe, Jr. Further, the forecast fixed costs were updated to include the anticipated impacts of currently pending pipeline rate cases, as well as to account for additional peaking supplies secured since the Companies' April 30, 2019 filings.

Q. Does your testimony also address the Companies' June 11, 2019 supplemental filing?

A. No. The supplemental filing was made to illustrate the potential impacts to the April 30, 2019 filing if KEDNY and KEDLI are unable to obtain incremental natural gas supplies from the NESE project. At this time, the Companies remain optimistic that the NESE project will be approved. Therefore, all corrections and/or updates discussed herein pertain to the April 30 base rate filing. The Companies will continue to update the parties on the status of the NESE project and implications, if any, to this rate filing.

Q. Which exhibits are you updating in your corrections and updates testimony?

A. My testimony includes the following exhibits that were prepared under my direction and supervision:

Exhibit __ (EDA-4CU)	KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage;
Exhibit __ (EDA-6CU)	KEDNY/KEDLI Forecast of Variable Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;

Exhibit __ (EDA-7CU)	KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;
Exhibit __ (EDA-8CU)	KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas;
Exhibit __ (EDA-9CU)	KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas.

Q. Please describe the impact of the updated gas sales forecast.

A. Overall, the impact of the updated gas sales forecast was minimal. The updated forecast increased the projected storage dollars by less than one percent, and forecast variable gas expense by approximately 1.5 percent. The marginal commodity cost of gas and estimated marginal capacity cost of gas were also minimally impacted, and in fact saw decreases of one percent and eleven percent, respectively. The changes to the forecast fixed costs are mainly due to the corrections and updates discussed further below.

Q. Please describe the changes included in the update to the forecast fixed costs in Exhibit __ (EDA-7CU).

A. The following three corrections and updates were made to the forecast fixed costs previously provided in my direct testimony:

- Exhibit __ (EDA-7) in my direct testimony inadvertently excluded the impact of pipeline rate cases currently pending before FERC. The updated exhibit is corrected to include $59 million per year of estimated pipeline rate increases, attributed mainly to the Transco and Texas Eastern rate cases. This correction will not increase rates in the Rate Year because it was excluded only in Exhibit __ (EDA-7), but included in the bill impacts submitted by the Rate Design Panels in the Companies' initial filings.
- Exhibit __ (EDA-7CU) also updates the estimates of city gate peaking reservation charges used in my direct testimony with actual costs from peaking agreements that have been executed since the initial filings;
- In the initial filings, Exhibit __ (EDA-7) inadvertently excluded estimated fixed costs associated with the existing Millennium, Union, and TransCanada contracts for some of the Data Years because the contracts do not contain evergreen provisions. However, the Companies reasonably expect that the contracts will be extended to maintain access to liquid supply points. Therefore, Exhibit __ (EDA-7CU) includes these costs for all three Data Years.

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Elizabeth D. Arangio

Case 19-G-0309
Case 19-G-0310

September 18, 2019

Case 19-G-0309
Case 19-G-0310

Rebuttal Testimony of Elizabeth D. Arangio

TABLE OF CONTENTS

I. Introduction 1

II. Purpose of Testimony 1

III. NESE Project 2

IV. Hedging 8

V. Supply Planning Process 9

VI. Conclusion 11

Case 19-G-0309
Case 19-G-0310

Rebuttal Testimony of Elizabeth D. Arangio

I. Introduction

Q. Please state your name and business address

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Elizabeth D. Arangio who previously submitted direct, supplemental, and corrections and updates testimony in these proceedings?

A. Yes, I am. The defined terms in my direct testimony have the same definitions here.

II. Purpose of Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my rebuttal testimony is to respond to the gas supply-related recommendations of the Department of Public Service Staff Gas Supply Panel ("SGSP"), City of New York ("CNY") Policy Panel, and Environmental Defense Fund ("EDF") witness Gregory Lander. Specifically, I will address the following topics:

(i) The CNY Policy Panel's comments and recommendations regarding the NESE project and the current gas supply constraint in Downstate New York;

(ii) The SGSP's recommendation regarding the Companies' gas hedging policies; and

(iii) The comments and recommendations of EDF witness Gregory Lander regarding the Companies' gas supply planning process.

III. NESE Project

Q. What is the current status of the NESE project?

A. Since my supplemental direct testimony, both the New York State Department of Environmental Conservation and the New Jersey Department of Environmental Protection have issued a Notice of Complete Application and concluded their respective public comment periods. The Companies continue to await the state agencies' approval of the necessary permits to proceed with the project. Although the Companies continue to remain cautiously optimistic that the NESE project will be approved, because the timing of approval is uncertain, the Companies have adjusted their filings to reflect the unavailability of the NESE project by the 2020-2021 winter season. This does not, however, preclude the possibility the NESE project could be approved and become available. If that occurs, the Company's position is that the Commission should consider the proposals and costs presented in the Company's direct and corrections and updates filings.

Q. Have you updated any exhibits to reflect the unavailability of the NESE project?

A. No. The exhibits submitted in my supplemental direct testimony will not differ meaningfully and provide an illustrative forecast of the gas costs in the Rate Year and Data Years under the assumption that the NESE project is not in service by the 2020-2021 winter season.

Q. The CNY Policy Panel (at 21-22) criticizes the Companies' supply planning process, contending that the Companies should have had an alternative plan in the event the NESE project was not approved and that the failure to do so was imprudent. Do you agree with these criticisms?

A. No, I do not. The CNY Policy Panel cites no analysis or evidence to support its claim aside from a blanket statement that the process must be deficient because the Company cannot serve incremental load absent approval of the NESE project. This criticism ignores the well-documented, comprehensive planning process undertaken by the Company for well over a decade.

In these filings, as well as past filings before the Commission and numerous stakeholder meetings, including the annual gas supply meeting with Staff and other parties, the Companies' detailed their long-term plan aimed at addressing growth in Downstate New York, of which the NESE project has

been a critical part. Since as early as 2009, the Companies planned for the construction of a metering and regulation station at Floyd Bennett Field ("FBF"), which provided 647,000 Dth/day of takeaway capacity. The FBF station, completed in 2015, required an act of Congress to enact the New York City Natural Gas Supply Enhancement Act of 2012. The Companies signed precedent agreements for two upstream projects to bring firm capacity through the FBF station, first through the Northeast Connector project that provided 100,000 Dth/day of capacity, and then the New York Bay Expansion project, providing another 115,000 Dth/day for a total of 215,000 Dth/day of capacity through the FBF station. The NESE project is the final piece that will bring an additional 400,000 Dth/day of pipeline capacity to the Downstate New York area.

At the same time, the Companies have also invested in large on-system projects to support the upstream capacity into the distribution system, including the Brooklyn-Queens Interconnect project and more recently the Metropolitan Reliability Infrastructure project. These capital projects, both upstream and on-system, have been a part of the Companies' system and supply plan and clearly communicated to all stakeholders through the years.

Other than an increased reliance on third-party peaking supply and local CNG stations, the Companies have no viable alternative to the NESE project in terms of traditional pipeline capacity – a fact that the CNY Policy Panel ignores. Furthermore, such stop-gap measures are short-term solutions and will not address the ongoing supply constraint. All other city gate stations are at their maximum capacity, and increased reliance on third party peaking supply creates a reliability risk of not being able to procure sufficient gas. The only other option would be to pursue construction of a new station and upstream pipeline expansion projects, which would not only impose significant costs but also face the same regulatory hurdles that the NESE project is currently experiencing.

Although pipeline capacity cannot be entirely displaced in the near future, the Companies have proposed several ways to reduce reliance on pipeline capacity. As presented by the Future of Heat Panel, the Companies proposed various decarbonization initiatives including renewable natural gas projects and enhanced energy efficiency programs. Furthermore, consistent with the Commission's goals, the Companies developed new non-firm demand response service classifications that will incentivize existing firm customers and new customers to choose interruptible service that will relieve peak day demand. The Companies, however, emphasize that such initiatives and

programs are part of a long-term plan to reduce carbon emission and do not provide a solution for the immediate shortfall of supply.

Q. The CNY Policy Panel recommends (at 22-23) that the Commission require the Companies to file additional testimony addressing how they will operate their systems, without a moratorium, if NESE is not approved. Do you agree?

A. No. The recommendation ignores the extensive record in this case. In these filings, the Companies filed two sets of testimony detailing the system investments required under scenarios where the NESE project is approved or not approved within the timeframe required. The Companies have also responded to numerous discovery requests from the parties to these rate filings, including the CNY, addressing system operation and needs under both scenarios.

Further, while the CNY Policy Panel highlights its proposed alternative demand side programs, it entirely ignores the Company's Future of Heat proposals, which are intended to promote demand side management and increase reliance on non-pipeline alternatives. My direct testimony also highlighted the Companies' efforts to procure indigenous and renewable natural gas. Although such supply and demand side programs are important,

such measures cannot replace the volume of natural gas that the Companies require from pipeline capacity. Said another way, there are currently no alternatives to the NESE project that can provide the same level of natural gas supply to the Companies' distribution system. Any other pipeline projects such as Con Edison's precedent agreement with Tennessee Gas Pipeline, to the extent they are available, will face the same regulatory hurdles as the NESE project. For all these reasons, there is no sound basis for the CNY Policy Panel's recommendation.

Q. The CNY Policy Panel recommends (at 25) that the Commission have Staff investigate the underlying causes of the moratorium, require the Companies to provide notice to customers of any deadlines associated with gas service for those who applied prior to May 15, 2019, and investigate whether the Companies are treating customers fairly through the process. Do you agree?

A. No. The Company has worked with Staff and its consultants for the past several months to support the Commission's ongoing analysis of the current supply constraints in Downstate New York in Case 19-G-0080. Staff's report on the investigation was due to be completed by July 1, 2019, but has been delayed.

Given the Commission's current on-going efforts, the Companies do not believe further investigations are necessary. The Companies have been working closely with Staff and its consultants to support the analysis and provided system models, engineering analyses, growth and demand forecasts, supply data, and other information. The information provided by the Companies demonstrates the extent of the current constraints and the Companies' inability to serve new customers in the absence of incremental supplies. Therefore, the CNY Policy Panel's recommendation is unnecessary and should be rejected.

IV. Hedging

Q. What did the SGSP recommend regarding the Companies' gas hedging policies?

A. The SGSP supported the Companies' recent change in hedging practices to incorporate financially settled Dominion South Point options and favoring up to 100 percent of options for the financial hedge component of the portfolios for the upcoming winter. The SGSP recommended (at 13) that the Companies be required to continue to monitor their hedging strategies on an ongoing basis and to review such changes as this with Staff during the annual winter supply review.

Q. Do you have any comment on the SGSP's recommendation?

A. The Companies acknowledge SGSP's recommendation to continue to monitor the hedging strategies, which is already a part of the Companies' formal hedging program. The Companies also commit to continue their current practice of reviewing any changes to their hedging practices with Staff during the annual winter supply review.

V. Supply Planning Process

Q. Please summarize EDF's recommendations regarding the Companies' current supply planning process.

A. EDF recommends the following:

1. Companies should be subject to an annual gas supply process, with discovery rights for intervenors and technical conferences as needed (at 4).
2. Companies should be required to submit a long-range plan, which would become the basis for evaluating future cost recovery (at 4).
3. Companies should submit additional categories of information to assist the Commission in its review of gas supply issues (at 5).

Q. Do you agree with EDF's recommendations?

A. No. As EDF notes (at 10-11), the Companies perform an annual gas supply review, which includes filing of various data supporting the Companies' long-

term supply plan as well as an in-person meeting with Staff and interested parties. Each year, the review is performed under a new case number and is available to the public. Interested parties can become a party to the proceeding as well as submit comments. The Companies do not believe that a formal evidentiary hearing is necessary to review the Companies' annual supply plan and will unnecessarily add time and cost with no significant incremental benefit.

Contrary to EDF's assertion, the annual winter supply review is a public process – the Companies file responses to Staff's data requests in the public docket, but with certain redactions for commercially sensitive and critical energy infrastructure information. Such redactions are necessary to ensure public safety and the competitive position of the Companies and are provided for exemption from public disclosure under the applicable Public Officers Law. The Companies do not believe that EDF's suggestions for additional information, technical conferences, and other procedural steps would provide a clear incremental benefit to the existing process compared to the incremental burden of cost and time.

Q. What does EDF recommend regarding the New York Facilities System ("NYFS")?

A. EDF recommends (at 19-20) that the Commission initiate a proceeding to formalize the planning assumptions of the NYFS and integrate into the more general downstate energy planning process of the NYFS members.

Q. Do you agree?

A. No. The Companies note that the Commission has reviewed various aspects of the NYFS through prior rate proceedings as well as other Commission proceedings, including the review of the NYFS agreement in Case 18-G-0318. The Companies do not believe that an additional proceeding or other formal procedure will add value to the existing venues for review of the NYFS.

VI. Conclusion

Q. Does this conclude your rebuttal testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Second Supplemental Testimony

of

Elizabeth D. Arangio

Case 19-G-0309
Case 19-G-0310

Dated: December 13, 2019

Q. Please state your name and business address.

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Elizabeth D. Arangio who previously submitted testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes, I am. The terms defined in my previous testimonies have the same meanings here.

Q. What is the purpose of your second supplemental testimony?

A. The purpose of my second supplemental testimony is to provide updates to certain exhibits previously submitted in my direct, supplemental, and corrections and updates testimonies. Specifically, the following exhibits were prepared under my direction and supervision:

Exhibit __ (EDA-1SS)	KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage;
Exhibit __ (EDA-2SS)	KEDNY/KEDLI Forecast of Variable Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;
Exhibit __ (EDA-3SS)	KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;

Exhibit __ (EDA-4SS)	KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas;
Exhibit __ (EDA-5SS)	KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas;
Exhibit __ (EDA-6SS)	The Companies' updated peak day supply/demand.

Q. What updates have been made to the exhibits?

A. The exhibits have been revised based on an updated gas sales forecast provided by Company Witness Theodore Poe, Jr. On November 26, 2019, the Commission issued an "Order Adopting and Approving Settlement" in Case 19-G-0678 ("November 26 Order"). The November 26 Order approved a settlement between the Companies and DPS Staff concerning the Companies' moratorium on new gas connections and provided that the Companies will lift the moratorium on new gas service at least until September 1, 2021. Consequently, the Companies updated their gas sales forecast to account for the addition of new customers. Please see the second supplemental testimony of Company Witness Theodore Poe, Jr. for a further discussion of the updated gas sales forecast.

Further, I updated the forecast fixed costs to include the impacts of recent pipeline rate cases, as well as to account for additional peaking supplies

secured since the Companies' corrections and updates filing on July 3, 2019.

Finally, please note that all the exhibits in this second supplemental filing reflect the Companies' strategy to meet forecast demand without NESE, and thus no NESE fixed or variable costs are included. Exhibit __ (EDA-6SS) reflects the latest peak day supply/demand balance and includes planned CNG and city gate peaking supplies that will be used to supply the forecast demand.

Q. How do the Companies plan to supply the incremental demands created by the increased number of customers?

A. The Companies developed a plan for the next two winters that includes expanded use of portable CNG and increased reliance on city gate peaking supplies. The Companies are also investigating further increases in CNG as needed. The Companies' construction and operation plans associated with the expanded use of CNG are discussed in more detail in the second supplemental testimony of the GIOP. The Companies will continue to procure city gate and CNG supplies to serve peak day and peak hour requirements for as long as they are needed. The planned increase in CNG

volumes, and continued acquisition of city gate peaking supplies, are reflected in Exhibit (EDA-6SS).

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Second Supplemental Direct Testimony of Elizabeth D. Arangio

INDEX OF EXHIBITS

Exhibit __ (EDA-1SS): KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage

Exhibit __ (EDA-2SS): KEDNY/KEDLI Forecast of Variable Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024

Exhibit __ (EDA-3SS): KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024

Exhibit __ (EDA-4SS): KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas

Exhibit __ (EDA-5SS): KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas

Exhibit __ (EDA-6SS): The Companies' updated peak day supply/demand

Exhibit ___ (EDA-1SS)

KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage

MR. MALONEY: Thank you, Your Honor.

BY MR. MALONEY: (Cont.)

Q. Ms. Arrangio, I take it you also sponsor several exhibits as part of your various pieces of testimony. Is it correct that with your direct testimony you sponsor exhibits that were marked initially as EDA 1 through 11 and have been subsequently marked Exhibits 104 through 114?

A. (Cont.) Yes.

Q. And as part of your supplemental testimony, did you sponsor exhibits that were initially marked as Exhibits EDA1S through EDA5S and were subsequently given the numbers -- Exhibit Numbers 201 through 205?

A. Yes.

Q. And as part of your corrections and updates, did you also sponsor exhibits that were labeled initially as Exhibits EDA4CU, EDA6CU, EDA7CU, EDA8CU, and EDA9CU and were subsequently given the designation as Exhibits 261 through 265?

A. Yes.

Q. Thank you. And do you also have before you exhibits that were sponsored as part of your second supplemental testimony that were labeled

EDA1SS through EDA6SS and were subsequently given the identification numbers 389 through 394?

A. Yes.

Q. And finally, as part of your second supplemental testimony, did you also sponsor a confidential version of Exhibit 5SS that has been given the identification number 672?

A.L.J. LEARY: Here in the exhibit?

MR. MALONEY: Here in the Exhibit 672.

A.L.J. LEARY: Thank you.

A. Yes.

Q. And were these exhibits prepared by you and under your supervision?

A. They were.

Q. And do you affirm that the information contained therein is true to the best of your knowledge?

A. Yes, it is.

MR. MALONEY: Thank you, Ms. Arrangio. Your Honor, Ms. Arrangio is available for cross examination.

A.L.J. COSTELLO: Okay. We have the Environmental Defense Fund.

CROSS EXAMINATION

BY MS. MURPHY:

Q. Good morning.

A. (Cont.) Good morning.

Q. So I'm going to start by asking you to refer to your rebuttal testimony at page nine.

A. Okay.

Q. So on page nine at line 20 your rebuttal testimony states, "As EDF notes, the companies perform an annual gas supply review, which includes filing of various data supporting the companies' long-term supply plan as well as an in-person meeting with staff and interested parties." Is that correct?

A. Yes.

Q. And you are referring here to the annual winter supply review process. Is that correct?

A. Yes.

Q. And does the New York Public Service Commission establish the parameters and requirements for the annual winter gas supply review?

A. The company responds to data requests issued by the staff and then the company meets with the staff following the submission of those data requests.

Q. And does the company's participation in the annual winter gas supply review incorporate any assessment of the company's greenhouse gas emissions reductions obligations?

A. The company looks at -- as part of the response to any of the data requests -- you know, starting with the company's forecast and any obligations that we're under for planning and all that would be incorporated in the company's responses to those data requests.

Q. But to -- to clarify, do those data request responses include an explanation of the greenhouse gas emissions resulting from the gas supply plan components?

A. I don't recall all of the questions specifically that were asked but if they -- I -- I don't recall if that specific question is part of the review, but if we are asked that we respond to any aspect of questions related to that.

Q. Perhaps I'll ask one more question to -- to clarify. The company would only provide information about the greenhouse gas emissions resulting from your supply plan if you were so asked by staff or the commission. Is that correct?

A. Yes. We respond to those questions as part of the -- the annual review, so if they were included in that, yes we'd respond specifically.

Q. So now if you could, please refer to page 14 of your direct testimony.

A. Yes. I'm there.

Q. So on page 14 of your direct testimony at line four, in response to a question about whether the companies planned to add incremental pipeline capacity, your testimony states, "Yes, due to continued customer interest in conversions from oil to natural gas for space and water heating as well as increased demand from existing customers and new construction in their service territories, the companies have continued to experience growth in the demand for natural gas. Is that correct?

A. Yes.

Q. And is that statement in reference to the Northeast Supply Enhancement Project Pipeline known as NESE?

A. Yes, on line one, right. Correct.

Q. Thank you. And so when the companies were originally planning on relying on the NESE project to satisfy their supply needs, was that reported in the annual winter supply review?

A. Yes. It would've been in response to questions for future portfolio additions.

Q. And sorry to make you jump around, if you could now please refer to your second supplemental testimony, which is the one that was filed December 13, on page three.

A. Okay. Sorry I have that one online. Page three, okay.

Q. So on page three at line four your testimony states, "Finally, please note that all the exhibits in this second supplemental filing reflect the company's strategy to meet forecast demand without NESE and thus no NESE fixed or variable costs are included." Is that correct?

A. Yes. That's what it says.

Q. So is it correct that the company's supply stack currently includes additional sources of gas supply to make up for the removal of the supply that would've been from NESE?

A. We have -- the supply stack exhibit that we -- are you referring to that in my testimony or just in general?

Q. In -- in general.

A. That's correct.

Q. And has the company conducted an analysis of the greenhouse gas emissions attributable to the NESE project compared to the greenhouse gas emissions attributable to the current updated supply profile?

A. Not that I'm aware of.

Q. And did the commissioner staff require -- request that the company conduct such an analysis?

A. Not that I'm aware of.

Q. Thank you. That concludes my cross examination.

A.L.J. COSTELLO: Okay. The Public Utility Law Project.

MR. BERKLEY: Thank you, Your Honor.

BY MR. BERKLEY:

Q. Good morning, Ms. Arrangio.

A. (Cont.) Good morning.

Q. I'm gonna start with your direct testimony.

A. Okay.

Q. And on page nine lines 11 through 18, specifically on line 15, you used the phrase force majeure. Could you explain what you mean by that here?

A.L.J. LEARY: Let's give some context. If you could just read that sentence, Mr. Berkley, so that the witnesses or the record give some context to the force majeure question.

MR. BERKLEY: Of course, Your Honor.

BY MR. BERKLEY: (Cont.)

Q. Reading the sentence that begins on line 13 and ends on line 15, "Assuming no ESCOs opt in for the Gulf Coast storage release, approximately 4.5 BCF (billion cubic feet) of Gulf Coast storage capacity, usually the entire KEDNY Washington storage service field, is utilized for off-system sales."

A.L.J. LEARY: Okay. Let me just -- I don’t have exactly that and it sounds like you're reading from your notes.

MR. MALONEY: I -- I was going to say --

A.L.J. LEARY: Let me just--

MR. BERKLEY: Yeah. I skipped -- I skipped a piece. Sorry.

A.L.J. LEARY: It's okay. Let me just read it if you don't mind.

MS. ARRANGIO: I was going to say mine doesn't say that.

A.L.J. LEARY: Let me just read into the record those two -- I'm going to go from line 11 to line 18. "The companies also have approximately 20 billion cubic feet (BCF) of Gulf Coast storage capacity. These storage fields are made available to energy storage companies, or ESCO's, as it states here, as part of the optional Gulf Coast storage release that occurs every April. Assuming no ESCO's opt in for the Gulf Coast storage release, approximately 4.5 BCF (billion cubic feet) of the total capacity is reserved for force majeure purposes. The remaining 15 BCF (billion cubic feet) of capacity, usually the entire KEDNY Washington storage service field, is utilized for off-system sales. If all ESCO's opt in, the capacity utilized

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas for off-system sales, or OSS, would be reduced accordingly." Go ahead Mr. Berkley. I hope that helped.

MR. BERKLEY: Thank you, Your Honor. I will try and read better in the next sections of the cross.

BY MR. BERKLEY: (Cont.)

Q. So returning to line 15, what is your definition of force majeure in this paragraph that was just read out loud?

A. (Cont.) Sure. So what we mean by force majeure is typically it's described as an act of God, something out of our control that we can't plan for.

Q. Thank you. Same page, same lines and this is referring to the phrase ESCO's opt in, which is on line 14. Could the company refuse the ESCO opt-in's and make that roughly 4.5 billion cubic feet of gas available for firm customers?

A. I do not believe so. I think the -- he storage option service is part of the company's customer choice program, and we're obligated to release that capacity should the escrows opt in for that service.

Q. Thank you. I'll ask some more questions about the customer choice program in a bit, but is it your understanding that there are no force majeure clauses in the contract or the ESCO's that would allow the company to choose not to give them that gas?

A. I'm sorry. I don’t understand the question. Sorry.

Q. Let me withdraw it for now. I'll come back to it later when we talk about customer choice.

A. Okay.

Q. Turning to page 11, lines 13 through 16. And the question is do the companies purchase supply from any local natural gas production? The answer was currently the Fresh Kills Landfill located in Staten Island is the company's only source of local production and accounts for less than 1% of the system throughput. The question is was the supply from the Fresh Kills Landfill one of the main reasons Staten Island was not included in the moratorium on gas hookups in 2019?

A. No. It's not.

Q. Could you briefly explain why Staten Island wasn't included in the moratorium?

A. If I recall correctly, it's because we had capacity, or there was capacity available, to feed the Staten Island system.

Q. Thank you. Moving on to page 12, lines two through three. The sentence says, "When the landfill is unable to self-supply the company's replace planned volumes with other resources in the portfolio." Could you please explain what sort of reasons the landfill would be unable to supply the companies?

A. I don't know specific reasons. But if they're not able to, you know, as it states here, we would replace those volumes with other resources in our portfolio.

Q. Thank you. Turning to lines eight through nine, which refer to an expected new biogas facility at Newtown Creek. Could you please clarify whether the Newtown Creek Biogas facility will be in service during the rate year?

A. It's expected to go into service in quarter one of 2020.

Q. Thank you. Turning to page 13, lines 13 through 20, and this relates to efforts that the companies are making to reduce pipeline capacity. Were any of the ways that the company explored to reduce reliance on pipeline capacity designed to evert or mitigate the projected moratorium of 2019?

A. I'm not sure I understand the question. Could you just perhaps rephrase it?

Q. Let me read the sentence that starts on line 15 and ends on 17.

A. Okay.

Q. "While pipeline capacity cannot be entirely displaced, the companies continue to explore ways to reduce reliance on pipeline capacity." My question is were any of the ways the company examined to reduce reliance on pipeline capacity designed to evert or mitigate the projected moratorium of 2019?

A. Well, we, you know, continue to deed the resources in the portfolio to serve customer requirements. So looking at the overall resources that we had, that's what we're talking about. If we -- if we still continue to need the resources, then we

continue to keep them in the portfolio to meet customer requirements.

Q. Thank you. Turning to page 14, this is lines 2 through 14. And I'll read to you from lines 12 through 14, which refer to a graph on page 15 of your testimony. And the sentence is, "The graph shows that without significant additional pipeline capacity as described below -- sorry -- as further described below, supply falls short of the demand." Could you explain what without specific additional pipeline capacity as further described below supply falls short of demand means?

A. Sure. So if you look at the graph on page 15, it's an oversimplified summary of the supply and demand balance that we have in the Downstate New York portfolio. So it shows the resources that we have under contract and then it shows the design day requirement, which is the black horizontal line. And it shows in the peach color, for those of you who have it in color, that at some point the demand exceeds the supply in the portfolio.

Q. Thank you. And were you referring just -- sorry. Let me restart that. Were you referring in your testimony or just now to the actual

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

physical capacity of the pipelines and assisted -- I'm sorry - associated infrastructure?

A. I'm referring to the portfolio of resources we have in the Downstate New York -- to serve the Downstate New York companies.

Q. And just to clarify, would that include both gas supply and actual infrastructure or is it one or the other?

A. This is -- my testimony is referring to the gas supply.

Q. Thank you.

A. In the associated infrastructure, not -- I just want to make clear, associated infrastructure upstream for those resources to be delivered.

Q. Thank you. And on this same topic, when you're talking about supply falling short of demand, were you referring only to firm gas supplies and not the spot market or the peaking market?

A. It would be all the resources in the portfolio that we have, so that includes some peaking supplies.

Q. Thank you. And were you referring only to local gas or to nonlocal gas also?

A. I'm not sure what you mean by local gas or nonlocal gas.

Q. I knew I shouldn’t have asked that question out of order. In your testimony, you refer to local gas from the Marcellus Shale Fields and nonlocal gas as being from further away destinations in the United States like Oklahoma, North Dakota and places like that.

MR. MALONEY: Can we get a specific reference to that?

MR. BERKLEY: I'll look for it, counsel. Like I said, it's out of order. So I'll look up the specific reference a little later in the questions.

A.L.J. LEARY: Let -- let's see if the witness can answer the question without the reference or point all of us to the reference in her testimony that discusses that.

BY MR. BERKLEY: (Cont.)

A. (Cont.)So in my testimony on page 21 of 45 perhaps we do talk about actual purchases from a specific year and perhaps that's what you're

talking about. But, in general, yeah, we need capacity on the pipelines to deliver to the citygate. So we can source the gas supplies based on the various pipeline contracts that we have both locally and then from the Gulf from Canada as well. So it would include all of the capacity we have and then what sourcing we can inject into that capacity and transport it to the company's distribution system.

Q. Thank you. Remaining on page 15, lines 1 through 12, specifically lines 11 and 12. If you refer -- I'm sorry. You state, "The companies have contracted for 100% of the incremental pipeline capacity that will become available via NESE." I'm sorry. The Northeast Supply Enhancement project known as NESE the rest of the time I ask that question. My question was did the companies have to make advance payments under the contract before the construction of the NESE pipeline would've begun?

A. No. We do not.

Q. Thank you. Turning to page 16 and this is beginning to ask what will be several questions about the moratorium as addressed in your testimony. On lines 16 and 17, referring to the potential that the NESE project would not go forward,

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

you state, "In that case, National Grid will have no choice but to impose a moratorium on new and additional gas service in affected areas to maintain system reliability." So my first question is what does additional gas service mean on line 17?

A. It would mean in this specific sentence just incremental load.

Q. Would it include load from customers, speaking hypothetically, that may have been off the system for some time as a result of a crisis like Superstorm Sandy or perhaps construction where the company had told them to turn the gas off for safety?

A. Well, so how we plan the portfolio, we internally receive a forecast of customer requirements and so that's how we build the portfolio. So whatever different aspects are built into the portfolio, that's what we plan, like I said, the gas supply portfolio for. So there would be you know a laundry list of perhaps items that would be included in the forecast that would either drive the forecast up or drive that down. So the net result of that is what we plan for for the resource portfolio.

Q. Thank you. On the same line, line 17, what does new service mean in the context of your testimony?

A. Again, the same response, the net. Whatever we see as incremental need in the forecast -- in the company's forecast.

Q. When the company puts together its forecast, does it take into account the potential loads added to the system by residential customers or businesses with whom it's been discussing construction over a period prior to the forecast?

A. I can't speak to the specifics of, you know, each component that's included into the forecast. Like I said, we see the total forecast for the gas supply requirements for the entire portfolio.

Q. Thank you, continuing to ask a few questions on the moratorium. Before imposing the moratorium, did the companies create any plans to triage between classes of customers when refusing new or additional service?

A. I don’t have the details of that if we did.

Q. So did the companies, to your knowledge, treat all customers within a class identically when refusing new or additional service?

A. I don’t have the details of that.

Q. Thank you. Moving back up the page to line four, you referred to something called an unacceptable risk of significant supply shortfalls and corresponding drops and system pressure. This is on lines four through six. What does that mean?

A. So when -- you're looking at new gas load without supply. So on a peak day we -- in the gas supply portfolio, we plan exactly to our design day requirement. We have no reserve margin built into our portfolio. So the forecasted customer requirements are the forecast -- the volumes that we plan to. So if there is -- as load grows, if we don’t have the supply to serve that, we can't meet that delta. We don’t have any reserve in the portfolio to do so.

Q. Thank you. And so there's no regulatory requirement to have a reserve margin?

A. No. In fact, we're not allowed to have a reserve margin.

Q. Turning to the concept of

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

unacceptable risk that is in your testimony on line four, was the unacceptable risk tied to only the 2019-2020 winter season? Or did it also -- was it also tied to the 2020-'21 winter season?

A. (Cont.) In this testimony, we did -- in this filing, we did not plan to have NESE in service for the '19-'20 winter.

Q. Thank you. So if the unacceptable risk was only tied to the '20-'21 winter, why did the company impose the moratorium in May of 2019, more than a year early?

A. In -- in summary, if we were not -- if we're not able to meet future customer requirements, you know, with the continued demand, that -- the addition of customers does not stop immediately. So we -- so because of that, we had to plan for not having any -- any incremental supply into the future. So this '19 -- 2019-2020, which is the winter we're in, was not part of -- would not have been impacted.

Q. Thank you. Turning to page 18 -- and I'll ask a question or two from the whole page. But for the moment, on lines six and seven -- and I'm going to stop at the comma on line seven. Your

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

testimony says, "The company's contract for quantities of gas to ensure sufficient supply to reliably meet design conditions." Were you able to find that?

A. Yes, I have it.

Q. Yeah, thank you. What does the term "design conditions" mean?

A. It would be -- we plan for -- in the gas pipe portfolio, we plan to meet a design day, as well as a design season. So that's when we have -- actually, for a design day it's an average of zero degrees. And for a design season, it's cold, very cold, temperatures across the November through March period. So we need to make certain that we have sufficient supplies to meet both scenarios, the design day as well as the design season.

Q. Thank you. Would that term also include the size and behavior of the physical plant and the number of customers the system was designed to serve?

A. I'm not sure what you mean by that question.

Q. Let me -- let me take it apart a little bit then. Is the system designed to serve a

certain number of customers?

A. The distribution system?

Q. Yes.

A. I can't speak to the details of the distribution system. When we plan the portfolio to meet the design day -- and we all work off of the same forecast, both the gas supply, and the resource portfolio, as well as the on-system planning.

Q. Thanks. I may come back to this. Same page, and I'm now looking at lines eight through 11, where you discuss term and spot contracts. And reading from line eight to 10, your testimony says, "A combination of term and spot contracts provides necessary flexibility with respect to volume, which allows the companies to respond to fluctuations in demand." Does the companies' use of such term and spot contracts allow the companies to respond to fluctuations leading to increases in demand?

A. Yes. The whole portfolio does, in fact.

Q. So it's not just to decreases in demand?

A. Correct, when looking at -- well, looking at the entire portfolio. We have to plan for

very cold days, very warm days. And then we have to plan, you know, within a month, we have to start the -- the planning process is such that we plan for the winter season actually for the year, November through October. And then we plan specifically for the November through March period.

And then we take it by, as the winter goes on, we look at it month by month and where we are in the portfolio, what weather is happening, where our storage facilities are with respect to what percent full they are. So we need to plan for the highest day forecasted for weather and the associated demand as well as the lowest. So we have to have a portfolio of resources that can allow us to do so. And term purchases, spot purchases, they help us to have that flexibility in the portfolio.

Q. Thank you. And I think I already know the answer, so let me ask this question this way. Is it fair to say that the company contracts for gas supplies in advance of the winter season?

A. Yes. Or the ability to call on those gas supplies, yes.

Q. Thank you. And -- and had the companies done so prior to filing this testimony in

April of 2019?

A. For what period?

Q. For the winter of 2019-2020.

A. No, we would not have done so by that time period.

Q. How many months in advance of a winter season does the company enter into contracts to meet fluctuations in demand?

A. So again, we look at -- typically we receive an updated forecast in June prior to the start of the winter season. So over the summer period, July right up through October, we plan different resources for the portfolio. So we'll have them all in place before the winter starts, but that can take all summer and all fall.

Q. Thank you. So is it fair to say that the companies had not contracted for gas supplies prior to the imposition of the moratorium, I'm sorry, gas supplies for the 2019-'20 winter before the imposition of the moratorium?

A. Well, so just -- I -- just to maybe explain. So we plan to meet the peak day, or the design day, excuse me, and the design season. So we have the -- typically we have what we're talking

about here, term contracts and spot contracts. For the most part, we already have the pipeline capacity. And so we look at different types of arrangements -- if we want to be able to have a call on gas supplies for a monthly period, which we call a baseload period, or day-to-day. So it depends on -- we would be looking for this upcoming winter and planning for the ’19 -- 2019-2020 winter prior to November 1st.

Q. Thank you. Do the companies have any regulatory requirements for advanced contracting or planning to have sufficient supplies to meet projected demand from firm customers for a specific winter season?

A. Well, I mean, off -- there are times that we do multi-year transactions. And we are -- we do file those transactions with the commission.

Q. So at some point, would the commission have -- or the Department of Public Service -- ask the company if it has sufficient supplies on hand for 2019-2020 winter?

A. Oh, sure. That’s part of the annual review that we were speaking about earlier, yes.

Q. Thank you. Moving on to page 20,

and I'll start with lines 18 through 22. And I'll actually read this paragraph. "Also, the companies consider options to replace long-haul capacity with shorter-haul capacity. For example, as supplies from the Marcellus Shale region became abundant and readily accessible, the companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada."

You refer to the company's choice not to renew expiring long-haul contracts. Could the companies have taken options on those supplies in case of an inability to meet customer demand, such as that that triggered the moratorium?

A. What this is referring to is actually upstream capacity that feeds into the capacity that delivers to the company's distribution system. So we're -- we're constantly looking to make certain that we have a least cost portfolio. So as part of that here, we were able to offset the cost for the long-haul with a different -- a shorter-haul, but still, that feeds into the distribution system. So we didn't give back any or turn back any capacity they -- that has the ultimate delivery to the

company's distribution system. It was the upstream feed into that capacity that we're -- that I'm referring to in the testimony here.

Q. So if the company had retained those contracts and entered into contracts for the short-haul gas, would there have been sufficient space in the system to fit the upstream capacity also? In other words, could -- could the company have delivered that gas to customers also and avoided the need to have a moratorium?

A. No. It's like a tree. So you have two limbs come in to -- to one. So you only have the capacity of the contract that delivers to the distribute -- distribution system. That's fixed.

And what we're talking about here is how we fed into that capacity. So no, we couldn't have. We weren't talking about -- we didn't give up any capacity to the company's citygates. It's just the feed into it and the source of the supply that we feed into it.

Q. Thank you. Turning to page 23, and I'm looking at the whole page, but specifically, lines 14 through 16. And let me read 14 through 16. "For example, on warm days and shoulder months, when

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

pipeline capacity is not fully utilized, the companies will seek to sell excess capacity to off-system customers." During the moratorium, did the company experience any excess capacity?

A. Because we plan for the peak day, oh, excuse me, the design day -- I'm sorry -- and the design season, if we don't experience design day weather or design season weather -- this is a bit of an oversimplification -- but then we do have excess resources. So in order to lower the cost of the portfolio, we engage in off-system sale transactions, and those credits go back to customers in the form of, you know, a credit to -- to the customer bill. So when we don't have design weather cold -- and -- and design day weather, we typically do have excess resources that we look to remarket in the marketplace to lower overall bills.

Q. Thank you. So during the moratorium period, which was announced in May and didn't end until the announcement of a settlement between the DPS and the commission and the company, did the company sell any excess capacity off-system?

A. I'm sure we did, yes. Because we always have excess capacity in the summertime.

Q. Thank you. Turning to page 31, and we're back to the customer choice program, and I'm focusing on lines 10 through 21. And you describe the customer -- the company's customer choice program provides customers the option to purchase their supplies from marketers. So my first question is, were any ESCO customers denied new or additional service during the moratorium? I'm sorry, you use the phrase "marketers." Were any marketer customers denied new or additional service during the moratorium?

A. I would not know the answer to that question.

Q. Would you know whether any ESCO, sorry, marketer customers -- that didn't sound as good as ESCO customers. Sorry, I -- that's why I keep saying that. Were any marketer -- marketers customers denied restoration of existing service during the moratorium?

A. I do not know the answer to that question.

Q. Thank you. Turning to page 32, and I'm focusing on lines 18 through 23. On lines 21 and 22 you state, "The companies have not experienced

any issues from implementation of the commission's mandatory release orders." Taking into account the moratorium declared on May -- I'm sorry, declared in May of 2019, would you still agree with that statement?

A. Well -- well, this statement talks about the company's customer choice program. So when that went into -- when that was implemented, some marketers had capacity, had their own capacity, to various companies' citygates. And -- and in fact, for our company's, we had a small, very small, volume, less than -- I think it was less than 500 dekatherms that we didn't plan for. So the company plans for the requirements of these customers as part of the customer choice program. And when these customers migrate to transportation, we release capacity to the customers' marketers so that they can use the same portfolio that the company has to serve these customers. So that's what that's referring to. And we no longer have anybody that's grandfathered, so we plan for the entire load that we then therefore, as those customers go to transportation, we release that capacity to their marketer.

Q. Thank you. Turning to page 44,

and I'm focusing on lines one through five, wherein you describe the percentage of the company's gas supply that is physically hedged. Do you know what percentage of other New York State gas companies' firm gas supply forecasted sales demand is physically hedged between November and March?

A. I'm sorry, what page are you on?

Q. Page 44 of your direct testimony.

A. Page 44, sorry.

Q. Sorry.

A. Okay, I'm sorry. Can you repeat the question again?

Q. Of course. And my apologies for not giving you the page number. And let me ask the question slightly differently. Do you know of any other New York State gas companies who physically hedge their firm gas supply forecasted sales demand between November and March?

A. I'm sorry, what lines are you referring to? Are you referring to the top, did you say?

Q. Yes. This is lines one through five. And let me -- let me try and restate this. And actually, let -- let me go to the second question

first. And let me see if I can bring you back to this with some foundation. You discuss on lines three and four that approximately 30% of the company's forecasted November through March firm sale demand is physically hedged. Is that correct?

A. Yes, as a result of the plan storage withdrawals.

Q. Is that 30% number an industry standard in New York State, to you knowledge?

A. This is reflective of the company's portfolio. So I'm not certain what other companies -- what other companies hedge with respect to their storage.

Q. Could the companies have increased their physical hedges to deal proactively with a moratorium?

A. Our physical hedges or our hedging program is designed to mitigate price volatility. So it deals with the capacity and the resources that we have in the portfolio.

Q. So when you're referring to -- I'll withdraw that. Let me move on to your supplemental direct testimony.

A. Okay.

Q. And if you could turn to page four, please.

A. Okay.

Q. Thank you. And I'm focusing on lines six through 14. And let me read pages, I'm sorry, lines nine through 14. "Yes. While the company's filings assumed the NESE project would be completed by the winter of 2020-2021, my direct testimony at lines 14 through 16 identified the possibility that the NESE project would not be available as planned and explained that the companies would not be able to prudently satisfy new or additional service requests if the NESE project were not placed in service." Do you still agree with that conclusion?

A. The what? That is was included in the testimony?

Q. Yes.

A. It was -- if what you're asking is that it was referred to, on pages 14 to 16, in my direct testimony?

Q. Yes.

A. Yes.

Q. Thank you. Are you familiar with

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

the settlement in case 19-G-0678 and the settlement order issued on November 26, 2019?

A. Which case was that? Sorry, I just want to make sure we're talking about the same one.

Q. That was the commission's investigation of the moratorium and subsequent settlement with the company.

A. Yes.

Q. Yes. Thank you.

A.L.J. LEARY: Sorry, just for clarification. Yes, you still agree with your testimony referenced on page four?

MS. ARRANGIO: Oh, I'm sorry. I was saying, yes, I am familiar with the settlement.

A.L.J. LEARY: Okay. I just wanted to make the record clear. You're familiar -- yes, you're familiar with the settlement. The question that's pending, I believe, Mr. Berkley, was whether the witness still agrees with her testimony on page four, lines --

MR. BERKLEY: 12 through 14, Your Honor.

A.L.J. LEARY: 12 through -- well,

roughly nine through 14.

MS. ARRANGIO: Yes, that my direct testimony talked about NESE perhaps not being in service. I think that was the question. And yes, I confirm that it does.

A.L.J. LEARY: No, that's not what I'm asking you.

MS. ARRANGIO: Oh, I'm sorry.

A.L.J. LEARY: I'm asking you whether you still agree with what's written on pages -- page four, lines nine through 14, specifically, "…the companies would not be able to prudently satisfy new or additional service requests if the NESE project were not placed in service"?

MS. ARRANGIO: Oh. Well, I think, you know, according to the company settlement, I think there -- there is a -- a short-term plan and a long-term plan. So for the near-term plan, the company has proposed, you know, again, a near-term plan and a longer-term plan. So we're committed to providing service under those plans agreed to in the settlement. Does that answer your question?

A.L.J. LEARY: No.

MS. ARRANGIO: Oh.

A.L.J. LEARY: I'm asking if you would -- if -- if this needs to be revised in any way as a result of the settlement. I thought that's where Mr. Berkley was going, and I have that -- I have a question myself. So -- so -- so if the NESE project would not be available as planned, and the companies would not be able to prudently satisfy new or additional service requests, do you still agree with that testimony?

MS. ARRANGIO: So I think since -- it -- when it was written in June, yes. Things have changed since that date. So I would have to say, yes, the company is committing to providing new and additional services in the event that NESE is not in service for the short-term and the long-term. So our settlement modifies those statements, yes.

A.L.J. LEARY: Thank you.

BY MR. BERKLEY: (Cont.)

Q. Thank you. So given that the settlement order is requiring the company to provide service to all of those customers that were denied service during the moratorium, would you characterize the company's actions to satisfy those requests as prudent or imprudent?

A. (Cont.) They would --

MR. MALONEY: I'm going to object. I think we went through a discussion of prudence and imprudence, and we -- at one time those words were sort of disallowed at this point earlier in the hearing.

A.L.J. LEARY: I think that's right. I -- I think that's right. We had a problem, because those are terms of art that kind of invoke a lot of perhaps legal conclusions, Mr. Berkley, that I don't think you intend to impute to this particular witness or the company. I mean, it's really -- the question of prudence, I mean, can you rephrase to get to the same place that you want to get to?

MR. BERKLEY: With all -- with all due respect to counsel, Your Honor, I'm just using the same phrase that the witness used. If the witness would like to change her phrase on lines 12 through 14 of page four, then I would be willing to follow-up on the basis of however else she described that process.

A.L.J. LEARY: Okay, let's lay a foundation, which you actually may have already done. But let's ask the witness what she meant by stating

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

at line 12, "that the companies" -- and 13 -- "that the companies would not be able to prudently satisfy new or additional service requests if the NESE project were not placed in service."

MS. ARRANGIO: Sure. So that we would -- I think earlier in some of the questions and answers, we talked -- I talked about how we plan to the company's forecast. And if we don't have incremental supplies to meet customer requirements, we wouldn't be able to do so.

BY MR. BERKLEY: (Cont.)

Q. So would it be fair to say that your use of the -- the word "prudently" on that line 12 relates to the ability or inability of the company to serve its customers?

A. (Cont.) From a gas supply portfolio, if we -- if we don't have the supply, we wouldn't, very simply, be able to meet the requirements.

Q. Thank you. I think I may come back to that in a bit. So same page, starting on line 16 and continuing through all of the next page, that is page five. On line three of page five -- let me back up and read that because it does start on the

previous page. Starting on -- on line 20 of page four, "Effective May 15, consistent with the commission's rules and regulations, and the terms of the companies' tariffs, and in consideration of public safety, the companies advised the commission and customers that they will not fulfill applications for new firm service connections or requests for additional firm load from existing customers in the affected areas of their service territories until the current supply constraints are addressed." On line three, what did you mean by "requests for additional firm load from existing customers"?

A. I'm generally talking about increased load here, talking about new service connections. And -- and as I said, we are the recipient of the forecast. And as part of that forecast, as I mentioned before, there are various inputs into that. And that would include, you know, they would include incremental usage from existing customers' expansion of existing facilities.

Q. So would that phrase, "additional firm load," include -- include customers who had been, for some reason, not taking gas for the system for some period of time?

A. Again, the -- the specifics of what goes into the forecast, it -- it's -- there are very many things that go into that. I can't speak to the details. We -- we get the net result of the load add it -- you know, the pluses and minuses that come out to the forecast. And we get the net -- net total that we plan for.

Q. Thank you. And you may have answered this earlier, but let me ask it again just in case. Did the companies lack the capacity to transport the projected additional gas necessary to meet projected increases and demand from firm customers?

A. For what? I'm -- I'm not sure what period you're talking about or --

MR. MALONEY: Yes. Also, counselor, what capacity are you talking about? Are you talking about upstream firm capacity? Are you talking about distribution system capacity? I believe Ms. Arrangio's already testified she doesn't -- she's not familiar with how the on-stream distribution system capacity is built and allocated. She is familiar with the upstream firm capacity part of the portfolio.

MR. BERKLEY: Thanks. I'll withdraw

that.

BY MR. BERKLEY: (Cont.)

Q. Could you move on to page six, please? And lines four through 14, you're discussing whether the -- the -- I'm sorry, the companies addition of incremental supplies to partially replace the supplies that would not be available if the NESE project was not placed in service. And I -- I changed the tense, but kept the original words. So my question is, were the companies' contracts for peaking supplies, which is what your testimony refers to, only planned to come online in the year 2021 that NESE was planned to come into service?

A. (Cont.) I'm sorry, I don't understand the question.

Q. Let me read a sentence that runs from line nine to line 13. "These incremental peaking supplies will enable the companies to provide service reliably to their existing firm customers, but are not sufficient to serve additional firm customer load in the affected areas of the company's service territory." And this in relation to your statement that the companies had procured an additional 261,000 dekatherms per day. So the question, I guess to ask

it slightly differently, is could the company have procured these or other supplies for the winter of 2019-2020? Sorry, companies.

A. (Cont.) The company had already procured resources for the 2019-2020 winter. These are resources that we planned, that we procured for the 2021 winter.

Q. Why were they procured for that 2020-2021 winter?

A. Because if the -- these are peaking supplies. So this is -- these are resources that the company issues a request for proposals, an RFP, for. And so this is not the company's capacity. We rely on third parties that can deliver to the company's citygates. So these are peaking supplies that we procured for the 2020-2021 winter in the event or knowing that this is for no -- for not having NESE on for the '19, '20, '21 winter. So procured those supplies now for next winter to make sure that we had those in the portfolio.

Q. Thank you. And was the purpose of contracting for the peaking supplies to meet a projected lack of sufficient gas to serve customers in the 2021 winter? I'm sorry, firm customers.

A. It was to make certain or to look at, if you look at exhibit EDA-5S as part of the -- my supplemental direct testimony here, it shows the supply-demand balance, including the 261,000 dekatherms. And we still have a net need over and above that for the 2021 -- 2020-2021 winter season.

Q. Thank you. Turning to page eight, and I'm interested in lines five through 10. As of June 2019, the date of this testimony, lines seven through eight, you assert that the companies are cautiously optimistic that the NESE project will -- will be approved. Did the companies have a plan B or -- to describe what plan B means, did you have any kind of plan in the event that the NESE project was not allowed to move forward?

A. I think I was -- answered some of those questions in -- in perhaps in direct testimony or in -- in a data response. So the -- this has been a plan that the companies had in place for -- I think starting in 2007, where the company procured what we entered into different, you know, varying contracts with Transco to provide incremental service to the company's citygate. We coupled those with on-system projects. And this NESE was the last piece of

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

capacity to provide incremental service to the company's distribution system.

So that was the company's, you know, plan that has been in place since, you know, over -- over 10 years now. So the company fully expected, you know, at that time when the plans were put into place for all of these capacity additions, et cetera, to go into place. So the company does not or did not have a plan B for whether or not the NESE project went into service. That's a very -- it's a very sizable project, 400,000 dekatherms a day.

So at the time, you know, the company signed up for -- is committed to the NESE project. The company -- we can't make further commitments to other projects in the event that the NESE project comes on. So at -- at that time -- and we did not have a plan B when we went forth with the NESE project.

Q. If you remember, you testified that the company's advanced contract for 100% of the NESE pipeline capacity, if it were built, didn't cost the company any money. Is that correct?

A. Well, you asked if we were -- had paid any money before that project comes in to

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

service.

Q. Yes. So are you agreeing? I'm sorry, I'm confused.

A. Well --

Q. Did -- did any money change hands or any valuta change hands between the company and the -- the entity that was -- that was going to build the NESE pipeline on the basis of the company's reservation of 100% of its capacity?

A. Not to date. Our commitments don't go until the project goes into service.

Q. Thank you. So couldn't the company have just used other money to provide a plan B, since it wasn't -- it hadn't committed any financial resources to NESE?

A. Well, although we didn't -- we don't pay for it until the project goes into service, when it goes into service we're obligated to pay for that. So if you have two plans, just for example, one for 400,000, and then we're to look for another pipeline project, if both of those projects go into service, we have commitments to pay for both of those projects. That's typically how that works.

So we would -- we wouldn't be able to

take gas under both projects. So we -- we can't commit to those two things at the same time.

Q. So is it your testimony that the company could not have entered any kind of an option arrangement with a third party to get a plan B if NESE was not to be constructed?

A. No. I'm -- what I was saying is, if we were looking for a one for one alternative to the NESE project, like another pipeline project, then that's what I was referring to. But these peaking supply contracts, we typically, you know, have typically engaged in those contracts from year to year. Sometimes, you know, maybe two years. But not long -- not as long as a pipeline contract commitment typically.

Q. Thank you. Are you aware whether there are any regulatory requirements that would have created a duty for the companies to have a plan B?

A. Not that I'm aware of.

Q. Thank you. And let us turn, finally, to your second supplemental direct testimony. And I'm -- I'm interested in lines 15 through 19 where you're discussing the November 26 settlement order between the companies and staff.

A.L.J. COSTELLO: What page number?

MS. ARRANGIO: What page?

MR. BERKLEY: I'm sorry, page two of the --

A.L.J. LEARY: Second supplemental?

BY MR. BERKLEY: (Cont.)

Q. Second supplemental testimony.

A. (Cont.) Lines 15?

Q. Through 19, yes.

A. Okay.

Q. Are you aware of the number of total customers that have been connected or reconnected to natural gas service in KEDNY-KEDLI's testimony -- I'm sorry -- territory since the November 26, 2019 order?

A. I'm not aware of the specific number, no.

Q. Thank you. Turning to page three and focusing on lines 11 through 15, and this is your discussion of how the companies planned to meet incremental demands caused by the increased number of customers, that is, increased because of the settlement order. My first question is, why did the companies choose not to take these measures to

satisfy incremental demands before the settlement order?

A. From my, you know, from a gas supply perspective, you know, the company looked at, you know, we looked at the supply-demand balance. And what I think the -- the expanded use of the CNG and reliance on peaking supplies, that's what we're looking at for the next two winters. There's a finite amount of peaking supply in the marketplace available, and also with what -- what we can take away on our own system from -- from the various receipt points and that interconnects with the interstate pipelines.

And the expanded portable CNG, although I -- I can't speak to the, you know, on-system specifics of that and the, you know, the engineering of it. But that will be a -- a sizable volume that we'll be looking at, you know, on those peak days that we need to rely on. And these plans came out as a result of -- of the settlement. I guess that's all I can say about that.

Q. Thank you. Is it fair to say that the company took no -- sorry, let me withdraw that and start over again. Is it fair to say that the

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

companies didn't analyze putting into service the incremental supplies that it has done under the -- the order, the settlement order? Yeah, that was a bad question. Let me try again.

A. Sorry.

MR. MALONEY: You could reask.

MR. BERKLEY: Yes. I'll -- I'll anticipate Mr. Norton.

BY MR. BERKLEY: (Cont.)

Q. Is it fair to say that the companies made no attempt to develop the -- a similar plan to that ordered by the commission in the settlement order before it imposed the moratorium?

A. (Cont.) No. I think the company really looked at what we had as a supply-demand balance before looking at this type of plan. I don't -- you know, this -- this will be a plan that we have to rely on, you know, third party supplies. We've contracted for some of them as -- as we just discussed. And you know, increased CNG usage. I -- I know we -- we currently have the use of CNG on our system. But what we're talking about for the -- the next two winters will be a significant increase over -- over what we have today. So we do employ CNG

today. But as I mentioned, the -- the next few years, this will be a significant increase over what we have in place already today.

Q. Is that significant increase one of the reasons that the company didn't explore that option instead of declaring a moratorium?

A. Well, I'm not sure we didn't explore it in the sense of -- so the CNG option is really part of our on-system. You know, this is -- is not -- they are not resources that we restreet -- that we receive from the upstream pipeline. So I don't know what degree we looked at incremental.

I know, as we have future customer projections before, whether NESE came on or not, we know that we have on-system CNG that we need to employ to meet those customer requirements already. So, you know, we're always looking at needing to meet customer requirements, whether or not, you know, that NESE was on or not, we knew we had until that came on or, you know, at that time. We knew we had on-system resources that we had to plan for to meet customer requirements on the peak day.

Q. Thank you. Generally speaking, are peaking supplies more expensive?

A. More expensive than what?

Q. If you purchase firm supply in advance of let's say six months or a year earlier than going to the peaking market at the let's say only three months before the winter.

A. Well, they're very -- they're different supplies. So again, because of the location, you know, w -- when we talk about purchasing the -- the term supplies and such that we discussed a little earlier today, those are for supplies into existing pipeline capacity that we have. And we typically always look at securing peaking supplies over, you know, over the winter period prior to the start of the winter. And simply because those supplies are the last resources that we call upon. And because we haven't updated -- we receive an updated customer requirements forecast each year sometime in the June period -- you know, we have to look at our analysis and make sure that we know what resources we have, what resources we need.

So we typically procure peaking supplies prior -- just prior to the start of the winter period as well. And again, they're usually for a set number of days for a set volume. And depending

on the pricing associated with those, it -- it really depends on typically the weather that you have and what -- what the -- the market is pricing gas supplies at for, you know, on those colder days. So it's a difference.

You know, the peaking supplies we contract for differently for than the -- the term supplies or spot supplies in the marketplace. It's really dependent on the market and what the pricing is at the time. But we lock into those supplies, and we lock in for a call on those supplies. So we don't set a fixed price for those. It's -- it's typically -- they're typically based on the market price at the time that you call upon those supplies.

Q. Thank you. Would it be fair to say that the plan developed by the company and by DPS to alleviate the moratorium will result in higher bill impacts for low-income customers and non-low-income customers?

MR. GOODRICH: So I'm -- I'm going to object to that question. It was phrased as, "Would it be fair to say the plan developed by the company and DPS," and DPS was not involved in developing the plan.

A.L.J. LEARY: Okay. With that caveat or revision, Mr. Berkley, is to the question, can you rephrase?

MR. BERKLEY: Yes, Your Honor.

A.L.J. LEARY: I -- I think Mr. Goodrich is implying that the company developed the plan. Is that correct?

MR. GOODRICH: Yeah. I just want to make sure that the record's clear that it was -- that DPS was not involved in developing the plan. That's all.

A.L.J. LEARY: Okay. But DPS agreed to that plan in the settlement order.

MR. GOODRICH: No. No.

A.L.J. LEARY: No. Okay.

MR. GOODRICH: The settlement order was -- was that the -- the company agreed to -- to end the moratorium and -- and various things. And one of the things the company agreed to do was develop a short-term plan and a long-term plan.

A.L.J. LEARY: Got it.

MR. GOODRICH: That staff, DPS, did not agree to those plans.

A.L.J. LEARY: Great. Can you revise

the question, Mr. Berkley?

MR. BERKLEY: I think so, Your Honor.

BY MR. BERKLEY: (Cont.)

Q. Would it be fair to say that the plan ordered by the commission in the settlement order will result in higher gas prices for residential customers in the company's service territories?

MR. GOODRICH: So again, I'm sorry, a commission did not order a plan. The -- the -- the settlement and the commission's order were that the company was going to move forward and connect customers as opposed to end -- end -- end the moratorium. The plan that I believe Mr. Berkley's referring to is probably the short-term plan that was something that the company was putting together, and I believe, filed after the -- the order came out as far as how it was going to meet demand over the next few winters. But the -- neither staff -- n -- neither the department nor the commission has -- has approved or ordered that plan to -- to -- per se.

MR. BERKLEY: Your Honor?

A.L.J. LEARY: Can you leave out the -- go ahead. I'm sorry, Mr. Berkley.

MR. BERKLEY: The title of the order is "Order Adopting and Approving Settlement."

A.L.J. LEARY: I -- I -- I understand --

MR. GOODRICH: Yes, but the plan was not in the settlement. The settlement said the company was going to develop a plan.

A.L.J. LEARY: So I understand Mr. Goodrich's clarification here. And I do want the record to be accurate. Do you want to ask whether as a result of the settlement order qua settlement order will result in higher bill impacts for low-income customers? Or something like that?

MR. BERKLEY: Yes, let me -- let me try and ask it that way, Your Honor. Thank you.

BY MR. BERKLEY: (Cont.)

Q. Under the plan that the company has engaged in after the commission's November order approving and adopting settlement, has -- under that plan, will low-income customers receive higher bill impacts, in other words, higher prices?

MR. MALONEY: I -- I'm going to object now. As compared to what?

A.L.J. LEARY: As -- as -- I -- I -- I

understand the question as compared to what they're seeing in their bills today. Is that what you're -- you're asking, Mr. Berkley?

MR. BERKLEY: Yes, before the implementation of the plan. So --

A.L.J. LEARY: Pre-plan --

MR. BERKLEY: Pre-plan --

A.L.J. LEARY: Post-plan rate.

MR. BERKLEY: -- gas prices.

A.L.J. LEARY: I'm going to -- I'm going to overrule the objection. The witness can answer.

A. (Cont.) Okay. I don't know what the bill impact -- I don't know what the bill impacts would be to those specific customers.

Q. (Berkley) Thank you. And -- and -- let me skip that question. Are you aware whether or not any of the companies' customers have complained to the companies' about their inability to get new or additional gas service during the moratorium?

A. I'm aware in general, yes.

Q. Would you know how many of such complaints were filed with the PSC through its QRS, I'm sorry, with DPS through its QRS/SRS system?

A. No.

A.L.J. LEARY: Can you spell those -- that acronym out, Mr. Berkley.

MR. BERKLEY: I knew you were going to ask that. The Quick Response System and Supplemental Response System. Did I get the second one right, Brandon? No?

MR. GOODRICH: No idea.

MR. BERKLEY: No idea. It's in part 12 of the commission's rules.

A.L.J. LEARY: Okay.

BY MR. BERKLEY: (Cont.)

Q. I'll ask that again.

A. (Cont.) Thank you.

Q. Are you aware how many complaints were filed with DPS in its complaint system?

A. No, I'm not.

Q. No. Thank you. No further questions, Your Honors.

A.L.J. COSTELLO: Okay, the companies?

MR. MALONEY: Can we take a 10-minute break?

A.L.J. COSTELLO: Sure.

A.L.J. LEARY: Oh, we're going to take

a break now anyway. You can use that break for -- not everybody needs to stay in the room. We're going to take a 10-minute break.

MR. MALONEY: Which is fine with us.

A.L.J. LEARY: If that's okay with you, and then you can use the time. You don't get a break. You don't get a break is my point. I apologize, but you don't. So we're going to go off the record.

A.L.J. COSTELLO: Good.

(Off the record 11:05 a.m.)

(On the record 11:18 a.m.)

A.L.J. COSTELLO: Companies have any redirect?

MR. MALONEY: Your Honors, we have no redirect. Thank you.

A.L.J. COSTELLO: Okay. Thank you. Ms. Arrangio, thank you for your testimony. You're excused.

MS. ARRANGIO: Thank you.

MR. GOODRICH: Your Honors --

A.L.J. LEARY: Yeah.

MR. GOODRICH: -- next on the list is, excuse me, I'm sorry. Next on the list is the DPS

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas Staff Gas Supply Panel. And the only cross for that was EDF, and they've withdrawn.

A.L.J. COSTELLO: Right.

MR. GOODRICH: However, since our Panel is here and -- and dressed up, as I put it to them in email, can -- can we call them to the stand? And we can get their testimony and whatnot on the record.

A.L.J. COSTELLO: Sure.

MR. GOODRICH: Thanks.

A.L.J. LEARY: Good move.

A.L.J. COSTELLO: Could you each please state your name and business addresses for the record?

MR. MAIORIELLO: Dave Maioriello. 3 -- Dave Maioriello, 3 Empire State Plaza, Albany, New York, 12333.

MS. PARK: Jennifer Park, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Please stand and -- and raise your right hand. Do you swear or affirm that the testimony you will p -- provide is the truth?

PANEL: Yes.

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

DAVE MAIORIELLO; SWORN

JENNIFER PARK; SWORN

A.L.J. COSTELLO: You may proceed.

DIRECT EXAMINATION

By MR. GOODRICH:

Q. Staff Gas Supply Panel, before you is a document entitled, "Prepared Testimony of Staff Gas Supply Panel," consisting of a cover page and 14 pages of questions and answers, dated August 2019. Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Is that on?

A. Yes.

Q. Do you have any changes or corrections to make to your prepared testimony?

A. No.

Q. Do you affirm that the information contained in the Panel's prepared testimony is true and correct to the best of your knowledge?

A. Yes.

Q. And if you are asked the same questions today under oath, would you answer them in

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

the same way?

A. Yes.

MR. GOODRICH: Your Honor, I ask that the prepared testimony of the Staff Gas Supply Panel be entered into the record as if given orally.

A.L.J. COSTELLO: That's granted. And at this point, the court reporter should insert DPS Gas Supply Panel direct testimony. #

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Gas Supply Panel

Davide Maioriello
Utility Engineering Specialist 3
Office of Electric, Gas and Water

Jennifer Park
Assistant Engineer
Office of Electric, Gas and Water

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Gas Supply Panel (Panel), please state your names, employer, and business address.

A. Our names are Davide Maioriello and Jennifer Park. We are employed by the New York State Department of Public Service (Department) and our business address is Three Empire State Plaza, Albany, New York 12223-1350.

Q. Mr. Maioriello, in what capacity are you employed by the Department?

A. I am a Utility Engineering Specialist 3 within the Gas and Water Rates and Supply Section of the Office of Electric, Gas and Water.

Q. Please summarize your educational and professional experience.

A. I attended Hudson Valley Community College and graduated with an Associate in Applied Science degree in construction technology. I continued my education at the SUNY Institute of Technology and, in May 1999, graduated with a Bachelor of Science degree in civil engineering technology. From February 2000 to February 2003, I was employed by SPEC Consulting, LLC as a Project Engineer. My work involved project management

and coordination, engineering, as well as various computer-aided design projects. From February 2003 to December 2005, I worked for the City of Albany Engineering Division where I was responsible for a number of tasks, including inspections of major roadway projects and other municipal construction projects. I also engaged in permit processing and utility work inspections. In December 2005, I joined the Department and, since that time, I have performed various engineering analyses of projects and associated capital budgets, operations and maintenance (O&M) expenses and programs, distribution system reliability needs, capacity and demand requirements, and review of various petitions and tariff filings of gas and electric utility companies in New York State.

Q. Please describe your duties in the Office of Electric, Gas and Water.

A. My responsibilities within the office focus on the analysis of natural gas utility policy issues, including distribution system reliability and planning, capacity asset management, gas purchasing practices, gas system

reliability, and issues related to the restructuring of the natural gas industry and use of natural gas in New York State.

Q. Have you previously provided testimony before the Public Service Commission (Commission)?

A. Yes. I have provided testimony with respect to a number of issues, including capital expenditures, materials and supplies, depreciation rates, sales forecasting, revenue decoupling mechanisms, revenue allocation, rate design, lost and unaccounted for gas, capital expenditure reconciliation mechanisms, retail access issues, reliability forecasting and supply planning, peaking supply and capacity issues, design day and winter forecasting, reliability improvements, gas network enhancement and reporting requirements. I have provided testimony in the following rate proceedings: Case 19-G-0666, Consolidated Edison Company of New York, Inc.; Cases 05-G-1494 and 14-G-0494, Orange and Rockland Utilities, Inc.; Cases 05-G-1635, 08-G-1392 and 15-G-0382, St. Lawrence Gas Company, Inc.; Cases 06-G-1185 and 16-G-0059, The Brooklyn Union Gas Company d/b/a

National Grid NY (KEDNY); Cases 06-G-1186 and 16-G-0058, KeySpan Gas East Corporation d/b/a National Grid (KEDLI); Cases 07-G-0141 and 16-G-0257, National Fuel Gas Distribution Corporation; Cases 08-G-0609, 12-E-0201, 12-G-0202 and 17-G-0239, Niagara Mohawk Power Corporation d/b/a National Grid; Case 09-G-0716, New York State Electric & Gas Corporation; Case 09-G-0718, Rochester Gas and Electric Corporation; Case 11-G-0142, Chautauqua Utilities, Inc.; and Case 14-G-0319, Central Hudson Gas & Electric Corporation.

Q. Ms. Park, in what capacity are you employed by the Department?

A. I am an Assistant Engineer and work in the Gas and Water Rates and Supply section in the Office of Electric, Gas, and Water.

Q. Please summarize your educational and professional background.

A. I graduated from University of Buffalo in May 2017 with a Bachelor of Science Degree in Chemical Engineering. In July 2017, I joined the Department of Public Service in the Office of Electric, Gas and Water.

Q. Please describe your duties in the Office of Electric, Gas and Water.

A. My responsibilities within the office have been focused in several aspects of utility ratemaking, including customer and volumetric forecasting, rate design, gas adjustment clause reconciliation, gas winter supply reviews, escrow account reviews, small water company rate case reviews, and water company transfer petition reviews.

Q. Have you previously testified before the Commission?

A. No.

Scope of Testimony

Q. What is the scope of your testimony in this proceeding?

A. Our testimony will address certain proposals made by KEDNY and KEDLI, together the Companies, related to the following gas program and supply topics: (1) Green Gas Tariff; and (2) Gas Hedging strategy.

Q. In this testimony, will you refer to, or otherwise rely upon, any information produced

during the discovery phase of this proceeding?

A. Yes. We will refer to, and rely upon, several responses to Information Requests (IRs), all of which are included in Exhibit__(SGSP-1).

Q. Are you sponsoring any other exhibits in support of your testimony?

A. No.

Q. What is the Rate Year in these proceedings?

A. The Rate Year is the 12-month period ending March 31, 2021.

Green Gas Tariff Offering

Q. What did the Companies propose with regard to a "Green Gas" tariff?

A. As discussed on page 31 of the pre-filed direct testimony of the Future of Heat Panel, the Companies propose a Green Gas Tariff offering to commence in April 1, 2021, or the year following the Rate Year, which the Companies refer to as Data Year 1. Under this offering, customers could voluntarily purchase renewable natural gas (RNG) for a portion or all their gas supply needs. The Companies state that different customers prefer different pricing structures,

where residential customers tend to prefer price certainty while non-residential customers prefer costs that scale as a percentage of use. In order to accommodate different customer preferences, the Companies propose a tiered system. Specifically, page 32 of the Companies' pre-filed Future of Heat Panel testimony provides four tiers: Low, Low-Middle, High-Middle, and High. The proposed tariff would provide customers with the option to choose a tiered level of RNG at a flat rate per month for residential customers, meaning those in service classes (SC) 1A, 1B, and 1B-DG, and either a flat rate or a percentage of monthly consumption for non-residential customers, meaning those in SCs 2, 3, 4, 7, and 21. Participation in the Green Gas Tariff would require residential customers to commit to remain in the program for a minimum of 12 months and non-residential customers to commit for 24 months.

Q. Explain how the Companies developed the tier structure and the rate associated with each tier

A. As stated in the response to IR DPS-365, the Companies proposed tier structure was initially

modeled based on the structure of a similar program that was successfully implemented at Vermont Gas Systems, Inc.

Q. Have the Companies provided an update to their tiered structure in their Corrections and Updates filing?

A. Yes, as shown on page six of the Companies' Future of Heat Panel corrections and updates testimony, the Companies propose to "update the residential tiers to include a flat-rate option that more closely tracks the 100 percent option for non-residential customers."

Q. Do the Companies propose any full-time equivalent employees (FTEs) associated with the Green Gas Tariff?

A. Yes, as shown on Exhibit_(FOH-13CU)in the Companies' Future of Heat Panel corrections and updates testimony, the Companies propose two FTEs to be hired in the Rate Year, one for KEDNY and one for KEDLI, to begin the process of implementing the green gas tariff program.

Q. Do the Companies propose to reconcile any over or under collections associated with RNG gas costs for the proposed Green Gas Tariff?

A. Yes.
Q. How?
A. The Companies propose to create a separate reconciliation for the RNG gas costs and include any monetized environmental credits to reduce the RNG costs for participating customers.
Q. Does the Panel support the proposed Green Gas Tariff?
A. Yes. We support the proposed program and believe that the tiered structure provides a simple subscription methodology for customers to participate along with pricing certainty that is easy for a customer to include in their decision making for budgeting purposes. However, we do have two recommendations.
Q. What do you recommend?
A. First, we recommend that the proposed FTEs be reduced from two to one FTE for the Rate Year. Second, we recommend that the Commission require KEDNY and KEDLI to reconcile gas costs and any associated gas credits as part of the existing gas adjustment clause mechanism.
Q. Why do you recommend reducing the FTEs associated with the green gas tariff program in

the Rate Year?

A. Since the program is new and will not be available to customers until Data Year 1, it is difficult to tell how many FTEs will be needed to administer the program. Therefore, we believe one FTE, half for each Company, will be sufficient to start with, for the preparation of the green gas tariff. Depending on how the program develops, and how many customers participate, the number of FTEs associated with the green gas tariff program may be adjusted in the future.

Q. Why are you concerned with the Companies' proposed separate reconciliation mechanism?

A. With the program just starting, and a lack of participation and RNG cost data, we are concerned that a separate RNG reconciliation could lead to bill impacts that may be detrimental to the initial participants and ultimately undermine their continued participation in the program.

Q. When should the Commission consider a separate reconciliation mechanism?

A. When the Companies have some track record of

forecasting the costs, so that the bill impacts of the reconciliation can be understood. Also, the Commission should consider the number of participating customers, across which the reconciliation would be spread.

Q. Why is it reasonable to have one mechanism to start the program?

A. With the proposed program being in its infancy, and that the supply and environmental benefits provided by the RNG supplies ultimately benefiting all customers, we recommend that any excess unsubscribed RNG supplies be included as part of the overall gas supply portfolio of all firm gas customers, including those that do not participate in the program. At this stage, any gas cost impacts to non-participating gas customers will be minimal and not have any material effect due to the small expected RNG volumes in the near future and the size of the current gas supply portfolio.

Gas Cost Hedging

Q. How do the Companies mitigate gas cost volatility associated with the gas supply

portfolio for customers?

A. As described starting on page 36 of the testimony of witness Elizabeth D. Arangio, the Companies use various methodologies to protect prices for approximately 50 percent, but no more than 60 percent, of forecast winter firm sales and to allow prices to fluctuate with the market for the remaining forecast winter sales. These methodologies are described in a diversity and volatility mitigation plan.

Q. Have the Companies recently changed their hedging practices?

A. Yes, as described starting on page 38 of witness Arangio’s testimony, the Companies have incorporated the use of options at Dominion South Point into the hedge portfolio and are favoring up to 100 percent of options for the financial hedge component of the supply portfolios for the upcoming winter. The Companies made these changes due to developed liquidity at the market supply point, lower option premiums and the downside protection provided by options.

Q. Do you support the use of 100 percent of options

as the Companies financial hedge component?

A. We support the Companies change because it provides an added level of diversification, which helps to mitigate gas cost volatility. We recommend that the Companies be required to continue to monitor their hedging strategies, including this one on an ongoing basis. Decisions such as this one should be reviewed with Staff during the annual winter supply review.

Q. What is the annual winter supply review?

A. In the annual winter supply review, Staff reviews the previous year’s forecasted gas supply portfolio costs versus the actual costs. Staff also reviews the gas cost forecast for the upcoming winter heating season and, among other things, Staff ensures that New York gas local distribution companies comply with the Commission Statement of Policy Regarding Gas Purchasing Practices issued in Case 97-G-0600 on April 28, 1998.

Q. Please explain what is contained in that Commission policy statement.

A. The Commission Statement of Policy Regarding Gas

Purchasing Practices establishes the requirement for LDCs to consider diverse price volatility mitigation strategies, among other things, as part of its gas purchasing practices to defuse price spikes and valleys for gas supplies.

Q. Do the Companies current practices comply with the requirements of the Commission’s policy statement?

A. Yes. Our review demonstrates that the Companies pricing diversity and volatility mitigation plan is in compliance with the requirements of the Commission’s policy statement.

Q. Does this conclude your testimony at this time?

A. Yes.

BY MR. GOODRICH: (Cont.)

Q. And Panel, also before you is an exhibit identified at the time of filing as SGSP-1. And this exhibit has now been marked for identification purposes as Exhibit 475. Was this exhibit prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. And do you affirm that the information contained in the exhibit just identified is true and correct to the best of your knowledge?

A. Yes.

Q. One moment, Your Honor. I was paranoid that I had forgotten an exhibit. That's every -- is that everything that you submitted in this case?

A. Correct.

MR. GOODRICH: Thank you. Your Honors, though there is none, the Gas Supply Panel is available for cross examination.

A.L.J. COSTELLO: Okay. Thank you. My understanding is that there is no cross examination for the panel, and we thank you. You're excused.

MR. MAIORIELLO: Thank you.

A.L.J. COSTELLO: And we'll go off the record.

(Off the record 11:21 a.m.)

(On the record 11:24 a.m.)

UNIDENTIFIED SPEAKER: You're back on the record.

A.L.J. COSTELLO: And my ask -- may I ask the gentleman seated furthest from me to start with stating your name and business address for the record?

MR. CHOW: Ranny Chow, 90 Church Street, New York, New York.

UNIDENTIFIED SPEAKER: What was your first name?

MR. CHOW: Ranny.

UNIDENTIFIED SPEAKER: Could you spell --

MR. CHOW: Ranny, R-A-N-N-Y.

MR. MERCURIO: Jonathan Mercurio, 90 Church Street, New York, New York.

UNIDENTIFIED SPEAKER: Can you spell your last name?

MR. MERCURIO: M-E-R-C-U-R-I-O.

UNIDENTIFIED SPEAKER: Thank you.

MR. PASNOW: Michael Pasnow, 3 Empire State Plaza, Albany, New York.

MR. THOMAS: Suresh Thomas, 90 Church Street, New York, New York.

UNIDENTIFIED SPEAKER: Could you spell your first name?

MR. THOMAS: S-U-R-E-S-H.

UNIDENTIFIED SPEAKER: Thank you.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you'll provide is the truth?

PANEL: Yes.

RANNY CHOW; SWORN

JONATHAN MERCURIO; SWORN

MICHAEL PASNOW; SWORN

SURESH THOMAS; SWORN

A.L.J. COSTELLO: You may be seated. Counselor, you may proceed.

MR. GOODRICH: Thank you, Your Honors.

DIRECT EXAMINATION

BY MR. GOODRICH:

Q. Staff Pipeline Safety Panel, before you is an exhibit entitled, "Prepared Testimony of the Staff Pipeline Safety Panel,"

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas consisting of a cover page and 96 pages of questions and answers dated August 2019. Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Do you have any changes or corrections to make to your prepared testimony?

A. No.

Q. Do you affirm that the information contained in the Panel's prepared testimony is true and correct to the best of your knowledge?

A. Yes.

Q. And has the Panel modified any of its positions stated in that testimony in response to any interrogatories asked in these proceedings?

A. Yes.

Q. Please explain.

A. In response to NG-47, we modified our position with regard to the damage prevention performance mech -- mechanisms in response to NG-49. We modified our position with regard to moving meters currently inside to locations outside of the service locations.

MR. GOODRICH: And Your Honors, I

understand that the companies will be offering the responses to interrogatories NG-47 and NG-49 as exhibits when they cross examine this panel.

A.L.J. LEARY: Are they in the record? Or are they going to be new exhibits added to the exhibit list?

MR. DECICCO: Your Honor, they'll be -- Phil Decicco for the company. They will be new exhibits added to the exhibit list.

A.L.J. LEARY: Okay.

BY MR. GOODRICH: (Cont.)

Q. So Panel, if you were to ask the same questions in your tes -- in your testimony today under oath, would you answer them the say way, excepting the modifications to your position set forth in response to interrogatories NG-47 and NG-49?

A. (Cont.) Yes.

MR. GOODRICH: Your Honor, I ask that the prepared testimony of the Staff Pipeline and Safety Panel be entered into the record as if given orally.

A.L.J. COSTELLO: That request is granted, and the court reporter should insert at this point DPS Staff Pipeline Safety Panel direct

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Pipeline Safety Panel

Suresh Thomas
Utility Supervisor

Jonathan Mercurio
Utility Engineering Specialist 3

Michael Pasinella
Utility Engineering Specialist 2

Ranny Chow
Assistant Engineer

Office of Electric, Gas and
Water

State of New York
Department of Public Service

Three Empire State Plaza
Albany, New York 12223-1350

90 Church Street, 4th Floor
New York, New York 10007

Q. Members of the Staff Pipeline Safety Panel, please state your names, employer and business addresses.

A. Our names are Suresh Thomas, Jonathan Mercurio, Michael Pasinella, and Ranny Chow. We are employed by the New York State Department of Public Service (Department). Our business addresses are Three Empire Plaza, Albany, New York 12223, and 90 Church Street, 4th Floor, New York, New York 10007.

Q. Mr. Thomas, what is your position at the Department?

A. I am employed as a Utility Supervisor in the Pipeline Safety Section of the Office of Electric, Gas, and Water.

Q. Please summarize your education and work experience.

A. I graduated from Mahatma Gandhi University in 1992 with a Bachelor of Science degree in Mechanical Engineering. I have been employed by the Department since July 2001 in the New York City Office. I have oversight responsibility for the pipeline safety section field Staff who audit pipeline operators for compliance with the

applicable State and Federal regulations to ensure pipeline safety. Prior to my employment at the Department, from 1999 to July 2001, I worked for the City of New York's Housing Preservation and Development Department as a housing inspector. From 1998 to 1999, I served as an engineer with Valence Technology, Inc., located in Henderson, Nevada. From 1995 to 1997, I worked as a project engineer for Lloyd Insulation Limited, providing oversight on thermal insulation projects related to the petrochemical and power plant industries. Lastly, from 1993 to 1994, I participated in a one-year apprenticeship program at a crude oil refinery.

Q. Have you previously provided testimony before the Public Service Commission (Commission)?

A. Yes. I provided testimony in various cases, including Case 02-W-1564, Sea Cliff Water Company; Case 03-W-1182, Heritage Hills Water Works Corporation; Cases 05-G-1494, 08-G-1398, and 14-G-0494, Orange and Rockland Utilities, Inc. (Orange and Rockland); Cases 13-G-0031, 13-S-0032, and 19-G-0066, Consolidated Edison

Company of New York, Inc. (Con Edison); Case 16-G-0058, KeySpan Gas East Corporation d/b/a National Grid (KEDLI); and Case 16-G-0059, The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY).

Q. Mr. Mercurio, what is your position at the Department?

A. I am employed as a Utility Engineering Specialist 3 in the Pipeline Safety Section of the Office of Electric, Gas and Water.

Q. Please summarize your education and work experience.

A. I graduated from Farmingdale State College, State University of New York in 2008 with a Bachelor of Science degree in Electrical Engineering Technology. I have been employed by the Department since 2009 in the New York City Office. I am familiar with Federal and State pipeline safety regulations, as well as with the operations of major gas utilities in New York State. My duties include reviewing proposed pipeline designs, reviewing proposed updates to utility operations and maintenance procedures, and reviewing proposed changes to Federal and

State pipeline safety regulations. In addition, I perform record and field audits of local distribution companies and interstate pipelines to ensure compliance with the Federal and state pipeline safety regulations. I also inspect construction activities of local distribution companies and interstate pipelines to ensure compliance with federal and state pipeline safety regulations.

Q. Have you previously provided testimony before the Commission?

A. Yes. I provided testimony in Case 10-E-0362, Orange and Rockland; and Cases 13-S-0032 and 16-G-0061, Con Edison.

Q. Mr. Pasinella, what is your position at the Department?

A. I am employed as a Utility Engineering Specialist 2 in the Pipeline Safety Section of the Office of Electric, Gas and Water.

Q. Please summarize your education and work experience.

A. I graduated from Clarkson University in 2009 with a Bachelor of Science degree in Civil Engineering. I have been employed by the

Department since December 2010. I am familiar with Federal and State pipeline safety regulations, as well as with the operations of major gas utilities in New York State. My duties include reviewing proposed pipeline designs and proposed updates to gas utility operations and maintenance procedures, processing petitions and waivers relating to code compliance matters, reviewing proposed changes to Federal and State pipeline safety regulations, and enforcement of probable violations relating to damage prevention.

Q. Have you previously provided testimony before the Commission?

A. Yes. I provided testimony in various cases, including Case 14-G-0319, Central Hudson Gas & Electric Corporation (Central Hudson); Case 15-G-0284, New York State Electric & Gas Corporation; Case 15-G-0286, Rochester Gas and Electric Corporation; Case 16-G-0058, KEDLI; Case 16-G-0059, KEDNY, Case; Case 16-G-0257, National Fuel Gas Distribution Corporation; Case 17-G-0239, Niagara Mohawk Power Corporation d/b/a National Grid (Niagara Mohawk); Case 18-G-

0068, Orange and Rockland; and Case 19-G-0066, Con Edison.

Q. Mr. Chow, what is your position at the Department?

A. I am employed as an Assistant Engineer in the Pipeline Safety Section of the Office of Electric, Gas and Water.

Q. Please summarize your education and work experience.

A. I graduated from the State University of New York at Buffalo in 2008 with a Bachelor of Science degree in Mechanical Engineering. I have been employed by the Department since March 2012. I am familiar with Federal and State pipeline safety regulations, as well as with the operations of the major gas utilities in New York State. My duties include conducting record, field, and construction inspections of local distribution companies and interstate pipeline operators to verify compliance with Federal and State pipeline safety regulations; conducting pipeline incident investigations and investigations of third-party damages; conducting investigations for safety-related

customer complaints; and reviewing proposed updates to utility operations and maintenance, and other programs.

Q. Have you previously provided testimony before the Commission?

A. No.

SCOPE OF TESTIMONY

Q. What is the purpose of the Staff Pipeline Safety Panel's testimony in this proceeding?

A. The purpose of our testimony is to address the proposals made by KEDNY and KEDLI (collectively the Companies) for safety performance measures in the areas of leak prone pipe (LPP) removal, leak management, damage prevention, emergency response, and compliance with the pipeline safety regulations. Our testimony will also address first responder training, residential methane detectors, the service line proceeding, New York City Local Law 152, outside meters, inactive accounts, methane leak pilot program, enhanced contractor safety inspections, buried vent lines, and annual reporting requirements.

Q. In this testimony, will the Panel refer to, or

otherwise rely upon, any responses to Information Requests (IRs) obtained during the discovery phase of this proceeding?

A. Yes, we will refer to, and have relied upon, several responses to IRs provided by the Companies. These responses are contained within Exhibit__(SPSP-1). We will refer to these responses by the designation given to them when sent, e.g., “DPS-100.”

Q. Does the Panel have any other exhibits?

A. Yes, we have seven additional exhibits. Exhibit__(SPSP-2) provides a breakdown of the high risk and other risk categories associated with Title 16 of New York Codes, Rules and Regulations (NYCRR) §§255 and 261, and 49 CFR §193. Exhibit__(SPSP-3), Exhibit__(SPSP-4), Exhibit__(SPSP-5), Exhibit__(SPSP-6), and Exhibit__(SPSP-7), are the 2013, 2014, 2015, 2016, and 2017 negative revenue adjustment letters, respectively. Exhibit__(SPSP-8) is a letter dated December 11, 2015 from the Deputy Director of the Office of Electric, Gas and Water communicating the pipeline safety performance measure reporting guidelines to the

Companies.

PIPELINE SAFETY PERFORMANCE MEASURES

Q. What is the purpose of pipeline safety performance measures?

A. The performance measures help to ensure reliable service by requiring local distribution companies (LDCs) to maintain focus on important safety areas. The performance measures for each respective LDC are derived from that company's actual historic performance levels, our knowledge of that company, and our experience with other LDCs across the State.

Q. Please explain the pipeline safety performance measures currently in place for the Companies.

A. Currently, the Companies have performance measures for LPP removal, leak management, damage prevention, emergency response, and compliance with the pipeline safety regulations. Pipeline Safety Staff has audited the Companies performance and will recommend disposition of any revenue adjustments.

Q. Do these performance measures have associated positive and negative revenue adjustments?

A. Both LPP removal and leak management have associated positive and negative revenue adjustments. The remaining performance measures only have associated negative revenue adjustments.

Q. What is the total basis point exposure for the Companies related to these adjustments?

A. Each Company is subject to a total adjustment of 150 basis points for negative revenue adjustments and a total of 15 basis points for positive revenue adjustments.

Q. Does the Panel recommend any modifications to these performance measures and associated positive and negative revenue adjustments?

A. Yes. We recommend positive revenue adjustments for leak management, damage prevention, and emergency response, removing the positive revenue adjustment for LPP removal, and the redistribution of the existing 150 negative revenue adjustments between performance measures.

Q. Please describe the Pipeline Safety Performance Measures Report.

A. The Pipeline Safety Performance Measures Report

is an annual report presented by Department Staff to the Commission. The report summarizes data, analyzes performance, and is used as a tool to track and identify LDC performance in the following areas of pipeline safety: damage prevention, emergency response, leak management, and compliance with the pipeline safety regulations.

Q. Where can the Pipeline Safety Performance Measures Report for recent years be found?

A. The Pipeline Safety Performance Measures Report can be found on the Commission's website. The most recent report can be found under 19-G-0298. That report contains data for calendar years 2014 through 2018.

LEAK PRONE PIPE REMOVAL

Q. What type of pipe is considered leak prone?

A. LPP generally includes cast iron, wrought iron, unprotected steel, and some early vintages of plastic pipe that can become brittle. Pursuant to the Direct Testimonies of KEDNY and KEDLI's Gas Infrastructure and Operations (GIOP) Panels, the Companies' population of LPP consists of

unprotected steel, cast and wrought iron, copper, and certain early vintages of plastic pipe including Aldyl-A.

Q. Please explain the importance of removing cast iron pipe.

A. In general, cast iron pipe is brittle, susceptible to graphitization, which is a form of corrosion, and has low beam strength. Beam strength refers to the amount of load a structure can withstand before it fails. The low beam strength of cast iron pipe means that the material can fail if it is subjected to increased loading from a force pushing up or down on it, or from a loss of ground support. This characteristic makes the material particularly susceptible to stresses from underground forces or disturbances. Such disturbances include, but are not limited to, ground settlement, freeze-thaw cycles, soil erosion or nearby excavation activities. The physical characteristics of cast iron pipe make it more prone to catastrophic failure than cathodically protected steel or plastic pipe. In addition, segments of cast iron pipe are

joined by hub joints, also known as bell and spigot joints. Packing material, such as jute, was used in the joints to prevent leakage when cast iron pipelines were first installed. Unlike steel welded joints, these cast iron joints develop leaks over time.

Q. How does the removal of cast iron pipe contribute to the safety of the natural gas system?

A. Cast iron pipe tends to be located in densely populated areas where there are many enclosed structures, subsurface structures, and continuously paved areas. In the event of a major leak or pipeline failure, these environmental circumstances may lead to greater volumes of below-ground gas migration, particularly under paved areas and through connected subsurface structures, which creates an increased risk for fires or explosions. Removing this pipe reduces the potential for incidents resulting from major leaks and failures, thereby improving public safety.

Q. What is meant by the term unprotected?

A. Unprotected means that the pipe lacks adequate

cathodic protection, thus rendering it susceptible to corrosion. Unprotected steel pipe often has inadequate or no coating, which makes efforts to cathodically protect such pipe ineffective and uneconomic.

Q. How do LPP removal programs improve the safety of natural gas systems?

A. Leaks on underground piping can create safety risks to the public and potentially lead to serious incidents, resulting in injuries to persons or property damage. Therefore, removing LPP reduces these safety risks.

Q. Please explain the importance of removing unprotected steel pipe.

A. Data collected by the United States Department of Transportation, Pipeline and Hazardous Materials Safety Administration (PHMSA) shows that corrosion is a leading cause of pipeline failures, and that unprotected steel is typically the pipeline material that is most susceptible to corrosion.

Q. What other benefits are associated with the removal of LPP?

A. Removing LPP should reduce the number of active

leaks, lead to a decline in leakage rates on the distribution system, and reduce the operating, maintenance, and overtime costs associated with surveying leaks and responding to emergency reports.

Q. Please describe the LPP removal component of the pipeline safety performance measures.

A. This component is designed to ensure that a company continues to proactively remove LPP from its system. Typically, a company would only remove LPP to ensure compliance with the pipeline safety regulations, to respond to significant customer complaints, or through coordinated municipal or state projects where a portion of the associated costs are shared.

Q. Do the Companies currently have LPP removal programs?

A. Yes.

Q. How do the Companies prioritize the removal of LPP?

A. The pipe to be removed from service is identified and ranked using a risk assessment model-based approach.

Q. What is a risk assessment model?

A. A risk assessment model prioritizes all segments of LPP according to attributes that pose the highest associated risk. The model uses several weighted factors, such as material type, diameter, pressure, date of installation, and leak history, to determine a ranking. This allows the Companies to provide a higher level of public safety by focusing its efforts on pipeline segments with higher risk. Although there are commonalities between the State's LDCs, it is important to note that the model has been tailored to the Companies unique mix of assets and may not be appropriate for other LDCs. A uniform approach cannot accommodate the characteristics that must be considered for each system.

Q. Do the Companies remove leak prone services in conjunction with their LPP programs?

A. Yes. When removing LPP, it is common to group mains and services together within a single project. These efforts provide the most cost-efficient approach towards removing LPP, in addition to minimizing the time a customer is without service.

Q. What current LPP removal requirements apply to KEDNY and KEDLI?

A. The Commission’s December 16, 2016 Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans in Cases 16-G-0058 and 16-G-0059 (2016 Rate Order), required KEDNY to remove a minimum of 50 and 55 miles of LPP in 2017 and 2018, respectively, and a cumulative minimum of 180 miles of LPP from 2017 through 2019. The 2016 Rate Order also required KEDLI to remove 105 miles of LPP in 2017, and 125 miles of LPP in 2018, and a cumulative minimum of 405 miles of LPP from 2017 through 2019. Should either of the Companies fail to meet these requirements, that Company would be subject to negative revenue adjustments.

Q. Are there currently positive revenue adjustments for LPP removal?

A. Yes. For each full mile, beginning with the second full mile of LPP removed above specified threshold levels, the Companies would earn a positive revenue adjustment of two basis points. For KEDNY, the thresholds were 57, 62, and 67 miles in calendar years 2017, 2018 and 2019,

respectively. For KEDLI, the thresholds were 117, 137 and 157 miles in calendar years 2017, 2018 and 2019, respectively. Each Company could earn a maximum of 10 basis points per calendar year.

Q. How did the Companies perform in 2017?

A. In their response to IR DPS-159, KEDNY and KEDLI explained that they removed 57.8 and 115.9 miles, respectively, of LPP in 2017.

Q. How did the Companies perform in 2018?

A. In their response to IR DPS-665, KEDNY and KEDLI stated that they removed 63.7 and 138.0 miles, respectively, of LPP in 2018.

Q. Were either KEDNY or KEDLI subject to any revenue adjustments for either 2017 or 2018?

A. Yes. For 2017, KEDNY earned a positive revenue adjustment of two basis points for its performance. For 2018, both KEDNY and KEDLI earned positive revenue adjustments of four basis points each.

Q. Did the Companies propose any modifications to their LPP programs?

A. Yes. KEDNY proposes to increase its LPP removals by five miles per year as it has

experienced an increase in leak rates, and KEDLI proposes to maintain its current level of LPP removals as its leak rates have stabilized. Both Companies propose cumulative LPP removal targets, to maintain positive and negative revenue adjustments, and to proactively remove pre-1974 high-density polyethylene services.

Q. At the Companies proposed removal rates, how long will it take to retire all LPP from their service territories?

A. In their response to IR DPS-158, KEDNY and KEDLI explain that they have approximately 1,509 and 3,515 miles, respectively, of LPP remaining on their systems. At the proposed removal rates, both Companies programs would be completed in approximately 19 years, or by 2038.

Q. How do the Companies LPP removal programs compare with those of other LDCs?

A. KEDNY and KEDLI have two of the larger inventories in terms of miles of LPP remaining in the State. Further, based on their current proposals, we anticipate KEDNY and KEDLI to be the last LDCs in the State to complete the removal of known LPP. Since the condition of

LPP deteriorates with time, while the corresponding risk increases, we would not recommend a reduction to either of the proposed removal rates. In addition, we note that, with inflation, the cost to customers for removal in later years will certainly be more than the removal of that same pipe now.

Q. Does the Panel agree with the annual and cumulative removal targets proposed by the Companies?

A. We agree with the proposed annual removal targets. As our testimony addresses a single rate year, cumulative targets are not necessary.

Q. Does the Panel agree with continuing the associated positive revenue adjustments for LPP removal?

A. No. Associated positive revenue adjustments are further discussed in the Staff Policy Panel Testimony.

Q. Does the Panel agree with the associated negative revenue adjustments for the LPP removal targets?

A. Yes. We agree each Company should incur negative revenue adjustments of 15 basis points

should that Company fail to meet its annual LPP removal targets.

Q. Why does the Panel consider the increase in exposure from the current eight to 15 basis points reasonable?

A. As previously discussed, the removal of cast iron, wrought iron, and unprotected steel piping enhances system safety and integrity, and reduces methane emissions from actively leaking pipe. For these reasons and based on the magnitude of the Companies' LPP removal programs, the corresponding increase in exposure is reasonable.

Q. Will the LPP removal targets affect the Companies risk assessment model?

A. No. We expect that KEDNY and KEDLI will continue to use their risk assessment models to rank segments of pipe for removal so that they remove the pipe that presents the greatest risk to the public before they remove lower-risk pipe. At the same time, we acknowledge that KEDNY and KEDLI should be given the flexibility to complete opportunistic removals, such as geographic area approaches, or in conjunction

with other entities such as municipal repaving projects. That said, overall risk reduction should remain the main driver of the removal program. In other words, if using the geographic area replacement approach, the areas removed should contain high-risk segments.

Q. Does the Panel have any other recommendations regarding the removal of LPP?

A. Yes. As the Companies discover additional LPP, they should identify strategies to accelerate the removal of this pipe, while maintaining the current schedule for completing their LPP programs. An example of this are the proactive removals of pre-1974 high-density polyethylene services.

Q. Are there any other conditions the Panel recommends the Commission impose on the Companies regarding these recommendations?

A. Yes. We recommend that both KEDNY and KEDLI submit annual reports to the Secretary of the Commission detailing their LPP removal progress. This report, at a minimum, should include material type, mileage, project location, rank of the segments removed at the time of removal

using the risk-based model, project cost, and a forecast of the scheduled LPP removal projects and their rank on the risk-based removal model for the upcoming calendar year. The report should also include a reconciliation of proposed versus actual removals. The Companies should be required to submit these reports no later than 45 days after the end of each calendar year.

Q. Should the LPP removal targets or the associated revenue adjustments expire?

A. No. The LPP removal targets and associated revenue adjustments should remain in effect until changed by the Commission.

LEAK MANAGEMENT

Q. Please explain the term leak management.

A. Leak management refers to a utility’s ability to monitor and repair existing and newly discovered leaks on its natural gas system.

Q. Do the Companies currently have a safety related target for leak management?

A. Yes.

Q. Does failure to meet the total leak management targets trigger negative revenue adjustments?

A. Yes. Should either KEDNY or KEDLI fail to meet their total leak management targets, that Company would be subject to a four basis point negative revenue adjustment.

Q. Please explain the total leak management target.

A. The total leak management target encompasses Type 1, Type 2A, Type 2, and Type 3 leaks as defined by 16 NYCRR §§255.811, 255.813, 255.815, and 255.817, respectively.

Q. How have the Companies performed in recent years with regards to the total leak management target?

A. In their responses to IRs DPS-114, DPS-124, and DPS-125, KEDNY and KEDLI state that they each met their respective calendar year-end targets from 2014 through 2018.

Q. Were there associated positive revenue adjustments for any of the previous five calendar years?

A. Yes. Beginning in calendar year 2017, KEDNY and KEDLI could each earn positive revenue adjustments as follows: one basis point for eliminating 50 of the highest volume emitting Type 3 leaks; two basis points for eliminating

100; three basis points for eliminating 150; four basis points for eliminating 200; and five basis points for eliminating 250.

Q. Did either Company earn positive revenue adjustments for either of the previous two calendar years?

A. In the response to IR DPS-666, KEDNY stated that it eliminated 261 and 205 of the highest volume emitting Type 3 leaks in calendar years 2017 and 2018, respectively, and KEDLI stated that it eliminated 303 and 275 of the highest volume emitting Type 3 leaks in calendar years 2017 and 2018, respectively. Thus, KENDY earned positive revenue adjustments of five and four basis points for 2017 and 2018, respectively, and KEDLI earned five basis points in each year.

Q. Did either of the Companies propose to update or modify their total leak management targets?

A. Yes. Both KEDNY and KEDLI proposed to reduce their total leak management targets by 150 and 750, respectively, per calendar year.

Q. Please explain what this means.

A. Currently, KEDNY and KEDLI have year-end targets of 3,250 and 8,500 or fewer total leaks,

respectively. The Companies propose year-end targets of 3,100 and 7,750 or fewer total leaks, respectively, for calendar year 2020.

Q. Did the Companies propose to update or modify the associated negative revenue adjustments?

A. Yes. The Companies propose that, should either KEDNY or KEDLI fail to meet its total leak management targets, that Company would be subject to a five basis point negative revenue adjustment.

Q. Did the Companies propose to update or modify the associated positive revenue adjustments?

A. No.

Q. Is there a difference between repairable and total leak backlogs?

A. Yes. As explained earlier, total leak backlogs encompass Type 1, Type 2A, Type 2, and Type 3 leaks. Repairable leak backlogs exclude Type 3 leaks because they are considered non-hazardous and are reasonably expected to remain that way.

Q. Do the Companies have repairable leak management targets?

A. Yes.

Q. Does failure to meet the repairable leak

management targets trigger negative revenue adjustments?

A. Yes. Should either KEDNY or KEDLI fail to meet its repairable leak management targets, that Company would be subject to an eight basis point negative revenue adjustment.

Q. How have the Companies performed in recent years with regards to the repairable leak management target?

A. In their responses to IRs DPS-114, DPS-124, and DPS-125, KEDNY and KEDLI explained that each Company met its respective calendar year-end targets for 2014 through 2018.

Q. Were there associated positive revenue adjustments for any of the previous five calendar years?

A. No.

Q. Did either of the Companies propose to update or modify their repairable leak management targets?

A. No.

Q. Did the Companies propose to update or modify their associated negative revenue adjustments?

A. Yes. The Companies proposed that, should either KEDNY or KEDLI fail to meet its repairable total

leak management target, that Company would be subject to a 10 basis point negative revenue adjustment.

Q. What does the Panel recommend?

A. We recommend that KEDNY and KEDLI reduce their total leak management targets by 150 and 750, respectively, per calendar year. However, the total leak management targets should be reduced in calendar year 2020 from 3,100 and 7,750, to 2,200 and 6,600, respectively.

Q. Why should the calendar year 2020 targets be reduced to the levels recommended?

A. In the response to IR DPS-114, KEDNY and KEDLI explained that they have averaged an approximate 316 and 823 reductions in their respective total leak backlogs in each of the previous five calendar years. In 2018, KEDNY and KEDLI reported that their total leak backlogs were 2,803 and 8,199, respectively. At the averaged reduction rates, and in accounting for an additional calendar year, 2019, the respective total leak management targets should be reduced in calendar year 2020 to 2,200 and 6,600.

Q. Does the Panel recommend repairable leak

management targets?

A. Yes. The Panel recommends that the repairable leak management targets for both Companies be reduced from 25 to 20. In response to IR DPS-114, KEDNY and KEDLI reported that in 2018, their backlogs of repairable leaks were 10 and zero, respectively.

Q. Why are the Panel's total and repairable leak management targets reasonable?

A. KEDNY and KEDLI use both historic and active leaks as weighted factors when prioritizing segments within its risk assessment model. Therefore, the removal of LPP alone should reduce leak rates. As more and more of the Companies' systems are comprised of newly-installed piping, reductions in leak inventory should be easier to attain. Also, periodic monitoring is required for all leaks on natural gas systems. This monitoring ensures that these leaks have not become an increased threat to public safety. Monitoring such leaks requires a physical visit to the location with work required to be performed by a qualified person using properly calibrated equipment. The

reduction in the amount of this work, in addition to the reduction of losses of natural gas due to leakage, would result in less costs for customers.

Q. What does the Panel recommend with regard to the associated negative revenue adjustments?

A. Each Company should be required to meet its annual repairable or total leak management targets on at least one day from December 26th through December 31st of the calendar year. If either Company fails to meet either its repairable or total leak management targets, we recommend that Company incur a negative revenue adjustment of 10 and five basis points, respectively.

Q. Why does the Panel consider the increase in combined exposure from the current 12 basis point to 15 basis points reasonable?

A. The Panel recommends increases to the negative revenue adjustments associated with leak management targets by three basis points to reflect the targets' importance and the corresponding risk unrepaired leaks pose to the public. The Companies' expeditious repair of

leaking pipelines enhances system integrity and safety and reduces methane emissions from the leakage. This reallocation is also consistent with recent performance measures for other LDCs.

Q. Why should the Companies be given from December 26th through December 31st of the calendar year to meet their leak management targets?

A. On an annual basis, the Companies strive to reduce their respective leak backlogs while new leaks are discovered daily. Therefore, a five-day threshold should be established during which each Company can meet its targets. This eliminates the potential for a target being missed due to an increase of discovered leaks on the last day of the calendar year. Should the repairable or total leak backlogs on any day from December 26th through December 31st fall below the established leak management targets, the Companies would be considered to have met the applicable target. In addition, only successful elimination of a leak will be considered a valid leak repair.

Q. What does this Panel mean by successful elimination of a leak?

A. Successful elimination of a leak means that the applicable company has either performed multiple inspections, two or more, to confirm that zero-percent gas-in-air readings have been maintained, or a recheck has been completed as set forth in the Commission's pipeline safety regulations.

Q. Are rechecks required for all leak types?

A. No. Pursuant to 16 NYCRR §255.819(a), upon completing a Type 1, Type 2A, or Type 2 leak repair, KEDNY and KEDLI must conduct a follow-up inspection between 14 and 30 days following the repair date to validate the repair. There is no follow-up inspection requirement for Type 3 leaks.

Q. Although not required by the Commission's regulations, should the Companies perform follow-up inspections for Type 3 leak repairs?

A. Yes, this Panel recommends that the Companies perform follow-up inspections for all Type 3 leak repairs and include these inspections in their procedures.

Q. How does the requirement that the elimination of a leak be confirmed as successful affect the

leak management measure?

A. If KEDNY or KEDLI repairs a leak or leaks within the final five working days of the calendar year, it would have, in theory, met its targeted levels for performance. However, until the follow-up inspection confirms successful elimination of the leaks, those leaks will remain in the backlog.

Q. Does the Panel recommend associated positive revenue adjustments for the leak management measure?

A. Yes. For every 100 additional leaks repaired beyond the total leak management targets, the Companies should earn two basis points. The incentive would not be pro-rated. For example, if one of the Companies' repaired 199 additional leaks, that would entitle that Company to a two-basis point positive revenue adjustment. Further, these positive revenue adjustments should be capped at six basis points, or 300 additional leak repairs.

Q. Should either of the leak management targets or the associated revenue adjustments expire?

A. No. These leak management targets and

associated revenue adjustments should remain in effect until changed by the Commission.

DAMAGE PREVENTION

Q. Please explain the term damage prevention.

A. Damage prevention refers to a utility's ability to minimize damage to its natural gas system facilities caused by excavation activities. Damage to a pipeline can result in the uncontrollable release of natural gas and could lead to an incident.

Q. Please describe the performance measures related to the prevention of excavation damage.

A. Under their existing rate plans established in the 2016 Rate Order, KEDNY and KEDLI are subject to performance targets that measure the Companies progress in minimizing damages to its underground pipeline facilities and encourage the Companies to continuously strive to improve its performance. These performance targets include metrics for damage caused specifically by mismarks and by the Companies and their contractors, as well as the total number of damages per 1,000 one-call tickets. The total

damage category includes damages caused by mismarks, by the Companies and their contractors, excavator error, and damages that occur when an excavator failed to notify the one-call notification system of its proposed excavation work.

Q. What is the one-call notification system?

A. The one-call notification system is an organization that establishes and carries out procedures to protect underground facilities from damage due to excavation and demolition. Pursuant to 16 NYCRR §753-3.1 of the Commission’s pipeline safety regulations, excavators are required to provide notice of their intent to excavate to the one-call notification system. There are two one-call notification systems in New York State; Dig Safely New York, and New York 811. Both Companies service territories are covered by New York 811. Both Dig Safely New York and New York 811 record the pertinent information from the excavator and transmit that information to the member utilities that may be affected by the proposed excavation work. The affected

utilities then mark the location of their underground facilities to put the excavator on notice and allow the excavator to take necessary precautions to avoid damaging the surrounding facilities. Each incoming call to Dig Safely New York and New York 811 generates multiple outgoing notices to the affected utilities, including, but not limited to, natural gas, electric, telecommunications, water, and sewer companies. The notice received by the utility from the one-call notification system is referred to as a one-call ticket.

Q. What is a mismark?

A. A mismark occurs when a utility fails to accurately mark the location of its underground facility in response to the one-call ticket. Consistent with the requirements of 16 NYCRR Part 753, and for the purpose of this measure, a mismark is considered as any instance where the markings are off by more than two feet. It also includes any instances where the utility fails to mark facilities in response to a properly requested one-call ticket.

Q. What are damages by Company and Company

contractors?

A. These are damages to Company facilities that are caused by Company personnel or by contractors who are directly working for that Company.

Q. How does prevention of excavation damage benefit public safety?

A. These damages often cause interruptions of service to customers, building evacuations, and road closures. Explosions, fires, injuries, and fatalities, while less frequent, have also occurred as a result of these types of damages. Therefore, reducing these types of damages improves public safety.

Q. Describe the Companies historic performance as it relates to damage prevention.

A. The Companies historic performance is well documented in the most recent Pipeline Safety Performance Measures Report, filed in Case 19-G-0298, which can be obtained from the Commission’s website. The damage prevention metrics are measured as a ratio of damages per 1,000 one-call tickets. For the previous five calendar years, 2014 through 2018, KEDNY and KEDLI averaged 0.40 and 0.55 for damages due to

mismarks, 0.02 and 0.08 for damages caused by the Companies or their contractors, and 1.65 and 2.18 for total damages, respectively. In 2018, KEDNY and KEDLI performed at the following levels: 0.75 and 0.79 for damages due to mismarks; 0.03 and 0.15 for damages due to the Companies or their contractors; and 2.56 and 2.79 for total damages per 1,000 one-call tickets, respectively.

Q. Why were the Companies performance levels for calendar year 2018 significantly higher than that of their historic five-year averages?

A. Pursuant to the recommendations of an operational audit focused on the accuracy and effectiveness of data submitted by nine New York State natural gas and combination natural gas and electric LDCs, filed in Case 13-M-0314, both KEDNY and KEDLI were to exclude refreshed and retransmitted one-call tickets from their respective ticket counts. This exclusion of refreshed and retransmitted one-call tickets went into effect in calendar year 2018 and was in conformance with the required reporting guidance. Thus, both Companies experienced

significantly worse for calendar year 2018 when compared with that of their historic five-year averages.

Q. Were the Companies required to report their damage prevention in conformance with the criteria established for the Pipeline Safety Performance Measures Report?

A. Yes. Under their existing rate plans established in the 2016 Rate Order, both KEDNY and KEDLI were to collect data for the computation of their respective damage prevention performance in accordance with the criteria established for the Pipeline Safety Performance Measures Report.

Q. Do the Companies currently have safety-related targets associated with damage prevention?

A. Yes.

Q. Have either KEDNY or KEDLI failed to meet any of their previously-established damage prevention targets?

A. Yes. In their response to IR DPS-134, KEDNY and KEDLI reported their respective 2018 performance as follows: 0.75 and 0.79 for damages due to mismarks; 0.03 and 0.15 for damages due to the

Companies or their contractors; and 2.56 and 2.79 for total damages per 1,000 one-call tickets. Under their existing rate plans established in the 2016 Order, the Companies damage prevention targets were as follows: 0.42 and 0.60 for damages due to mismarks; 0.09 and 0.13 for damages due to the Companies or their contractors; and 1.87 and 1.87 for total damages per 1,000 one-call tickets, respectively. Thus, both KEDNY and KEDLI failed to meet their targets for mismarks and total damages, and KEDLI failed to meet its target for damages due to the Company or its contractors.

Q. Did the Companies incur negative revenue adjustments for their failure to meet the respective damage prevention targets?

A. Yes. For their performance in 2018, KEDNY and KEDLI incurred negative revenue adjustments of 14 and 18 basis points, respectively.

Q. Did the Companies propose to update or modify their damage prevention targets?

A. Yes. The Companies propose to consolidate the damage prevention measure by eliminating the targets for mismarks and for damages due to the

Companies or their contractors, to increase the associated negative revenue adjustments for the total damage targets, and to implement associated positive revenue adjustments for superior performance. In addition, both Companies requested one additional full-time equivalent advisor to educate and train municipal personnel, excavators, and contractors.

Q. Does the Panel support the elimination of the damage prevention targets?

A. Yes. We support the elimination of both damage prevention targets for mismarks and for damages due to the Companies or their contractors.

Q. Is the consolidation of the damage prevention measures consistent with other LDCs?

A. Yes. The damage prevention metrics have evolved to reflect the continued success in reducing overall damages. As the total number of damages for each category diminishes, having a meaningful comparison of minor fluctuations is not beneficial. The Commission adopted joint proposals with similar consolidations in recent cases, specifically, in the Commission’s March

15, 2018 Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans in Case 17-G-0239 for Niagara Mohawk; June 14, 2018 Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan in Case 17-G-0460 for Central Hudson; and March 14, 2019 Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans in Case 18-G-0068 for Orange and Rockland.

Q. What does the Panel recommend regarding damage prevention metrics for the Companies?

A. We recommend restructuring the damage prevention metric to focus on the total damage prevention target with associated negative and positive revenue adjustments as follows: for a total damage performance rate per 1,000 one-call notifications greater than 2.50, the Companies would incur negative revenue adjustments of 20 basis points; for a rate between 2.25 and 2.50, the Companies would incur negative revenue adjustments of 10 basis points; for a rate between 2.00 and 2.25, the Companies would incur negative revenue adjustments of five basis points; for a rate between 1.50 and 2.00, the

Companies would not incur revenue adjustments; for a rate between 1.25 and 1.50, the Companies would earn positive revenue adjustments of five basis points; and for a rate less than 1.25, the Companies would earn positive revenue adjustments of 10 basis points. In reporting their performance on these damage prevention measures, the Companies should continue to comply with the Pipeline Safety Guidance provided in Exhibit__(SPSP-8).

Q. Why does the Panel consider the increase in exposure for damage prevention from 18 basis points to 20 basis points reasonable?

A. We recommend an increase to the negative revenue adjustments associated with damage prevention by two basis points to encourage the Companies to enhance their respective damage prevention programs. As we previously explained, these damages often cause interruptions of service to customers, building evacuations, and road closures. Explosions, fires, injuries, and fatalities, while less frequent, also have occurred as a result of these damages. Therefore, reducing these types of damages

improves public safety. This re-allocation is also consistent with recent re-allocations for other LDCs.

Q. Why are the Panel's recommended targets reasonable?

A. Given the total volume of one-call notifications in the Companies' respective service territory, and the inherent risks associated with excavation work, public safety should be of the utmost importance. In addition, as the Companies remove older LPP, damages due to mismarks should fall due to more updated and accurate records.

Q. Are damages due to mismarks, the Companies, and their contractors within KEDNY's and KEDLI's control?

A. Yes.

Q. Are total damages also within the Companies control?

A. Not entirely. Specifically, damages caused by failure to notify New York 811, sometimes referred to as no-calls, and/or unsafe excavator practices are not within the control of the Companies. However, the Companies can minimize

these damages by influencing excavator activity through education and outreach efforts, by continuing to bill excavators for repair costs when the excavator is at fault, and by referring problem excavators to Pipeline Safety Staff for enforcement purposes. In addition, KEDNY and KEDLI should consider developing best practices, in conjunction with trade associations or other organizations focused on protecting underground facilities, such as the Common Ground Alliance.

Q. Are damages due to no-calls a component of the total damage prevention measure?

A. Yes. Damages due to no-calls are instances where the excavator failed to provide notice of intent to excavate to the one-call notification system, and thus no ticket was generated. This measure, which is a part of the total damages calculation, provides insight into the general awareness of excavators of the one-call notification system.

Q. How does Pipeline Safety Staff assist LDCs with enforcement of damage prevention?

A. For nearly two decades, Pipeline Safety Staff has conducted an enforcement program, which

includes the collection of penalties for violating the Commission's damage prevention regulations. In 2007, this program was expanded to include instances of damage due to no-calls. Damages due to no-calls are the most straightforward instances of violations of 16 NYCRR Part 753. The enforcement of such damages requires minimal LDC participation, results in greater excavator awareness, and can lead to a lower rate of damage to underground pipeline facilities. In addition, when promptly notified and resources are available, Pipeline Safety Staff provides aid to LDCs by performing on-site investigations.

Q. Should either of the damage prevention targets or the associated revenue adjustments expire?

A. No. These damage prevention targets and associated revenue adjustments should remain in effect until changed by the Commission.

EMERGENCY RESPONSE

Q. Please describe the emergency response performance measures applicable to KEDNY and KEDLI.

A. Pipeline safety regulations require each utility to provide a monthly report of the total number of reports received, and responded to, in 15-minute intervals during normal business hours, weekdays outside of normal business hours, weekends, and holidays. These measures evaluate utility response to gas leak, odor, and emergency reports generated by the public and non-Company personnel. Staff and LDCs jointly established the following standards for the emergency response performance measures: respond to 75 percent of all gas leak, odor, and emergency reports within 30 minutes; respond to 90 percent of all gas leak, odor, and emergency reports within 45 minutes; and respond to 95 percent of all gas leak, odor, and emergency reports within 60 minutes.

Q. What is the significance of the emergency response performance measure?

A. Leaks on inside piping, improperly operated or installed appliances, and gas migration into a building from leaks on outside piping all present risks to the public. The Companies recognize this and dispatch personnel on a

priority basis in response to reports of gas leaks, odors, or emergencies. All LDCs are required to maintain a log of such reports and track the elapsed time between the dispatch and arrival of qualified personnel responding to the scene. As the LDC's response times increase, so does the potential for an incident or safety related condition to the public. Therefore, it is imperative that LDCs minimize their response times to gas leaks, odors, or emergency reports.

Q. Do the Companies currently have targets for their emergency response performance?

A. Yes. Both KEDNY and KEDLI are subject to the aforementioned standards for the emergency response performance measures developed by Staff and the LDCs. Failure to meet the 30, 45, or 60-minute measures results in negative revenue adjustments of six, four, and two basis points, respectively. The Companies can request exclusions of gas leak, odor, and emergency reports resulting from mass area complaints, major weather related occurrences, or major equipment failures, or instances where one of the Companies receives 20 reports or more within

any two-hour period and for the duration of such mass area event.

Q. How have the Companies performed in their emergency response efforts?

A. For each of the previous five calendar years, 2014 through 2018, the Companies exceeded the established minimum performance levels.

Q. Specifically, in 2018, how did the Companies perform in their emergency response efforts?

A. In their amended response to IR DPS-132, KEDNY and KEDLI explained that, respectively, they responded to 80.0 and 77.2 percent of gas leak, odor, and emergency reports within 30 minutes, 95.2 and 95.9 percent of gas leak, odor, and emergency reports within 45 minutes, and 98.5 and 99.7 percent of gas leak, odor, and emergency reports within 60 minutes.

Q. Do the Companies propose to update or modify their emergency response targets?

A. No. Both KEDNY and KEDLI propose to maintain the existing emergency response targets.

Q. Do the Companies propose to update or modify the associated revenue adjustments?

A. Yes. Both KEDNY and KEDLI propose to increase

the associated negative revenue adjustments from six, four, and two basis points to 12, eight, and five basis points, and to institute positive revenue adjustments for exceeding the 30-minute target.

Q. Please explain the Companies' proposed positive revenue adjustments.

A. KEDNY and KEDLI propose that they earn the following positive revenue adjustments: two basis points for responding to 86.00 through 87.99 percent of gas leak, odor, and emergency reports within 30 minutes; four basis points for responding to 88.00 through 89.99 percent of gas leak, odor, and emergency reports within 30 minutes; and six basis points for responding to 90.00 percent or greater of gas leak, odor, and emergency reports within 30 minutes.

Q. What does the Panel recommend with regard to targets for this measure?

A. We recommend that both KEDNY and KEDLI continue to be required to respond to 75 percent, 90 percent, and 95 percent of all gas leak, odor, and emergency reports within 30, 45, and 60 minutes, respectively.

Q. What does the Panel recommend with regard to the Companies' ability to exclude certain mass area events from this measure?

A. The Companies should be allowed to continue to request exclusions of gas leak, odor, and emergency reports resulting from mass area events, major weather-related occurrences, and major equipment failure. For each requested exclusion, the Companies must provide notification to safety@dps.ny.gov within seven days of such event. Pipeline Safety Staff will consider these exclusions on a case-by-case basis and will respond to the Companies indicating whether it concurs with the request.

Q. What does the Panel recommend with regard to associated negative revenue adjustments for failing to meet the emergency response measures?

A. We agree with the Companies proposal to increase the basis point exposure as follows: failure to meet the 30, 45, or 60-minute targets would trigger negative revenue adjustments of 12, eight, or five basis points, respectively.

Q. Why does the Panel consider the increase in exposure from 12 basis points to 25 basis points

reasonable?

A. We recommend an increase to the negative revenue adjustments associated with emergency response of 13 basis points to reflect the importance of the Companies' response to gas leak, odor and emergency reports. Timely assessment of leaks on inside piping, improperly operated or installed appliances, and gas migration into a building from leaks on outside buried piping, improves both public and system safety.

Q. Why are these emergency response targets reasonable?

A. As mentioned above, shortening response times to gas leak, odor, and emergency reports reduces the potential for incidents or safety-related conditions.

Q. Does the Panel recommend positive revenue adjustments to encourage further improvement of the Companies' emergency response times?

A. Yes. The Panel agrees with the Companies proposals. Specifically, the Companies should have the opportunity to earn positive revenue adjustments as follows: two basis points for responding to 86.00 through 87.99 percent of gas

leak, odor, and emergency reports within 30 minutes; four basis points for responding to 88.00 through 89.99 percent of gas leak, odor, and emergency reports within 30 minutes; and six basis points for responding to 90.00 percent or greater of gas leak, odor, and emergency reports within 30 minutes.

Q. Is this positive revenue adjustment consistent with that of other LDCs?

A. Yes. This proposal was first introduced by Staff in Case 17-G-0239 for Niagara Mohawk, then again in Case 17-G-0460 for Central Hudson, Case 18-G-0068 for Orange and Rockland. The Commission adopted joint proposals including consistent positive revenue adjustments in each of those cases. Staff also recommended similar positive revenue adjustments in Case 19-G-0066 for Con Edison. That case is still pending.

Q. Why does Staff only recommend a positive revenue adjustment for the 30-minute target?

A. By their very nature, the 30, 45, and 60-minute emergency response targets are all related; as such, an improvement to the 30-minute performance would directly result in

improvements to both the 45 and 60-minute emergency response targets.

Q. Should either of the emergency response targets or the associated revenue adjustments expire?

A. No. The emergency response targets and associated revenue adjustments should remain in effect until changed by the Commission.

COMPLIANCE WITH THE PIPELINE SAFETY REGULATIONS

Q. Where are the Commission’s pipeline safety regulations located?

A. The Commission’s pipeline safety regulations are located within 16 NYCRR Part 255. There are also relevant statutory provisions in the General Business Law and the Public Service Law.

Q. How are violations identified?

A. Pipeline Safety Staff conducts periodic audits, investigates incidents involving natural gas and liquified natural gas facilities, and investigates complaints related to natural gas and hazardous liquid pipeline systems. Typically, when Pipeline Safety Staff discovers an instance of non-compliance with the Commission’s pipeline safety regulations, a

compliance meeting is held with the affected company to detail the code section(s) related to the instance(s) of non-compliance.

Q. What is the purpose of the compliance meeting?

A. The compliance meeting is an opportunity for the affected company to provide information clarifying any deficiencies found by Staff. Such information might include explanations or records that were not available at the time of the audit or investigation.

Q. How long does the affected company have to provide this information?

A. The affected company is required to provide this information within five business days of the compliance meeting. After this period, Pipeline Safety Staff reviews the information available and issues a formal letter detailing the specifics of the violations with respect to the underlying regulations.

Q. Is the Panel generally concerned with any instances of non-compliance with the pipeline safety regulations?

A. In general, yes, this Panel is concerned that any number of violations could indicate a lack

of control, issues with quality assurance, or a culture that is willing to accept non-compliance with the pipeline safety regulations as the cost of doing business.

Q. Do the Companies currently have mechanisms to incent compliance with the pipeline safety regulations?

A. Yes. In its Order Adopting Terms of a Joint Proposal issued on June 13, 2013 in Case 12-G-0544, the Commission instituted a compliance mechanism for KEDNY. That mechanism applied for calendar years 2013 through 2016. In the 2016 Rate Order, the mechanism was updated for KEDNY, and a corresponding mechanism was instituted for KEDLI. That mechanism applied for calendar years 2017 through 2019.

Q. Please generally describe the mechanism.

A. KEDNY and KEDLI incur a negative revenue adjustment for each instance, referred to as an occurrence or violation, of non-compliance with the pipeline safety regulations. The amount of the negative revenue adjustment varies based on whether the non-compliance regards a high risk or other risk provision of the regulations.

Q. What is the difference between a violation and an occurrence?

A. Pipeline Safety Staff uses the term occurrence to describe the number of instances that a specific violation takes place, but for purposes of this compliance measure, there is no difference between the terms violation and occurrence. Historically, negative revenue adjustment letters noted the violation of a specific requirement and identified the total number of occurrences that requirement was violated. The term violation is commonly referred to in discussions and is widely understood within the pipeline industry. These words are and can be used interchangeably.

Q. How does the Panel account for violations for which the Companies cannot provide a record?

A. Pipeline Safety Staff considers it a violation when requested Company records are not provided or are found to be inaccurate.

Q. How are violations categorized?

A. Violations are categorized as either high risk or other risk and are based on the likelihood of risk to public safety. High risk refers to code

requirements that, if not followed, lead to a greater likelihood of an adverse impact on public safety relating to loss of life or property. A breakdown of code sections by risk category is provided in Exhibit__(SPSP-2).

Q. How did the Companies perform in 2013?

A. Pursuant to the negative revenue adjustment letter, contained in Exhibit__(SPSP-3), for 2013, Pipeline Safety Staff cited KEDNY for 161 high risk and 285 other risk violations, which resulted in a negative revenue adjustment of 15 basis points. In 2013, KEDLI was not subject to a non-compliance measure or associated adjustments.

Q. How did the Companies perform in 2014?

A. Pursuant to the negative revenue adjustment letter, contained in Exhibit__(SPSP-4), for 2014, Pipeline Safety Staff cited KEDNY for 63 high risk and 56 other risk violations which resulted in a negative revenue adjustment of 30 basis points. In 2014, KEDLI was not subject to a non-compliance measure or associated adjustments.

Q. How were the 2013 and 2014 non-compliance

adjustments treated in the 2016 Rate Order?

A. In the 2016 Rate order, the 2013 and 2014 non-compliance negative revenue adjustments were deferred for later use to offset costs associated with safety-related programs.

Q. How did the Companies perform in 2015?

A. Pursuant to the negative revenue adjustment letter, contained in Exhibit__(SPSP-5), for 2015, Pipeline Safety Staff cited KEDNY for 37 high risk and 23 other risk violations which resulted in a negative revenue adjustment of 37 and four-ninths basis points. In 2015, KEDLI was not subject to a non-compliance measure or associated adjustments.

Q. How did the Companies perform in 2016?

A. Pursuant to the negative revenue adjustment letter, contained in Exhibit__(SPSP-6), for 2016, Pipeline Safety Staff cited KEDNY for 16 high risk and 28 other risk violations, which resulted in a negative revenue adjustment of 18 and one-ninths basis points. In 2016, KEDLI was not subject to a non-compliance measure or associated adjustments.

Q. How did the Companies perform in 2017?

A. Pursuant to the negative revenue adjustment letter, contained in Exhibit__(SPSP-7), for 2017, Pipeline Safety Staff cited KEDNY for 15 high risk and 8 other risk violations, which resulted in a negative revenue adjustment of five and eight-ninths basis points. For 2017, Pipeline Safety Staff cited KEDLI for one high risk and zero other risk violations, which resulted in a negative revenue adjustment of one-quarter of one basis point.

Q. How did the Companies perform in 2018?

A. The non-compliance data from 2018 is currently under review by Pipeline Safety Staff.

Q. How should KEDNY and KEDLI use the incurred revenue adjustments?

A. As previously mentioned, KEDNY and KEDLI have earned a total of 15 and 14 basis points in positive revenue adjustments and incurred a total of 120 and four-ninths and 18 and one-quarters basis points in negative revenue adjustments, respectively. The net balance of the revenue adjustments should be used to offset costs associated with pipeline safety-related programs; such as residential methane detectors,

low-pressure main valve installations, and the mapping of instrument and regulation control lines. The offset costs associated with pipeline safety-related programs are further discussed in the Staff Gas Infrastructure and Operations Panel Testimony. We also recommend that KEDNY be required to file a plan for the use of excess negative revenue adjustment funds for pipeline safety programs within 90 days of a Commission order in this proceeding. Prior to filing this plan, KEDNY should be required to consult with Pipeline Safety staff. The plan should detail how the excess funds are to be spent and over what period.

Q. Are there any potential penalties resulting from the Companies non-compliance with the pipeline safety regulations?

A. Yes. Regarding KEDLI, the November 15, 2018 Order Instituting Proceeding and to Show Cause, issued in Case 18-G-0094, identified multiple instances of non-compliance with the cathodic protection requirements for its Northport Facility. Regarding both KEDNY and KEDLI, the July 12, 2019 Order Instituting Proceeding and

to Show Cause, issued in Case 17-G-0317, identified multiple instances of non-compliance with operator qualifications, performance, and inspection requirements with respect to work completed by Company and contractor personnel.

Q. Do the Companies propose to update or modify their compliance mechanism?

A. Yes. Both KEDNY and KEDLI propose to adjust the structure and reduce the total potential negative revenue adjustments from 100 basis points to 75 basis points per year.

Q. What does the Panel recommend?

A. The Panel agrees with the concept of modifying the structure of the mechanisms and with the reduction in the negative revenue adjustments from 100 to 75 basis points per year.

Q. Why does the Panel consider the decrease in exposure from 100 to 75 basis points reasonable?

A. The decrease in exposure is due to the re-allocation of basis points to more appropriately balance the Companies' exposure among the pipeline safety performance measures. In this case, the negative revenue adjustments associated with compliance should decrease by 25

basis points which is also consistent with those of other LDCs.

Q. Please explain the Panel’s recommendations for high risk violations found during the record audits.

A. The first five high risk violations would not subject the affected Company to negative revenue adjustments; from the sixth to the twentieth high-risk violations, the affected Company would incur one-half of one basis point per violation; and for each high-risk violation greater than 20, the affected Company would incur one basis point per violation.

Q. Please explain the Panel’s recommendations for other risk violations found during the record audits.

A. For the first 15 other risk violations, the affected Company would not be subject to negative revenue adjustments. For each other risk violation greater than 15, the affected Company would incur one-fourth of one basis point per violation.

Q. Please explain the Panel’s recommendations regarding high risk violations found during

field audits and investigations.

A. For field audits and investigations, the affected Company would incur negative revenue adjustment of one-half of one basis point for each of the first 20 high-risk violations; for each high-risk violation in excess of 20, the Companies would incur negative revenue adjustments of one basis point per violation

Q. Please explain the Panel’s recommendations regarding other risk violations found during field audits and investigations.

A. For all other risk violations identified during field audits and investigations, the affected Companies would incur negative revenue adjustments of one-fourth of one basis point per violation.

Q. Does the Panel recommend associated positive revenue adjustments for the non-compliance measure?

A. No. We do not recommend associated positive revenue adjustments for this performance measure. These audits and investigations are focused on compliance with the minimum pipeline safety requirements. Therefore, the Companies

should not be compensated for satisfying their obligation to comply with minimum pipeline safety requirements.

Q. Does the Panel recommend a cap for the associated negative revenue adjustments for record audit violations of a given code section?

A. Yes. We recommend a ten violation cap for repeated record audit violations of each of the code sections identified in Exhibit__(PSP-2).

Q. Does this mean that Pipeline Safety Staff would not pursue enforcement if there were more than 10 violations of a given code section?

A. No. We consider more than 10 record audit violations of a single code section to be gross non-compliance with regulatory requirements, which would necessitate additional action. Should either Company incur more than 10 record audit violations of a single code section, that Company should be required to file, in these proceedings, a remediation plan that explains how it will address and resolve the compliance issues. This plan should include dates by which all cited violations will be brought into compliance or, where appropriate, when remedial

actions will be taken to mitigate recurrence. The filing should be made within 90 days of Pipeline Safety Staff's audit letter. Such gross non-compliance could be subject to separate enforcement actions by the Commission.

Q. If KEDNY or KEDLI fails to comply with their implementation plan, how should those violations be treated?

A. Should either KEDNY or KEDLI fail to comply with their implementation plan, those violations of a given code section in excess of 10 should count toward the negative revenue adjustments under the compliance measure.

Q. Does the Panel recommend a violation cap for repeated field audit violations of a particular code section?

A. No. Field audit and investigation violations are typically discovered when Pipeline Safety Staff witnesses the Company performing an action or task. Notwithstanding Pipeline Safety Staff's presence, violations of the minimum pipeline safety regulations or the Companies' procedures still occur. Pipeline Safety Staff expects that the Companies' performance is at

its best while being observed. Therefore, any field audit or investigation violation could indicate a lack of competency in performing the action or task.

Q. Why does the Panel recommend that the Companies file and execute an implementation plan for more than 10 record audit violations of a single code section?

A. The compliance mechanism provides a financial disincentive for non-compliance with the pipeline safety regulations. If the financial disincentive is insufficient to induce compliance, such that a company violates a specific code section more than 10 times, then additional measures are necessary to change the performance. It is critical for the Commission to be able to address all violations of its pipeline safety regulations given the potential for serious harm. The implementation plan would provide a specific mechanism to further this goal and may avoid the need for formal penalty actions against the Companies for each record audit violation. The Commission, however, always has the authority to pursue a penalty

action to promote regulatory compliance notwithstanding any financial disincentive or reporting requirement that also may be in place.

Q. Should either the compliance mechanism or the associated revenue adjustments expire?

A. No. The compliance mechanism and associated revenue adjustments should remain in effect until changed by the Commission.

FIRST RESPONDER TRAINING

Q. What enhancements did the Companies propose to their first responder training programs?

A. In their response to IR DPS-674, KEDNY and KEDLI propose to hire two full-time equivalent instructors to conduct the “Train the Trainer” and “Hands-On Training” programs within their respective service territories. The instructors and the Companies’ subject matter experts will work with the New York State Association of Fire Chiefs and the Fire Department of New York to refine first responder training procedures for natural gas emergencies.

Q. Does the Panel have any recommendations with respect to the Companies’ first responder

training programs?

A. Yes. The Companies must maintain continual efforts to train first responders.

Q. How do continual efforts benefit first responders and, ultimately, customers?

A. First responders would be better informed on how to respond prior to an incident. Continuing these efforts would reinforce the actions first responders should take, and those to avoid, during an incident and would educate first responders on how to safely operate certain field devices. Familiarity with equipment should reduce response time and minimize the potential for escalated incidents by reducing errors made in a haste to render a situation safe. Also, improved training would enhance the safety of the first responders by providing more situational awareness of their surroundings and the effects of their actions.

Q. What does the Panel recommend the Companies do to expand and continue these efforts?

A. We support the Companies continued efforts in their first responder training programs, and further recommend that they adopt the principles

of the Pipeline Emergency Responders Initiative (PERI). Also, the Companies should conduct regular drills with local fire departments and municipalities and provide, under this case, an annual report detailing its progress including, but not limited to, the date and times of the drills, who was in attendance, what topics were reviewed, and any applicable recommendations.

Q. What is PERI?

A. PERI is a training program used to prepare fire departments to prevent and manage pipeline incidents.

Q. To date, have the Companies incurred any costs associated with PERI?

A. No. This is a newer initiative and the Companies have not yet incurred any associated costs.

RESIDENTIAL METHANE DETECTORS

Q. What do the Companies propose for their residential methane detector programs?

A. On pages 43 through 44 of the Gas Safety Panel’s direct testimony, KENDY and KEDLI propose to annually distribute approximately 54,000 and

36,000 residential methane detectors, respectively. The Companies intend to target customers with room meter sets, low income customers, multi-unit residences, and customers served by two and four-inch, cast iron piping. The Companies included the costs for these programs in the revenue requirements but state that they are willing to fund these programs through the negative revenue adjustment deferrals.

Q. Do the Companies plan to install these methane detectors for selected recipients?

A. No. The Companies propose to purchase and distribute the methane detectors.

Q. What does the Panel recommend?

A. We support the proposed residential methane detection programs and the use of negative revenue adjustment deferrals to mitigate rate impacts. The Companies should coordinate their efforts with other planned work to minimize associated costs. The Panel further recommends that the Commission direct the Companies to submit an annual report indicating, at a minimum, deployment strategies and associated

costs to date.

Q. Does the Panel have any other recommendations regarding residential methane detectors?

A. Yes. We support the use of unspent research and development funds to offset the costs associated with deploying advanced residential methane detectors.

Q. What are advanced residential methane technologies?

A. Advanced residential methane technologies devices capable of communicating with the utility’s equipment.

Q. Do the Companies currently poses the infrastructure capable of supporting advanced residential methane technologies?

A. No. The Companies would respectively budget the unspent research and development funds to develop an implementation plan to install required infrastructure compatible with that of advanced residential methane technologies. We recommend that the Commission direct the Companies to submit their plans within 90 days of the Commission’s order setting rates in these proceedings.

SERVICE LINE PROCEEDING

Q. What is the service line proceeding?

A. In its April 2, 2015 Memorandum and Resolution Adopting Gas Safety Regulation Amendments, issued in Case 14-G-0357, the Commission adopted a new definition of gas service line, set forth in 16 NYCRR §255.3(a)(43). Under this new definition, service lines extend to the outlet of each customer's meter, or at the connection to a customer's piping, whichever is further downstream. This includes meters located inside a building and those inside each apartment within a larger apartment building.

Q. What are the impacts of this definition change?

A. The primary impact of this change relates to the Companies obligations to inspect piping inside all buildings up to the gas meter, and to add training, qualification, and testing for individuals who make repairs on inside gas piping, upstream of the meter.

Q. Did the Companies address the impacts from the service line proceeding in their testimony?

A. Yes. On pages 17 through 20 of the Gas Safety Panel's direct testimony, KEDNY and KEDLI

requested additional operational and maintenance funding related to their obligation to perform leakage survey inspections, atmospheric corrosion inspections, necessary repairs resulting from the inspections, and additional full-time equivalent inspectors.

Q. When should the Companies perform these additional inspections?

A. To reduce costs, KEDNY and KEDLI should complete most of the inspections during their normal course of business and during dedicated visits for other purposes, such as service reconnections. For some locations with inside meters, multiple attempts to inspect may be required due to the Companies inability to access the building.

Q. Have the Companies started these inspections?

A. Yes. As required by the April 20, 2017, Order Establishing Statewide Inspection Schedules and Procedural Requirements and the August 3, 2017, Order Granting in Part Petitions for Rehearing, Reconsideration and Clarification in Case 15-G-0244, KEDNY and KEDLI began their leakage surveys and atmospheric corrosion inspections.

The Commission's orders required natural gas utilities to begin inspections of inside piping in business district locations as of April 20, 2017. In their response to IR DPS-675, KEDNY and KEDLI explained that they have inspected 100 percent of their baseline inspections within business districts.

Q. What does the Panel recommend?

A. While the Companies met their statutory requirements to perform leakage survey and atmospheric corrosion inspections within business districts, we do not view the associated costs to access inside services within business districts reasonable. This Panel recommends that KEDNY and KEDLI submit a procedure by which they will automatically apply a provision (for example, after one refusal, two scheduled and then cancelled appointments, or if they do not respond to notifications). In applying the provision, the Companies would reduce the costs associated with revisiting the same address on multiple occasions. The amounts collected under the penalty provision should be used to offset the cost of service line

inspections. In addition, we recommend that KEDNY and KEDLI coordinate inside service line inspections with other activities. This would include, but not be limited to, meter replacements, removal of inactive meters, and complaint investigations.

NEW YORK CITY LOCAL LAW 152

Q. What is New York City Local Law 152?

A. In 2016, the New York City Council passed Local Law 152 (Local Law 152), which requires owners of buildings in New York City, other than those classified in occupancy group R-3, to perform periodic inspections of their buildings' natural gas piping. Occupancy group R-3 building are primarily residential buildings with no more than two dwelling units. The natural gas piping inspected under Local Law 152 would include the jurisdictional inside piping from the service line definition change. The inspections required by Local Law 152 would include examinations for atmospheric corrosion and testing for leakage.

Q. What is the status of Local Law 152?

A. The New York City Department of Buildings (DOB), which is the agency with primary enforcement responsibility, issued proposed rules on November 30, 2018. Comments on the new rules were due on January 2, 2019. As it stands, the proposed rules have several issues that would, if adopted, require duplicative inspections, one performed to meet requirements on the building owner and a second one performed to meet requirements on the utility pursuant to State and Federal regulations, based on the definition of service line. This outcome would be inefficient, burdensome, and more expensive to customers.

Q. Explain the issues with the current rule as identified by the Panel.

A. There are three issues with the DOB rules as proposed which prevent the inside inspections required by building owners from satisfying the Companies' obligations to perform inspections under the new service line definition. The first issue is that the DOB regulations do not require the plumber performing the inside inspections to meet minimum training and

qualification standards that require persons performing these inspections to be operator qualified and to be part of a drug and alcohol testing program; both of which are requirements of the pipeline safety regulations. Without those requirements, the inspections performed by the plumber would not be valid for the inspections required of the Companies.

Q. What does this Panel mean when it uses the term plumber?

A. We refer to licensed master plumbers or individuals under the direct and continuing supervision of licensed master plumbers, as stated in Local Law 152 §28-318.3.

Q. What is the second concern with the current proposed rules?

A. The proposed rules do not explicitly require inside inspections for natural gas piping in tenant spaces, for example, within apartments. If the proposed rules do not require inspections within tenant spaces, the inside piping associated with meters inside apartments would not be inspected by the plumber and would require a separate inspection by the Companies.

Q. What is the third issue with the current proposed rules?

A. The timeframes proposed by the DOB do not align with the inspection timeframes prescribed by the pipeline safety regulations. Local Law 152 requires inspections once every five years but allows for modifications to the timeframe by the Commissioner of the DOB if the Public Service Commission adopts an alternate inspection schedule. In contrast, the rules proposed by the DOB require an inspection once every five years with no option or opportunity for modification. KEDNY and KEDLI perform leakage surveys in business districts on an annual basis and in non-business districts every three years. As a result, the DOB inspections will be scheduled on a different cycle than those inspections required to be performed by the Companies.

Q. Given these issues, what does this Panel recommend?

A. The Panel recommends that KEDNY and KEDLI continue to work with DOB to further align the proposed rules implementing Local Law 152 with

the inspection requirements of the service line definition change. If the proposed rules are adopted as issued without modification, we recommend that KEDNY and KEDLI be required to file a plan, within 60 days of issuance of the DOB's final rule or within 60 days of the Commission's order setting rates in these proceedings, whichever is later, detailing how the Companies will work with building owners to encourage the use plumbers who can perform the inspections in compliance with the requirements of the pipeline safety regulations. This could include providing equipment that electronically transmits inspection and survey data to the Companies, which may encourage building owners to hire licensed master plumbers that are operator qualified and under a drug and alcohol testing program, as the most efficient way to complete all inspections of the building. Further, KEDNY and KEDLI should conduct an education campaign to explain to building owners the benefits of having a single inspection performed, including the reduced burden on building owners and their tenants with respect

to the coordination and scheduling of inspections. Insofar as the DOB's rules are not final, the amount the Companies can collect in rates for the inspections should be a placeholder with a reconciliation to a new estimate once the final rules are issued.

OUTSIDE METERS

Q. What are the benefits of outside meters?

A. The primary benefit is that outside meters are safer than inside meters. In the event of an emergency, outside meters allow utility personnel and/or emergency responders to quickly shut off the flow of gas without entering the building and without having to locate the curb valve. An additional benefit is that a utility does not require access to the premises in order to perform the required leakage surveys and atmospheric corrosions inspections.

Q. Do the Companies have programs to move meters outside?

A. Yes, pursuant to the 2016 Rate Orders, KEDNY and KEDLI are required to move meters outside congruent with any planned service line

replacements, new service installations, or under other circumstances that offer the customer and the Companies the opportunity to relocate the meter outside.

Q. What does the Panel recommend?

A. We recommend that the Companies be required to move meters outside whenever they perform any service line replacements, either by insertion or direct burial. This would be for any reason, for example, as part of LPP removals or to repair a leak. In addition, KEDNY and KEDLI should also move meters outside whenever it performs service transfers, which is when it moves an existing service line to a newly installed pipeline. The Panel recommends that the Companies be required to submit an annual report indicating the amount of service work it performed in that year and the number of meters it moved outside as part of that work.

Q. Why does the Panel recommend this?

A. When the Companies perform service work, they have a construction crew on site and often are already interrupting service to the customer. Moving the meter outside at this time is the

most convenient, cost effective, and practical time to do so. Since the Companies replace a leak-prone service with modern materials, it is unlikely it will return to perform additional work in the near future. Therefore, the Companies should avail themselves of the opportunity to relocate the meter outside at the time of replacement.

INACTIVE ACCOUNTS

Q. In their filing, did the Companies address inactive accounts, or natural gas accounts without a customer of record?

A. Yes. On pages 22 through 24 of the Gas Safety Panel's direct testimony, the Companies proposed to implement a 30-day inactive account process, phased in over three years. KEDNY and KEDLI are requesting 108 and three full-time equivalent positions, respectively, in order to implement the inactive account programs. These positions are spread across the areas of "field operations, CMS, call center, legal, and back office".

Q. Please explain the reasons accounts remain

inactive.

A. There are many reasons that accounts remain inactive, including the Companies' inability to gain access, the referral of accounts to field operations for physical disconnection, and the legal replevin process used to secure access to the customer's premises.

Q. Should all meters and service regulators be installed in a readily-accessible location?

A. Yes. Pursuant to 16 NYCRR §255.353(a), all meters and service regulators are required to be installed in a readily-accessible location and be protected from corrosion and other damage, including any vehicular damage that may be anticipated.

Q. How much natural gas usage is associated with inactive accounts?

A. In their response to IR DPS-670, KEDNY and KEDLI stated, from 2014 through 2018, 1,980,919 and 90,362 hundredths of cubic feet (CCF) were consumed by 28,490 and 1,575 inactive accounts, respectively.

Q. Does the Panel consider these volumes lost and unaccounted for natural gas?

A. Since these volumes are accounted for by inactive accounts, they should not be considered lost and unaccounted for gas. The Companies are responsible for properly accounting for these volumes and should adjust the lost and unaccounted for gas calculation in the Companies' annual reconciliation of costs so that customers do not pay for these volumes.

Q. How do the pipeline safety regulations treat the abandonment or inactivation of facilities?

A. The Commissions' regulation, 16 NYCRR §255.727(d), requires one of following: closing the valve to prevent the flow of gas with a locking device or other means designed to prevent the opening of the valve by persons other than those authorized by the operator; install a mechanical device or fitting that will prevent the flow of gas in the service line or in the meter assembly; or physically disconnect the building's or unit's piping from the gas supply and seal the open pipe ends.

Q. Is there a timing requirement to commence the abandonment or deactivation of a facility?

A. While not explicitly defined within the pipeline

safety regulations, it is important to note that the regulations do not recognize an active service without a customer, and it is a requirement for LDCs to commence this process as soon as practicable. The regulations do require each segment of pipeline that becomes unsafe to be replaced, repaired, or removed from service, and specifically requires certain action take place when service is discontinued.

Q. What does the Panel recommend with respect to inactive accounts?

A. This Panel recommends that KEDNY and KEDLI be required to take immediate and continuous action for inactive accounts pursuant to the Commissions' regulation, 16 NYCRR §255.727(d). While we are supportive of the Companies efforts to date, pursuant to 16 NYCRR §255.353(a), the Companies are required to install all meters and services in a readily-accessible location that is protected from corrosion and other damage, including any vehicular damage that may be anticipated. Therefore, when the Companies encounter any meters or services that are not readily-accessible, it should be required to

take immediate and continuous action to remedy these conditions to comply with the Commission's regulation.

METHANE LEAK PILOT PROGRAM

Q. Please explain the methane leak pilot program.

A. On pages 45 through 46 of the Gas Safety Panel's direct testimony, the Companies propose: to explore the use of cavity ring down spectroscopy (CRDS) to enhance their leak repair programs; "to hire a contractor to operate a Google car with CRDS technology for use in areas previously identified as possible high leak concentration areas to aid in prioritizing for repair possible large emitters within those areas"; and to hire an incremental full-time equivalent natural gas distribution engineer.

Q. What would be the responsibilities of the full-time equivalent engineer?

A. In their response to IR DPS-280, the Companies explain that the full-time equivalent engineer would be responsible to engage a contractor, manage the contract, manage field investigations and repairs, investigate other leak emission

technologies, and to generate reports.

Q. Are CRDS devices approved for use in New York State?

A. According to the Department of Public Service Pipeline Safety Program website, CRDS devices are not yet approved for leakage surveys or leakage investigations within New York State. The website is available at: http://www3.dps.ny.gov/W/PSCWeb.nsf/All/F8539043E0B4A5698525841F005A64E4?OpenDocument.

Q. What does the Panel recommend?

A. While this Panel is supportive of the research and development of CRDS-like technologies, and the use of a contractor for data acquisition, the Panel is not supportive of the full-time equivalent engineer to manage the methane leak pilot program.

Q. Please explain why this Panel is not supportive of the full-time equivalent engineer.

A. At this point in time, it would be inappropriate for customers to pay for the additional full-time equivalent engineer to manage a methane leak pilot program for an unapproved leak detection device.

Q. How should the methane leak pilot program be managed?

A. The methane leak pilot program should be incorporated into the Companies' research and development efforts.

Q. Does the Panel have further recommendations?

A. Yes, the Panel further recommends that the Commission direct the Companies to submit an annual report on the progress of this pilot. This annual report, at a minimum, should include: a listing of the highest emitting leaks that were discovered by the CRDS device; leak rankings; leak locations; leak identification numbers; and relative methane emissions.

ENHANCED CONTRACTOR SAFETY INSPECTIONS

Q. What are enhanced contractor safety inspections?

A. On pages 20 through 22 of the Gas Safety Panel's direct testimony, the Companies propose to enhance the contractor safety inspections program by assigning one inspector to every contractor crew. Currently, a single inspector is assigned to cover multiple crews by rotating through work schedules.

Q. What would be the responsibilities of the proposed full-time equivalent inspectors?

A. These safety and compliance inspectors would perform on-site inspections including field verification of operator qualifications, ensuring compliance with applicable pipeline safety regulations and procedures, and ensuring safe work practices.

Q. What do the Companies propose for their enhanced contractor safety inspection programs?

A. On pages 20 through 22 of the Gas Safety Panel's direct testimony, KENDY and KEDLI propose to hire 125 and 74 incremental full-time equivalent inspectors, respectively. These full-time equivalent inspectors would be phased-in with the program over three years.

Q. Will these safety and compliance inspectors be performing duties other than those described?

A. Yes. In their response to IR DPS-676, the Companies stated that "there may be occasions, where the workload is light, that the supervisors and inspectors may perform functions outside of their normal duties." The Companies estimated that approximately five-percent of the

inspectors' time would be dedicated to other duties.

Q. What does the Panel recommend?

A. This Panel is supportive of the enhanced contractor safety inspection programs. However, the requested full-time equivalent positions for KEDNY and KEDLI should be reduced by five-percent to 118 and 70, respectively, to reflect the estimated workload.

BURIED VENT LINES

Q. Please explain the buried vent line programs.

A. On pages 35 through 36 of the Gas Safety Panel's direct testimony, the Companies "estimate that approximately five-percent of their respective high-pressure services with inside meter sets have regulator vent lines that are buried underground. Buried vent lines are more susceptible to corrosion and damage because the vent pipes are not coated and do not have cathodic protection. Corroded and/or damaged vent lines pose an increased risk for gas migration into a building or regulator over-pressurization caused by water intrusion and/or

freeze-ups caused by extreme weather.

Q. Were there any recent incidents where a buried vent line could have caused an over-pressure situation?

A. Yes. “In February 2016, KEDNY experienced an incident on its system that involved a buried vent line that was found to be corroded and could have contributed to an over pressure condition due to freezing water in the regulator.”

Q. What do the pipeline safety regulations require with regard to cathodic protection of buried pipelines?

A. Pursuant to 16 NYCRR §255.455(a), “Except as provided in subdivisions (b), (c), and (f) of this section, each buried or submerged pipeline installed after July 31, 1971, must be protected against external corrosion. (1) It must have an external protective coating meeting the requirements of section 255.461 of this Part. (2) It must have a cathodic protection system designed to protect the pipeline in its entirety, installed and placed in operation within one year after completion of

construction."

Q. Are there cathodic protection regulations for buried pipelines installed before August 1, 1971?

A. Yes.

Q. Do the Companies know when the identified buried vent lines were installed?

A. No, not at this time. In response to IR DPS-678, the Companies stated that "[r]egulator vent lines are installed at the time of the original service and/or service regulator. The Companies have not identified installation dates of the buried vent lines that have been identified through the inside service line inspections to date."

Q. Is this response acceptable?

A. No. Vent lines are a critical component to the safe and reliable distribution of natural gas through pipelines. Should a regulator experience an over pressure situation, the vent line would transport the excess natural gas to a safe location; typically, venting into the atmosphere outside of the building. Should a vent line, or particularly a buried vent line

become compromised - either through damage, or corrosion - the potential for an over pressure situation exists.

Q. Are the Companies required to maintain maps or records of service lines?

A. Yes. Pursuant to 16 NYCRR §255.603(c), “Each operator shall establish and maintain the maps of its transmission lines and distribution mains and maps or records of its service lines as necessary to administer its operating and maintenance plan.”

Q. Are buried vent lines included within the definition of a service line?

A. Yes. Pursuant to 16 NYCRR §255.3(a)(43), service line is defined as “the piping, including associated metering and pressure reducing appurtenances, that transports gas below grade from a main or transmission line to the outlet of the customer meter or at the connection to a customer’s piping, whichever is further downstream where a meter is located within the building; if a meter is located outside the building, the service line will be deemed to terminate at the outside of the

building foundation wall."

Q. What does the Panel recommend?

A. At this time, the Panel does not view the Companies' efforts with regards to buried vent lines as acceptable. At a minimum, buried vent lines should have already been identified by the Companies, not as a part of their inside service line inspection programs, but as a part of their operating and maintenance plans. Thus, any requested funding for the buried vent line programs should be borne by the shareholders and not the customers due to the Companies failures to adequately address the vent lines in the past.

ANNUAL REPORTING REQUIREMENTS

Q. Does the Panel recommend any other conditions that the Companies be required to meet related to the Panel's safety-related performance measure recommendations?

A. We recommend that the Companies be required to submit annual reports on their performance related to LPP removal, leak management, damage prevention, and emergency response by February

28th of the following calendar year. We recommend that this report be required annually until changed by the Commission.

Q. Does this complete the Panel’s testimony at this time?

A. Yes.

BY MR. GOODRICH: (Cont.)

Q. Panel, also before you are exhibits identified at the time of filing as SPSP-1 through SPSP-8. These have now been marked for identification purposes as Exhibits 481 through 488, respectively. Were those exhibits prepared by you or under your direct supervision?

A. (Cont.) Yes.

Q. And do you affirm that the information contained in the exhibits just identified is true to the best of your knowledge?

A. Yes.

MR. GOODRICH: Your Honors, the Staff Pipeline Safety Panel is now available for cross examination.

A.L.J. COSTELLO: Okay, Mr. Decicco?

CROSS EXAMINATION

BY MR. DECICCO:

Q. Good morning, Panel. This morning, I’m going to ask you several questions about the Panel’s pre-filed testimony, specifically, the proposed gas safety performance measures and your proposed adjustments to the company’s gas safety programs. But before I get into some of those

specific items, I just want to ask one or two general questions on the topic of pipeline safety. Does the Panel agree that there has been an increased focus in the national gas industry on gas pipeline safety over the last decade?

A. (Panel) Yes.

Q. And has this focus led to new safety regulations and mandates from both federal regulators as well as at the commission?

A. That que -- is that in general? Or are there specific --

Q. Yeah. And -- and -- and we're going to get into some of the specifics. But in general, would you say that there's sort of been a -- a proliferation of regulations at both the federal and state level around gas safety over the last several years?

A. Yes.

The panel agree these new regulations and mandates have generally increased the utilities' work and reporting requirements in the area of gas safety.

MR. GOODRICH: So objection. I mean, I think, speaking about this generally is -- is overly

broad and vague. And -- and what -- what exact answer is the panel going to provide that's -- that's useful? I think, it should focus on particular items.

MR. DECICCO: Yeah, fair enough. I'll withdraw the question.

BY MR. DECICCO: (Cont.)

Q. So beginning on page nine of the panel's testimony, there is a discussion of the proposed pipeline safety performance measures. And before I get into what the staff has proposed, is the panel aware of the company's position that pipeline safety performance measures are not necessary or appropriate in a one-year litigated case?

A. Can you repeat the question, please?

Q. Sure. I -- I'm just asking you if you're aware of the company's position in its testimony that -- that gas safety performance measures are not necessary in -- in, or appropriate, in a one-year litigated case.

A. Yes.

Q. On page ten of the panel's testimony with the question beginning on line six, the panel indicates that it supports pipeline safety

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

performance measures with 150 basis points of potential negative revenue adjustments. Is that correct?

A. That's correct.

Q. And is it the panel's position that these gas safety performance measures would apply to the companies in the rate year?

A. Can you repeat the question?

Q. Sure.

A. Because you're stating a rate year, and I'm getting hung up on -- these are set on a calendar year basis, so I just wanted to confirm that.

Q. Sure. And we'll get into that in -- in -- in a second. So let me withdraw it, and we -- we'll get back to it. Does the panel know exactly how much a basis point is worth at each of the companies?

A. The exact dollar amount, no.

Q. Okay. If I said it was approximately 330,000 at KEDNY and 215,000 at KEDLI, does that sound about right? Or sound around in that ballpark?

A. For what year are we talking

about?

Q. For the – for the rate year.

A. Can you repeat those numbers?

Q. Sure. For KEDNY, about $330,000. And for KEDLI, $215,000.

A. Approximately, yes.

Q. Okay. So in round numbers, 150 basis points is worth approximately $49.5 million at KEDNY and $32 million at KEDLI.

A. Assuming your math is correct, then yes.

Q. Okay, I appreciate that. And the panel also proposes a set of positive revenue adjustments as part of its performance metrics proposal. Is that correct?

A. That’s correct.

Q. Okay. And the panel proposes a maximum of 22 basis points of positive revenue adjustments. Is that correct?

A. That’s correct.

Q. Okay. Does the staff panel agree that the current set of gas safety performance measures were agreed to in the last negotiated rate settlement in cases 16G0058 and 0059?

A. Can you repeat the question please?

Q. Sure. Does the panel agree that the current set of gas safety performance measures – so the measures currently in – strike that. So the gas safety performance measures in the company's current rate plan, those were agreed to in the last negotiated rate settlement in cases 16G0058 and 16G0059.

A. Yes.

Q. Okay. And in those cases, the companies agreed to be subject to performance measures with -- with potentially 150 basis points of negative revenue adjustments as part of a comprehensive settlement. Is that correct?

MR. GOODRICH: So I would just state that that order, you know, speaks for itself.

MR. PASNOW: Can you repeat the question?

BY MR. DECICCO: (Cont.)

Q. Sure. And -- and I'm just asking if it's your understanding that the companies agreed to be subject to those performance measures as part of a comprehensive settlement.

A. Yes.

Q. Now last week, the staff policy panel testified in these proceedings that it was the staff policy panel's understanding that staff's proposed rate allowance in this case for the rate year includes the resources needed to achieve the proposed targets in the pipeline safety panel's recommended performance measures. And I'm -- I'm asking this panel. Has the panel considered staff's proposed rate allowance for capital investments and operating expenses in setting the proposed gas safety performance targets?

A. Can you repeat the question, please?

Q. Sure. Has this panel considered staff's proposed rate allowance for capital investments and operating expense in setting the proposed gas safety performance targets?

A. Can you repeat that one more time?

Q. Sure. I -- I'm just going to repeat the same question again. So has this panel considered whether staff's proposed rate allowance for capital investments and operating expense is

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

sufficient to meet the performance targets proposed in your gas safety performance measures?

A. Is it somewhere in our testimony where we reviewed the capital and operational expenses associated with specific programs?

Q. I -- I don't believe it is. And that's -- that's why I'm asking.

MR. GOODRICH: So I'm gonna object that this is, you know -- again this is like the earlier question. It's -- it's vague in that it's just very generalized to all of CapEx and all of OpEx, which is pretty much the entire case, as opposed to pointing to specific programs related to safety that the -- the panel would've reviewed requests with regard to those -- those particular programs.

A.L.J. LEARY: We're going to overrule your objection to the extent that the panel can answer this question about whether they considered -- and I think there were actually two questions in here, because I wrote them both down. Whether they considered the rate allowance for capital expenditures, or CapEx, as Mr. Goodrich referred to it, and operating expenses, or OpEx, in setting the

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

performance standards.

And then the thing that you added in the second question was whether those were sufficient. So it goes to the issue of what you considered in filing your direct testimony. So let's take the first question first, which is did you consider the rate allowance for capital expenditures and operating expenses in setting the performance standards in your testimony. Say your name first, please.

MR. THOMAS: Suresh Thomas. Our focus was mainly on the compliance requirements based on regulations and established performance metrics in the past.

BY MR. DECICCO: (Cont.)

Q. So just to be clear, is -- is that a no that you didn't consider the -- the rate allowance for -- for capital investments and -- and operating expense?

A. (Thomas) Again, our focus was mainly on the programs, not exactly on the dollar amounts.

Q. And I'm not trying to be difficult, but just to be clear. The answer to that -

- the question then is no, you didn't consider.

A.L.J. LEARY: I think – excuse me, Mr. Decicco. I think that you have an answer where the focus was.

MR. DECICCO: Okay.

A.L.J. LEARY: I think the record will be clear about what that answer means. You don't have to recharacterize the panel's testimony.

MR. DECICCO: Fair enough. I -- I'll move on.

BY MR. DECICCO: (Cont.)

Q. To the panel's knowledge, has the commission ever ordered gas safety performance measures with associated negative or positive revenue adjustments outside of a rate settlement?

MR. GOODRICH: So objection. I mean, this really goes to a legal argument about the commission ordering these in -- in litigated cases. And the panel is -- is speaking to the policy need for it -- for these -- these measures. If the company wants to argue about the legal ability to do it, we can do so in brief. And then the other any past rate orders are -- are, you know, in the public record and available for use in brief already.

MR. DECICCO: Yeah. Your Honors, this panel refers to several other cases in -- in making points about the various performance measures that they're adopting, including what the commission has approved in other cases. I'm just asking the question whether they're aware of the commission having ever ordered performance -- gas safety performance measures in a litigated case. I'm not asking them to draw a legal conclusion. Just whether they're aware of it being ordered.

A.L.J. LEARY: Okay. I understand both points. I do not think that this asks for a legal conclusion, and I have no doubt that the company will indeed argue what Mr. Goodrich predicts that they will argue. I am in complete agreement with that. But let's go to the testimony and build perhaps, Mr. Decicco, from where they discuss prior cases and other cases, et cetera. And then ask the question of their awareness of cases in which the matter was litigated and the commission in a litigated case ordered the performance mechanism. So if you could do a two-step process here for the panel?

MR. DECICCO: Sure. Yeah.

A.L.J. LEARY: That would make the

record a little more clear.

MR. DECICCO: And -- and I will do that. And I will do that in the context of the -- some of the specific metrics that we are -- we are talking about.

BY MR. DECICCO: (Cont.)

Q. So why don't I move to the -- to the topic of gas safety reporting? And now outside of the performance measures --

A.L.J. LEARY: Excuse me, Mr. Decicco. I was -- I was overruling Mr. Goodrich's objection. You're free to ask that question, and the panel's required to answer it. I was simply suggesting that, for purposes of the record, tying it somehow to their testimony. You are free to ask them if they know of the commission ordering in a litigated case these kind of, or any, performance measures. So I don't mean to cut you off from that. We -- we overruled that objection.

BY MR. DECICCO: (Cont.)

Q. So then I -- I will ask the panel again. To the panel's knowledge, has the commission ever ordered gas safety performance measures with associated negative or positive revenue adjustments

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

outside of a rate settlement?

A. (Pasnow) Not to our knowledge, no.

Q. Okay, thank you. With regards to gas safety reporting, outside of the performance measures and the reporting requirements in the company's current rate plan, are the companies subject to reg -- regulatory reporting requirements in New York for various gas safety performance metrics?

A. Can you repeat that question please?

Q. Sure. So putting aside what the companies agreed to in their last -- last rate plan, are gas utilities in New York subject to gas safety performance reporting requirements in New York State?

A. And you're asking that other than what is in the current orders or joint proposals?

Q. Correct.

A. Yes.

Q. And in your testimony starting on -- on page ten line 22, you described the -- the pipeline safety performance measures report. Can you briefly describe what that report is?

A. Yes. I'm just going to read right from our testimony. "The pipeline safety performance measures report is an annual report presented by the department staff to the commission. The report summarizes data, analyzes performance, and is used as a tool to track and identify LDC performance in the following areas of pipeline safety: damage prevention, emergency response, leak management, and compliance with the pipeline safety regulations."

Q. And the results are compiled by staff annually and reported to the commission. Is that -- is that correct?

A. Both annually, yes.

Q. Okay. And this reporting happens, again, regardless of whether the companies have agreed to a set of performance metrics in their rate plans.

A. Yes.

Q. Okay. So based on this mandatory reporting, do you agree that staff and the commission have the ability to monitor the company's safety performance with or without performance measures in their rate plan?

A. In terms of monitor --

monitoring, yes.

Q. And would the panel agree that the commission can take action in the form of a penalty proceeding if the safety performance of a gas utility fails to meet industry standards or is otherwise imprudent?

MR. GOODRICH: So objection. Because reference to the penalty proceeding requires -- there's -- there's specific requirements in the law, and that -- so that is a legal -- that is asking for a legal conclusion as to when a penalty proceeding under 25 and 25A are actually available.

MR. DECICCO: I'll withdraw and ask it in a different question.

BY MR. DECICCO: (Cont.)

Q. Does the -- the gas safety staff in New York have the -- the ability to recommend to the commission that a penalty proceeding be brought if it observes safety performance that is -- that is substandard or fails to meet the commission's regulations?

MR. GOODRICH: So just again, there are specific requirements -- legal requirements as to when a -- a penalty action is -- is able to be

brought. And that is -- is a -- while that is a -- as far as being able to recommend one to the commission, that is -- that is something that obviously I -- I -- I would concede that pipeline safety staff would be involved in. However, they themselves could not recommend a penalty action. It would have to be done in concert with DPS counsel as far as whether one could even -- you know, whether it fits within the penalty action provisions of the statute. And -- and that's really what this is going at.

A.L.J. LEARY: Mr. Decicco, did you ask whether the panel can recommend or gas safety -- gas safety personnel from the department can recommend to the commission --

MR. DECICCO: Yeah.

A.L.J. LEARY: -- to undertake a penalty action? What was --

MR. DECICCO: Your Honors, I'm trying to get to the simple point of whether the commission has the ability to enforce gas safety regulations outside of performance metrics.

A.L.J. LEARY: Why don't you ask that question?

MR. DECICCO: I will ask that

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

question.

A.L.J. LEARY: Is that --

MR. GOODRICH: But I would -- I would object to that. I mean, it's the same thing. He's getting at can the commission, you know, bring a penalty action. And this panel can't talk to the specifics of the penalty action statutes and what may or may not be necessary there, as opposed to what may or may not fall under the -- the metrics in -- in the rate case.

A.L.J. LEARY: Mr. Decicco, if this is something that goes to the commission's authority, this panel does not have testimony before us now about the commission's authority.

MR. DECICCO: Yeah.

A.L.J. LEARY: Asking them if they know is out -- really outside of the scope of, you know, really who -- who's testifying today. So if it's already in the law of what the commission's authority is, you and Mr. Goodrich and anybody else in this room could read those laws differently. That's what lawyers do. They all read things differently. So --

MR. DECICCO: Yeah. Your Honors --

A.L.J. LEARY: I don't think it's an appropriate question for this panel. I think Mr. Goodrich, you've sort of strayed into this, do you know what's in the law? Do they have the authority kind of thing? And I don't see this panel as the -- the appropriate group of witnesses to respond to that question.

MR. DECICCO: Your Honor, I'll withdraw the question and we can move on.

BY MR. DECICCO: (Cont.)

Q. I believe you said before that in New York, gas safety performance is measured on a calendar year basis. Is that -- is that correct?

A. That's correct.

Q. Is the panel recommending calendar year metrics in this rate case?

A. We are.

Q. Is it the panel's position that its proposed metrics would apply in calendar year 2020, the current year?

A. That is correct.

Q. Okay. Does the panel agree that the companies need to plan, either through capital investments or their operating work plans, to achieve

more stringent performance targets in the pipeline safety performance measures?

A. Can you repeat the question?

Q. Sure. Does the panel agree that the companies need to plan, either through their capital investments or through their operating work plan, to achieve more stringent performance targets in the pipeline safety performance measures?

MR. GOODRICH: So objection to more stringent is vague and subject to -- to discussion -- to interpretation. If counsel could be more specific?

MR. DECICCO: So -- okay. I can use a word other than stringent. So I -- I'll re -- rephrase the question.

BY MR. DECICCO: (Cont.)

Q. Does the panel agree the companies need to plan, again, through either their capital plans or their operating work plans, to achieve performance targets that are more difficult or more challenging than those in their current rate plan?

A. This question is in general, correct?

Q. Correct.

A. Yes.

Q. And does the panel agree that if the commission were to adopt the performance targets in these rate cases that the targets would be finalized when the commission issues a final order in this proceeding?

A. Yes.

Q. Okay. And under your proposal for calendar year 2020 performance measures, would you agree that the companies would need to be planning or working against those targets now?

A. Can you repeat the question, please?

Q. Sure. Under your proposal to -- to implement performance measures that start in 2020, which is the year we're currently in obviously, would you agree that the companies need to start planning to achieve those performance targets, and actually working to achieve those performance targets now in the current year?

A. (Thomas) This panel's proposal was aware through our testimony that it was filed last year. So since then, the company was aware of those requirements. So it should have been planned a

little earlier, not just now onwards. We are already into the second month of the year.

Q. So is it the panel's position that the companies should start -- should be planning to meet your proposed performance measures before they are finalized in an order, if they are finalized in an order?

A. Knowing what was proposed in our testimony, yes.

Q. Let me ask it a different way. Would you agree that setting more challenging performance targets or -- or finalizing more challenging performance targets in the middle of a year would put the companies at a significant disadvantage in terms of achieving those targets?

MR. FORST: Objection. Can counsel clarify what he means by significant challenge?

MR. DECICCO: Does the panel understand what a significant challenge means?

A.L.J. LEARY: Let's just see here.

MR. DECICCO: I mean, these are just words, English words.

A.L.J. LEARY: Yeah. No, it's okay. So let's -- let's get -- what I struggle with is the

middle of the year. Are you suggesting the middle of the year would be the point in time when the commission is anticipated to act in the litigated case?

MR. DECICCO: Co -- correct, Your Honor. And -- and the question I'm asking is if the targets are in fact finalized in June or July or later, we don't know at this point, whether the company would be disadvantaged in trying to achieve a calendar year performance target that's set seven or more months into the year.

A.L.J. LEARY: I think the -- I think the panel can answer that question.

BY MR. DECICCO: (Cont.)

A. (Pasnow) Can you repeat the question one more time?

Q. Sure. So does the panel agree that setting more challenging performance targets in the middle of the year -- potentially June or July when -- when an order is finalized in this proceeding -- would put the companies at a disadvantage in terms of achieving those new targets?

A. (Thomas) Can you tell us about the challenges you are talking about?

Q. Sure. So to the extent that -- that -- that a performance target -- let's use the example of -- of leak prone pipe. To the extent that the company is being asked to deliver more leak prone pipe miles, and it finds out about that definitively in June or July, would the panel agree that it is now a challenge to try to make up that target because it -- it wasn't final until the middle of the year?

A. The example you gave, I don't think it will be challenging because that's what the company proposed in your testimony.

Q. So what about with respect to -- to leak rates or leak backlog? If the company was to be ordered to -- to reduce its backlog to a certain level, and that was only set in June or July, would you agree that the company would be at a disadvantage having not known what the target was until the middle of the year?

A. Regarding the example you just mentioned about the leak backlog, knowing the company's performance as of the year end, I do not think it is such a big challenge for the company to attain those targets that are proposed by staff.

Q. Why don't we just turn to some of

the specific metrics? Beginning on page 11, the staff panel discusses a leak prone pipe metric. In the company's current rate settlement, the parties agreed to both positive and negative revenue adjustments associated with the removal of leak prone pipe from the company's system. Is that correct?

A. (Pasnow) That was for the current metrics, correct?

Q. Correct.

A. Then yes, there were PRAs and NRAs.

A.L.J. LEARY: So it's important that we not use acronyms. So state what -- I believe you're referring to a positive revenue adjustment and a negative revenue adjustment. Just use those words in your testimony so the record is more clear.

MR. DECICCO: (Cont.)

A. Yes, both positive and negative revenue adjustments.

Q. And the companies have met the minimum removal targets in each year and earned positive incentives for KEDNY in 2017 and for KEDNY and KEDLI in 2018. Is that correct?

MR. GOODRICH: I'm sorry. Could you

just repeat that?

BY MR. DECICCO: (Cont.)

Q. Sure. And I -- I'm -- I would turn your attention to page 18 of your testimony. And the question again is the companies have met the minimum removal targets in each year of the current rate plan and earned positive incentives for KEDNY in 2017 and for KEDNY and KEDLI in 2018. Is that correct?

A. That's correct.

Q. In this case, the panel supports a negative revenue adjustment only. Is that correct?

A. That's correct.

Q. And the panel also supports increasing the potential negative revenue adjustment from the current eight basis points to 15 points – 15 basis points. Is that correct?

A. (Pasnow) As it was proposed by the company, yes.

Q. In supporting that position in terms of the increased risk for basis points, is the panel aware that the gas – excuse me. Is the panel aware that staff's gas infrastructure and operations panel is proposing a $14 million reduction to the

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

company's forecast for its main replacement program?

A. We're now aware of the exact dollar amount, but we are aware of it.

Q. With regard to the positive metric, does the panel agree that positive incentives are good tools for driving performance outcomes?

A. Yes.

Q. And in fact, the panel supports new positive incentives in both the damage prevention and emergency response categories in this case. Is that correct?

A. And leak management, yes.

Q. And the members of this – and members of this panel, I should say, have supported a positive incentive for leak prone pipe in the company's prior rate case. Is that correct?

A. That is correct.

Q. Okay. Has the commission set a target for removing leak prone pipe in New York State?

A. Can you repeat the question, please?

Q. Sure. And I'm just asking you if the panel is aware whether the commission has set a

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

target for removing leak prone pipe in New York State.

A. Is the target a timeframe or a specific number of miles?

Q. Either one, if -- if you're aware.

A. (Thomas) We don't know the exact timeframe, but there was a prior expectation to commission order -- order of commission proceeding that the leak prone pipe in the state need be removed in a 20-year kind of timeframe. Okay.

Q. Thank you. And starting on page 14 of your testimony, the panel spends a fair amount of time describing why it's important to -- to remove leak prone pipe, including safety considerations and -- and leak reductions, and even to some extent cost reductions. Is that correct?

A. (Pasnow) That's correct.

Q. Okay. And -- and you also testified that the companies have two of the larger leak prone pipe inventories, in terms of miles, in New York State. Is that correct?

A. Correct.

Q. So for all these reasons, does

the panel agree that there are significant custom -- customer benefits to removing leak prone pipe from the company's systems?

A. Yes.

Q. And would that panel agree that a positive incentive for leak prone pipe removal would help drive performance in terms of the company's leak prone pipe performance?

A. Yes. And I believe the staff policy panel did address a positive revenue adjustment associated with leak prone pipe.

Q. Let's -- let's turn to leak management. In the company's current rate settlement, the parties agreed to both negative and positive revenue adjustments associated with leak management. Is that correct?

A. That's correct.

Q. And -- and there are two components to the metric. There is a total leak metric and a workable or repairable leak metric. Is that correct?

A. That's correct.

Q. Could you just, for the benefit of the -- the folks that don't deal with this every

day, just explain generally what the total leak metric and what the workable or repairable metric measures?

A. The commission's regulations classify leaks as either type 1, 2A, 2, or 3. Typically, types 1, 2A, and 2 are considered repairable. They have monitoring requirements on a periodic basis. They're required to be repaired on a certain basis. All differing -- differing based on the severity of the leak. Type 3 leaks are typically considered to be non-hazardous. So when we're discussing the leak management metric, repairable are the first three, and which are type 1, 2A, and 2. And total would also include those three plus type 3 as well.

Q. Type 3, just for clarity. Type 3 leaks are also called non-hazardous leaks. Is that correct?

A. Typically referred to as non-hazardous, yes.

Q. Thank you. And in, under the current rate plan, the companies will incur a negative rate adjustment for failing to reduce their backlog of total leaks against set targets as well as

workable or repairable leaks. Is that correct?

A. That is correct.

Q. And they also have the ability to earn an incentive or a positive revenue adjustment for hitting certain targets in total leak reduction. Is that correct?

A. Can you say that again?

Q. Sure. And -- and the companies also have the ability to earn an incentive or a positive revenue adjustment for eliminating a certain number of leaks each year from their backlog.

A. I believe the metric was set up for the reduction of the highest emitting leaks, which would essentially be the same. It wouldn't be additional leaks, just to clarify.

Q. Okay. And now in this case, the panel supports the leak management metric, but proposes different baselines or starting points for the total leaks, and more stringent targets for reparable leaks. Is that correct?

A. That is correct.

Q. Could you explain the panel's position with regards to the total leak metric on the different baseline?

A. On page 28 of our testimony, line 13, in the response to IRDPS114, KEDNY and KEDLI explain that they have averaged an approximate 316 and 823 reductions in the respective total leak backlogs of each of the previous five calendar years. In 2018, KEDNY and KEDLI reported that their total leak backlogs were 2,803 and 8,199 respectively. At the average reduction rates and in accounting for the additional calendar year of 2019, the respective total leak management targets or the baselines, as you were referring to them as, we recommend in our panel should be reduced in calendar year 2020 to 2,200 and 6,600.

Q. So, for the total leak metrics, the panel's proposing to lower the baseline or the starting point by 900 leaks and 1,150 leaks at KEDNY and KEDLI respectively. Is that correct?

A. Can you say those numbers again?

Q. Sure, 900 leaks at KEDNY and 1,150 at KEDLI.

A. In accounting for the additional year, yes.

Q. Okay. And for the workable or repairable target, the panel proposes going from a --

a performance target of 25 leaks in the backlog to 20 leaks in the backlog at year end. Is that correct?

A. That is correct.

Q. The panel also supports increasing the negative revenue adjustment from a total potential of 12 basis points to 15 basis points. Is that correct?

A. As it was proposed by the company, yes.

Q. Did the commission's regulations specifically address the total number of leaks in a backlog? I mean, is there any specific target in the regulations in terms of either total leaks or workable leaks?

A. In terms of maintaining a backlog, no.

Q. Do type 3 leaks, or non-hazardous leaks, as we said need to be repaired under the commission's regulations?

A. Repaired? No. However monitored in additional inspection activities, yes.

Q. Okay. Given the more challenging targets that the company is setting – excuse me, that the panel is setting – as prepared to the company's

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas proposal. Does the panel propose to allow the companies to recover any additional costs to eliminate leaks on its system?

A. Can you explain what you mean by more challenging?

Q. Sure. So I think in the example of repairable leaks or workable leaks, the panel is proposing to go from a target of 25 to a target of 20 leaks a year at backlog. Would the panel agree that that is a more challenging target?

A. In general, yes. However, based on the company's current performance which is beating both of those targets or exceeding both of those targets, it would be reasonable to set that target where it is.

Q. Let me ask. Would the panel agree that the number of leaks received, for instance, can vary year over year depending on various factors such as weather?

A. Among others, yes.

Q. Okay. I am just asking a question. The panel is not proposing that the companies be allowed to recover additional money in order to achieve these targets.

A. Can you repeat that one more time?

Q. Sure. I'm just asking a simple question. The panel is not proposing that the companies be allowed to recover incremental money to -- to meet these proposed performance targets.

A. It is not in our testimony, no.

Q. The panel also recommends on page 32 of its testimony that to successfully eliminate a type 3 or non-hazardous leak, for purposes of this metric. The company should be required to perform an inspection or a recheck of that leak. Is that correct?

A. Yes.

Q. And -- and could the panel explain what a -- a recheck is?

A. On page 32 starting with line nine, pursuant to 16NYCRR section 255.819A, upon completing a type 1, 2A, or 2 leak repair KEDNY or KENLI must conduct a follow-up inspection between 14 and 30 days following the repair date to validate said repair.

Q. And the -- the panel acknowledges those additional follow-up inspections are not

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

required to close out a type three leak under the commission's regulations. Is that correct?

A. That is correct.

Q. Okay. Has the panel done any assessment of the additional work that would be required to perform rechecks on the company's type three leak repairs?

A. Type three leaks that the company's repairing are using dollars allocated for leak repairs -- the total number of leaks that are allocated in place. On top of that, the company's allowed adjustments, if they exceed those targets set in this regulation. From the safety standpoint, we are to make sure that any leaks repaired are solid repairs. Those repairs are not coming back as another leak in the future. For that reason, for all those reasons, those leaks type three leaks repair need to be rechecked.

Q. And I'm not disputing the reason as to why this panel thinks it's a good idea. I'm just asking whether the panel has considered the incremental work associated with doing additional rechecks or inspections of type 3 leaks.

A. We do understand there will be

additional work to go and do the rechecks.

Q. Does the panel support additional funding to perform these additional inspections or these additional rechecks?

A. (Thomas) Can you repeat that part of the question again?

Q. Sorry, yes. Sure. The question is whether the panel supports additional funding to perform the incremental inspections that you just discussed.

MR. GOODRICH: So, I'm going to object as this is sort of speculative. Can-can company counsel point to any specific proposal for specific costs that it -- it would seek recovery of?

MR. DECICCO: With respect, the panel just testified that this would involve additional work. And I'm asking whether or not the panel would also then support recovery of the cost to perform that additional work.

MR. GOODRICH: But I'm asking if the company has actually proposed any additional work with any specific additional costs associated with it.

MR. DECICCO: Okay. It's not our

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas proposal. It's the panel's proposal. That's why I'm asking.

A.L.J. LEARY: No. I think Mr. Goodrich is asking whether there's anything quantitative or qualitative that the company has presented to either this panel or DPS staff generally that would address the additional work needed to do these inspections, as well as what the additional funding requirement would be.

MR. DECICCO: Your Honor, again, I believe I asked the panel whether, again it's the staff's proposal, not the company's proposal. I'm asking whether, in their view, they've first considered if there was incremental work. I believe the testimony was that there was. And then whether they've considered the cost of that or would support recovery of the cost of that.

MR. GOODRICH: But -- but then you move onto would the panel support recovery of cost. And it's -- it's very general.

MR. DECICCO: Okay.

MR. GOODRICH: In this, you know, rate case we're dealing with what are specific costs. And what is the specific incremental work? Because we

have even, the panel said it generally yes, there would be more work. But it's not to say that some of it's not already being done.

MR. DECICCO: Yeah okay.

MR. GOODRICH: Like there's a whole review that would be required there.

MR. DECICCO: Yeah. And again, it's not the company's per panel. It's the staff's. So why don't I ask the question?

BY MR. DECICCO: (Cont.)

Q. Has the staff done any assessment of the incremental work that the companies would be required to do in order to perform these additional rechecks that it is proposing?

A. We have not performed an assessment for the associated costs with those additional inspections.

Q. Okay, thank you. Your Honors, I probably have another 45 minutes of questions. Perhaps this would be a logical place for us to break for lunch.

A.L.J. LEARY: I think that's a good idea. Is that okay? So let's take. It's not 12:20 by my watch. Let's go with 1:10 come back. We're off the

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas record.

A.L.J. COSTELLO: Record, and we have a -- a few additional appearances since this morning we made.

MS. WHEELOCK: Laurie Wheelock with the Public Utility Law Project.

MR. FUNG: Justin Fung, City of New York.

A.L.J. COSTELLO: Thank you. And I don't know -- Mr. Maloney, will you be dealing with the --

MR. MALONEY: Sure.

A.L.J. COSTELLO: --exhibit?

MR. MALONEY: Thank you, Your Honor. As -- as you know, the other day we had some discussion about a proposed exhibit from the Staff Finance Panel that was put in on redirect. It was, I believe, Exhibit 660. And the company and staff have discussed certain things, and I've passed around a document entitled, "Fitch Ratings: Fitch Affirms Brooklyn Union" --

A.L.J. LEARY: Excuse me.

MR. MALONEY: I'm sorry.

A.L.J. LEARY: 661. Hearing Exhibit

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

661.

MR. MALONEY: I'm sorry, Your Honor, 661. "Fitch Affirms Brooklyn Union and KeySpan Gas East IDRs at B -- Triple B Plus, Outlook Stable," dated 20th of February 2020. And the company has agreed to waive any further cross or discovery on the document that has been marked as Exhibit 661. And it's our understanding that no party has an objection to this being marked for identification and received into the record at such time as we do that. So --

A.L.J. COSTELLO: Okay, we'll mark this as Exhibit 673 for identification. And just to give them an opportunity, no objections? No. Hearing none, okay, we can --

MR. MALONEY: Thank you, Your Honors.

A.L.J. COSTELLO: Thank you. And we'll go back to the cross examination.

MR. DECICCO: Thank you, Your Honor. And -- and I will try to move through this a little quicker, because although getting cooler, it is still -- still pretty hot.

CROSS EXAMINATION

BY MR. DECICCO: (Cont.)

Q. So I -- I want to talk about the

-- the -- the damage prevention metric, which the Panel discusses on page 34 of its testimony. And as you explain, damage prevention metric measures the company's ability to minimize instances of damage to its system caused by excavation activities. Is that generally accurate?

A. (Pasnow) Generally, yes.

Q. Okay. And it's measured based on the instances of damage in different categories per 100 one call tickets. And a one call ticket is a notification to the utility of excavation work in their territory. Is that correct?

A. Per 1,000 one call tickets, but yes.

Q. Okay. Oh, thank you. And I just want to -- and this cap -- in this issue, just focus on one concept. And that is, what is the Panel's understanding of what's meant by a "refresh" -- or I believe the company uses the term "relocate" -- one call ticket?

A. Bear with me a second. In one of our exhibits, labeled SPSP-8, I believe it is marked as Exhibit 488, if I'm not mistaken, within that exhibit, retransmits or refreshes are defined as any

one call ticket which has the same requesting party and location of the proposed scope of work. And retransmits or refreshes should be excluded from the total performance measure.

Q. Okay. For each refresh or relocate ticket, to your knowledge, are the companies required to respond to that ticket, and go out in the field, and then remark their facilities?

A. They are, yes.

Q. Okay. So in terms of the company's response, it is effectively the same activities that are required as opposed to receiving an original one call ticket?

A. Generally, yes.

Q. Okay. Does the Panel agree that the damage prevention performance targets in the current joint proposal adopted by the commission in the 2016 rate cases includes refresh or relocate tickets in the total ticket count used to calculate the performance measures?

A. Answering your question in two parts. The first is a -- as it's being reported or as it was being reporting by the companies, yes, they were being included. However -- and the second part

to the answer is, the original intent was for them not to be included. This difference was brought to light -- and we mention this on our -- in our testimony on page 38 -- regarding an operational audit that focused on the accuracy and effectiveness of the data submitted by several of the New York State local distribution companies. And this was brought to light as a result of that audit.

Q. And -- and I appreciate that response, and I -- that's helpful. I -- I guess, just to the question though. Is -- was -- was the targets that were set in the 2016 cases, the current performance metrics, did the -- did the baseline of tickets include those relocate or -- or refresh tickets?

A. Yes, for those targets, yes.

Q. Okay. Does the Panel know what percentage of the company's mark out -- the company's mark out tickets are relocates?

A. We do not have that data currently available to us.

Q. Okay. Now I -- I'd ask you, the company's testified at page 38 of its rebuttal testimony, line nine, "that analysis of statewide

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

date -- data shows that relocates make up more than 50% of the down state ticket volumes." Does the Panel have any reason to dispute that relocates make up more than 50% of the down state ticket volumes?

A. Approximately speaking, yes. It's -- it's roughly in that ballpark.

Q. And does the Panel -- well, let me ask this. Is the Panel, in this case, recommending to remove those relocate or refresh tickets from the company's performance metrics?

A. We are, yes.

Q. Okay. And does the Panel agree that adjusting the ticket count to exclude 50% of the ticket volume e -- e -- effectively 50% from the denominator would necessitate adjustments to the performance targets?

A. Can you repeat the question, please?

Q. Sure. And -- and maybe I can say it a little clearer. Would you agree that removing 60% of the ticket count from the denominator used to calculate the company's performance, would you agree that that would require an adjustment to the performance targets used to measure the damage

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

prevention performance?

A. Yes.

Q. Okay. And has the company, excuse me, has the Panel in this case proposed to adjust the targets based on that new ticket count?

A. It has.

Q. Okay. I'd like to show you -- and I'm going to refer to the 2018 Pipeline Safety Performance Measures Report that was produced in case 19-G-0298, dated July 13, 2009. Is -- is this document referred to in your testimony at all?

A. It is.

Q. Okay. And is this is a document that -- that -- that staff produces?

A. It is.

Q. Okay.

MR. DECICCO: And Your Honors, I don't intend to mark this, because this is a public document. But I have copies for anyone that's interested in seeing it.

A.L.J. LEARY: Can you provide one to us or two to us?

MR. DECICCO: (Cont.)

Q. Is -- is it fair to say Panel,

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

this is the -- the annual report that we discussed previously in your testimony where the staff gathers and analyzes gas safety data for -- for New York gas companies?

A. (Cont.) Yes.

Q. Okay. Looking at Appendix B, page 37 of the report, the top table on this -- I'll wait for you to get there.

A. Go ahead.

Q. Yeah. In the top table entitled, "Number of One Call Tickets," if you look at -- at the far right column, 2018, and compare that to the column 2017 for grid Long Island, which is KEDLY, and grid New York, which is KEDNI, is it fair to say that the reported ticket volume for KEDLY dropped from 185,313 to 126,872, or approximately 60,000 tickets? Is that correct?

A. When comparing those two years, yes.

Q. Okay. And at KEDNI, the ticket count dropped from 283,474 in 2017 to 128,359 in 2018, or more than fif -- 155,000 tickets. Is that accurate?

A. That's accurate.

Q. Okay. And is that a result of -- of the company or the reporting removing the refresh or relocate tickets from the ticket count?

A. It is.

Q. Okay. And just to sort of show the impact of that, I'd ask that you look at page 38 of the report. And -- and -- and the tables here are discussing or -- or -- or showing what's called damages due to, on the top, no calls, which is effectively, as I understand it, damage that results from the fact that the excavator did not call the company or request a mark out. Is that correct?

A. That is correct.

Q. Okay. So looking at the first table, which is sort of the raw number of damages due to no calls, NG Long Island, which is KEDLI, when from 124 in 2017 to 111 in 2018. Is that correct?

A. Yes.

Q. And N Grid New York, which is KEDNY, went from 47 to 41. Is that correct?

A. That is correct.

Q. But when looking at the table below, the damage prevention, although the company's performance actually improved in terms of the raw

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

numbers, is it fair to say that the reported performance in terms of statistically or the ratio actually deteriorated significantly?

MR. GOODRICH: I object to characterizing -- the characterization as significant. I mean, the numbers can speak for themselves. And you can ask what the numbers are, but --

MR. DECICCO: (Cont.)

Q. So I'll ask you what the numbers are. What is the -- what does the performance say with regards to KEDLY in 2017 as opposed to 2018?

A. (Cont.) 0.67 for '17 for 2017, and 0.87 for 2018 for KEDLY.

Q. And would those -- those numbers suggest that -- that KEDLY actually -- that their -- that its performance deteriorated in 2018, not withstanding the fact that the number of -- of -- of instances of damage were actually lower?

A. It was suggest that, yes.

Q. And for KEDNI -- I -- I don't want you to read them. It's -- would you agree it went from 0.17 to 0.32 between 2017 and 208?

A. They did.

Q. And would that suggest that KEDNI's performance was actually twice as bad as it was in 2018 as it was in 2017, roughly?

A. Approximately, yes.

Q. Okay.

A. And -- and I ask you these questions to make the point that the reduction in the ticket volume will significantly impact the way in which the company's performance is reported. Is that correct?

A. It will change it, yes.

Q. Okay. But in this example, even though both companies' performance actually improved in terms of raw numbers, the reported ratio was -- was significantly worse. Is that -- is that correct? Again, I use the term "significantly" -- was worse.

A. Yes.

Q. Okay. And so before when you said, "the Panel has adjusted its targets or its proposed targets based on the phenomenon of -- of removing the relocate tickets," have you actually done an analysis to determine what these numbers should be based on the fact that the -- that the total ticket volume has gone down substantially?

A. Can you repeat the question again?

Q. Sure. In setting your targets in -- in this proceeding, has the Panel considered the fact that the relocate tickets will not be reported going forward and adjusted its targets accordingly?

A. We've considered that, yes.

Q. And as a result, have you lowered the targets at both companies to reflect the fact that there's 50% less tickets in the ticket count?

A. I'm getting hung up on the word "lowered."

Q. Sure.

A. Technically, they're erased, because the performance would be higher. But yes, to answer that question.

Q. Okay. So moving on to the l -- last of the metrics, the gas regulations performance metric. This metric measures instances of non-compliance identified during staff's field and records audits. Is that correct?

A. That's correct.

Q. Okay. Among the changes proposed to the gas regulations performance metrics, the Panel

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas recommends eliminating a 10 occurrence cap per code section for violations identified during field audits. Is that correct?

A. That's correct.

Q. Okay. Does the current performance measure, the current metric for gas regulation performance, does that include a 10 occurrence cap per code section for field audits?

A. The current metrics include a 10 cap, yes.

Q. Okay. And does the Panel understand what -- what that cap is intended to -- to do?

A. Yes.

Q. Okay. And can you explain what your understanding of the purpose of the cap is?

A. Yes. The -- the cap itself would serve -- just trying to find -- in our testimony, we have it written out. So I'm just trying to find that right now. Essentially, the first 10 violations found of a particular code section would count toward this metric. Any violations of that particular code section in excess of 10 could be considered within a remediation plan, because we consider more than 10

gross non-compliance with the regulatory requirements. Therefore, additional action by the companies would need to be taken.

Q. Yeah. And -- and -- and -- and feel free to disagree. But is -- is the cap intended to avoid the situation where one instance with multiple occurrence sort of creates a -- a -- an unfair, an unjust result in the sense that it could be many, many violations for effectively issue. Is that -- is that the intent of the cap?

A. Can you state the question again, please?

Q. Sure. And -- and I was just -- I was just asking if the intent of the cap was to avoid the situation where one issue creates multiple violations and -- and would result in a significant penalty, and -- and -- and that would be effectively, you know, unfair.

MR. GOODRICH: Objection. I -- unfair is very subjective.

MR. DECICCO: Fair enough, yeah.

BY MR. DECICCO: (Cont.)

Q. Is it your understanding that cap is intended to avoid the situation where one issue

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas creates multiple violations that are subject to a penalty or more than 10 violations that are subject to a penalty?

A. (Cont.) I believe the intent of the cap was to allow the companies to take a proactive look at a particular regulation. Yes, to potentially avoid penalties in the sense of over penalizing the companies where a remediation plan could be a better course of action.

Q. Okay. Thank you. And -- and does the Panel agree generally that the company's performance in this area has improved over the last several years?

MR. GOODRICH: Sorry, could, excuse me, could counsel please define in this area?

MR. DECICCO: Yeah. So in the metric that were talking about, the -- the gas regulations performance metric.

A.L.J. LEARY: Just let me -- are you -- when you say the last several years, are you comparing the last three-year rate plan that was subject to a joint proposal or going beyond -- before that?

MR. DECICCO: Yeah, I think over the

term of the current rate plan would be a fair period of time to make that --

A.L.J. LEARY: Okay.

MR. DECICCO: -- observation.

BY MR. DECICCO: (Cont.)

A. (Pasnow) I'm getting hung up on the word "improved." Because it -- it's the Panel's stance that any number of violations against the -- the minimum safety requirements of the commission's regulations, that there shouldn't be any.

Q. Fair -- fair enough.

A. So to say "improved" is -- is difficult for me to get to that concept in my head.

Q. Sure. Well, according to you testimony on page 60, "staff cited KEDLI for a total of one violation in 2017," which I believe is lowest in the state. Is that correct?

A. I don't have that information available to me to answer that question.

Q. Okay. Would you agree that one violation in 2017 is better than KEDLI's performance in -- in prior years?

A. It was less, yes.

Q. Okay. Given the company's

performance, is there any reason to believe that eliminating a cap would drive further performance improvements?

A. Can you restate the question one more time, please?

Q. Sure. And this is my last question on metrics, if it makes you feel any better. I said, given -- given the company's recent performance, is there any reason to believe that eliminating the cap will drive further performance improvements?

A. It may.

Q. The Panel agrees that it may. Okay. Let's move on to the adjustments or the proposed adjustments to the companies' gas safety programs. And I'd like to first start -- talk about the inside service line inspection program. Beginning on line -- beginning, excuse me, on page 73 of the Panel's testimony. And in that testimony, the Panel discusses the company's obligation to inspect inside piping at premises with inside meters after the commission adopted a new definition of a gas service line. Is that correct?

A. That's correct.

Q. Does the Panel agree that operating in a heavily urban environment, like New York City, is more challenging in terms of performing inside service inspections than in -- in less urban areas of -- of the state?

A. Not necessarily. With the equipment having to be readily accessible, it -- it -- the expectation should be that it should be accessed or -- or assessed either in an urban or a non-urban environment.

Q. Okay, would the Panel agree three are -- are typically more inside meters in urban environments than in less urban areas of the state?

A. Generally, yes.

Q. Okay. And does the Panel agree generally that accessing facilities in the type of multiunit buildings you'll find in urban environments can present challenges in terms of -- of gaining access?

A. It can present challenges. However, they're supposed to be readily accessible.

Q. Okay. Is -- is the Panel's view that a meter located inside the house is not readily accessible in every case or inside the premise, I

should say, is not readily accessible in every case?

A. On page 84 of our testimony, we -- we -- we referenced the specific requirement of 16NYCRR part two 55 535A, "All meters and service regulators are required to be installed in a readily accessible location and be protected from corrosion and other damage, including any vehicle -- vehicular damage that may be anticipated."

Q. And -- and I guess the question is, is it the Panel's view that a meter located, for example, in the basement of a premise is -- is readily accessible or -- or -- or is not readily accessible I guess is a better way of -- of asking the question.

A. As compared to what?

Q. Being outside.

A. Generally speaking, yes.

Q. So it's the Panel's view that every inside meter is not considered readily accessible under this regulation?

MR. GOODRICH: Objection. I don't think -- I mean, I think your prior question asked about comparatively more or less readily accessible. And then you jumped to a categorical.

MR. DECICCO: Well, in fairness, I think it got answered comparatively.

BY MR. DECICCO: (Cont.)

Q. But my question was whether every inside meter under your -- under the Panel's view would be considered not readily accessible.

A. (Cont.) As we previously stated, it is our expectation that the equipment is readily accessible.

Q. I -- I'm going to move on. On page 75 of the Panel's testimony, you recommend the companies automatically apply a financial penalty when customers do not provide access to -- to the companies' facilities. Is that correct?

A. Can you repeat the question one more time? I'm sorry.

Q. Sure. On page 75 of the Panel's testimony, you recommend that the companies automatically apply a financial penalty when a custom -- when customers do not provide access. Is that correct?

A. Yes.

Q. Okay. Does the Panel agree this approach could result in additional customer

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

complaints and negatively impact the company's customer satisfaction ratings?

A. I'm going to break that question into two parts. The first is the complaints. The companies already have an outreach program that should be geared towards making the customers aware of this provision. In terms of the second part of this question, the satisfaction reports, or -- is that what it is? We can't speak to those reports, because I -- I have not performed an analysis of those reports. So I don't know whether it would negatively affect those or not.

Q. Okay. Moving onto New York City local law 52, which is discussed on page 76 of the panel's testimony, one – excuse me. New York City local law 152. And as the panel discusses, this law requires periodic inspections of gas facilities located inside buildings. And it applies to – to building owners. Is that correct?

A. That is.

Q. Does the panel agree that local law 152, as currently proposed, would not satisfy the commission's requirements for service line inspections?

A. According to our response in NG-74, I'm not sure whether that's on the record or not. Staff by blind safety panel agrees that local law 152 inspections would not satisfy the company's obligations to inspect all gas service lines per its recently adopted change in definition.

Q. Okay. Does the panel know whether there are any forthcoming changes to local law 152 that would impact whether these – these inspections satisfy service line inspections?

A. We do not.

Q. Okay. So for that reason, I – I would ask. Does the panel agree that there is really no basis for setting the company's rate allowance for service line inspections as a placeholder as proposed on page 81 of your testimony?

A. That is correct.

Q. I believe we discussed outside meters, so I'm going to move on. With regards to the enhanced contractor inspect -- inspection program on page 89, the panel testifies that it generally supports the company's enhanced contractor inspect -- inspection program, excuse me, or proposes a 5% reduction to the number of proposed FTEs. Is that

correct?

A. That's correct.

Q. Okay. And could you explain the basis for your proposed 5% reduction to the number of FTEs?

A. Yes. In the company's response to IRDPS-6767, the company stated that there may be occasions where the workload is light. That the supervisors and inspectors may perform functions outside of their normal duties. And that the company's estimated approximately 5% -- I'm being told that was a wrong cite to IR. It was 676. My apologies.

Q. No worries. And to be clear, that that is included in SPSP1, which has been marked as Exhibit 481 for identification.

A. Yeah. And just to complete the sentence, the company's estimated that approximately 5% of the inspector's time would be dedicated to other duties.

Q. And is staff taking the -- the position that that 5% you referenced, that other work to be performed, would be sort of imprudently incurred or otherwise not benefitting customers?

A.L.J. LEARY: Let's -- let's stay with the ban on imprudent --.

MR. DECICCO: Yeah okay. I'll -- I'll rephrase. I'll rephrase the question -- rephrase the question.

BY MR. DECICCO: (Cont.)

Q. Is staff taking the position that that other work – the 5% that's proposing to be disallowed -- would not be benefitting customers?

A. Not for the particular purpose of as it was proposed by the companies.

Q. But would you agree that that – that 5% of the time that – that the company identified could be used to be doing other productive work that would be benefitting customers?

MR. GOODRICH: Objection. It's the company's burden to demonstrate that -- that its FTEs are doing -- doing business work. Doing – doing properly recoverable work. If they've said the 5% not going to be used for this purpose, then it's up to the company to demonstrate what it is. Not for staff to make assumptions as to what it could be.

MR. DECICCO: Whoa. Let me ask this question this way. Is it the panel's understanding

that that 5% of the time would be spent doing something that did not benefit customers?

A.L.J. LEARY: Hold on a second. Is there anything in the record that shows this 5% and what they're doing so that we can -- the panel can evaluate, as could we, whether it benefits customers or not?

MR. DECICCO: Your Honor, the panel's proposing a disallowance of those costs. I just wanted to understand what the basis of their -- of their disallowance is. I believe the testimony is that they're relying on an IR. But we said that they would be doing other work. And -- and I don't -- didn't know that doing other work was necessarily a reason for disallowance. But I wanted the panel's understanding on -- on that.

A.L.J. LEARY: Well, it's -- the problem I'm having is that other work in DPS676 is not identified or defined in order for the panel to say, oh yeah, that that benefits customers or not. I mean they could be. I mean, I -- I -- I just personally don't know. So I'm -- I'm just having an issue. If you could bring something in from the record that would clarify that 5% and ask the panel

about that work, I just don't see it in DPS676.

MR. DECICCO: Your Honor, I'll move on to another -- another topic.

BY MR. DECICCO: (Cont.)

Q. With regards to -- to the issue of buried vent -- buried vent lines, excuse me, on page 91 the panel recommends disallowance for the company's proposal for a proactive program to remediate vent lines that have become buried. Is that correct?

A. It is.

Q. Okay. Does the panel agree that since the companies have been performing additional service -- service line inspections, there have been more cases to identify issues like buried vent lines in the service territory?

A. Can you repeat the question one more time, please?

Q. Sure. I said, the question was, I was asking if the panel agrees that since the companies have been performing additional inside service inspections there have been more occasions to identify conditions like buried vent lines.

A. The regulations contained in

16NYCRR part 255 455A, it states except as provided in subdivisions B, C, and F of this section, each buried or submerged pipeline installed after July 31, 1971 must be protected against external corrosion. And then it goes into additional requirements there. When asked in IRDPS-678, the company stated that regulator vent lines are installed at the time of the original service and/or service -- at the time of the original service and/or service regulator. The companies have not identified installation dates of the buried vent lines that have been identified through the inside service line inspections to date. So when you're asking the questions about them being identified currently through this newer program, it is the panel's stance that these should have already been identified by the companies.

Q. And I don't want to belabor this point, but I -- the question just being, do you agree that performing more service line inspections presents more opportunity to discover buried vent lines?

MR. THOMAS: I think over the years, through leak surveys working on the service line every single year or every three or five-year cycle,

the company was performing these inspections. It should have been identified a long time ago. Not by this new requirement of inside a service line inspection came into effect.

BY MR. DECICCO: (Cont.)

Q. Okay. Does the panel agree that a vent line can become buried through no fault of the company's?

A. (Pasnow) It -- it's possible. However, with the periodic inspections outside of what is being proposed for the inside service line inspections, these -- these situations should have had already been identified as well.

Q. And -- and would you agree that once identified, the company is obligated to remediate these conditions?

A. Correct.

Q. Okay. And this is the last topic, I promise, because it is getting hot again. So on page 87 of the panel's testimony, staff discusses the enhanced high emitter methane detection program. And is it the panel's view that -- that you generally support this program?

A. The panel's pretty clear on page

88 line 12. While this panel is supportive of the research and development of CR -- CRDS, which stands for Cavity Ring Down Spectroscopy like technologies and the use of a contractor for data acquisition, the panel is not supportive of a full-time equivalent engineer to manage the methane leak pilot program.

Q. So -- so just to sort of paraphrase that, the panel supports the program. It doesn't support the com -- what the company has requested in terms of -- of funding for the program. Is that fair?

MR. GOODRICH: Objection. I think he explained what staff's testimony was that they supported the program but did not support an FTE.

MR. DECICCO: Okay.

MR. GOODRICH: Sorry.

BY MR. DECICCO: (Cont.)

Q. Does the panel support the use of technology to identify high emitting leaks?

A. For the research and development of technologies such as those, yes.

Q. Okay. On page 89, the panel suggests starting on line 3, the methane leak pilot program should be incorporated into the company's

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas

research and development efforts. Is that correct?

A. That's correct.

Q. Did the panel review the company's research and development rate allowance to determine whether there's sufficient funding for this program?

A. Can you repeat the question, please?

Q. Sure. Did the panel review the company's research and development rate allowance to determine whether there was sufficient funding for this program?

A. No. Those rate expenses were not specifically reviewed.

Q. Okay. Thank you. That's my last question. I would like to enter in two exhibits into the record starting with the panel's response to NG47. That is the date of the response is November 22, 2019. Respondent is Mike Panz -- Mike Panzella (phonetic spelling) utility engineer, damage prevention. And the subject is damage prevention performance.

A.L.J. COSTELLO: We'll mark that as Exhibit 674 for identification.

MR. DECICCO: Thank you. Panel, was this information request prepared by you or under your supervision?

MR. PASNOW: It was. Oh, it was. Sorry.

BY DECICCO: (Cont.)

Q. And is this a true and accurate copy of the information request?

A. It appears to be, yes.

Q. Okay. And as we discussed before in this response, and I -- and I -- and I believe you discussed with Mr. Goodrich when you started testifying. Is this the response in which the panel clarifies its testimony to say that in fact the companies did not incur a negative revenue adjustment with respect to their damage prevention performance? I'm sorry, yes.

A. It was.

Q. Okay. And in Section D of this response, is this where the panel proposed new targets with respect to the company's damage prevention performance?

A. It is.

Q. And can you tell me, looking at

DI? It says for rate -- for rate -- for a rate per 1,000 one-call notifications greater or 2.50, the companies would incur negative revenue adjustments of 20 basis points. Do you know how generally these targets compare to the company's current performance targets?

A. Yes.

Q. Are they? We didn't want to use the word challenging before. But are they -- are they more difficult to achieve than the company's current performance targets?

A. With excluding refreshes and retransmit tickets, yes. They would be more difficult.

Q. Okay. So were -- were we to pull out the -- the approximately 50% before, we said 50% of relocated refresh tickets we discussed before, these targets would be more challenging for the company to achieve?

A. Say that again, please.

Q. Yeah. I -- I just repeated your answer. I'm going to -- I'm going to withdraw the question. The second and my last order of business here is infor -- information request NG49. I think

that is the one, right? Okay. And again, the response -- the date of the -- the date of the response is November 22, 2019. And I'd ask the panel. Was this exhibit prepared by you under your supervision?

A. Yes.

Q. And is this a true and accurate copy of the information request?

A. It is.

Q. Your Honors, I'd -- I'd ask for a hearing exhibit number for what has been marked as NG49.

A.L.J. COSTELLO: That will be Exhibit 675.

MR. DECICCO: Great, thank you. And I have nothing further for the pan -- panel. Thank you very much for your attention.

A.L.J. COSTELLO: Mr. Goodrich, do you need a moment?

MR. GOODRICH: We would like to take a few minutes. If we could have ten? And I understanding it is hot, and everyone wants to be finished.

A.L.J. COSTELLO: Ten minutes?

MR. GOODRICH: Sure.

A.L.J. COSTELLO: It's -- it's 2:01 now. So…

A.L.J. LEARY: Record? DPS staff counsel, do you have further questions on redirect?

MR. DECICCO: We have no redirect, Your Honor.

A.L.J. LEARY: All right. We're going to go off the record. Oh, I'm sorry. Thank you very much to the panel. And you are excused. I have a couple things. This is off the record now.

19-G-0309/19-G-0310 - 2-24-20 - Brooklyn Union Gas STATE OF NEW YORK

I, JANET WALLRAVIN, do hereby certify that the foregoing was reported by me, in the cause, at the time and place, as stated in the caption hereto, at Page 1 hereof; that the foregoing typewritten transcription consisting of pages 3691 through 4197, is a true record of all proceedings had at the hearing.

IN WITNESS WHEREOF, I have hereunto subscribed my name, this the 2nd day of March, 2020.

JANET WALLRAVIN, Reporter